Introduction to Mathematical Probability

University of Chicago - STAT 25100

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Contents

Preface

These are lecture notes that I initially wrote when I taught STAT 25100: Introduction to Mathematical Probability at the University of Chicago in the fall of 2021. The main source of inspiration for the some of the content and style of this document is Ramon van Handel's lecture notes for the course ORF 309: Probability and Stochastic Systems at Princeton University.

I gratefully acknowledge Owen Fahey, Jeff Guo, Zewei Liao, Shailinder Mann, Christian Paz, Nicholas Zhang, and more anonymous students (through their comments on the Ed Discussion forum) for pointing out typos and errors in earlier versions of this document and the homework associated to the class. Please be warned, however, that since this document was not thoroughly reviewed, it is likely that typos and errors still remain.

Chapter 1

Introduction

1.1. What is Probability?

Before starting with the actual material of this course, we should take some time to introduce what mathematical probability is about in informal terms. There are two aspects of the title "Introduction to Mathematical Probability" that require an explanation, namely:

(1) What is probability?

(2) What makes certain probabilities mathematical?

We begin by answering the first question.

Many experiments in everyday life and science are what we could call deterministic. That is, experiments whose outcomes are always the same predictable result. A simple example of a deterministic experiment in physics would be as follows:

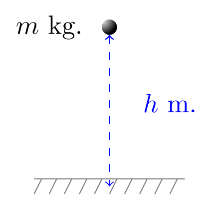

Figure 1.1. The deterministic experiment of dropping a ball with a known mass from a known height in a vacuum.

Example 1.1 (Dropping a ball)**.** Suppose that we drop a ball with a known mass (say, m kilograms) from a known height (say, h meters) in a complete vacuum (i.e., there is no friction, wind, etc. to take into account). If we measure the amount of time that it takes for the ball to reach the ground once it is dropped, then we will always get the same predictable result. Furthermore, using Newton's laws of motion, we can predict how long it will take for the ball to reach the ground.

In sharp contrast to this, probability is concerned with the study of experiments whose outcomes involve a significant degree of unpredictability. There are many different circumstances that give rise to unpredictable experiments, and there are many different phenomena that are at the origin of this uncertainty. To illustrate this, consider the following two examples:

Example 1.2 (Applying to college)**.** One example of a random experiment that you are all intimately familiar with is that of applying to college. The possible outcomes of this experiment are to be admitted, waitlisted, or rejected (at least when the results first come out).

Unless one is involved in some kind of college admission fraud scheme, it is not always possible to predict the outcome of applying to a given college with 100% accuracy. In order to pull this off, you would at least need to have access to the file of every applicant as well as the minds of the members of the selection committee, so as to predict what will be their assessment of your file and how that compares to the other applicants competing against you.

This is of course not practically possible. Moreover, even if someone did somehow obtain this information, it is not clear that they would be able to put it to much use given the complexity of the task. Consequently, it is more helpful to think about this experiment in terms of the likelihood of each possible outcome, rather than attempt to predict exactly what is going to happen.

Example 1.3 (Tossing dice)**.** A classical example of a random experiment is that of watching a dealer at a casino cast a six-sided die on a gambling table. In this case, the outcome is a number from 1 to 6, which represents the face pointing upwards after the die lands on the table and stops rolling.

At first glance, the experiment of tossing a die seems like it should be deterministic: In similar fashion to the act of dropping a ball in a vacuum, if we know the mass of the die, its instantaneous velocity, rotation, and position relative to the table once it leaves the dealer's hand, etc. (i.e., if we have enough information), then in principle we can use Newton's theory to predict exactly how it will land. However, there are good reasons why this is not how we analyze games of chance that involve dice.

Even in the best of circumstances, predicting the toss of a die with 100% accuracy will involve calculations that are not practical to carry out in the small amount of time between the die being cast and it falling on the table. Moreover, the toss of a die is an extremely chaotic process; that is, minute changes in the initial conditions of the die as it leaves the dealer's hands will typically result in drastically different outcomes. Measuring the initial condition of the die to the necessary accuracy in a fraction of a second is of course completely out of the question. Furthermore, if you place your bet before the die is even cast, then this whole discussion is moot, because the observation of how the die is tossed happens after any decision that would rely on the outcome of the experiment is made. In light of these circumstances, the only thing that we can practically do is conceive of this experiment as being random.

Some of you may find it curious that both examples that I have just provided are not inherently random. Indeed, one can easily argue that these are in fact deterministic experiments for which we simply lack the knowledge, tools, or competence to analyze. In both cases, we can imagine that an omniscient supercomputer might be able to always predict the outcome with 100% accuracy, provided it has enough information. In fact, the question of whether or not there actually exists experiments that are inherently random is something that philosophers and scientists debate about to this day. Here, by an inherently random experiment, we mean one such that even an entity that knows everything that could possibly be known about the present state of the universe and has infinite computational power would not be able to predict its outcome with 100% accuracy.

That said, regardless of whether the universe is ultimately completely deterministic or has some inherent randomness, the study of probability can easily be motivated by the following observation: Our competence in understanding the universe (as impressive as it is compared to what it was hundreds or thousands of years ago) is still very limited, at least when compared to the highly chaotic and complex nature of many problems facing us in everyday life and science. As a consequence, probabilistic thinking is used to great success in an impressive number of scientific disciplines. For instance:

Example 1.4 (Chemistry and physics)**.** Probability is fundamental to our understanding of modern chemistry and physics, both from the purely theoretical point of view (e.g., wave functions in quantum physics; understanding temperature and pressure in terms of random collisions of microscopic particles) and the modelling point of view (i.e., certain chemical and physical systems are modelled using randomness to simulate complex impurities or disorder that is not amenable to current computational tools).

Example 1.5 (Biology and medicine)**.** The human body is an extraordinarily complex system whose behavior can be influenced by a large variety of genetic and environmental factors. Thus, the human body's reaction to certain treatments or drugs is often best understood from the point of view of populations rather than individuals. In particular, when developing new treatments (such as vaccines), their safety and effectiveness is typically assessed in large clinical trials wherein it is assumed that a patient's reaction to the treatment is random; risks and benefits of experimental treatments are quantified using efficacy rates and side effect rates.

Example 1.6 (Economics and finance)**.** The price of commodities and the behavior of consumers at a large scale and across long periods of time are notoriously difficult to predict exactly. Hence random models are ubiquitous in economics, finance, risk management, and insurance.

We could go on with many more examples. Whatever is your favorite subject of inquiry, it is likely that randomness plays an important role in its understanding.

1.2. What is Mathematical Probability?

Now that we have a basic grasp of what probability is about and why it is important, we can ponder what is mathematical probability, as opposed to regular probability.

In short, the appellation “mathematical probability” serves to distinguish between the intuitive assessments of probability that we all carry out in our daily lives and the more quantitative nature of the probabilistic analyses that are required in science. To illustrate this, we once again use some examples:

Example 1.7 (Applying to college)**.** When trying to assess the probability of being admitted to a given college, most people will not carry out precise and technical computations. Instead, most rely on their intuition to get a general feeling of how likely an offer of admission might be. In practice, this often looks like this: For any given college, one can collect data that helps assess its competitiveness, such as admission rate, average SAT scores of admitted students, prestige and reputation, etc. Then, using this, one typically classifies colleges into vague categories, such as safety, match, and reach. The latter indicates a vague belief about the likelihood of each possible outcome.

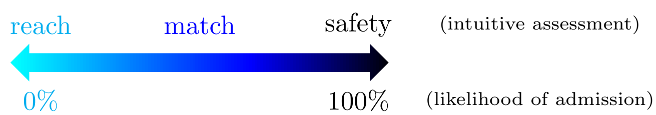

Figure 1.2. Non-quantitative and intuitive classification of the likelihood of of college admissions.

While this non-quantitative assessment seems appropriate to analyze experiments like applying to college, there are many situations where this is woefully inadequate. There are many situations where the stakes are too high for such handwaving, and when differentiating between a very good or very bad outcome relies on precise assessments of probability.

Example 1.8 (Safe and effective vaccine)**.** Suppose that you are part of a team that is developing a new vaccine. Before your vaccine can be approved for mass deployment, you need to make sure that it is both safe and effective. It is widely understood that this is assessed in clinical trials by giving the vaccine to a number of people, and then waiting to see what happens. That being said, what kinds of results, exactly, do we need to see before we decide that a vaccine is safe and effective? How many individuals, exactly, do we need to involve in a trial in order to be confident in our findings? For this, vague and non-quantitative intuitions about what does or does not feel right are not good enough.

In this context, what mathematical probability provides is a rigorous framework that helps us reason about uncertainty in a way that is structured, logical, and methodical. The ability to back-up our probability assessments with rigorous and quantitative analyses gives us the power to make more accurate predictions. A mathematical understanding of probability is, among other things, what allows people to do the following:

(1) Test new treatments for all kinds of diseases and figure out which treatments are actually the most effective.

(2) Design games in casinos that are very slightly biased in favor of the house (so that it consistently makes money in the long run), but that are still seemingly fair enough to attract gamblers.

(3) Design insurance policies that charge just enough money to be competitive, yet not promptly go out of business in the event that many people file claims simultaneously (this, among other things, relies on the ability to accurately predict the frequency of very rare but catastrophic events and their costs).

Therefore, in both academia and industry, a facility with the mathematics of probability has very clearly demonstrated itself to be highly valuable, and thus highly sought after.

Chapter 2

The Foundations of Mathematical Probability

In this chapter, our aim is to introduce the machinery that forms the basis of the mathematical theory of probability. Given that the entirety of the theory that we will build in this course relies on the notions introduced in this chapter, it is of crucial importance that you develop a good familiarity with the latter.

2.1. The Sample Space

As explained in the previous chapter, the outcome of a random experiment cannot be predicted with 100% accuracy. Instead, the best that we can do is assign probabilities to the possible outcomes of the experiment, which are quantities that reflect our degree of confidence that certain outcomes will occur. In order to be able to do this, we must of course know what are all the possible outcomes of the experiment:

Definition 2.1 (Sample space)**.** The sample space of a random experiment, which is typically denoted by Ω (i.e., the capital greek letter omega), is a set that contains every possible outcome of the experiment.

This definition suggests that in order to analyze random experiments mathematically at the most basic level (i.e., classify all of their possible outcomes), we need to spend some time developing a basic vocabulary with which to describe collections of objects. This is the first purpose of this section. After we have developed these basic tools, we will look at examples of how to use them to define the sample spaces of various random experiments.

2.1.1. Sets and n-Tuples.

Definition 2.2 (Set)**.** A set is an unordered collection of distinct elements.

There are many basic sets that you may have encountered before in your mathematical studies. For instance:

Example 2.3. It is customary to use $\mathbb{N}$ to denote the set of positive integers, $\mathbb{Z}$ to denote the set of all integers (positive, negative, and zero), $\mathbb{Q}$ to denote the set of rational numbers (i.e., all fractions of two integers), and $\mathbb{R}$ to denote the set of real numbers (i.e., rational and irrational numbers, such as π, e, $\sqrt{2}$, etc.)

More generally, we denote a set using curly brackets, namely, $\{$ and $\}$. For example, if A is the set of integers from one to six, then we can write

$$A = \{1, 2, 3, 4, 5, 6\}. \tag{2.1}$$

Although we have written A's elements in increasing order, this was only for cosmetic reasons and purely arbitrary. Since the elements of a set are unordered, it is equivalent to write the set in (2.1) as

$$A = \{3, 1, 5, 6, 4, 2\}.$$

Because a set is a collection of distinct elements, $\{1, 1, 2\}$ is not a set; the element 1 appears twice in $\{1, 1, 2\}$, which is not allowed by definition.

In the example in (2.1), we wrote the set A by exhaustively enumerating all of its elements. However, this method cannot be used if a set is extremely large or infinite. In some cases, we can get around this difficulty if there is an obvious pattern in the elements of the set. For example, if B is the set of integers from one to one million, then we can write

$$B = \{1, 2, 3, \ldots, 1\,000\,000\},$$

with the understanding that the pattern of incrementing integers by one in the sequence $1, 2, 3$ is continued all the way up to one million. More generally, the set of all positive integers (which is of course infinite) can similarly be written as

$$\mathbb{N} = \{1, 2, 3, \ldots\}.$$

In situations where no obvious pattern of enumeration of a set is available, we can instead write a set by specifying the properties that its elements satisfy. This leads us to the following:

Notation 2.4 (Set by property specification)**.** Suppose that C is the set containing all sets of two numbers from one to six (in other words, a set with two elements taken from the set A in (2.1)). Every element in C can be written as a set of the form $\{i, j\}$, where i and j are distinct integers from one to six. In formal mathematical language, we can write this as

$$C = \big\{\{i, j\} : i, j \in A\big\}. \tag{2.2}$$

In the above, the symbol : is understood as the mathematical equivalent of "such that," and the symbol $\in$ is understood as the mathematical equivalent of "is an element of." Thus, (2.2) is a way to write the english sentence "C is the set containing every set of the form $\{i, j\}$ such that both i and j are elements of the set A" in formal mathematical language. More generally, writing a set using property specification can be done as

$$\{\text{type of object} : \text{property that must be satisfied}\}.$$

Next, we discuss ordered collections:

Definition 2.5 (n-tuple)**.** Let $n \in \mathbb{N}$ be a positive integer. An n-tuple is a collection of n elements arranged in a particular order.

n-tuples are typically denoted with the usual parentheses, i.e., (and). An example of a 3-tuple is

$$T = (1, 2, 3).$$

Note that, since the order matters here, the above 3-tuple is not equal to $(2, 3, 1)$. Although the two 3-tuples contain the same elements, they are not in the same order. In general, we can write an n-tuple as

$$(1^{\text{st}} \text{ element}, \ 2^{\text{nd}} \text{ element}, \ 3^{\text{rd}} \text{ element}, \ \ldots \ , \ n^{\text{th}} \text{ element}).$$

Unlike a set, an n-tuple may contain the same element multiple times (e.g., $(1, 1, 2)$ is a 3-tuple distinct from $(1, 2)$).

Remark 2.6. Sets and n-tuples are not the only structures that are used in mathematics to describe collections of objects. For instance, we could introduce multisets,[1] which are defined as unordered collections of objects where repetitions are allowed, and thus interpolate between sets and n-tuples. In this course, we settle with sets and n-tuples since the latter are sufficient for our purposes.

2.1.2. A Few Examples of Sample Spaces. We now discuss a few examples of sample spaces that illustrate some of the subtleties that can be involved in specifying collections of outcomes.

Example 2.7 (Applying to college)**.** In the previous chapter, we discussed the random experiment of applying to college. The sample space of such an experiment could be

$$\Omega = \{a, r, w\},$$

where a stands for accepted, r stands for rejected, and w stands for waitlisted.

Example 2.8 (Draw a two-card hand)**.** Suppose that we have a standard 52-card deck of playing cards. In a standard deck, each card has a value from one to thirteen, and a suit from the four standard suits, namely, clubs ($\clubsuit$), diamonds ($\diamondsuit$), hearts ($\heartsuit$), and spades ($\spadesuit$). In formal mathematical language, we can write the set of all cards in a standard deck as

$$SD = \big\{(v, s) : v \in \{1, 2, 3, \ldots, 13\} \text{ and } s \in \{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\}\big\}; \tag{2.3}$$

in words, a card is an ordered 2-tuple (v, s) where the first element v is the value of the card and the second element s is its suit (e.g., the ace of spades is $(1, \spadesuit)$, the queen of hearts is $(12, \heartsuit)$, etc.).

Suppose that we perform the experiment of picking a two-card hand from the deck at random. Here, we assume that the only thing that matters is which cards we have in our hand; we do not care about the order in which the two cards were picked. Moreover, since every card in SD is unique, we cannot have the same card twice in our hand. These two requirements are the same as the rules that define a set. Thus, a good sample space for this experiment is the set of every set with two elements taken from SD:

$$\Omega = \big\{\{c_1, c_2\} : c_1, c_2 \in SD\big\}. \tag{2.4}$$

[1] See the Wikipedia page.

Note that, because the hand $\{c_1, c_2\}$ is a set, this automatically implies (by definition of a set) that the two cards c_1 and c_2 are distinct and that their order does not matter. Thus, we do not need to add these requirements as an extra condition in the definition of Ω.

Example 2.9 (Two cards in order)**.** Suppose that we perform the experiment of picking two distinct cards from a standard deck, and that this time we care about the order in which the two cards are drawn. In this case, we can define

$$\Omega = \big\{(c_1, c_2) : c_1, c_2 \in SD \text{ and } c_1 \neq c_2\big\}.$$

In this sample space, we must add the condition that $c_1 \neq c_2$, because in general 2-tuples are allowed to contain the same element twice. Moreover, unlike the previous example, the 2-tuples $\big((1,\spadesuit),(12,\heartsuit)\big)$ and $\big((12,\heartsuit),(1,\spadesuit)\big)$ are considered two distinct elements of Ω even though they amount to the same hand. $\big((1,\spadesuit),(12,\heartsuit)\big)$ is the outcome that the ace of spades is drawn first and the queen of hearts is drawn second; $\big((12,\heartsuit),(1,\spadesuit)\big)$ is the same hand drawn in the opposite order.

Example 2.10 (Two cards with replacement)**.** Suppose that we perform the experiment of drawing one card from a standard deck, then putting that card back into the deck, and finally picking a second card from the deck. Like the previous example, we assume here that the order is important. However, given that the first card is placed back into the deck before picking the second one, it is possible that we pick the same card twice. Thus, a good sample space in this scenario is

$$\Omega = \big\{(c_1, c_2) : c_1, c_2 \in SD\big\}.$$

Every example of sample space that we have seen so far has been finite. That being said, there is no reason why a sample space has to be finite. Here is an example of an infinite sample space:

Example 2.11 (Waiting for your grade)**.** You have just completed the final exam for one of your courses. We assume that the amount of time (measured in hours) until your final grade is posted online is random. In principle, your grade could be posted at any moment after you finish your exam (as instructors and professors do not always submit them before the deadline). Thus, we could define

$$\Omega = \{t \in \mathbb{R} : t \geq 0\},$$

that is, the set of all nonnegative real numbers. In this set, every nonnegative real number t represents the outcome that it takes exactly t hours for your final grade to be posted. We assume that the outcome of this experiment cannot be a negative number, as this would mean that your final grade was submitted before you even finish the exam.

2.2. Events

Now that we have introduced some tools that help us characterize the possible outcomes of random experiments, we develop the mathematical machinery that allows us to ask questions and make statements about random experiments. This leads us to the notion of events.

2.2.1. Definition and Examples of Events. Informally, an event is a question that can be answered unambiguously by yes or no once a random experiment is performed. The definition of an event in formal mathematical language is as follows:

Definition 2.12 (Event)**.** Consider a random experiment with sample space Ω. An event A for this random experiment is a subset of Ω. That is, A is a set such that every element in A is also an element of Ω. In general, we denote the fact that a set A is a subset of some other set B as $A \subset B$. In particular, if A is an event, then $A \subset \Omega$.

In order to understand how this mathematical definition captures the intuitive meaning of an event that I stated above, we look at a few examples:

Example 2.13 (Draw a two-card hand)**.** Suppose that you perform the random experiment of drawing two unordered and distinct cards from a standard deck (i.e., Example 2.8), which we recall has the sample space

$$\Omega = \big\{\{c_1, c_2\} : c_1, c_2 \in SD\big\}.$$

Once this is done, I could ask you the following:

$$\text{“\textit{Does your hand contain the ace of spaces?}”} \tag{2.5}$$

This is a question that can be answered unambiguously by yes or no once the experiment is performed; hence it is an event. Mathematically, we define this event as the subset $A \subset \Omega$ containing every two-card hand that have the ace of spades:

$$A = \Big\{\big\{(1, \spadesuit), c_2\big\} : c_2 \in SD\Big\}.$$

That is, A is the subset of Ω containing every outcome for which the answer to the question in (2.5) is yes.

Example 2.14 (Waiting for your grade)**.** Suppose that you perform the random experiment of waiting for your final grade to be posted after taking the final exam (i.e., Example 2.11). This has the sample space

$$\Omega = \{t \in \mathbb{R} : t \geq 0\}.$$

After this experiment is performed, I could ask you the following:

“Was your final grade posted within 72 hours of finishing the exam?”

Mathematically, we represent this event as the subset $A \subset \Omega$ containing every outcome for which the answer to this question would be yes, namely:

$$A = \{t \in \mathbb{R} : 0 \leq t \leq 72\}.$$

We end with some standard terminology:

Notation 2.15. We say that an event $A \subset \Omega$ has occurred if the outcome of the experiment is contained in the set A. Otherwise, we say that A has not occurred.

2.2.2. Unions, Intersections, and Complements. As it turns out, what makes events interesting is not just that they allow to formalize the notion of asking simple questions about outcomes of random experiments: Once we have translated basic questions about experiments into subsets of Ω, we can then use various notions in set theory to combine these basic questions into increasingly complex questions. For this, we need a few definitions:

Definition 2.16 (Intersection)**.** Let $A, B \subset \Omega$ be two events. We define the intersection of A and B, denoted $A \cap B$, as the set

$$A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}.$$

In words, this is the subset of Ω that contains the elements that are both in A and B. In the english language, the event $A \cap B$ can be translated as the question "did both A and B occur?"

This definition is a perfect opportunity to introduce a very useful tool in probability called the Venn diagram.Venn diagrams consist of simple visual representations of sets. For example, a Venn diagram of the intersection of two events would be Figure 2.1 below. Therein, we have represented the sample space Ω by a large

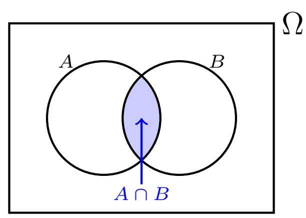

Figure 2.1. The intersection of two events A and B.

rectangle. In this picture, the space inside the rectangle serves as a graphical representation of the possible outcomes that the experiment can take. We can therefore think of events as being represented by subsets of that rectangle, such as the two circles labelled A and B in Figure 2.1. Then, we can represent the intersection of these two events as the region that is contained in both A and B, thus giving a very compelling geometric intuition for what $A \cap B$ represents. As you will gradually see throughout this chapter and later, Venn diagrams are very useful in probability, both as a means of understanding various definitions and results intuitively, and as a guide to formal computations.

Notation 2.17 (Multiple intersections)**.** It is possible to define the intersection of more than two sets. Given some events $A_1, A_2, A_3, \ldots$, we can define the intersection

$$A_1 \cap A_2 \cap A_3 \cap \cdots$$

as the set that contains the outcomes that are in A_1, and in A_2, and in A_3, and so on. That said, when we have a large (possibly even infinite) collection of events, we use the short hands

$$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n \qquad \text{and} \qquad \bigcap_{i \geq 1} A_i = A_1 \cap A_2 \cap A_3 \cap \cdots .$$

The next set-theoretic definition that we introduce is the union:

Definition 2.18 (Union)**.** Let $A, B \subset \Omega$ be two events. We define the union of A and B, denoted $A \cup B$, as the set

$$A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}.$$

In this definition, it should be clearly emphasized that the "or" is not mutually exclusive. Thus, in words, $A \cup B$ is the subset of Ω that contains the elements that are in A, the elements that are in B, and the elements that are in both A and B. In the english language, the event $A \cup B$ can be translated as the question "did at least one of A or B occur (or both)?" See Figure 2.2 below for a Venn diagram of the union.

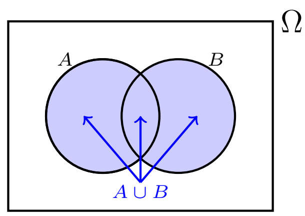

Figure 2.2. The union of two events A and B.

Notation 2.19 (Multiple unions)**.** In similar fashion to intersections, there exists a shorthand to express the union of a large number of sets. Given some events $A_1, A_2, A_3, \ldots$, we define the shorthand

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n \qquad \text{and} \qquad \bigcup_{i \geq 1} A_i = A_1 \cup A_2 \cup A_3 \cup \cdots.$$

The third and last set-theoretic notion that we introduce is as follows:

Definition 2.20 (Complement)**.** Let $A \subset \Omega$ be an event. The complement of A, denoted A^c, is the set

$$A^c = \{\omega \in \Omega : \omega \notin A\}.$$

In words, this is the subset of Ω that contains the elements that are not in A. In the english language, the event A^c can be translated as the question "did A not occur?" See Figure 2.3 below for a Venn diagram of the complement.

With the notions of intersection, union, and complement in hand, it is possible to express an impressive variety of complicated combinations of events. For example, consider the following:

Example 2.21 (Combination of three events)**.** Suppose that we have three events $A, B, C \subset \Omega$, and that we are interested in outcomes where at least one of A or B occurs, but not C. Mathematically, we can represent this using the intersection, union, and complement as the event

$$(A \cup B) \cap C^c.$$

See Figure 2.4 below for a Venn diagram of this combination of events.

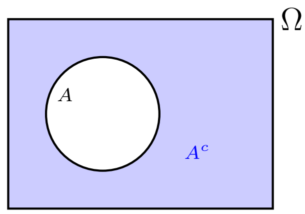

Figure 2.3. The complement of an event A.

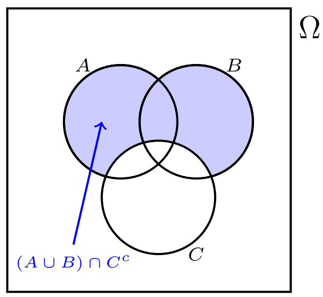

Figure 2.4. Venn diagram of $(A \cup B) \cap C^c$.

2.3. The Probability Measure

In the previous sections, we have developed the language with which we can describe the possible outcomes of random experiments, as well as ask unambiguous yes/no questions about these outcomes. We now arrive at the point where we discuss how to quantitatively assess the likelihood of outcomes and events. This is what the probability measure is about.

Informally speaking, the probability measure, which we denote by $\mathbf{P}$, is a function that assigns to any event $A \subset \Omega$ a real number $\mathbf{P}[A]$ from zero to one. In english, we call $\mathbf{P}[A]$ "the probability that A occurs." This number is meant to be an assessment of how likely A is to occur, ranging from impossible when $\mathbf{P}[A] = 0$ to certain when $\mathbf{P}[A] = 1$. Otherwise, if $\mathbf{P}[A] = p$ for some number $0 < p < 1$ that is in between zero and one, then this indicates that there is some uncertainty regarding whether or not A will occur; the closer p is to one, the more confident we are that it will occur, and vice versa.

The precise meaning of $\mathbf{P}[A] = p$ for some number $0 \leq p \leq 1$, both theoretically and intuitively, will be explored in details in this course. That being said, before discussing such things, we give a formal definition of the probability measure. In this formal definition, we state three properties that probability measures are assumed to satisfy, called the axioms of probability. As their name suggests, these axioms are not something that we prove about probability measures. Instead, they are properties that are considered to be so self-evident that we assume they are true without proof. Without further ado, here is the definition in question:

Definition 2.22 (Probability Measure)**.** The probability measure $\mathbf{P}$ is a function that assigns to every[2] event $A \subset \Omega$ a real number $\mathbf{P}[A]$, called the probability of A, which satisfies the following three conditions:

(Axiom 1) $0 \leq \mathbf{P}[A] \leq 1$.

(Axiom 2) $\mathbf{P}[\Omega] = 1$.

(Axiom 3) Suppose that the events $A_1, A_2, A_3 \ldots$ are mutually exclusive, that is, for every $i \neq j$, the intersection $A_i \cap A_j$ is empty. We denote this by $A_i \cap A_j = \varnothing$, where $\varnothing$ is the symbol for the empty set; see Figure 2.5 for a Venn diagram. In words, this means that no two events in the collection $A_1, A_2, A_3, \ldots$ can occur simultaneously. Then, the probability measure must satisfy

$$\mathbf{P}\left[\bigcup_{i \geq 1} A_i\right] = \sum_{i \geq 1} \mathbf{P}[A_i].$$

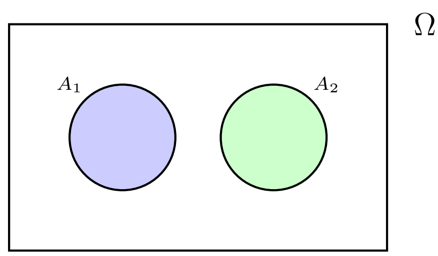

Figure 2.5. The events A_1 and A_2 have no outcome in common; in other words, they are mutually exclusive. By Axiom 3, the probability that at least one of the these events occur (i.e., $\mathbf{P}[A_1 \cup A_2]$) is the sum of the probabilities of both events (i.e., $\mathbf{P}[A_1] + \mathbf{P}[A_2]$).

With this definition stated, a number of important remarks are in order:

2.3.1. Every Event vs. Most Events. In the definition of probability measure, I added a footnote to the claim that $\mathbf{P}$ assigns a probability to every event, saying that I actually mean most events. Indeed, when the sample space is a very large infinite set (such as the nonnegative real numbers in Example 2.11), some subsets of Ω can be so pathological and weird that insisting that $\mathbf{P}$ must be defined on *every* subset of Ω *and* satisfy the three axioms leads to logical paradoxes.

When studying some of the more sophisticated and abstract mathematical results in probability, the subtleties involved with these logical paradoxes become important. Hence, if you ever take graduate-level courses in probability (typically advertised as "measure theoretic probability theory[3]"), then you will discuss a more detailed definition of the probability measure that formulates more precisely what are these pathological sets that we need to avoid.

However, we will never encounter these kinds of events in this course. Thus, for all practical purposes, you can ignore this remark for the remainder of this quarter

[2] Here, I really mean "most events." More on this technical point later.

[3] See, for instance, the Wikipedia article on measure theory.

and be safe in the knowledge that, from the point of view of this course, it amounts to little more than abstract nonsense.

2.3.2. Intuition and Common Sense. There are two things that I mentioned before defining probability measures that have yet to be addressed, namely:

(1) The intuitive interpretation of the statement that $\mathbf{P}[A] = p$ for some event $A \subset \Omega$ and number $0 \leq p \leq 1$.
(2) The three axioms of probability measures are so self-evidently true that it is reasonable to assume that they hold without proof.

As it turns out, these two notions are closely related, in the sense that once we have a good intuition for the meaning of the probability measure, then the claim that the three axioms are self evident becomes more compelling.

Arguably the simplest way to intuitively understand probabilities is to think of them as frequencies: Suppose, for instance, that we carry out the experiment of flipping a coin with two sides called heads and tails. Most people would agree that the probability that the coin lands on heads is one half. But what does this mean in practice? One way to interpret this statement would be to say that, when flipped, the coin will land on heads roughly half of the time. More specifically, if we flip the coin a thousand times and compute the empirical frequency

$$\frac{\text{number of times (out of 1 000) the coin lands on heads}}{1\,000},$$

then this should be approximately equal to one half. More generally, if we repeatedly perform any random experiment a large number of times, then for any event A related to that experiment, it should be the case that

$$\mathbf{P}[A] \approx \frac{\text{number of times that } A \text{ occurs}}{\text{number of times that we perform the experiment}}. \tag{2.6}$$

As compelling as they are from the intuitive point of view, empirical frequencies are too ambiguous to form the basis of a rigorous mathematical theory. In particular, if we perform a random experiment many times and then compute the fraction of experiments wherein a certain event occurs, then this itself will be a random quantity. For instance, if we flip a coin one thousand times, then the number of heads in these thousand flips will not always be exactly 500. Thus, an empirical frequency alone is not enough to give a completely unambiguous answer to "what is the probability that the event A occurs?" The answer to that question should be a fixed nonrandom number, which is what $\mathbf{P}[A]$ is supposed to represent.

In this context, probability measures and the axioms of probability can be viewed as an attempt to introduce unambiguous nonrandom quantities that formalize some of the properties that empirical frequencies satisfy. To see why this is the case, suppose that we perform the same random experiment with sample space Ω a large number of times. Then:

(1) For any event $A \subset \Omega$,

$$0 \leq \frac{\text{number of times that } A \text{ occurs}}{\text{number of times that we perform the experiment}} \leq 1.$$

Indeed, the number of times that A occurs is a number between zero (in which case the ratio above is zero) and the total number of times the experiment is performed (in which case the above ratio is one).

(2) Since the sample space Ω contains by definition every possible outcome of the experiment, the event Ω will occur every single time that we perform the experiment. Consequently,

$$\frac{\text{number of times that } \Omega \text{ occurs}}{\text{number of times that we perform the experiment}} = 1.$$

(3) If A and B are mutually exclusive, then

$$\frac{\#\text{ at least one of } A \text{ or } B \text{ occur}}{\#\text{ experiments}} = \frac{\#\ A \text{ occurs}}{\#\text{ experiments}} + \frac{\#\ B \text{ occurs}}{\#\text{ experiments}},$$

where # is a shorthand for "number of times." Indeed, since A and B cannot occur simultaneously, the number of outcomes such that at least one of A or B occur is the same as the sum of the number of outcomes where A occurs with the number of outcomes where B occurs. Because of the mutual exclusivity, no outcome will be counted twice in this sum.

These three properties are nothing more than the axioms of probability reformulated in the context of empirical frequencies. The axioms serve to ensure that our theoretical probabilities, whatever they are, also satisfy the same three properties.

2.3.3. How to Assign Probabilities - Modelling vs. Inference. In light of the previous remark, we now understand that probability measures are a way to formalize some of the intuitive properties that empirical frequencies satisfy in a rigorous setting. At this point, however, a pressing question remains: How, exactly, does one go about assigning the values of the probability measure? If $\mathbf{P}[A]$ is not an empirical frequency, then what is it? As it turns out, answering this question is what mathematical probability is all about. In practice, this often takes one of the following two forms:

The first type of problem is what we could call modelling. In such a problem, we begin by making assumptions about the values of the probability measure on a collection of elementary events, say, $A_1, A_2, A_3, \ldots$. That is, we assign numbers between zero and one to the probabilities $\mathbf{P}[A_1], \mathbf{P}[A_2], \mathbf{P}[A_3], \ldots$. Then, using the axioms of probability, we study what these assumptions imply about the probabilities of various more complicated events, such as $A_1 \cup A_2$, $(A_1 \cup A_2) \cap A_3^c$, etc.

The second type of problem, which in a way is the opposite of modelling, is what we could call inference. Inference is the process of using our observations, data, experiments, etc. in order to learn something about the probability measure. Here, the usefulness of mathematical probability lies in the ability to make precise and quantitative statements on what we can infer about the probability of an event from a given set of empirical data. In particular, inference allows to formalize the relationship between empirical frequencies and theoretical probabilities that we expressed earlier as

$$\mathbf{P}[A] \approx \frac{\text{number of times that } A \text{ occurs}}{\text{number of times that we perform the experiment}}.$$

The above definitions of modelling and inference problems may seem abstract to you at this point. On the one hand, starting from the next chapter, we will begin our study of one of the simplest instances of a modelling problem with the uniform measure. That example should help illustrate how modelling looks like in practice. On the other hand, while most of what we will do in this course will be modelling problems, we will also discuss one of the most fundamental results in probability regarding inference, namely, the law of large numbers.

2.3.4. Other Self-Evident Properties. The three axioms of probability measures were justified by the claim that they are nothing more than self-evident properties. We then provided intuition for this claim by arguing that these same three properties are satisfied by empirical frequencies, the latter of which can be viewed as experimental approximations of the probability measure (e.g., (2.6)). At this point, it is natural to ask: Of all the self-evident properties that probability measures should satisfy, why did we only assume the three that are listed in Definition 2.22? For instance, consider the following properties:

Proposition 2.23. *Let* $\mathbf{P}$ *be a probability measure.*

(1) (**Monotonicity.**) *If* $A \subset B$*, then* $\mathbf{P}[A] \leq \mathbf{P}[B]$.

(2) (**Complement rule.**) *For every event* A*, one has* $\mathbf{P}[A^c] = 1 - \mathbf{P}[A]$.

(3) (**Empty event.**) $\mathbf{P}[\varnothing] = 0$.

(4) (**Inclusion-exclusion formula.**) *For every events* $A, B \subset \Omega$*, one has*

$$\mathbf{P}[A \cup B] = \mathbf{P}[A] + \mathbf{P}[B] - \mathbf{P}[A \cap B].$$

Why did we not include these properties as additional axioms of probabilities measures? After all, each of these properties can be argued to be just as self-evident as the three axioms in Definition 2.22 using empirical frequencies. The answer, which is very far from obvious, is that the entirety of the mathematical theory of probability, including the four properties stated in Proposition 2.23, can be proved to be logical consequences of the three axioms. This discovery is often accredited to Andrey Kolmogorov, who showed in a very influential 1933 monograph[4] that the basics of the mathematical theory of probability can be derived in its entirety using only the three axioms in Definition 2.22.

To give an example of how this works, consider the claim that if $A \subset B$, then $\mathbf{P}[A] \leq \mathbf{P}[B]$. In order to understand why this should be true, it can be instructive to draw a Venn diagram, as shown in Figure 2.6 below. Looking at this picture, it becomes immediately apparent that, since A is completely contained in B, every element in B is either inside of A (i.e., the red region in Figure 2.6) or inside of B without being inside of A (i.e., the blue region in Figure 2.6). Thus, the probability of B can only be bigger than that of A; the difference between $\mathbf{P}[A]$ and $\mathbf{P}[B]$ comes from the probability of an outcome in the blue region in Figure 2.6.

In order to turn this intuition into formal mathematics, we argue as follows: We can write the event B as the union

$$B = A \cup (B \cap A^c),$$

[4] Kolmogoroff, A. Grundbegriffe der Wahrscheinlichkeitsrechnung. Reprinting of the 1933 edition. Springer-Verlag, Berlin-New York, 1973.

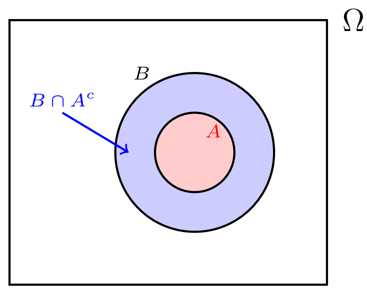

Figure 2.6. If $A \subset B$, then we can write B as the disjoint union $A \cup (B \cap A^c)$.

where the events A and $B \cap A^c$ are clearly mutually exclusive. Therefore,

$$\mathbf{P}[B] = \mathbf{P}[A \cup (B \cap A^c)] \tag{2.7}$$
$$= \mathbf{P}[A] + \mathbf{P}[B \cap A^c] \qquad \text{(Axiom 3; } A \text{ and } (B \cap A^c) \text{ are disjoint)} \tag{2.8}$$
$$\geq \mathbf{P}[A] + 0 \qquad \text{(Axiom 1; } \mathbf{P}[E] \geq 0 \text{ for every event } E) \tag{2.9}$$
$$= \mathbf{P}[A], \tag{2.10}$$

which concludes the proof.

Remark 2.24. The proof that we have just carried out provides yet another compelling illustration of the power of Venn diagrams. At first glance, the abstract formulas in (2.7)–(2.10) may seem a bit daunting. However, if we take the time to carefully justify each step using Figure 2.6, then the proof becomes much more intuitive and obvious. Whenever you are asked to contend with abstract probabilities and events (in the homework for this class or in any other context), it is a good habit to draw a Venn diagram of what you are doing to guide your steps.

If you ever take an advanced course in probability, then you will likely spend a lot of time proving increasingly complex results from the three axioms in the same amount of detail as done here. Given that the focus of this course is more on computations and intuition, most of the results that we discuss in this class will not be proved, or at least not proved with full details. That being said, it is nevertheless interesting to know that everything that we will do ultimately rests on this very parsimonious and elegant foundation.

Chapter 3

The Uniform Measure

At the end of the previous chapter, we discussed two types of problems that arise in mathematical probability, which we called modelling and inference. We explained therein that modelling problems consist of making some assumptions about what the probability measure looks like on a collection of basic events, and then seeing what we can compute about more complicated events from these assumptions. In this chapter, our purpose is to look at a first example of this process, namely, the uniform measure. As we will soon see, the uniform measure is arguably one of the simplest probability measures imaginable. However, despite its simplicity, it can still give rise to some very nontrivial and interesting mathematical problems. Without further ado, let us now define the uniform probability measure.

3.1. The Uniform Measure and Counting Problems

Definition 3.1. Let $\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\}$ be a finite sample space containing n elements, where n is an arbitrary positive integer. The uniform probability measure on Ω is such that

$$\mathbf{P}[\omega_i] = \frac{1}{n} \qquad \text{for every } 1 \leq i \leq n. \tag{3.1}$$

In words, the uniform probability measure can be defined for any random experiment whose sample space only contains a finite number of outcomes, and every possible outcome has exactly the same probability, namely, one over the total number of possible outcomes.

Remark 3.2. The uniform measure is called uniform because every outcome has the same probability. In fact, it can be proved from the three axioms (I will leave the proof of this observation as a simple exercise) that the uniform measure as defined in (3.1) is the only probability measure that assigns the same probability to every outcome in Ω.

Remark 3.3. If we have a random experiment that has an infinite number of possible outcomes, then the uniform measure as we have defined it here does not make sense. Indeed, in order for every outcome to have the same probability in this case, we would need to divide by infinity!

The uniform measure is a useful model in probability whenever we expect that there is some form symmetry in the mechanism that determines outcomes, making it so that all outcomes are equally likely. Here are some classical examples:

Example 3.4 (Flip an Unbiased Coin)**.** Suppose that we flip an unbiased coin with two sides labelled heads and tails. The sample of space of this experiment is

$$\Omega = \{h, t\},$$

with h representing heads and t representing tails. Since the coin is unbiased, we expect that it should land on each side with equal probability; hence we assume that this experiment is modelled by the uniform measure

$$\mathbf{P}[h] = \mathbf{P}[t] = \frac{1}{2}.$$

Example 3.5 (Cast a Fair Die)**.** Suppose that we cast a fair six-sided die. The sample space of this experiment consists of the six faces of the die

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Because the die is fair, we expect that it should land on each face with equal probability; hence we assume that this experiment is modelled by the uniform measure

$$\mathbf{P}[i] = \frac{1}{6} \qquad \text{for every } 1 \leq i \leq 6.$$

Example 3.6 (Draw a five-card hand)**.** Suppose that we draw a five-card hand from a well-shuffled deck of cards. The sample space for this experiment is

$$\Omega = \big\{\{c_1, c_2, c_3, c_4, c_5\} : c_i \in SD \text{ for } 1 \leq i \leq 5\big\},$$

where we recall that SD is the set of cards defined in (2.3). Since the deck is well shuffled, we expect that every possible five-card hand should occur with equal probability; hence we assume that this experiment is modelled by the uniform measure

$$\mathbf{P}[\{c_1, c_2, c_3, c_4, c_5\}] = \frac{1}{\text{number of five-card hands}}. \tag{3.2}$$

In all of these examples, if feels reasonable to assume that we are under the uniform probability measure; if a die is fair, if a coin is unbiased, and if a deck of cards is well shuffled, then there is no reason to expect that any particular outcome should be more likely than any other outcome. That being said, the third example involving cards emphasizes the main difficulty of the uniform measure.

You will note that, in equation (3.2), I have not explicitly stated what is the probability of every five-card hand. In order to compute this explicitly, we need to be able to count the number of elements in Ω, which in this case is the number of ways to draw a set of five distinct cards from a standard deck. Thus, in order to be able to actually compute the uniform measure, we need, at the very least, to be able to count the number of elements in various sets. In fact, with a simple argument, we can reduce the computation of the probability of any event whatsoever with the uniform measure to a counting problem:

Notation 3.7. Let A be any set. We use $\#(A)$ to denote the number of elements in the set A.

Proposition 3.8. *Let Ω be a finite sample space and $\mathbf{P}$ be the uniform probability measure. For every event $A \subset \Omega$, it holds that*

$$\mathbf{P}[A] = \frac{\#(A)}{\#(\Omega)}.$$

Proof. Let us enumerate the elements of the event A as

$$A = \{a_1, a_2, \ldots, a_{\#(A)}\}$$

(here, the enumeration ends at $\#(A)$ because this is the number of elements in A). We can instead write A as the disjoint union

$$A = \{a_1\} \cup \{a_2\} \cup \cdots \cup \{a_{\#(A)}\},$$

and thus

$$\begin{aligned}
\mathbf{P}[A] &= \mathbf{P}[a_1] + \mathbf{P}[a_2] + \cdots + \mathbf{P}[a_{\#(A)}] && \text{(Axiom 3)}\\
&= \underbrace{\frac{1}{\#(\Omega)} + \frac{1}{\#(\Omega)} + \cdots + \frac{1}{\#(\Omega)}}_{\#(A) \text{ times}} && \text{(uniform measure)}\\
&= \frac{\#(A)}{\#(\Omega)},
\end{aligned}$$

which concludes the proof. □

In conclusion, every possible computation of probabilities involving the uniform measure can be reduced completely to a counting problem, namely, counting how many elements are in the event A and the sample space Ω. Here is a simple example:

Example 3.9 (Sum of Two Dice)**.** Suppose that we cast two fair six-sided dice. The sample space for this experiment is

$$\Omega = \{(i,j) : 1 \leq i, j \leq 6\},$$

where i represents the outcome of the first die, and j represents the outcome of the second die. Suppose that $\mathbf{P}$ is the uniform probability measure. If we define

$$A = \text{“is the sum of the two dice equal to 7?”} = \{(i,j) \in \Omega : i + j = 7\},$$

then what is $\mathbf{P}[A]$? By Proposition 3.8, we know that

$$\mathbf{P}[A] = \frac{\#(A)}{\#(\Omega)} = \frac{\#\big(\{(i,j) \in \Omega : i + j = 7\}\big)}{\#\big(\{(i,j) : 1 \leq i, j \leq 6\}\big)}.$$

In order to carry out this counting problem, it can be useful to represent the outcomes of Ω in a table. Looking at the top of Figure 3.1, we see that by enumerating and counting all possible outcomes, $\#(\Omega) = 36$. Next, if we calculate the sum of the two dice for every possible such outcome (as is done on the bottom of Figure 3.1), then it is easy to manually count that there there are six outcomes that give a sum of 7. Therefore, we conclude that

$$\mathbf{P}[A] = \frac{6}{36} = \frac{1}{6}.$$

Figure 3.1. Possible outcomes of tossing two dice (top) and the sum of their two faces (bottom).

	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

$$\left(\begin{array}{c||c} & j \\ \hline i & (i,j) \end{array}\right)$$

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$$\left(\begin{array}{c||c} & j \\ \hline i & i+j \end{array}\right)$$

This particular example might leave you with the impression that computing with the uniform measure is easy: Everything reduces to counting outcomes. However, this conceptual simplicity is deceptive. Indeed, as it turns out, sample spaces and events do not have to be that complicated to induce rather tricky counting problems. Here are a two illustrative examples of this:

Example 3.10 (Birthdays)**.** Suppose for simplicity that there are only 365 days in a year (i.e., Month+Day, excluding February 29), and that in any group of people, the birth dates of individuals is uniformly distributed (i.e., any configuration of birthdays among the individuals in the group is equally likely). What is the probability that, in a group of 23 people, at least two people share the same birthday?

Example 3.11 (Full House)**.** Suppose that we draw a five-card hand from a standard deck of cards, that is, a set of five unordered and distinct cards. We assume that the deck is well-shuffled, so that every possible five-card hand is equally likely. What is the probability that our hand is a full house? That is, a hand that contains three cards with one value, and two cards with another value; for example

$$\big\{(1,\heartsuit),(1,\spadesuit),(1,\diamondsuit),(6,\spadesuit),(6,\clubsuit)\big\}$$

is a full house, as it contains three aces and two sixes.

If you spend some time thinking about these two random experiments, then you will no doubt come to the conclusion that the number of elements in both their sample spaces and the events we are interested in cannot be counted as easily as in Example 3.9. In particular, the sample spaces involved contain so many elements (more than 2 500 000 for the number of 5-card hands and more than $(1\,000\,000\,000)^6$ for the number of birthdays) that an exhaustive enumeration of them similar to Figure 3.1 is completely out of the question.

3.2. Counting Techniques

Examples 3.10 and 3.11 motivate the development of techniques for counting the number of elements in sets that are more sophisticated than exhaustively enumerating every element. For this purpose, in this section we introduce a number of counting techniques. Then, in Section 3.3, we apply these techniques to solve the problems posed in Examples 3.10 and 3.11.

3.2.1. Mutually Exclusive Unions - The Sum Rule. The first and most straightforward counting technique that we discuss is the sum rule:

Proposition 3.12 (Sum Rule)**.** *Let $A_1, A_2, \ldots, A_k$ be mutually exclusive sets (that is, for every $i \neq j$, one has $A_i \cap A_j = \varnothing$). Then,*

$$\#\left(\bigcup_{i=1}^{k} A_i\right) = \sum_{i=1}^{k} \#(A_i).$$

The sum rule is more or less completely obvious. In order to visualize it, it suffices to look at a simple example:

Example 3.13. Let $A = \{1, 2, 3\}$ and $B = \{5, 6, 7, 8\}$. Then, $\#(A) = 3$ and $\#(B) = 4$. Because these two sets have no element in common, the number of elements in their union

$$A \cup B = \{1, 2, 3, 5, 6, 7, 8\}$$

is equal to $\#(A) + \#(B) = 7$. Indeed, no element is counted twice if we add the number of elements in A to the number of elements in B.

The above application of the sum rule is admittedly not very impressive. However, the sum rule can still be very useful in nontrivial situations. More specifically, the sum rule can be very powerful when used to break down a difficult counting problem into a series of simpler counting problems. We will see an example of this in the next subsection.

3.2.2. n-Tuples - Tree Diagrams and the Product Rule. Let $A_1, A_2, \ldots, A_n$ be a collection of sets (which may or may not be equal to one another). Suppose that we are interested in counting the number of elements in the set of n-tuples such that for each $1 \leq i \leq n$, the i^{th} element is taken from the set A_i. That is, we want to count the number of elements in the set

$$A = \big\{(a_1, a_2, \ldots, a_n) : a_i \in A_i \text{ for every } 1 \leq i \leq n\big\}. \tag{3.3}$$

For this, we have the product rule:

Proposition 3.14 (Product Rule)**.** *If A is the set in* (3.3)*, then*

$$\#(A) = \#(A_1) \cdot \#(A_2) \cdots \#(A_n).$$

We can first test this out with a familiar example:

Example 3.15 (Number of Cards)**.** Recall the definition of the set of cards in a standard deck:

$$SD = \big\{(v, s) : v \in \{1, 2, 3, \ldots, 13\} \text{ and } s \in \{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\}\big\}.$$

Since there are 13 elements in the set of values $\{1, 2, 3, \dots, 13\}$ and 4 elements in the set of suits $\{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$, we conclude by the product rule that

$$\#(SD) = 13 \cdot 4 = 52,$$

as is well known of a standard deck.

A good way to convince yourself that the product rule is correct in general is to use a graphical enumeration technique called the tree diagram. In order to have a tree diagram that is not too large, we look at an example of a set that has fewer elements than the set of all standard cards:

Example 3.16 (Formal outfit)**.** Suppose that you are trying to choose an outfit for a formal event. You have

(1) two pairs of formal pants, call them p_1 and p_2;
(2) three pairs of formal shoes, call them s_1, s_2, and s_3; and
(3) two formal blazers, call them b_1 and b_2.

The set of all possible outfits you could construct from this is

$$O = \big\{(p, s, b) : p \in \{p_1, p_2\},\ \ s \in \{s_1, s_2, s_3\},\ \text{and } b \in \{b_1, b_2\}\big\};$$

that is, the set of 3-tuples where the first element is a pair of pants, the second element is a pair of shoes, and the third element is a blazer. As per the product rule, we know that the number of elements in this set is $2 \cdot 3 \cdot 2 = 12$. In order to visualize why this is the case, we can look at Figure 3.2 below. The process of drawing this

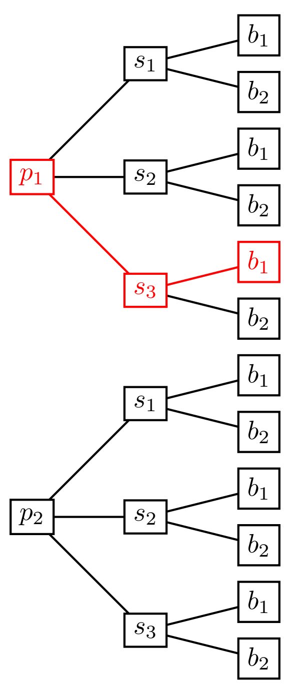

Figure 3.2. Outfit tree diagram.

diagram can be explained in words as follows: We begin by drawing two nodes (on the left of the tree) that represent the number of ways to choose a pair of pants. Next, for each possible choice of pants, you can choose any of your three pairs of shoes. Thus, we draw three nodes (in the middle of the tree) connected to each of the possible ways to choose the pants. Finally, for every possible combination of one pair of pants and one pair of shoes, you can choose one of the two blazers; this is represented by the nodes on the right of the tree.

In the tree diagram thus obtained, each branch (i.e., a triple of pants, shoes, and blazer that are connected) represents one of the possible ways to assemble an outfit. For instance, the branch drawn in red in Figure 3.2 represents the 3-tuple (p_1, s_3, b_1). A visual inspection of the way in which this diagram branches out at every step makes the product rule very intuitive.

Before moving on to the next counting principle, we look at an example that combines the sum and product rules. This will help illustrate the claim I made earlier that the sum rule can be useful to break down complex counting problems into simpler ones.

Example 3.17 (Die and coins)**.** Suppose that we perform the following random experiment: Firstly, we cast a six-sided die. Secondly, if the result of the die is equal to the number k (where $1 \leq k \leq 6$), then we flip a coin (which lands on either heads or tails) k times in a row, assuming that the order of the coin flips matters. The sample space for this experiment is as follows:

$$\Omega = \big\{(k, c_1, \ldots, c_k) : 1 \leq k \leq 6, \text{ and } c_1, \ldots, c_k \in \{h, t\}\big\}.$$

Here, the outcomes of the die and the coin flips are represented in an n-tuple; the first element k represents the outcome of the die, and the following elements $c_1, \ldots, c_k$ represent the outcomes of the k coin flips in order.

What makes this sample space complicated is that it consists of n-tuples of different sizes, depending on what is the outcome of the die. This is not just a simple set of n-tuples of fixed size, as was the case in the statement of the product rule. However, and this is where the sum rule comes in handy, we can write Ω as a union of mutually exclusive sets that fit the product rule. More specifically, for every $1 \leq k \leq 6$, let $A_k \subset \Omega$ denote the event that the outcome of the die was equal to k. That is,

$$\begin{aligned}
&A_1 = \big\{(1, c_1) : c_1 \in \{h, t\}\big\}\\
&A_2 = \big\{(2, c_1, c_2) : c_1, c_2 \in \{h, t\}\big\}\\
&\cdots\\
&A_6 = \big\{(6, c_1, c_2, \ldots, c_6) : c_1, c_2, \ldots, c_6 \in \{h, t\}\big\}.
\end{aligned}$$

Clearly, these events are mutually exclusive, as the result of the die cannot be two numbers simultaneously. Moreover, we have that

$$\Omega = \bigcup_{k=1}^{6} A_k,$$

because the result of the die must be a number between one and six; hence these events account for every possibility in our random experiment. Thus, by the sum rule,

$$\#(\Omega) = \sum_{k=1}^{6} \#(A_k).$$

The reason why applying the sum rule here is useful is that, unlike Ω, the events A_k are precisely in the form of the sets counted by the product rule. Indeed, by the product rule we have

$$\#(A_k) = 1 \cdot \underbrace{2 \cdot 2 \cdots 2}_{k \text{ times}} = 2^k,$$

as we have only one way to choose the result of the die (it must be equal to k on the event A_k), and then 2 ways to choose the outcome of each following coin flip. Then, using a calculator, we conclude that

$$\#(\Omega) = \sum_{k=1}^{6} 2^k = 126.$$

3.2.3. Distinct k-Tuples - Permutations. Let A be a finite set, and suppose that we are interested in the set of k-tuples (where $k \leq \#(A)$) containing distinct elements from A. That is, the set

$$B = \big\{(a_1, a_2, \ldots, a_k) : a_i \in A \text{ for } 1 \leq i \leq k \text{ and } a_i \neq a_j \text{ for all } i \neq j\big\}. \tag{3.4}$$

In words, this consists of choosing k distinct elements from the set A, and then putting them in a specific order. In order to state our result regarding the number of elements in such a set, we need some notations:

Notation 3.18. Let n be a nonnegative integer. We denote the number $n!$, which is called "n factorial," as

$$n! = \begin{cases} 1 & \text{if } n = 0, 1 \\ n \cdot (n-1) \cdots 2 \cdot 1 & \text{if } n \geq 2 \end{cases}.$$

In words, $n!$ is the product of every integer from 1 to n, with the convention that $0! = 1$. Let k be a nonnegative integer such that $k \leq n$. We define the number ${}_nP_k$, which we will call "n-permute-k," as

$${}_nP_k = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

Proposition 3.19 (n-permute-k rule)**.** *If B is the set defined in (3.4), then*

$$\#(B) = {}_{\#(A)}P_k.$$

In other words, for any $k \leq n$, the quantity ${}_nP_k$ counts the number of ways to choose a k-tuple of distinct elements taken from a set containing n elements.

Much like the product rule, the n-permute-k rule can be justified using tree diagrams. For this, we consider a simple example:

Example 3.20 (One president and one vice president)**.** Suppose that a company is looking to fill two leadership positions; one position of president, and one position of vice president. After carefully examining the applicant pool, the hiring committee settles on a shortlist of four individuals, which we label 1, 2, 3, and 4. The set of all possible ways to fill the two position using these four candidates is as follows:

$$C = \big\{(p,v) : p,v \in \{1,2,3,4\} \text{ and } p \neq v\big\}.$$

Here, p represents the individual who will be the president, and v represents the individual who will be the vice president. Since these two positions are distinct, the individuals filling the two roles cannot be the same; hence the condition $p \neq v$.

Thanks to the n-permute-k rule, we know that the number of elements in this set is $4 \cdot 3 = 12$. In order to illustrate how this works, consider the tree diagram in Figure 3.3 below. The process of drawing this tree is similar to what we did for the

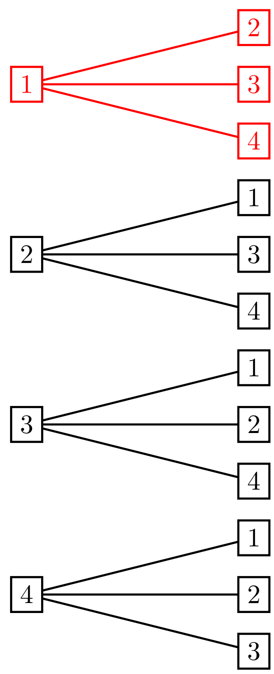

Figure 3.3. President (left) and vice president (right) tree diagram.

product rule, but with one crucial difference: We begin by drawing four nodes on the left of the tree, representing the four possible ways in which we can choose who is the president. Next, for each of these four choices, we connect additional nodes on the right to represent the possible choices of vice president. However, unlike in the product rule, the elements that we are allowed to choose in this second step depend on who was selected as president. For instance, if we look at the branches in the tree corresponding to the case where candidate number 1 is president (drawn in red in Figure 3.3), then the nodes corresponding to the choice of vice president only include candidates 2, 3 and 4. More generally, any two nodes that are connected in the tree cannot be the same number.

If we were to generalize the intuition of the tree diagram to the statement in Proposition 3.19 in counting the number of elements in the set

$$B = \big\{(a_1, a_2, \ldots, a_k) : a_i \in A \text{ for } 1 \leq i \leq k \text{ and } a_i \neq a_j \text{ for all } i \neq j\big\},$$

then we can explain the relation

$$\#(B) = {}_{\#(A)}P_k = \#(A)(\#(A)-1)(\#(A)-2)\cdots(\#(A)-k+1)$$

as follows:

$$\begin{aligned}
\#(A) &= \# \text{ ways to choose } a_1 \in A\\
(\#(A)-1) &= \# \text{ ways to choose } a_2 \in A \text{ s.t. } a_2 \neq a_1\\
(\#(A)-2) &= \# \text{ ways to choose } a_3 \in A \text{ s.t. } a_3 \neq a_1, a_2\\
&\cdots\\
(\#(A)-k+1) &= \# \text{ ways to choose } a_k \in A \text{ s.t. } a_k \neq a_1, a_2, \ldots, a_{k-1}.
\end{aligned}$$

3.2.4. Subsets - Combinations. We now arrive at the last counting technique that we introduce in this chapter: Let A be a finite set, and suppose that we are interested in counting the number of subsets of A with k elements (where $k \leq \#(A)$). That is, we want to count the number of elements in

$$S = \big\{\{a_1, a_2, \ldots, a_k\} : a_i \in A \text{ for } 1 \leq i \leq k\big\}. \tag{3.5}$$

Here, we recall that, by definition of a set, this automatically implies that the elements a_i must all be different from one another. In order to state our result regarding the number of elements in this set, we once again introduce some notation:

Notation 3.21. Let n and k be nonnegative integers such that $k \leq n$. We define the number ${}_nC_k$, which we will call "n-choose-k," as

$${}_nC_k = \frac{{}_nP_k}{k!} = \frac{n!}{(n-k)!k!}.$$

Proposition 3.22 (n-choose-k rule)**.** *If S is the set defined in (3.5), then*

$$\#(S) = {}_{\#(A)}C_k.$$

In other words, for any $k \leq n$, the quantity ${}_nC_k$ counts the number of ways to choose a subset of k elements from a set containing n elements.

The n-choose-k rule can be justified with a very elegant argument often called "strategic over-counting." In short, this strategy consists of deliberately counting too many elements, and then dividing the count by the right quantity to account for this. To illustrate how this works in practice, we consider an example:

Example 3.23 (Two software engineers)**.** Suppose that a company has two open software engineer positions. We assume that the two positions are indistinguishable from one another. The hiring committee are considering four candidates, which we label 1, 2, 3, and 4. The set of all possible ways to fill the positions is

$$D = \big\{\{e_1, e_2\} : e_1, e_2 \in \{1, 2, 3, 4\}\big\}.$$

Here, e_1 and e_2 are the two candidates who will get the job; because the two positions are indistinguishable and must be filled by distinct individuals, we can represent the possible outcomes as subsets $\{e_1, e_2\} \subset \{1, 2, 3, 4\}$. According to the n-choose-k rule, the number of elements in this set is ${}_4C_2 = \frac{4 \cdot 3}{2} = 6$.

This can be illustrated by a tree diagram, as done in Figure 3.4. We construct

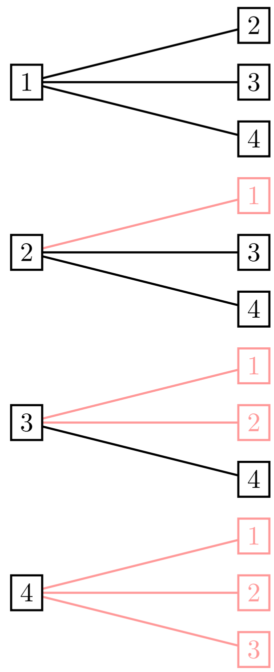

Figure 3.4. Software engineers tree diagram; we remove the red branches to correct the strategic over-counting.

the tree corresponding to the present example by doing a strategic over-counting argument: Let us pretend for the moment that the order in which the candidates get the job offers matter. For instance, we could pretend that e_1 is a senior software engineer position, and that e_2 is an intern position. If that were the case, then the number of ways to fill the two positions would be 12; this can be justified with exactly the same argument as in Example 3.20, which yields the tree diagram in Figure 3.4.

However, in doing so, we have over-counted the problem. We do not actually care about the order of the software engineers e_1 and e_2. Therefore, to account for this, we go through each branch in the tree one by one, and remove branches corresponding to pairs of applicants that have appeared before in the tree. If we go through this process from top to bottom in Figure 3.4, then the branches that we end up removing are in red. After doing this, we now see that only 6 branches survive, corresponding to the six possible ways to choose the two software engineers.

The general n-choose-k rule stated in Proposition 3.22 can be justified with a similar strategic over-counting argument. If we want to count the number of elements in the set

$$S = \big\{\{a_1, a_2, \ldots, a_k\} : a_i \in A \text{ for } 1 \leq i \leq k\big\},$$

then we can first pretend that the order of elements matters. Thus, we instead count the number of elements in the set

$$B = \big\{(a_1, a_2, \ldots, a_k) : a_i \in A \text{ for } 1 \leq i \leq k \text{ and } a_i \neq a_j \text{ for all } i \neq j\big\},$$

which is $\#(B) = {}_{\#(A)}P_k$. This is an over-count, because for any choice of the elements $a_1, \ldots, a_k$, we are counting every permutation of those elements in a different order as distinct elements. Therefore, in order to correct for this over-counting, it suffices to divide $\#(B)$ by the number of ways to permute k elements in different orders, which is ${}_kP_k = k!$. In conclusion,

$$\#(S) = \frac{\#(B)}{k!} = \frac{{}_nP_k}{k!} = {}_nC_k.$$

3.2.5. Techniques vs Results. Given their fundamental importance in counting problems, it is certainly a good idea to try to remember the four counting results that were formally stated in this section, namely, the sum, product, n-permute-k, and n-choose-k rules. However, in the interest of not missing the forest for the trees, I would also encourage you to keep in mind that the product, n-permute-k, and n-choose-k rules are all consequences of two very important ideas, which can be informally stated as follows:

(1) **Tree diagrams.** Graphical representations of a systematic enumeration of a set, wherein the branching out of nodes clearly highlight the multiplicative nature of counting n-tuples (e.g., the product and n-permute-k rules).

(2) **Strategic over-counting.** Deliberate over-count of a set by neglecting one or several constraints (which makes the counting problem easier), followed by a division to remove elements that appear too many times (e.g., the n-choose-k rule can be obtained from the n-permute-k rule by first neglecting the requirement that elements in a subset are not ordered, and then dividing by $k!$ to correct for this).

Indeed, as you will no doubt come to appreciate throughout this course, most interesting counting problems do not fit neatly into only one of the four counting rules stated in this section. Consequently, it pays off to understand the underlying mechanisms at the origin of those four rules, so as to be able to interpolate from them to solve more complicated problems.

3.3. A Return to the Two Examples

As promised at the beginning of the previous section, we now solve the two problems posed in Examples 3.10 and 3.11.

3.3.1. Solution of Example 3.10. The problem posed in this example was the following: What is the probability that, in a group of 23, at least two people share the same birthday? For this, we also made the assumption that there are only 365 birthdays (excluding Feb. 29), and that every possible assignment of 23 birthdays is equally likely. A sample space for this situation could be as follows:

$$\Omega = \big\{(b_1, b_2, \ldots, b_{23}) : b_i \in \{1, 2, 3, \ldots, 365\} \text{ for all } 1 \leq i \leq 23\big\}.$$

Here, we assume that we have numbered the individuals in the group from 1 to 23, and that b_i represents the birthdate of the i^{th} individual, where we have numbered all possible birthdays from 1 to 365. The event that we are interested in is this:

$$A = \big\{(b_1, b_2, \ldots, b_{23}) \in \Omega : b_i = b_j \text{ for at least one pair } i \neq j\big\}.$$

Since we assume that the probability measure is uniform, this means that

$$\mathbf{P}[A] = \frac{\#(A)}{\#(\Omega)}.$$

On the one hand, the number of elements in this sample space is straightforward to count: Ω in this example is exactly the kind of set whose number of elements is counted by the product rule. Applying the latter to this situation, we get that

$$\#(\Omega) = 365^{23}.$$

On the other hand, $\#(A)$ is rather more complicated. As it turns out, however, there is a way to get around this issue. If you recall Proposition 2.23 in our discussion of the axioms of probability measures, one of the properties that I claimed was a logical consequence of the axioms was the complement rule:

$$\mathbf{P}[A^c] = 1 - \mathbf{P}[A] \qquad \text{equivalently} \qquad \mathbf{P}[A] = 1 - \mathbf{P}[A^c].$$

As it turns out, the complement of A is a much simpler set than A itself:

$$A^c = \big\{(b_1, b_2, \ldots, b_{23}) \in \Omega : b_i \neq b_j \text{ for every } i \neq j\big\}.$$

In particular, this fits exactly in the n-permute-k rule:

$$\#(A^c) = {}_{365}P_{23}.$$

Thus, we conclude that

$$\mathbf{P}[A] = 1 - \mathbf{P}[A^c] = 1 - \frac{\#(A^c)}{\#(\Omega)} = 1 - \frac{{}_{365}P_{23}}{365^{23}},$$

which, with a calculator, can be checked to be approximately equal to 0.507.

The fact that $\mathbf{P}[A] > 0.5$ in this example is a famous result in elementary probability called the birthday paradox.[1] The idea behind calling this a paradox is that many people find it counterintuitive that you only need 23 people in a group for it to be more likely than not that at least two individuals share a birthday. That being said, it is important to remember that the computation that we carried out here is only valid under the assumptions that

(1) no one has Feb. 29 as their birthday; and

(2) every possible assignment of birthdays in a group is equally likely.

Thus, the extent to which you take the claim that $\mathbf{P}[A] \approx 0.507$ seriously depends on the extent to which you believe that these two assumptions accurately reflect reality. Such questions are discussed in more details in Section 3.4, which is an optional bonus section for your enrichment and personal interest.

[1] See, e.g., the Wikipedia page with that same name.

3.3.2. Solution of Example 3.11. We recall that, in this example, the sample space is the set of all five-card hands

$$\Omega = \big\{\{c_1, \ldots, c_5\} : c_i \in SD \text{ for } 1 \leq i \leq 5\big\},$$

and we are interested in the event

$$A = \big\{\{c_1, \ldots, c_5\} \in \Omega : \{c_1, \ldots, c_5\} \text{ is a full house}\big\}.$$

If we assume that every hand is equally likely, then we know that

$$\mathbf{P}[A] = \frac{\#(A)}{\#(\Omega)}.$$

On the one hand, the sample space Ω is exactly of the form that is counted by the n-choose-k rule. Because $\#(SD) = 52$, we know that

$$\#(\Omega) = {}_{52}C_5 = 2\,598\,960.$$

On the other hand, $\#(A)$ is much more complicated. Recall that a full house of a triple of cards with the same value, together with a pair of cards with the same value. Thus, if we write the i^{th} card in our hand as $c_i = (v_i, s_i)$, where v_i is the value and s_i is the suit, then we can write A in a more detailed way as follows:

$$A = \Big\{\big\{(v, s_1), (v, s_2), (v, s_3), (w, t_1), (w, t_2)\big\} \\ : v, w \in \{1, \ldots, 13\} \text{ and } s_i, t_i \in \{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\}\Big\}.$$

Let us take a moment to carefully parse what we are doing here:

(1) v is the common value of the triple, and w is the common value of the pair. Although this is not stated explicitly, we know that $v \neq w$, because for every number $1 \leq k \leq 13$, there are only four cards with the value k in SD. If $v = w$, then this means that we have five cards with the same value in a set of cards, which is impossible.

(2) The s_i's are the suits of the cards in the triple, and the t_i's are the suits of the cards in the pair. Once again, even though this is not explicitly mentioned, it must be the case that $s_i \neq s_j$ and $t_1 \neq t_2$ because otherwise we would have the same card multiple times in our hand, which would violate the definition of a set. However, it is possible that $s_i = t_j$ for some $1 \leq i \leq 3$ and $1 \leq j \leq 2$.

This does not fit exactly into any of the four counting rules that we have defined above. However, with a sequence of clever manipulations, it is possible to break down this problem into simpler problems that do fit the four rules.

To see this, consider the set

$$A' = \Big\{\big((v, w), \{s_1, s_2, s_3\}, \{t_1, t_2\}\big) \\ : v, w \in \{1, \ldots, 13\},\ v \neq w,\ \text{and } s_i, t_i \in \{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\}\Big\}.$$

In words, elements of A' consist of 3-tuples wherein

(1) the first (v, w) element is a 2-tuple of distinct values;

(2) the second element $\{s_1, s_2, s_3\}$ is a set of 3 suits; and

(3) the third element $\{t_1, t_2\}$ is a set of 2 suits.

It is not too difficult to see that the sets A and A' contain exactly the same number of elements. Indeed, the elements of these two sets both serve to uniquely identify a full house: Just like in the set A, in A' the numbers v and w identify the value of the triple and pair respectively (these are ordered because the full house wherein the triple has value v and the pair has value w is not the same as the full house wherein the triple has value w and the pair has value v), the s_i's identify the suits of the triple, and the t_i's identify the suits of the pair.

The purpose of reformulating the set A in the form of A' is that the latter set does fit into the product rule: It is a set of 3-tuples, where each element is taken from different sets without constraints. Thus, if we denote the sets

$$\begin{aligned} A'_1 &= \{(v,w) : v,w \in \{1,\ldots,13\},\ v \neq w\} \\ A'_2 &= \big\{\{s_1,s_2,s_3\} : s_i \in \{\clubsuit,\diamondsuit,\heartsuit,\spadesuit\}\big\} \\ A'_3 &= \big\{\{t_1,t_2\} : t_i \in \{\clubsuit,\diamondsuit,\heartsuit,\spadesuit\}\big\}, \end{aligned}$$

then we know by the product rule that

$$\#(A) = \#(A') = \#(A'_1)\cdot\#(A'_2)\cdot\#(A'_3).$$

Next, we now that A'_1 can be counted by the n-permute-k rule, and that A'_2 and A'_3 can be counted by the n-choose-k rule. Because there are 13 values and 4 suits,

$$\#(A) = \#(A'_1)\cdot\#(A'_2)\cdot\#(A'_3) = {}_{13}P_2 \cdot {}_4C_3 \cdot {}_4C_2 = 3\,744.$$

We therefore conclude that

$$\mathbf{P}[A] = \frac{\#(A)}{\#(\Omega)} = \frac{3\,744}{2\,598\,960} \approx 0.0014.$$

Remark 3.24. In closing this section, I want to make a final remark regarding the argument that was used to prove that $\#(A) = 3\,744$. Indeed, the way that we did this was not to count the elements of A directly, but instead *reformulate* the elements of the set A into a different set A' which is easier to analyze.

This idea, which we could call the bijection trick, is very important in the business of counting the number of elements in sets. Whenever you are faced with such a problem, you should always keep the question “Is there a way to reformulate this set into one which can be analyzed using the basic counting techniques?” in your bag of tricks. You will have the opportunity to practice this type of thinking in the homework.

3.4. Is the Uniform Measure a Good Model? (Bonus)

In this chapter, we have performed a variety of interesting probabilistic computations under the assumption that the probability measure is uniform. In particular, every conclusion that we draw from the calculations performed in this chapter (including the “fact” that you only need 23 people in a group for it to be more likely than not that two birthdays are shared, and the probability of a full house) are only valid if this assumption is true. It is therefore natural to ask:

Is the uniform probability measure actually a good model for the experiments that we analyzed in this section (or any other model)?

Answering this question turns out to be surprisingly difficult. In the remainder of this section, we briefly discuss the legitimacy of uniformity in three examples that we discussed in this chapter.

3.4.1. Dice. Assuming that the outcome of casting dice is uniform is probably the least controversial of every model in this chapter (apart from coin flips). If the dice are manufactured using a reasonably uniform material and they are thrown in good faith, then there is no obvious reason to expect that the outcome should be anything but uniform. The uniformity of a die is also something that, in principle, can be tested empirically: One can toss the die a large number of times and then check that each outcome comes up with more or less the same frequency. The youtube videos "Fair Dice (Part 1)" and "Fair Dice (Part 2)" by Numberphile provide an interesting—albeit very informal—discussion of the subtleties involved in this topic.

3.4.2. Cards. If we think about the uniformity of drawing cards, then it is clear that the extent to which a hand is uniformly random depends on how the deck was shuffled before the hand is drawn and/or the process that is used to draw the hand. For instance, if we draw a hand by picking the first five cards from a deck that is fresh out of the box (i.e., ordered from aces to kings, and spades to hearts), then we will always get exactly the same hand, which is not random at all.

Interestingly enough, the question of how well should a deck be shuffled in order to mix it uniformly has been studied in the scientific literature, in papers by Aldous and Diaconis, Bayer and Diaconis, and Johansson (among others). If you are interested in a completely nontechnical/informal explanation of some of the conclusions of these papers, you can take a look at the youtube video "The Best (and Worst) Ways to Shuffle Cards" by Numberphile, wherein Persi Diaconis (co-author of two of the papers cited above) is interviewed on that question.

3.4.3. Birthdays. In the case of birthdays, it is far more difficult to determine the extent to which the uniform assumption is legitimate. For one thing, in our calculation we dismissed February 29 as a possible birth date. The reason for this is obvious: Since leap years only occur once every four years (with some rare exceptions), it stands to reason that this birthday should be less common than other birthdays. Thus, if we decide to include February 29 in our calculations, then we are forced to do one of the following two things:

(1) Assume that being born on February 29 is as common as any other day. This seems unreasonable, unless we assume that we are only looking at birthdays of people who were all born on the same leap year (say, the year 2000—more on this later).

(2) Take into account that February 29 is less common than other birthdays, which means that we cannot use the uniform measure anymore (in particular, the calculation that we performed in Section **??** is not applicable).

The extent to which removing February 29 from consideration affects the realism of this problem is not immediately obvious.

In order to understand just how less common February 29 is as a birth date, we can look at statistics on the frequency of births on any day of the week. For this purpose, we use data on the number of births in the US on every day from January 1 2000 to December 31 2014 (sourced from the Social Security Administration, see this link). If we compile this data in a plot, then we obtain Figure 3.5. Therein, we have assigned to each date in the year a number from 1 to 366 (this is including February 29) in order from January 1 to December 31; these numbers are on the x-axis of the plot in Figure 3.5. Then, for each of these dates, the value on the y-axis represents the cumulative number of births in the US on that date in the years 2000-2014.

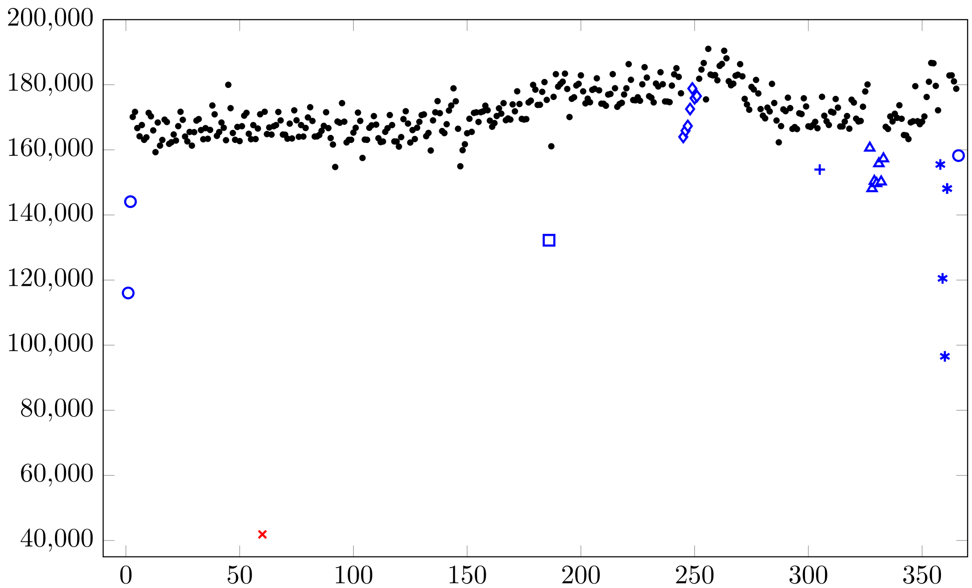

Figure 3.5. Number of births for each date (numbered from 1 to 366) in the US from 2000 to 2014 (inclusively). ∘ = Jan. 1 & 2 and Dec. 31; × = Feb. 29; □ = Jul. 4; ⋄ = Labor Day (Sep. 1–7); + = Oct. 31; △ = Thanksgiving (Nov. 22–28); ∗ = Dec. 23–26.

This plot makes it very clear just how rare it is to have February 29 as a birth date; on the plot, the number of births for that day is represented by a red x. That said, Figure 3.5 clearly shows that February 29 is not the only rare birth date (at least in the US). Indeed, as is made clear by the legend provided in the caption below Figure 3.5, it is much rarer for people to be born on holidays, such as January 1 or July 4. Therefore, it is legitimate to ask: if we felt compelled to exclude February 29 from consideration in Example 3.10 in order to ensure the uniformity of birthdays, should we also exclude other holidays? Apart from that, most other birthdays (marked as black dots) appear to be somewhat uniformly distributed, except for the fact that birth dates seem to be slightly more common in between days 190 and 275 (which roughly corresponds to June–September).

As a final remark, even if the near-uniformity of non-holiday birth dates in Figure 3.5 makes you confident in the legitimacy of the uniformity assumption in Example 3.10, we must still remain very careful about how the sample of people whose birthdays we are comparing was selected. In order to illustrate this, consider the plot in Figure 3.6. This uses the same data that was used to compile Figure

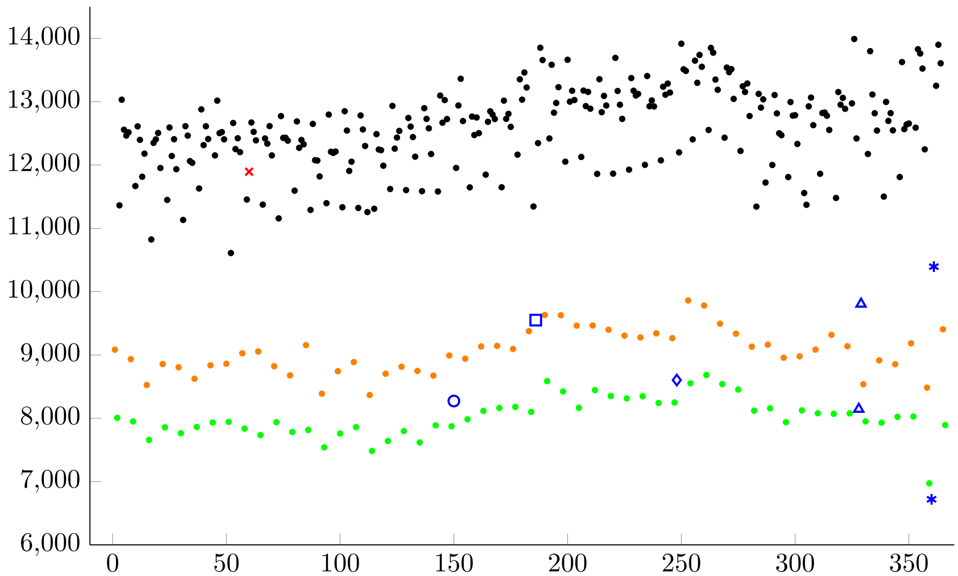

Figure 3.6. Number of births for each date (numbered from 1 to 366) in the US in the year 2000. • = Saturdays; • = Sundays; × = Feb. 29; ◦ = Memorial Day (May 29, 2000); □ = Jul. 4; ⋄ = Labor Day (Sep. 4, 2000); △ = Thanksgiving & Black Friday (Sep. 23 and 24, 2000); $*$ = Dec. 25 & 26.

3.5, but instead of combining births from 2000 to 2014, Figure 3.6 only includes births in the year 2000. In that plot, we see certain similarities with Figure 3.5—for instance, birth dates that coincide with holidays seem once again much less common. However, there are substantial differences as well. Most notably:

(1) Since 2000 was a leap year, the number of births on February 29 on that year looks fairly close to the typical number of births for any other week day.

(2) Birth dates on Saturdays and Sundays are very noticeably less common than most week days.

What this hints at is the following: Even if you believe that Figure 3.5 shows that birth dates in the population as a whole (which includes many different birth years) are fairly uniform, you have to be careful when dealing with probabilities involving small groups (such as 23 people, as in Example 3.10). Indeed, if you have a group of people who were all born the same year—such as 23 college students in the same graduating class—then you will have to somehow take into account that Saturdays and Sundays are much less common as birth dates.

In light of all of these observations, the extent to which one should trust the assumption that birthdays in a given group are uniform and its consequence

$$\mathbf{P}[\text{at least two share a birthday in a group of 23}] \approx 0.507$$

is not a matter of triviality. I will leave whether or not one should accept this computation as approximately correct as an open question for you to ponder. In any case, this course is mostly concerned with developing the ability to perform probability calculations under various assumptions, rather than testing the validity of these assumptions with empirical data. Nevertheless, the data in this section serves as a good cautionary tale to the effect that one should always remain appropriately skeptical when trying to apply simple mathematical models to complex real-world situations.

Chapter 4

Conditioning and Independence

In the previous two chapters, we have begun developing the basics of probability theory with the definitions of the sample space, events, and the probability measure. Then, we have studied what is arguably the simplest probabilistic model, which is the uniform probability measure. As you have seen in the previous chapter, and now also in the first homework, just with these few notions there are already some interesting and nontrivial mathematical problems that arise. However, there are still some fundamental problems involving probabilities that fall outside of the theory that we have built up to this point.

Stated succinctly, one of the main features that is currently missing from our theory is a mechanism that can adequately describe evolution and interactions in random experiments. This leads us to the notion of conditional probability, which is what this chapter is about. Before providing a formal definition of conditional probability, let us spend some time explaining what this notion is attempting to capture in informal terms.

4.1. Evolution and Interactions in Informal Terms

The probability model that we have developed up to now involves a single random experiment whose possible outcomes are recorded in a sample space Ω. Once the experiment is performed, we assume that a single outcome $\omega \in \Omega$ is selected at random. Using the notion of events, we can ask various yes/no questions about this outcome. The probability measure allows us to assess the likelihood of these events. This formalism is appropriate to describe simple experiments that are over in an instant (or in a relatively short amount of time), such as tossing coins, casting dice, drawing hands of cards, etc.

However, many random experiments that are of interest in science and everyday life take very long periods of time to occur. As such an experiment unfolds, it is sometimes possible to make partial observations that enable us to update our initial assessments of the probabilities of outcomes of the experiment. To make this more concrete, consider the following example:

Example 4.1. Suppose that we attempt to predict who will run for president of the United States in 2024, and which party will win the presidency. We could try to determine the probability of each outcome of this experiment right now, but it will be difficult to convince ourselves that any assessment that we make at this time can be very meaningful. There are still way too many unknowns for this, and 2024 is too far into the future.

However, as we gradually approach November 2024, things will happen, such as revelations and scandals, political movements, important rulings, poll results, etc. The political landscape will evolve, and as this happens, so will our assessments of the probabilities of events associated with that experiment. In order for us to know how our probability assessments should change as we witness the political landscape evolving, we have to understand how the events that we are witnessing in between now and election day interact with the probability that any given candidate runs or a given party wins.

In short, in both science and everyday life, much of probabilistic thinking concerns asking how we should use new information to update our current beliefs about the probabilities of events. The mechanism that we use to carry this out must rely on understanding how various events interact with one another. That is, if we know or assume that one particular event occurs, then how does this affect the probability of every other event?

4.2. Conditional Probability

In formal mathematics, the notion of evolution or interaction as described in the previous section is captured by the conditional probability:

Definition 4.2 (Conditional probability)**.** Let $A, B \subset \Omega$ be two events, assuming that $\mathbf{P}[B] > 0$. The conditional probability of A given B, which is denoted by $\mathbf{P}[A|B]$, is defined as

$$\mathbf{P}[A|B] = \frac{\mathbf{P}[A \cap B]}{\mathbf{P}[B]}. \tag{4.1}$$

In words, we can interpret $\mathbf{P}[A|B]$ as "the probability that A occurs once we observe that B has occurred." The requirement that $\mathbf{P}[B] > 0$ in this definition can be justified both from the mathematical and conceptual points of view. On the one hand, $\mathbf{P}[B] > 0$ is required for the formula (4.1) to make sense (i.e., not have a division by zero). On the other hand, $\mathbf{P}[B] > 0$ means conceptually that it is possible that the event B occurs. Thus, if $\mathbf{P}[B] = 0$, then we cannot ever expect to "observe that B has occurred."

There are several different ways to argue that (4.1) is the only sensible mathematical definition of conditional probability. Much like when we defined probability measures, one way to argue that this is the right definition is by invoking empirical frequencies. To give a brief reminder: Consider a random experiment whose outcomes are contained in Ω. If we carry out that same experiment a large number of times, then for every event $A \subset \Omega$, we should have the approximate equality

$$\mathbf{P}[A] \approx \frac{\text{number of times that } A \text{ occurs}}{\text{number of times that we perform the experiment}}.$$

We can give a similar common sense interpretation to conditional probabilities. Suppose that we want to assess the probability that A occurs, having already observed that B has occurred. To do this, we first repeat the experiment many times. However, this time we are only interested in what happens when the event B occurs, and so we discard all the outcomes where B did not occur. In practice, this might look something like Figure 4.1. In that picture, we imagine that the dots

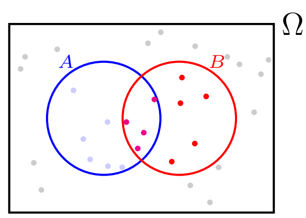

Figure 4.1. Empirical frequency approximation of the conditional probability. Since we assume that B occurs, we discard everything else. The new probability of A under this observation is the probability that A occurs together with B, divided by the probability that B occurs in the first place.

in the sample space correspond to the outcomes that were obtained after performing the experiment a large number of times. Once we discard all outcomes not in B, we are left with the red and magenta dots. Then, we approximate $\mathbf{P}[A|B]$ by looking at the fraction of the remaining outcomes (that is, those in B) for which the event A also occurs; namely, the number of magenta dots divided by the number of magenta and red dots:

$$\begin{aligned}\text{(4.2)}\quad \mathbf{P}[A|B] &\approx \frac{\#\text{ times } A \text{ and } B \text{ occur}}{\#\text{ times } B \text{ occurs}} \\ &= \frac{\left(\dfrac{\#\text{ times } A \text{ and } B \text{ occur}}{\#\text{ times perform experiment}}\right)}{\left(\dfrac{\#\text{ times } B \text{ occurs}}{\#\text{ times perform experiment}}\right)} \approx \frac{\mathbf{P}[A \cap B]}{\mathbf{P}[B]}.\end{aligned}$$

This is consistent with our definition of conditional probability.

In order to make the definition of conditional probability more concrete, consider the following example:

Example 4.3 (Sum of two dice given the second)**.** Suppose that we cast two fair dice. We write the sample space as

$$\Omega := \big\{(i,j) : 1 \leq i, j \leq 6\big\}$$

and assume that $\mathbf{P}$ is the uniform probability measure. Consider the events

$$A = \text{“Is the sum of the two dice equal to 5?”} = \big\{(i,j) \in \Omega : i + j = 5\big\},$$

$$B = \text{"Does the second die land on 5?"} = \big\{(i,j) \in \Omega : j = 5\big\},$$

and

$$C = \text{"Does the second die land on 4?"} = \big\{(i,j) \in \Omega : j = 4\big\}.$$

What are $\mathbf{P}[A]$, $\mathbf{P}[A|B]$, and $\mathbf{P}[A|C]$?

We begin with $\mathbf{P}[A]$. Since we assume that the probability measure is uniform, this probability is $\frac{\#(A)}{36}$. There are four outcomes $(i,j) \in \Omega$ such that $i + j = 5$, namely,

$$A = \big\{(1,4), (2,3), (3,2), (4,1)\big\}. \tag{4.3}$$

Thus, $\mathbf{P}[A] = \frac{4}{36} = \frac{1}{9}$.

Next, we consider $\mathbf{P}[A|B]$ and $\mathbf{P}[A|C]$. Before computing these conditional probabilities using the formula in (4.1), however, it can be instructive to take a moment to ponder what these two conditional probabilities should look like using only the conceptual meaning of the notion. Both events B and C provide information on the outcome of the second die. It stands to reason that this knowledge should influence the probability that the sum of both dice is equal to 5. More specifically:

(1) If the event B occurs (i.e., the second die lands on 5), then the sum of the two dice can only be greater than 5. Thus, in this case, the event A should be impossible, which means that $\mathbf{P}[A|B]$ should be zero.

(2) If the event C occurs (i.e., the second die lands on 4), then the sum of the two faces can be equal to 5 (i.e., if the first die lands on 1). Given that a proportion of 8/9 outcomes of the two dice give a sum different from 5, and that the first die landing on 1 is not very unlikely, this knowledge seems like it should increase the probability that A occurs. Thus, it is natural to expect that $\mathbf{P}[A|C] > \mathbf{P}[A]$.

Let us now carry out these computations using the formula (4.1). For this, we note that we can write the sets

$$\begin{aligned}
B &= \big\{(1,5), (2,5), (3,5), (4,5), (5,5), (6,5)\big\},\\
C &= \big\{(1,4), (2,4), (3,4), (4,4), (5,4), (6,4)\big\},\\
A \cap B &= \varnothing,\\
A \cap C &= \big\{(1,4)\big\}.
\end{aligned}$$

Thus, by definition of conditional probability and the uniform probability measure,

$$\mathbf{P}[A|B] = \frac{\mathbf{P}[A \cap B]}{\mathbf{P}[B]} = \frac{\#(A \cap B)/36}{\#(B)/36} = \frac{\#(A \cap B)}{\#(B)} = \frac{0}{6} = 0,$$

as we had suspected. Similarly,

$$\mathbf{P}[A|C] = \frac{\mathbf{P}[A \cap C]}{\mathbf{P}[C]} = \frac{\#(A \cap C)/36}{\#(C)/36} = \frac{\#(A \cap C)}{\#(C)} = \frac{1}{6},$$

which is also consistent with our intuition since $\mathbf{P}[A|C] = \frac{1}{6} > \frac{1}{9} = \mathbf{P}[A]$.

4.3. Independence

Apart from providing a means of updating one's belief about the likelihood of events once new information is obtained, conditional probability can be viewed as a tool to analyze how various events interact with one another. That is, the sizes of the distances between the probabilities

$$\big|\mathbf{P}[A|B]-\mathbf{P}[A]\big| \qquad \text{and} \qquad \big|\mathbf{P}[B|A]-\mathbf{P}[B]\big|$$

quantify how much influence B has on the occurrence of A and vice versa. For example, if we let A, B, and C be defined as in Example 4.3, then the fact that

$$\big|\mathbf{P}[A|B]-\mathbf{P}[A]\big| = \left|\frac{1}{6}-\frac{1}{9}\right| = \frac{1}{18} \qquad \text{and} \qquad \big|\mathbf{P}[A|C]-\mathbf{P}[A]\big| = \frac{1}{9}$$

serves to quantify the extent to which observing the value of the second die changes the probability of the sum being equal to 5.

Notation 4.4. Here, we recall that

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

is used to denote the absolute value of some number x, so that the distance between two numbers x and y on the real line is equal to $|x-y|$.

This notion raises an interesting question: What happens if

$$\big|\mathbf{P}[A|B]-\mathbf{P}[A]\big| = 0 \qquad \text{and} \qquad \big|\mathbf{P}[B|A]-\mathbf{P}[B]\big| = 0?$$

This question leads to the notion of independence:

Definition 4.5. We say that two events A and B such that $\mathbf{P}[A], \mathbf{P}[B] > 0$ are independent if

$$\mathbf{P}[A|B] = \mathbf{P}[A] \qquad \text{and} \qquad \mathbf{P}[B|A] = \mathbf{P}[B], \tag{4.4}$$

or, equivalently, if

$$\mathbf{P}[A \cap B] = \mathbf{P}[A]\mathbf{P}[B]. \tag{4.5}$$

(**Note.** Proving that (4.4) and (4.5) are equivalent amounts to a simple exercise using the definition of conditional probability; I encourage you to carry it out to familiarize yourself with the definition.)

In words, A and B are independent if observing that A occurs does not impact the probability that B will occur, and vice versa.

Example 4.6 (Two hands in between a reshuffle)**.** Suppose that we carry out the following sequence of steps:

(1) Shuffle a deck of cards.
(2) Pick the first five cards from the deck, and record the hand that is thus obtained.
(3) Put the cards back into the deck and reshuffle it.
(4) Pick the first five cards from the deck, and record the hand that is thus obtained.

In short, we pick two five-card hands, making sure to reshuffle the deck in between. Suppose that we define the events

$$A = \text{“Is the first hand a full house?”}$$

and

$$B = \text{“Is the second hand a full house?”}$$

Given that the deck was reshuffled between picking the two hands, it stands to reason that the result of either hand should have no impact whatsoever on the result of the other hand. Thus, in this scenario it seems reasonable to assume that A and B are independent.

While the conditions (4.4) and (4.5) are equivalent from the mathematical point of view, it is still interesting to keep them both in mind. On the one hand, the conditional probability definition (4.4) makes the conceptual meaning of independence very clear: observing one of the two events has no impact at all on the probability of the other event. On the other hand, (4.5) is arguably more useful in practical applications; while this may not be immediately obvious to you at this time, you will see throughout this course as you do more and more exercises that (4.5) is ubiquitous in computations involving independence.

As a final notion in this section, it turns out that independence can be extended to any number of events:

Definition 4.7. We say that events $A_1, A_2, \ldots, A_n \subset \Omega$ such that $\mathbf{P}[A_i] > 0$ are independent if for any $1 \leq i \leq n$ and $j_1, j_2, \ldots, j_k \neq i$, one has

$$\mathbf{P}[A_i | A_{j_1} \cap A_{j_2} \cap \cdots \cap A_{j_k}] = \mathbf{P}[A_i].$$

Equivalently, $A_1, A_2, \ldots, A_n \subset \Omega$ are independent if for every choice of distinct $1 \leq i_1, i_2, \ldots, i_k \leq n$, one has

$$\mathbf{P}[A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}] = \mathbf{P}[A_{i_1}]\mathbf{P}[A_{i_2}] \cdots \mathbf{P}[A_{i_k}]. \tag{4.6}$$

Remark 4.8. It is natural to wonder why we need to introduce an additional definition for the independence of multiple events. For instance, if we know that

(1) A is independent of B,
(2) B is independent of C, and
(3) A is independent of C,

then does this necessarily mean that A, B, C are independent? Perhaps surprisingly, the answer is no. While we will not further elaborate on this point here (as you will study this phenomenon in an upcoming homework), this counterexample illustrates the fact that one has to be a bit careful when dealing with independence.

We finish this section with an example that highlights an interesting connection between independence and the uniform probability measure, as well as illustrates how (4.5) and (4.6) are often used in practice:

Example 4.9 (Independence and uniformity)**.** Suppose that we are interested in a random experiment that can give finitely many possible outcomes, let us say the elements in the set

$$A = \{\omega_1, \omega_2, \ldots, \omega_n\}.$$

Suppose that, for whatever reason, we feel confident that the outcome of this experiment is uniform, so that the probability of observing any individual outcome ω_i is $\frac{1}{n}$, i.e., one over the total number of outcomes. For instance, this situation could describe the tossing of a fair coin (in which case $A = \{h, t\}$), the tossing of a fair die (in which case $A = \{1, 2, 3, 4, 5, 6\}$) or assigning a birthday uniformly (in which case $A = \{1, 2, \ldots, 365\}$ if we omit Feb. 29).

Now fix some integer k, and suppose that we perform the aforementioned experiment k times in a row. A good sample space for this new experiment is

$$\Omega = \{(a_1, a_2, \ldots, a_k) : a_i \in A\},$$

where for each $1 \leq i \leq k$, a_i represents the outcome of the i^{th} time we performed the experiment. Since the outcome of each individual a_i is uniform on the set A, does this automatically mean that the outcome of the multiple experiments is uniform on the set of k-tuples in Ω?

To answer this question, we first set up some notations: For the remainder of this example, we use the shorthand

$$\{T_i = \omega\} = \big\{(a_1, a_2, \ldots, a_k) \in \Omega : a_i = \omega\big\}; \tag{4.7}$$

in words, this is the event that the i^{th} trial of the experiment gives the outcome ω. By the product rule, we know that $\#(\Omega) = n^k$. Thus, if we want to argue that the experiment under consideration is uniform, then we have to show that for any outcome $(a_1, a_2, \ldots, a_k) \in \Omega$, one has

$$\begin{aligned} &\mathbf{P}[(a_1, a_2, \ldots, a_k)] \\ &\qquad = \mathbf{P}[\{T_1 = a_1\} \cap \{T_2 = a_2\} \cap \cdots \cap \{T_k = a_k\}] = \frac{1}{\#(\Omega)} = \frac{1}{n^k} \end{aligned} \tag{4.8}$$

If we assume that successive trials of the experiment are independent, in the sense that the events

$$\{T_1 = a_1\}, \{T_2 = a_2\}, \cdots, \{T_k = a_k\}$$

are independent, then (4.8) is true. Indeed, in such a case we have that

$$\begin{aligned} &\mathbf{P}[\{T_1 = a_1\} \cap \{T_2 = a_2\} \cap \cdots \cap \{T_k = a_k\}] \\ &\qquad = \mathbf{P}[T_1 = a_1]\mathbf{P}[T_2 = a_2] \cdots \mathbf{P}[T_k = a_k] = \frac{1}{n^k}. \end{aligned}$$

In general, however, there is no reason to expect that

$$\mathbf{P}[\{T_1 = a_1\} \cap \{T_2 = a_2\} \cap \cdots \cap \{T_k = a_k\}]$$

should be equal to n^{-k}. Indeed, if the result of (say) the i^{th} trial somehow influences the outcome of the other trials, then the probability measure on the k trials $(a_1, \ldots, a_k)$ need not be uniform.

Remark 4.10. The above example provides additional justification or context for some of the uniform models that we have studied in the previous chapter. For example, the assumption that the result of tossing two dice

$$\big\{(d_1, d_2) : 1 \leq d_1, d_2 \leq 6\big\}$$

is uniform can be justified with the idea that the result of each individual die (i.e., d_1 and d_2) is uniform on $\{1, 2, 3, 4, 5, 6\}$, and that the result of the two dice are independent of one another. In most situation this is perfectly reasonable: Unless someone is somehow cheating, then there is no reason to expect that the result of one die toss should somehow influence the other.

Remark 4.11. The above example is also a good opportunity to discuss some standard so-called "abuses of notation" in probability. Up to this point, we have always been very thorough and explicit in the notations used to define sets and events. For the most part, this is a good thing, because a fully rigorous mathematical treatment of probability is not possible without unambiguous definitions and notations.

However, as you become more acquainted with the basic theory, you may start to notice that there is often some redundancy in the set theory notation. In such cases, set theory notation can become more of a burden. That is, involving tedious writing without providing more clarity. As a remedy to this situation, mathematicians working in probability have come up with a number of notational shorthands to denote various probabilistic objects. In the previous example, I have used several instances of that:

On the one hand, we had the notation (4.7), wherein $\{T_i = \omega\}$ denotes the set of k-tuples $(a_1, \ldots, a_k)$ in Ω such that $a_i = \omega$. This is an abuse of notation because, although we use the bracket notation $\{\}$, the statement "$\{T_i = \omega\}$" taken literally is not correct notation for a set. Without the additional context that we are considering a sample space made of elements of the form $(a_1, \ldots, a_k)$ and that T_i refers to the outcome of the i^{th} trial, a_i, the statement "$T_i = \omega$" alone does not constitute an unambiguous identification of mathematical objects. In the specific context of the above example, however, its meaning is obvious once clarified.

On the other hand, we have used the notation $\mathbf{P}[T_i = \omega]$ to mean the probability of the event denoted $\{T_i = \omega\}$, and we have used $\mathbf{P}[(a_1, \ldots, a_k)]$ to mean the probability of the event that contains the single element $\{(a_1, \ldots, a_k)\}$. Technically, probability measures are actually only defined on events, which are sets. That being said, the notations

$$\mathbf{P}[\{T_i = \omega\}] \qquad \text{and} \qquad \mathbf{P}[\{(a_1, \ldots, a_k)\}]$$

are widely considered to be untidy because of the overabundance of brackets. Thus, since removing the curly brackets does not make the expression unintelligible, many prefer to use expressions like $\mathbf{P}[T_i = \omega]$ and $\mathbf{P}[(a_1, \ldots, a_k)]$.

Going forward, we will increasingly use these two abuses of notation without further comment.

4.4. The Law of Total Probability

Up until now, we have been using the conditional probability as a means of updating probabilities of events once new information is obtained, and as a method to characterize the lack of interactions between events, the latter of which led to the notion of independence. In contrast to that, in this section we study the usefulness of conditional probability as a computational tool that can be used to substantially reduce the complexity of certain problems. These considerations lead us to what is by far one of the most important (i.e., to make sure that you understand and remember for exams) computational tools in all of probability, which is the law of total probability:

Proposition 4.12 (Law of total probability)**.** *Let $B_1, B_2, \ldots, B_n \subset \Omega$ be a collection of events that satisfy the following three conditions:*

(1) $\mathbf{P}[B_i] > 0$ *for every* $1 \leq i \leq n$,

(2) $B_i \cap B_j = \varnothing$ *whenever* $i \neq j$, *and*

(3) $\bigcup_{i=1}^{n} B_i = \Omega$.

Then, for any event $A \subset \Omega$, one has

$$\mathbf{P}[A] = \sum_{i=1}^{n} \mathbf{P}[A|B_i]\mathbf{P}[B_i]. \tag{4.9}$$

There are three things that we should discuss regarding this result, namely: Why is it true? What does it mean intuitively/conceptually? What makes it so useful (i.e., why did I say that it is by far one of the most important computational tool in probability)? We now answer these three questions.

4.4.1. Proof of the Law of Total Probability. The proof of the law of total probability is intimately connected to a result in the first homework, which was called the case-by-case property therein. Indeed, the latter states that, under the conditions stated in the law of total probability, one has

$$\mathbf{P}[A] = \sum_{i=1}^{n} \mathbf{P}[A \cap B_i]. \tag{4.10}$$

As explained in the hint to that problem in the homework, this property is most easily understood when accompanied by an illustration, such as in Figure 4.2 below. Once (4.10) is established, we obtain (4.9) by a more or less trivial manipulation

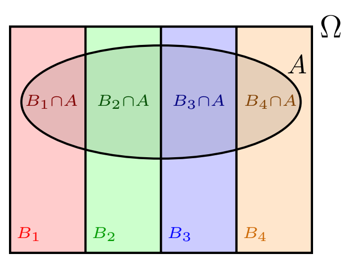

Figure 4.2. Partition of Ω and A according to the B_i's. This makes it abundantly clear that A can be written as a union of mutually exclusive events, obtained by intersecting A with the B_i's.

that follows directly from the definition of conditional probability:

$$\mathbf{P}[A|B_i]\mathbf{P}[B_i] = \frac{\mathbf{P}[A \cap B_i]}{\mathbf{P}[B_i]}\mathbf{P}[B_i] = \mathbf{P}[A \cap B_i].$$

4.4.2. Interpretation of the Law of Total Probability. Now that we understand why the law of total probability is true from the purely mathematical point of view, it is worth taking the time to parse the statement from the intuitive point of view. Indeed, being comfortable with the conceptual meaning of the result will help you develop the ability to recognize when and how to apply it in practice.

The first thing to discuss are the assumptions that we have made on the events B_i. The assumptions in question state that the B_i's form a partition of the sample space, meaning that

(1) every outcome in Ω is contained in at least one of the B_i's (because their union is the sample space), and
(2) an outcome cannot be contained in two distinct B_i's simultaneously (because they are mutually exclusive).

Graphically, this can be illustrated as in Figure 4.2. We see that the sample space therein is partitioned into four distinct regions labelled B_1 to B_4 that have no overlap; hence every outcome in Ω is contained in no more and no less than one of the events B_i. With this said, looking back at the statement of the theorem, we see that (4.9) claims that probability of any event A can be written as the sum of the conditional probabilities of A given all the events B_i, which are themselves multiplied by the probability of B_i. Conceptually, we can think of this as a way to formalize a case-by-case analysis of the problem of computing $\mathbf{P}[A]$ according to the partition given by the sets B_i.

More specifically, suppose that we look at one of the events B_i, say for instance B_1. If we could somehow observe that B_1 has occurred, then the probability of A would become the conditional probability $\mathbf{P}[A|B_1]$. However, unless we actually make the observation that B_1 has occurred, then we cannot simply assume that it has and declare that $\mathbf{P}[A] = \mathbf{P}[A|B_1]$. Therefore, in order to account for every possibility, we must also ask ourselves: What if, instead of B_1, it is B_2 that occurs? Or, what if it is B_3 that occurs, and so on...

In this context, the law of total probability accounts for every possible scenario regarding which B_i will occur: In (4.9), we combine the conditional probabilities of A for every possible outcome of which B_i occurs. Moreover, each contribution of the conditional probability $\mathbf{P}[A|B_i]$ in (4.9) is multiplied by the probability that B_i occurs, namely $\mathbf{P}[B_i]$. This can be explained by the fact that the probability that we are allowed to replace $\mathbf{P}[A]$ by $\mathbf{P}[A|B_i]$ is the probability that B_i occurs in the first place.

4.4.3. Why is the Law of Total Probability Useful? In order to concretize the intuitive description of the total probability that I just gave, as well as showcase its usefulness, we now look at two examples that illustrate how to use it and how not to use it.

In order to get a sense for this, we look once again at the statement of the law of total probability, which is as follows:

$$\mathbf{P}[A] = \sum_{i=1}^{n} \mathbf{P}[A|B_i]\mathbf{P}[B_i].$$

If we want to use this tool to compute $\mathbf{P}[A]$, then it better be the case that the probabilities $\mathbf{P}[A|B_i]$ and $\mathbf{P}[B_i]$ are actually easier to compute than $\mathbf{P}[A]$ itself. Otherwise, we are simply wasting our time writing $\mathbf{P}[A]$ as some more complicated expression involving a (possibly) large sum. To give an example of how a mindless application of the property can be useless, we consider the following scenario:

Example 4.13 (Full house and aces)**.** Consider the experiment of drawing a five-card hand from a standard deck, assuming that $\mathbf{P}$ is the uniform probability measure. Consider the events

$$A = \text{“is the hand a full house?”}$$

and

$$B = \text{“does the hand contain at least one ace?”}$$

By the law of total probability, we can write

$$\mathbf{P}[A] = \mathbf{P}[A|B]\mathbf{P}[B] + \mathbf{P}[A|B^c]\mathbf{P}[B^c];$$

indeed, B and B^c are clearly mutually exclusive, and their union is the whole sample space. However, it is not clear why one would want to write the probability of A in this way. Indeed, if you spend a bit of time thinking about it, then you will note that the probabilities $\mathbf{P}[A|B]$, $\mathbf{P}[B]$, $\mathbf{P}[A|B^c]$, and $\mathbf{P}[B^c]$ are not any easier to compute than $\mathbf{P}[A]$ itself. Thus, this “application” of the total probability rule only serves to make an already challenging problem even more difficult.

In sharp contrast to this, let us now consider a simple scenario where the law of total probability is actually useful:

Example 4.14 (Die and coins)**.** Recall the experiment wherein we begin by casting a six-sided die, and if the result of the die is equal to k, then we flip a coin k times in a row, assuming that the order of the coin flips matters. A good probability space for this is

$$\Omega = \big\{(d, c_1, \ldots, c_d) : 1 \leq d \leq 6, \text{ and } c_1, \ldots, c_d \in \{h, t\}\big\},$$

where d represents the outcome of the die and $c_1, \ldots, c_d$ represent the outcome of the d coin flips in order. We assume that the results of the die toss and coin flips are fair and independent in the following sense:

(1) For every $1 \leq k \leq 6$ and $i \geq 1$,

$$\mathbf{P}[d = k] = \frac{1}{6} \qquad \text{and} \qquad \mathbf{P}[c_i = h] = \mathbf{P}[c_i = t] = \frac{1}{2}.$$

In words, the result of the die and each coin flip are uniform.

(2) For every number $1 \leq k \leq 6$ and set A, the events

$$\{d = k\} \qquad \text{and} \qquad \{(c_1, \ldots, c_k) \in A\}$$

are independent. In words, apart from determining the number of times that the coins are flipped, the result of the die does not otherwise influence the results of the coin flips.

(3) For any sequence of heads/tails $\ell_1, \ldots, \ell_k \in \{h, t\}$, the events

$$\{c_1 = \ell_1\}, \ldots, \{c_k = \ell_k\}$$

are independent. In words, the results of successive coin flips are independent of one another.

Suppose that we play a game of chance based on this experiment, with the following rule: You win the game if the (random) number of coin tosses that come up heads is strictly greater than the number of coins that come up tails. If we denote the event

$$W = \text{“do you win the game?”}$$

then what is $\mathbf{P}[W]$? What makes this problem tricky is that there are many different scenarios that might lead to victory, and it might not be immediately obvious at first glance how to analyze them all systematically.

Whenever faced with a complex problem of this form, it is a good reflex to ask yourself the following question: “Is there a particular feature of the experiment's outcome that, if you could know it in advance, would simplify the analysis of the problem?” If the answer to that question is yes, then it is quite likely that applying the probability rule where the evens B_i represent the possible outcomes of the “particular feature” mentioned in the previous question.

To make this concrete, we consider asking this question with the specific scenario in this example: What makes this problem tricky is that we don't know in advance how many heads are required to win the game. For instance, if the coin is flipped only once, then one heads is enough to win; conversely, if the coin is flipped three times, then one flip is not enough. In particular, if we could know in advance what will be the result of the die, then we would know exactly how many heads are required to win, which would simplify the problem. For instance, if I tell you in advance that the result of the die will be three, then you know that you need two heads or more to win. Thus, it is probably a good idea to use the law of total probability to write

$$\mathbf{P}[W] = \sum_{k=1}^{6} \mathbf{P}[W|d = k]\mathbf{P}[d = k]. \tag{4.11}$$

We are allowed to to this because the events $\{d = 1\}, \{d = 2\}, \ldots, \{d = 6\}$ satisfy the assumptions of the law of total probability: Each of them have a positive probability of $\frac{1}{6}$, they are mutually exclusive, and their union is equal to the whole sample space Ω.

In order for (4.11) to really pay off, it must be the case that the computations of $\mathbf{P}[W|d = k]$ and $\mathbf{P}[d = k]$ are easier than that of $\mathbf{P}[W]$ directly. This turns out to be the case: On the one hand, by assumption that the die is fair, we know that $\mathbf{P}[d = k] = \frac{1}{6}$ for any choice of k; hence (4.11) becomes

$$\mathbf{P}[W] = \frac{1}{6} \sum_{k=1}^{6} \mathbf{P}[W|d = k].$$

On the other hand, the computation of $\mathbf{P}[W|d = k]$ should be made simpler by the fact that $d = k$ identifies the number of coins that are flipped. For example, we can interpret

$$\begin{aligned}\mathbf{P}[W|d=3] &= \mathbf{P}[\text{at least two coins (out of 3) are heads}|d=3] \\ &= \mathbf{P}[\text{at least two coins (out of 3) are heads}] \\ &= \frac{{}_3C_2+1}{2^3} = \frac{1}{2},\end{aligned}$$

where the first equality comes from the fact that $d = 3$ means three coins are flipped, the second equality comes from the assumption that the result of coin flips (apart from how many coins are flipped) is independent of the result of the die, and the third equality follows from the fact that the uniformity of individual coin flips and the independence of distinct flips implies that the result of the three flips are uniform on all 3-tuples (c_1, c_2, c_3). More generally, we have that

$$\begin{aligned}\mathbf{P}[W|d=1] &= \frac{1}{2} \\ \mathbf{P}[W|d=2] &= \frac{1}{4} \\ \mathbf{P}[W|d=3] &= \frac{1}{2} \\ \mathbf{P}[W|d=4] &= \frac{{}_4C_3+1}{2^4} = \frac{5}{16} \\ \mathbf{P}[W|d=5] &= \frac{{}_5C_3+{}_5C_4+1}{2^5} = \frac{1}{2} \\ \mathbf{P}[W|d=6] &= \frac{{}_6C_4+{}_6C_5+1}{2^6} = \frac{11}{32}.\end{aligned}$$

Putting everything together, we conclude that

$$\mathbf{P}[W] = \frac{1}{6}\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{2}+\frac{5}{16}+\frac{1}{2}+\frac{11}{32}\right) = \frac{77}{192} \approx 0.401.$$

The above example is emblematic of the usefulness of the law of total probability, in the sense that is serves as a general blueprint of how to successfully apply the latter in practice. The general process of doing so could be summarized as follows:

(1) Identify a partition of events $B_1, B_2, \ldots, B_n \subset \Omega$ such that, if we knew in advance which of the B_i's occurred, then the problem would become much simpler to analyze.
(2) Compute $\mathbf{P}[A|B_i]$ and $\mathbf{P}[B_i]$ (which will be easier than $\mathbf{P}[A]$).

Going forward in these notes, the homework, and exams, you will encounter a number of situations where the law of total probability naturally arises.

4.5. Bayes' Rule

In this last section in our chapter on conditional probability, we study a deceptively simple property called Bayes' rule:

Proposition 4.15 (Bayes' rule)**.** *If $A, B \subset \Omega$ are such that $\mathbf{P}[A], \mathbf{P}[B] > 0$, then*

$$\mathbf{P}[A|B] = \mathbf{P}[B|A] \cdot \frac{\mathbf{P}[A]}{\mathbf{P}[B]} \tag{4.12}$$

The proof of this formula amounts to a triviality: By definition of conditional probability, we have that

$$\begin{aligned}\mathbf{P}[A|B] = \frac{\mathbf{P}[A\cap B]}{\mathbf{P}[B]} = \frac{\mathbf{P}[A\cap B]}{\mathbf{P}[B]}\cdot 1 = \frac{\mathbf{P}[A\cap B]}{\mathbf{P}[B]}\cdot\frac{\mathbf{P}[A]}{\mathbf{P}[A]}\\ = \frac{\mathbf{P}[A\cap B]}{\mathbf{P}[A]}\cdot\frac{\mathbf{P}[A]}{\mathbf{P}[B]} = \mathbf{P}[B|A]\cdot\frac{\mathbf{P}[A]}{\mathbf{P}[B]}.\end{aligned}$$

Therefore, from the purely mathematical point of view, this is not an extraordinarily profound result. That being said, Bayes' rule nevertheless turns out to be very useful in a number of problems. We now present two examples of this.

The first example is one that illustrates the sometimes confusing or counterintuitive nature of the difference between $\mathbf{P}[A|B]$ and $\mathbf{P}[B|A]$:

Example 4.16. Suppose that you are developing a new test to detect a disease. We denote the events

$$D = \text{“Does the patient have the disease?”}$$

and

$$T = \text{“Does the patient test positive?”}$$

We assume the following:

(1) Data complied over several decades by the CDC suggests that $\mathbf{P}[D] = 0.01$.
(2) Experimental data with patients who were already known to have the disease suggests that $\mathbf{P}[T|D] = 0.9$.
(3) Experimental data with a control group of patients known not to have the disease suggests that $\mathbf{P}[T|D^c] = 0.09$.

If a patient that was not previously known to either have or not have the disease tests positive, then what is the probability that they actually have the disease?

At first glance, many people feel that the answer should be 0.9. Indeed, $\mathbf{P}[T|D] = 0.9$ seems to indicate that the test is 90% accurate at detecting the presence of the disease. However, this is not the answer to the question being asked: What the question is asking for is $\mathbf{P}[D|T]$. Given that the problem statement gives us the opposite conditional probability as an assumption, this is a perfect opportunity to use Bayes' rule:

$$\mathbf{P}[D|T] = \mathbf{P}[T|D]\cdot\frac{\mathbf{P}[D]}{\mathbf{P}[T]} = 0.9\cdot\frac{0.01}{\mathbf{P}[T]}.$$

We note that $\mathbf{P}[T]$ is not given directly by the assumptions. However, we can obtain it by the law of total probability:

$$\begin{aligned}\mathbf{P}[T] &= \mathbf{P}[T|D]\mathbf{P}[D] + \mathbf{P}[T|D^c]\mathbf{P}[D^c]\\ &= \mathbf{P}[T|D]\mathbf{P}[D] + \mathbf{P}[T|D^c]\big(1-\mathbf{P}[D]\big)\\ &= 0.9\cdot 0.01 + 0.09\cdot 0.99\\ &= 0.0981.\end{aligned}$$

Putting everything back together, we conclude that

$$\mathbf{P}[D|T] = 0.9\cdot\frac{0.01}{0.0981}\approx 0.1,$$

which is substantially smaller than 0.9.

The second example illustrates that, in some situations, although we want some conditional probability $\mathbf{P}[A|B]$, the probabilistic model that we are considering is such that it is instead much more natural to compute $\mathbf{P}[B|A]$:

Example 4.17 (Determining the coin)**.** Consider the following game of chance: Suppose that I have two coins, say, c_1 and c_2. The first coin, c_1, is fair, that is, it comes up heads or tails with probability $\frac{1}{2}$ when flipped. The second coin, c_2, is biased in favor of tails, so that

$$\mathbf{P}[c_2 = h] = \frac{1}{3} \qquad \text{and} \qquad \mathbf{P}[c_2 = t] = \frac{2}{3}.$$

The game has the following steps:

(1) At the beginning of the game, I choose one of my two coins (without telling you the result) uniformly at random. That is, if c is the coin that I choose, then we have that
$$\mathbf{P}[c = c_1] = \mathbf{P}[c = c_2] = \frac{1}{2}.$$
(2) Once I have chosen one of my two coins, I flip it 10 times independently and record the sequence of heads/tails thus obtained.
(3) Then, I tell you the sequence of heads/tails, and your objective is to use this information alone to guess which of my two coins I chose in the beginning.

A good sample space for this experiment could be

$$\Omega = \big\{(c, F) : c \in \{c_1, c_2\},\ F = (f_1, \ldots, f_{10}),\ f_i \in \{h, t\}\big\}$$

where c represents the coin that I chose, and F represents the resulting sequence of 10 coin flips. As per the assumptions above (i.e., the independence of the coin flips and the odds of both coins), we have that

$$\mathbf{P}[F = (f_1, \ldots, f_{10})|c = c_1] = \frac{1}{2^{10}};$$

and if we let $\#_h(f_1, \ldots, f_{10})$ denote the number of heads in the sequence $f_1, \ldots, f_{10}$ and $\#_t(f_1, \ldots, f_{10})$ denote the number of tails, then

$$\mathbf{P}[F = (f_1, \ldots, f_{10})|c = c_2] = \left(\frac{1}{3}\right)^{\#_h(f_1,\ldots,f_{10})} \left(\frac{2}{3}\right)^{\#_t(f_1,\ldots,f_{10})}.$$

Suppose that we play one round of this game, and the sequence of heads/tails from the 10 coin flips is the following:

$$F = (h, h, t, h, t, t, h, t, t, t). \tag{4.13}$$

Given that there are more tails than heads in this sequence, you decide to guess that the coin was c_2 (i.e., the coin biased towards tails). What is the probability that your guess is correct given the observation in (4.13)? That is, what is

$$\mathbf{P}[c = c_2|F = (h, h, t, h, t, t, h, t, t, t)]?$$

Answering this question directly is a bit difficult, especially when compared to the opposite conditioning. That is, it is much easier to compute the probability of a particular sequence of heads/tails if we know which coin is being used, than the reverse. Thus, this is a perfect opportunity to use Bayes' rule:

$$\begin{aligned}&\mathbf{P}[c = c_2 | F = (h,h,t,h,t,t,h,t,t,t)]\\&\qquad = \mathbf{P}[F = (h,h,t,h,t,t,h,t,t,t) | c = c_2] \frac{\mathbf{P}[c = c_2]}{\mathbf{P}[F = (h,h,t,h,t,t,h,t,t,t)]}.\end{aligned}$$

Since there are 4 heads and 6 tails in the sequence (h,h,t,h,t,t,h,t,t,t), if we assume that we are using the second coin, then we have that

$$\mathbf{P}[F = (h,h,t,h,t,t,h,t,t,t) | c = c_2] = \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^6 \approx 0.001084.$$

By assumption, the coin c_2 was chosen with probability $\frac{1}{2}$; hence

$$\mathbf{P}[c = c_2] = \frac{1}{2}.$$

The only quantity that remains to be computed is

$$\mathbf{P}[F = (h,h,t,h,t,t,h,t,t,t)].$$

For this, the law of total probability will come in handy, since computing the probability of a sequence of flips is easy once we know which coin was used. Thus,

$$\begin{aligned}&\mathbf{P}[F = (h,h,t,h,t,t,h,t,t,t)]\\&= \mathbf{P}[F = (h,h,t,h,t,t,h,t,t,t) | c = c_1]\mathbf{P}[c = c_1]\\&\qquad\qquad + \mathbf{P}[F = (h,h,t,h,t,t,h,t,t,t) | c = c_2]\mathbf{P}[c = c_2]\\&= \frac{1}{2^{10}} \cdot \frac{1}{2} + \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^6 \cdot \frac{1}{2}\\&\approx 0.001030.\end{aligned}$$

Putting everything together, we therefore obtain that

$$\begin{aligned}\mathbf{P}[c = c_2 | F = (h,h,t,h,t,t,h,t,t,t)] &= \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^6 \cdot \frac{\frac{1}{2}}{\frac{1}{2^{10}} \cdot \frac{1}{2} + \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^6 \cdot \frac{1}{2}}\\&\approx 0.5260.\end{aligned}$$

Consequently, we see that it is indeed wise to guess that $c = c_2$ because that guess is more likely than not to be correct. That said, it is interesting to note that the odds of the guess being incorrect are still very close to 50%.

In summary, the ability to successfully apply Bayes' rule in practice can be essentially reduced to the following:

(1) Develop the reflex to ask "Is the problem asking to compute a conditional probability that is the opposite of what is given and/or natural to compute?"

(2) If that is the case, remember that Bayes' rule expresses the relationship between "opposite" conditional probabilities as

$$\mathbf{P}[A|B] = \mathbf{P}[B|A] \cdot \frac{\mathbf{P}[A]}{\mathbf{P}[B]}.$$

(3) When applying Bayes' rule, keep in mind the law of total probability. More specifically, if you are given and/or can easily compute $\mathbf{P}[B|A]$, $\mathbf{P}[B|A^c]$, and $\mathbf{P}[A]$, then you can write

$$\mathbf{P}[B] = \mathbf{P}[B|A]\mathbf{P}[A] + \mathbf{P}[B|A^c]\mathbf{P}[A^c].$$

With this "algorithm" in hand, you will be in a good position to solve many problems that involves the use of Bayes' rule.

4.6. Two Additional Remarks on Bayes' Rule (Bonus)

While we have now covered all of the material that I wanted to discuss regarding Bayes' rule (and more generally conditional probability), in this last bonus section I provide two additional remarks on the latter. No aspect of this section is required to be able to solve the problems involving Bayes' rule in this course.

That being said, if you have the time and intend to continue studying probability and statistics after this course, then I encourage you to take a look. Indeed, it is no exaggeration to say that, despite its trivial mathematical simplicity, Bayes' rule is one of the most fundamental ideas in all of probability and statistics. While this section is by no means an exhaustive explanation of why that is, my hope is that the discussion herein will encourage you to think more deeply about the result.

4.6.1. Reconciling Example 4.16 With Intuition. Example 4.16 is a very well-known computation in elementary probability that is notorious for being counterintuitive. The specific numbers in the example are taken from a quiz that was given to a number of gynecologists as part of a training session.[1] As it turns out, only 21% of the test takers correctly identified that $\mathbf{P}[D|T] \approx 0.1$, and 47% stated that $\mathbf{P}[D|T] = 0.9$. Thus, the subtleties involved with Bayes' rule is something that even the most specialized and highly trained individuals can find deeply counterintuitive.

If you also initially felt that the answer should have been 0.9, or still have difficulty making sense of the actual answer intuitively, then it is worth taking some time to understand why $\mathbf{P}[D|T]$ and $\mathbf{P}[T|D]$ are so different from one another in this scenario. Indeed, armed with Bayes' formula, one can come to the realization that the conclusion of Example 4.16 is actually not surprising in the least, provided that one pays attention to the right details.

By replicating the applications of Bayes' and the law of total probability in Example 4.16, we have the relationship

$$\mathbf{P}[D|T] = \frac{\mathbf{P}[T|D]\mathbf{P}[D]}{\mathbf{P}[T|D]\mathbf{P}[D] + \mathbf{P}[T|D^c]\big(1 - \mathbf{P}[D]\big)}. \tag{4.14}$$

In order for a fraction of two numbers to be very small, the numerator must be much smaller than the denominator (i.e., you must divide a number by a much bigger number). In this context, the fact that the probability (4.14) is small can be explained by observing that

$$\mathbf{P}[T|D^c]\big(1 - \mathbf{P}[D]\big) = 0.09 \cdot 0.99 = 0.0891; \tag{4.15}$$

[1] Gigerenzer G, Gaissmaier W, Kurz-Milcke E, Schwartz LM, Woloshin S. Helping Doctors and Patients Make Sense of Health Statistics. Psychol Sci Public Interest. 2007 Nov;8(2):53-96. doi: 10.1111/j.1539-6053.2008.00033.x. Epub 2007 Nov 1. PMID: 26161749.

is much bigger than

$$\mathbf{P}[T|D]\mathbf{P}[D] = 0.9 \cdot 0.01 = 0.009. \tag{4.16}$$

Intuitively speaking, then, what is the significance of the disparity between (4.15) and (4.16)? This can be answered by the law of total probability. Indeed, the latter implies that we can write

$$\mathbf{P}[T] = \mathbf{P}[T|D]\mathbf{P}[D] + \mathbf{P}[T|D^c]\big(1 - \mathbf{P}[D]\big).$$

Thus, the two probabilities in (4.15) and (4.16) represent the two distinct scenarios that can lead to a positive test in a patient, namely:

(1) The first scenario, which is accounted for by $\mathbf{P}[T|D]\mathbf{P}[D]$, is that a patient tests positive because they actually have the disease (i.e., $\mathbf{P}[D]$) and the test correctly determined this to be the case (i.e., $\mathbf{P}[T|D]$).

(2) The second scenario, which is accounted for by $\mathbf{P}[T|D^c]\big(1 - \mathbf{P}[D]\big)$, is that a patient does not actually have the disease (i.e., $\mathbf{P}[D^c] = 1 - \mathbf{P}[D]$), but the test nevertheless gives a false positive result (i.e., $\mathbf{P}[T|D^c]$).

In this context, the fact that (4.15) is much bigger than (4.16) states that most positive results actually come from false positives.

As a final remark, we note that most positive results being false positives does not contradict that the test is very good at being positive when disease is actually present (i.e., $\mathbf{P}[T|D] = 0.9$). The fact that most positive results come from false positives can be explained by noting that the proportion of people without the disease is overwhelming (i.e., 99% of the population) and that, while somewhat infrequent, false positives still do happen from time to time (i.e. 9% chance). Thus, even though the test will be positive 90% when administered to a patient with the disease, the probability that someone has the disease in the first place is only 1%. Hence the opportunities to observe true positives are very infrequent, and in fact negligible when compared to the opportunities of observing a false positive from a patient without the disease.

4.6.2. Why Does Bayes' Rule Arise Naturally in Computations? Looking back at Example 4.16, some of you may be left with the impression that it is a bit contrived. At first glance, it may seem that the only reason why we needed to use Bayes's rule was that the problem statement gave us $\mathbf{P}[T|D]$, whereas what we were asked for was instead $\mathbf{P}[D|T]$. However, the fact that Example 4.16 gave us the opposite conditional probability from what we actually wanted was not merely an artificial barrier to make for a trickier exercise. Instead, there is a fundamental reason for this—which explains some of the importance of Bayes' rule—that can be succinctly expressed as follows:

Remark 4.18. In many situations, the conditional probabilities that we want will be the opposite of what we can practically test empirically, or naturally compute using probabilistic modelling.

In order to explain this remark, we consider a very general setting: Suppose that we are trying to understand some type of phenomena, and we formulate a hypothesis about it. For example, one phenomena could be the result of a medical test, and a hypothesis would be that a given patient has the disease. Given that we do not know in advance whether or not the hypothesis is true, we can view it as an event in a random experiment, wherein

$$H = \text{“Is the hypothesis true?”}$$

Suppose that, somehow, we formulate an initial assessment of the likelihood that the hypothesis is true. That is, we have an initial belief as to what $\mathbf{P}[H]$ is. Then, as we obtain new information, we can update our prior belief about the likelihood of the hypothesis being true. If O is a new observation that we make following some experiment, then we want to update our prior belief as

$$\mathbf{P}[H] \mapsto \mathbf{P}[H|O],$$

that is, the probability of H now becomes the conditional probability of H given the new information contained in O. The problem with this, however, is that $\mathbf{P}[H|O]$ is typically not very natural to compute using probabilistic modelling or to infer in experiments. It is instead $\mathbf{P}[O|H]$ that is more natural.

Example 4.17 serves as a perfect illustration of this. The natural assumptions to make in that model are about the fairness and independence of coin flips. Once this is done, computing the probability of a given sequence of coin flips is a matter of triviality. In comparison, computing the probability that a certain coin was used given a sequence of flips directly from the model is much less obvious. More generally, the business of computing the probability of certain observations given certain hypotheses is essentially what probabilistic modelling is all about (e.g., the probability of observing a full house under the hypothesis that the hand is uniformly random, or the probability of observing that the sum of two dice is equal to 7 under the hypothesis that the dice are fair independent, etc.)

As you progress through this course (and after), I encourage you to keep this in mind whenever you solve a problem that requires the use of Bayes' rule; you will likely also come to the conclusion that Remark 4.18 very often describes such problems well.

Chapter 5

Discrete Random Variables, Expected Value, and Variance

In the previous chapters, we have thoroughly developed the basic mathematical theory of probability using events and probability measures. With these notions, we are now able to formalize the following processes:

(1) Ask unambiguous yes/no questions about the outcomes of experiments (i.e., defining events).

(2) Assign probabilities to the occurrences of various events.

(3) Understand how the probability of one event evolves once we observe that another event has occurred.

With this theory in hand, in the examples in lectures and the homework problems, we were able to answer a number of interesting question with a degree of confidence and specificity that would otherwise be impossible.

In this chapter, we begin developing some of the more sophisticated aspects of probability theory, which go beyond computing probabilities of individual events. This brings us to the notion of random variables, as well as the expected value and the variance. As you will gradually appreciate throughout this chapter and the remainder of the course, these notions are crucial in the analysis of more subtle questions involving random experiments.

5.1. A Motivating Example

Before we begin laying out the notions studied in this chapter, we look at an example of a question that motivates their introduction:

Example 5.1 (To play or not to play)**.** Consider the following game of chance:

(1) You must pay \$1 to play one round of the game.

(2) Once you pay the fee, a dealer casts two fair and independent dice.

(3) If the two dice land on the same number, then you win that round and earn \$5. If the two dice land on different numbers, then you lose that round and earn no money.

Is it to your financial benefit to play this game?

At first glance, it may seem that this question is too vague to answer. Unlike most of the problems that we studied so far, the above example is not asking us to compute the probability of a clearly defined event. However, the problem with attempting to answer this question with the tools that we currently have at our disposal is much more fundamental than this lack of precision.

More specifically, if you spend some time thinking about it, you will no doubt come to the conclusion that solving Example 5.1 is not simply a matter of computing the probability of one or several events. We could, for instance, compute the probability that you win or lose one round of the game, but this (or any other probability, for that matter) does not take into account the crucial element of the game, namely: The amount of money that you must pay to play one round, and the amount of money that you earn if you win.

To further illustrate this point, consider the following two modifications of the game in Example 5.1:

(1) You must pay \$1 to play one round of the game, and if the two dice land on the same number you win \$0 (i.e., nothing; you may as well lose).

(2) You must pay \$1 to play one round of the game, and if the two dice land on the same number you win \$1 000 000.

The probability of winning or losing in both of these games is precisely the same. However, it is clearly not to your advantage to play the first game, and clearly to your advantage to play the second: With the first game, you are guaranteed to lose one dollar irrespective of whether or not you "win." With the second game, you can keep paying \$1 until the two dice eventually land on the same number, at which point you will get a huge payout of a million dollars that will almost certainly make up for what you spent. Thus, the difficulty involved with Example 5.1 is that the amount that you earn if you win (namely, \$5) is both low and high enough to make the long-term profitability of playing the game ambiguous.

So then, how could one go about solving Example 5.1? If not by computing the probabilities of events, then how? In practice, one could attempt to answer this question empirically or with computer simulations as follows:

(1) Simulate the outcome of casting two dice a large number of times n. This will give you a sequence of n 2-tuples

$$(d_1,e_1),(d_2,e_2),\ldots,(d_n,e_n), \tag{5.1}$$

where for every number $1 \leq k \leq n$, the 2-tuple (d_k,e_k) represents the result of the k^{th} toss of two dice. d_k is the result of the first die, and e_k the result of the second.

(2) For each outcome obtained in (5.1), record what would have been your profit if you had played the game at that round. This will give you a sequence of

numbers

$$p_1, p_2, \ldots, p_n,$$

where for each $1 \leq k \leq n$, the number p_k represents the profit you would have realized if playing the game at that round, namely:

$$p_k = \begin{cases} \$4 & \text{if } d_k = e_k \\ -\$1 & \text{if } d_k \neq e_k. \end{cases}$$

Indeed, you make a profit of \$4 (i.e., your earning of \$5 minus your expenditure of \$1) if you win the round, and you simply lose the \$1 fee if you lose the round.

(3) Compute the empirical average profit over all games:

$$\frac{p_1 + p_2 + \cdots + p_n}{n}. \tag{5.2}$$

If this average is positive, then we conclude that it is to your advantage to play the game. Otherwise, if the average is negative, then we conclude that it is not to your advantage.

Indeed, if the average is positive, then this would seem to indicate that you stand to make money by playing the game in the long run, and otherwise if the average is negative the you stand to lose money.

One drawback of the above method is that the quantity (5.2) is itself random. In particular, it is possible that if you perform the experimental procedure multiple times, then the empirical average (5.2) will in one instance be positive, and in another instance be negative. Consequently, in the same spirit as the correspondence between empirical frequencies and the probability measure hinted at in (2.6), it would appear that answering questions such as Example 5.1 relies on our ability to define a theoretical construct that captures the notion of average. This is one of the main objectives of this chapter, and leads us to the notions of random variables, the expected value, and the variance.

5.2. Discrete Random Variables and Their Distributions

Before we can define a theoretical notion of average, we must introduce what it is that we are computing an average of. For this, we introduce random variables.

5.2.1. Discrete Random Variables.

Notation 5.2 (Countable infinity)**.** We say that a set A is countably infinite if it contains infinitely many elements that can be exhaustively enumerated in an ordered list of the form

$$\omega_1, \omega_2, \omega_3, \ldots \tag{5.3}$$

that is, with a first element, a second element, a third element, and so on.

For instance, the set of positive integers

$$\mathbb{N} = \{1, 2, 3, \ldots\}$$

is countably infinite, as its elements can obviously be exhaustively enumerated in the usual increasing order. Another example of a countably infinite set would be the set of all integers $\mathbb{Z}$ (both positive and negative), which we can write in an ordered list as

$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \ldots\}.$$

While writing the integers in this manner is arguably not quite as natural as

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\},$$

it makes the fact that $\mathbb{Z}$ is countably infinite more obvious.

Lastly, if an infinite set cannot be exhaustively enumerated in an ordered list, then we say that it is uncountably infinite. An example of such a set would be the set of all real numbers $\mathbb{R}$ (i.e., all integers, fractions, and irrational numbers like e, π, and $\sqrt{2}$). Thanks to a classical argument due to famed mathematician Georg Cantor,[1] no matter how hard you try, you will never be able to come up with a list of the form (5.3) that accounts for every real number.

Definition 5.3 (Discrete random variable)**.** Consider a random experiment with sample space Ω. A random variable X is a function that assigns to every possible outcome $\omega \in \Omega$ of the experiment an output $X(\omega) \in \mathbb{R}$ that is a real number. We say that X is a discrete random variable if its image (i.e., the set of all possible outputs that it can give) is finite or countably infinite.

Example 5.4 (Profit)**.** Consider the random experiment described in Example 5.1. A good sample space for this experiment is

$$\Omega = \big\{(d, e) : 1 \leq d, e \leq 6\big\},$$

where d represents the result of the first die, and e represents the result of the second die. Consider the random variable

$$X\big((d, e)\big) = \begin{cases} 4 & \text{if } d = e \\ -1 & \text{if } d \neq e. \end{cases}$$

In words, for any possible outcome of the toss of two dice, this random variable outputs your profit from playing one round of the game in Example 5.1 with those tosses. That is, if you win, then your profit is the \$5 prize minus the \$1 fee, and if you lose, then your profit is $-\$1$. This is clearly a discrete random variable, since its possible outputs are only 4 and -1.

From the perspective of defining theoretical averages, random variables are the objects whose averages we want to compute. That is, random variables allow to assign numerical values to the outcomes of a random experiment. This is very often necessary in order to compute any average, since the average of objects that are not numbers does not quite make sense. For instance, we cannot compute the "average" of 5-card hands or of heads/tails coin tosses, as these quantities are not numbers. Before moving on to other things, a remark is in order:

Remark 5.5 (Continuous random variable)**.** It is possible for certain random variables to have an image that is uncountably infinite. For instance, if the random variable X represents the amount of time (measured in fractions of hours) until you receive your final grade after exams, then presumably the set of all possible

[1] The argument in question is known as "Cantor's diagonal argument;" see, e.g., the Wikipedia page with the same name.

outputs of X could be the numbers in the interval $[0,\infty)$. More generally, random variables with uncountable images give rise to the notion of "continuous" random variables. Given that a mathematical analysis of continuous random variables involves a number of delicate technicalities, we postpone our treatment of the latter to a later chapter (at which time you will have had the opportunity to become acquainted with the basic theory of random variables).

5.2.2. Distributions. In order to distinguish between random variables and compute their averages, we need to understand the probabilities that they output different numbers. This leads us to the notion of distributions. Before that, we introduce one more standard abuse of notation:

Notation 5.6 (Shorthands)**.** Let X be a random variable defined on some sample space Ω, and let $x \in \mathbb{R}$ be a real number. Consider the event "Will the random variable X output the number x?" Mathematically, we can write this event as

$$\{\omega \in \Omega : X(\omega) = x\},$$

that is, the set of all outcomes $\omega \in \Omega$ to which the random variable X assigns the output value x. Most mathematicians working in probability find these kinds of expressions a bit cumbersome. Thus, it is much more common to use the shorthand

$$\{X = x\} = \{\omega \in \Omega : X(\omega) = x\}.$$

Moreover, when we write $\mathbf{P}[X = x]$, it should be understood that what we actually mean is

$$\mathbf{P}[X = x] = \mathbf{P}\big[\{\omega \in \Omega : X(\omega) = x\}\big].$$

Definition 5.7 (Range and distribution)**.** Let X be a discrete random variable. The range of X, denoted R_X, is the set

$$R_X := \big\{x \in \mathbb{R} : \mathbf{P}[X = x] > 0\big\}.$$

In words, the range is the set of all possible outputs of the random variable, since if $\mathbf{P}[X = x] = 0$, then we will never observe that X has given the output x.

The distribution of X is the set of probabilities

$$\big\{\mathbf{P}[X = x] : x \in R_X\big\}.$$

In words, this consists of the probabilities that X is equal to x, for every number x that is one of the possible outputs of X.

Example 5.8 (Sum of two dice)**.** Consider the experiment of casting two fair and independent dice, with sample space

$$\Omega = \big\{(d_1, d_2) : 1 \leq d_1, d_2 \leq 6\big\}.$$

Consider the random variable

$$X\big((d_1, d_2)\big) = d_1 + d_2,$$

which represents the sum of the two dice. What is X's distribution?

As per the above definition, we first have to figure out what is the range of X, that is, the set of all possible values. For this, we refer back to Figure 3.1, wherein we had represented graphically all of the possible outcomes of the sum of two dice. By examining the bottom table therein, we conclude that $R_X = \{2, 3, 4, \ldots, 12\}$.

Figure 5.1. Possible outcomes of tossing two dice (top) and the sum of their two faces (bottom).

x	2	3	4	5	6	7	8	9	10	11	12
$\mathbf{P}[X = x]$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Next, we must compute $\mathbf{P}[X = x]$ for every $x \in R_X$. As per Example 4.9, we know that our assumption that the two dice are fair and independent means that the probability measure on Ω is uniform. Thus, $\mathbf{P}[X = x]$ is equal to the number of outcomes $(d_1, d_2) \in \Omega$ such that $d_1 + d_2 = x$, divided by 36. Looking once again at the bottom of Figure 3.1, we obtain the distribution of X in Figure 5.1.

While the table in Figure 5.1 provides the distribution of the sum of two dice, it is arguably not the best way to visualize it. In order to do this, it is often a good idea to draw a histogram representation of the distribution, as is done in Figure 5.2. In that plot, we see that the values in the range are enumerated on the x-axis.

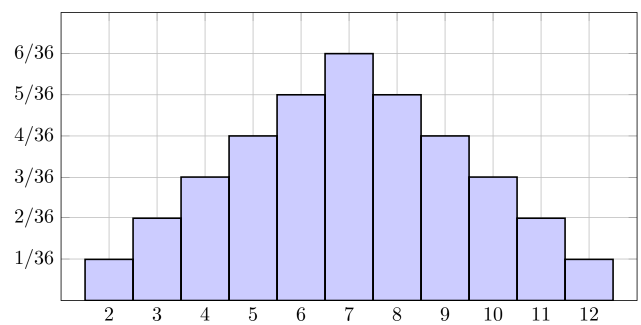

Figure 5.2. Histogram representation of the distribution of the sum of two dice.

Moreover, for every number in the range, there is a bin whose height represents the probability that the random variable outputs that number. While Figures 5.1 and 5.2 contain exactly the same information, the latter's interpretation is more immediately obvious. Indeed, with such a graphical representation, we immediately infer which outputs are more likely than others, and to what extent.

In a slightly different direction, histogram plots of distributions can also be used to shed light on the terminology "distribution." Indeed, since the range R_X contains all of the possible outcomes of some random variable X, we have that

$$\sum_{x \in R_X} \mathbf{P}[X = x] = 1.$$

That is, the probability that at least one of the possible outcomes occurs is 1. Thus, the process of drawing a histogram plot, such as Figure 5.2, can be imagined as the procedure illustrated in Figure 5.3 below. That is:

Step 1. We begin with a bin whose total length is equal to 1, which corresponds to the fact that the probabilities of every possible outcome of any random variable sums to 1 (top of Figure 5.3).

Step 2. Then, we "distribute" parts of this bin to each possible outcome of the random variable. The length of the bin associated to an outcome x corresponds the the probability $\mathbf{P}[X = x]$ (bottom of Figure 5.3).

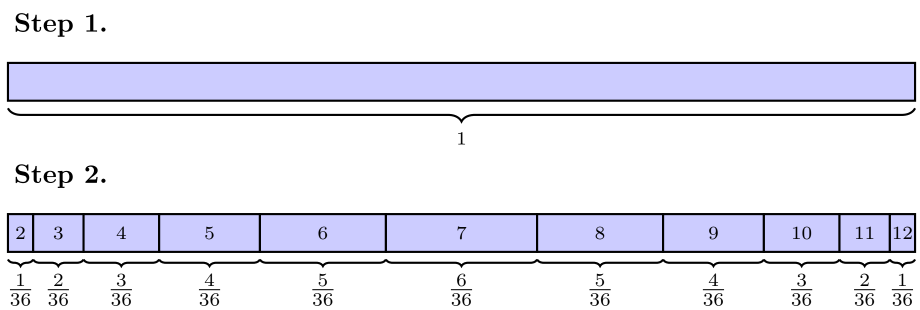

Figure 5.3. "Distributing" probabilities to every outcome of the sum of two dice. The length of the bin corresponding to some outcome $x \in R_X$ is equal to the probability $\mathbf{P}[X = x]$.

Thus, in probability theory, different distributions correspond to different ways of assigning probabilities to numbers in a range. In closing this section, we look at two examples of random variables that have the same range as the sum of two dice, but a different distribution.

Example 5.9 (Uniform distribution)**.** Let Y be a random variable with range $R_Y = \{2, 3, \ldots, 12\}$ and distribution $\mathbf{P}[Y = y] = \frac{1}{11}$ for every $y \in R_Y$. We call this the uniform distribution on $\{2, 3, \ldots, 12\}$ (since each number in the range has the same probability), and the latter can be plotted as in Figure 5.4 below.

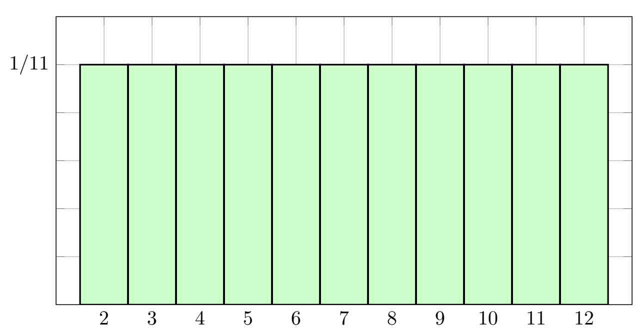

Figure 5.4. Histogram representation of the uniform distribution on $\{2, 3, \ldots, 12\}$.

Definition 5.10 (Uniform distribution)**.** Uniform distributions similar to the above example can be defined on any set of numbers. For instance, if a and b are two integers such that $a < b$, then we say that a random variable U has the uniform distribution on $\{a, a+1, \ldots, b\}$, which we denote by $U \sim \text{Unif}\{a, b\}$, if the following holds: $R_U = \{a, a+1, \ldots, b\}$ and $\mathbf{P}[U = u] = \frac{1}{b-a+1}$ for every $u \in R_U$ (i.e., there are $b - a + 1$ elements in R_U, each of which has the same probability).

Example 5.11 (Staircase distribution)**.** Let Z be a random variable with range $R_Z = \{2, 3, \ldots, 12\}$ and distribution $\mathbf{P}[Z = z] = \frac{z-1}{66}$ for all $z \in R_Z$ (it can be checked with a calculator that the sum of all these probabilities is in fact one). We call this the staircase distribution, a name which is easily explained by looking at the shape of its histogram plot in Figure 5.5 below.

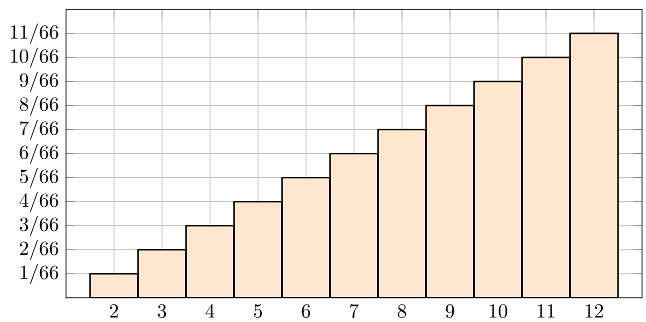

Figure 5.5. Histogram representation of the so-called staircase distribution on $\{2, 3, \ldots, 12\}$.

5.3. Expected Value and Variance

5.3.1. Definition of Expected Value. We are now in a position to define a theoretical notion of average. The definition is as follows:

Definition 5.12 (Expected Value)**.** Let X be a discrete random variable, and let f be a function on the real numbers. We define the expected value of the random variable $f(X)$, denoted $\mathbf{E}[f(X)]$, as

$$\mathbf{E}[f(X)] = \sum_{x \in R_X} f(x)\mathbf{P}[X = x]. \tag{5.4}$$

Note that, unlike $f(X)$, which is a random number whose value cannot be known before performing the experiment, $\mathbf{E}[f(X)]$ is a non-random constant.

Before discussing any example, we should take a moment to convince ourselves that the expected value does in fact capture our intuitive notions of what a theoretical average should be like. In order to do this, consider the following scenario: Suppose that we carry out a large number (say, n) of trials of some random experiment with sample space Ω. Suppose that $\omega_1, \omega_2, \ldots, \omega_n \in \Omega$ are the outcomes that we thus obtain. Given some random variable X on this sample space, we obtain from this a sequence of numbers $x_1, x_2, \ldots, x_n \in R_X$ via

$$x_i = X(\omega_i), \qquad 1 \le i \le n.$$

That is, the x_i's are the outputs given by the random variable X associated to the results ω_i of the random experiment. The theoretical average of $f(X)$ for some function f should, in some sense, be close to the empirical average

$$\frac{f(x_1)+f(x_2)+\cdots+f(x_n)}{n}. \tag{5.5}$$

At this point, in order to see the connection between (5.4) and (5.5), we apply a clever rearrangement: We claim that we can rewrite (5.5) as follows:

$$\sum_{x\in R_X} f(x)\frac{\#\big(\{1\leq i\leq n : x_i = x\}\big)}{n}. \tag{5.6}$$

Indeed, to go from (5.5) to (5.6), we have simply regrouped the terms $f(x_i)$ according to which number in R_X the x_i's are equal to. To give a specific example of this procedure in action, we can write

$$\begin{aligned}
&\frac{f(1)+f(3)+f(2)+f(2)+f(1)+f(3)+f(2)}{7}\\
&=\frac{\big(f(1)+f(1)\big)+\big(f(2)+f(2)+f(2)\big)+\big(f(3)+f(3)\big)}{7}\\
&=f(1)\cdot\frac{2}{7}+f(2)\cdot\frac{3}{7}+f(3)\cdot\frac{2}{7}
\end{aligned}$$

With (5.6) in hand, we can now justify (5.4) by referring back to the intuitive correspondence between empirical frequencies and the probability measure in (2.6), which, in the present context, gives us that

$$\mathbf{P}[X=x]\approx\frac{\#\big(\{1\leq i\leq n : x_i = x\}\big)}{n}.$$

More generally, the formula in (5.4) can be though of as an average in the sense that we are summing over all possible outputs that the random variable X can take (i.e., its range R_X), then then weighting each of these outputs by the probability that they occur.

Now that you hopefully have a good grasp of what the expected value is supposed to mean, we can go back to the example that motivated its development earlier in this chapter:

Example 5.13 (To play or not to play)**.** Recall the game of chance introduced in Example 5.1. In Example 5.4, we defined a random variable that keeps track of what we are actually interested in for this game, namely, your profit after playing one round. More specifically, we had the sample space

$$\Omega=\big\{(d,e) : 1\leq d,e\leq 6\big\},$$

where d represents the result of the first die and e represents the result of the second die, and the profit random variable was defined as

$$X\big((d,e)\big)=\begin{cases}4 & \text{if } d=e\\ -1 & \text{if } d\neq e.\end{cases}$$

Clearly, the range of this random variable is $R_X = \{-1, 4\}$. Moreover, we can easily compute its distribution:

$$\mathbf{P}[X = -1] = \mathbf{P}[d \neq e] = \frac{30}{36} = \frac{5}{6}$$

and

$$\mathbf{P}[X = 4] = \mathbf{P}[d = e] = \frac{6}{36} = \frac{1}{6},$$

where we have used the assumption that the two dice are fair and independent to compute $\mathbf{P}[d \neq e]$ and $\mathbf{P}[d = e]$.

With all of these elements into place, we can finally answer the question posed in Example 5.1, namely, is it wise to play this game? To answer this, we compute the expected value of X:

$$\mathbf{E}[X] = \sum_{x \in R_X} x\,\mathbf{P}[X = x] = (-1) \cdot \mathbf{P}[X = -1] + 4 \cdot \mathbf{P}[X = 4] = -\frac{5}{6} + \frac{4}{6} = -\frac{1}{6}.$$

In words, on average, you stand to lose one sixth of a dollar (approximately 16.7 cents) every time you play one round of this game. Thus, it is probably not wise to play the game.

Before moving on, a couple of remarks:

Remark 5.14. The game described in Example 5.1 is typical of what games of chance designed by casinos look like: The odds favor the house, but are still close enough to appear somewhat fair. The prospect of winning \$5 if the two dice are the same appear enticing, as \$5 is five times the cost of one round and two dice being equal is not all that unusual. However, this is just barely not enough to make the game profitable for the player on average. The fact that basic (i.e., non-quantitative) intuition fails to provide a compelling answer to Example 5.1 provides yet more evidence for the usefulness of developing a rigorous theory of probability.

Remark 5.15. At this time, it may not be clear to you what exactly we can infer from the fact that $\mathbf{E}[X] = -\frac{1}{6}$ in Example 5.13. That is, what does this imply about the profits that a casino can expect to make in the long run with such a game, or conversely, the amount of money that a player can expect to lose in the long run? Answering this relies on making the relationship between the theoretical expected value and empirical averages such as (5.6) more precise. That is, if we actually go through the process of playing many rounds of the game in practice, what kinds of quantitative predictions about our cumulative profit can be made from the knowledge that $\mathbf{E}[X] = -\frac{1}{6}$? This will be clarified when we discuss the law of large numbers in a future chapter.

5.3.2. Linearity of the Expected Value. The computation in (5.13) might leave you with the impression that the expected value and its computation in practice amounts to a trivial extension of what we have been doing so far. That is, the expected value is nothing more than a sum of probabilities of various events, which are themselves computed using the theory developed in the previous chapters. However, this is not quite the case.

As it turns out, the expected value satisfies a number of interesting properties, which have the perhaps surprising consequence that it is often possible to compute the expected value of a random variable without computing its distribution! Consequently, the expected value can sometimes provide some information on a random variable even in cases where the distribution is too difficult to compute. One of the key properties that makes this possible is the linearity of the expected value:

Proposition 5.16 (Linearity of the expected value)**.** *Let X and Y be two discrete random variables, and let $a \in \mathbb{R}$ be a nonrandom constant. Then,*

$$\mathbf{E}[X+Y] = \mathbf{E}[X] + \mathbf{E}[Y]$$

and

$$\mathbf{E}[aX] = a\mathbf{E}[X].$$

While this result can be proved rigorously using the axioms of probability, we will not do so here. In order to convince yourself that the linearity of expected value makes sense intuitively, I encourage you to think about its interpretation from the point of view of empirical averages, such as (5.6). That is, what happens if you compute the empirical average of the sum of two random variables, or of a random variable multiplied by a constant.

Example 5.17 (The sum of one million dice)**.** Let us use $m = 1\,000\,000$ as a shorthand for the number one million (in order to keep equations tidy). Consider the experiment of tossing one million fair dice, with sample space

$$\Omega = \big\{(d_1, d_2, d_3, \ldots, d_m) : 1 \leq d_i \leq 6 \text{ for all } 1 \leq i \leq m\big\}.$$

As usual, d_i is the result of the i^{th} toss. Suppose that we are interested in the sum of these one million dice, that is, the random variable

$$X = d_1 + d_2 + d_3 + \cdots + d_m.$$

What is $\mathbf{E}[X]$?

If we apply the definition of expected value naively, then we have to compute

$$\mathbf{E}[X] = \sum_{x \in R_X} x\,\mathbf{P}[X = x].$$

However, this requires us to compute the distribution of X, that is, for any number x, we have to figure out what is

$$\mathbf{P}[d_1 + d_2 + d_3 + \cdots + d_m = x].$$

If it is not immediately apparent to you that this is an extraordinarily tedious task, then I challenge you to try to compute a number of these probabilities, keeping in mind that you will have to compute millions of them.

The linearity of the expected value allows us to sidestep this difficulty entirely. Indeed, we note that the result of each individual die d_i is itself a random variable. It is a function from the sample space, which outputs a real number representing the result of one particular die. Thus, X is in fact a sum of random variables!

Applying linearity, we therefore get that

$$\begin{aligned}\mathbf{E}[X] &= \mathbf{E}[d_1 + d_2 + d_3 + \cdots + d_m] \\ &= \mathbf{E}[d_1] + \mathbf{E}[d_2 + d_3 + \cdots + d_m] \\ &= \mathbf{E}[d_1] + \mathbf{E}[d_2] + \mathbf{E}[d_3 + \cdots + d_m] \\ &\cdots \\ &= \mathbf{E}[d_1] + \mathbf{E}[d_2] + \mathbf{E}[d_3] + \cdots + \mathbf{E}[d_m].\end{aligned}$$

With this in hand, we only need to know what is the expected value of the individual dice, but this is trivial: Since the dice are fair, their ranges are the numbers from 1 to 6, and their distribution on these numbers is uniform:

$$\mathbf{E}[d_i] = \sum_{k=1}^{6} k\mathbf{P}[d_i = k] = \sum_{k=1}^{6} \frac{k}{6} = \frac{7}{2}.$$

Consequently,

$$\mathbf{E}[X] = m \cdot \frac{7}{2} = \frac{7\,000\,000}{2} = 3\,500\,000.$$

Thus, the expected value can provide some information about the random variable even without computing the distribution! I encourage you to keep this in mind as you go through exercises asking you to compute the expected value of various random variables: In many such cases, you might come to realize that computing the distribution would be extraordinarily difficult. This further cements the usefulness of the expected value as a theoretical concept.

5.3.3. Definition of Variance. You might be wondering why the definition of the expected value in (5.4) is stated for any function of a random variable $f(X)$ rather than simply X itself. The reason for this is that it is sometimes interesting to compute the expected value of various functions of the random variable under consideration. One of the main examples of this is the variance:

Definition 5.18 (Variance)**.** Let X be a random variable. We define the variance of X, denoted $\mathbf{Var}[X]$, as

$$\mathbf{Var}[X] = \mathbf{E}\big[(X - \mathbf{E}[X])^2\big] = \mathbf{E}[X^2] - \mathbf{E}[X]^2.$$

Remark 5.19. The fact that we have the equality

$$\mathbf{E}\big[(X - \mathbf{E}[X])^2\big] = \mathbf{E}[X^2] - \mathbf{E}[X]^2$$

is a simple consequence of the linearity of the expected value (you should try to establish this equality yourself, as a simple exercise).

In order to understand what the variance means conceptually, it is arguably most helpful to look at the expression

$$\mathbf{Var}[X] = \mathbf{E}\big[(X - \mathbf{E}[X])^2\big]. \tag{5.7}$$

This expression states that the variance measures the average of the random variable $(X - \mathbf{E}[X])^2$. Recall that for any real number $\mu \in \mathbb{R}$, the function $f(x) := (x - \mu)^2$ is a parabola that points upward, with a minimum at μ (see, e.g., Figure 5.6 below). In particular, the function

$$f(x) = (x - \mathbf{E}[X])^2$$

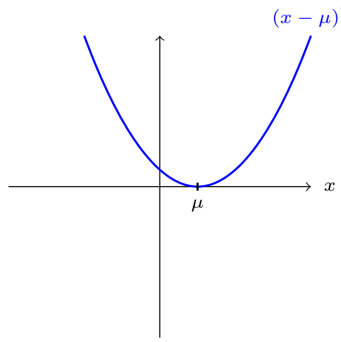

Figure 5.6. Plot of the parabola $(x-\mu)^2$.

increases as x moves away from $\mathbf{E}[X]$. Thus, the variance is a measure of far X will be, on average, from its expected value $\mathbf{E}[X]$. Stated another way, the variance measures how random a random variable is, in the following sense:

(1) If $\mathbf{Var}[X] \approx 0$, then $X \approx \mathbf{E}[X]$. Thus, before the experiment is even performed, we can already guess that X's value will be close to $\mathbf{E}[X]$.[2]

(2) Conversely, if $\mathbf{Var}[X]$ is very large, then X's output is more unpredictable in the sense that X will on average be quite far from $\mathbf{E}[X]$.

While the business of quantifying the degree of randomness of a random variable may not seem particularly interesting to you to at this point, this notion will turn out to be of massive importance when we discuss the law of large numbers.

Remark 5.20 (Why does the variance have a square?)**.** Illustrative examples for the conceptual meaning of the variance will be provided in just a moment. However, before we do this, there is one last thing that we should address regarding the variance. In the previous paragraph, I argued that the variance is used to quantify the average distance between a random variable and its expected value. But then, this begs the question: Why did we define the variance as

$$\mathbf{Var}[X] = \mathbf{E}\big[(X - \mathbf{E}[X])^2\big],$$

with a square over $X - \mathbf{E}[X]$? Why don't we instead define the variance as

$$\mathbf{E}\big[|X - \mathbf{E}[X]|\big], \tag{5.8}$$

i.e., the expected value of the actual distance between X and $\mathbf{E}[X]$ (recall the definition of the absolute value $|x|$ in Notation 4.4), or as

$$\mathbf{E}\big[f(X - \mathbf{E}[X])\big],$$

where f is any other function that increases as its input gets farther away from zero? If our ultimate objective is to design a measure of how far a random variable is from its expected value, then it may seem at first glance contrived and unnatural to use the variance instead of the much more straightforward (5.8).

[2]In the extreme case where $\mathbf{Var}[X] = 0$, then we have that $X = \mathbf{E}[X]$ (this is not too difficult to prove). Since $\mathbf{E}[X]$ is a non-random constant number, then a variance of zero implies that the "random variable" X is in fact not random at all!

The reason why the variance is typically used instead of (5.8) is not for conceptual reasons, but instead practical concerns. This is connected to the fact that the variance can also be written as

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2. \tag{5.9}$$

As it turns out, the formula (5.9) is typically much easier to use in practice than (5.8) or even (5.7). Thus, although (5.8) and the variance more or less contain the same information, actually computing (5.8) is typically much harder in practice (we will see some examples of this in the sequel).

As a final note, in certain situations when one is concerned with the units that a random variable quantifies, the square in the variance might be undesirable. For instance, if X represents an amount of money in dollars (as it did in Example 5.4), then $\mathbf{Var}[X]$ is a quantity in dollars squared, which is a bit weird. In order to get around this issue but still retain the computational advantage offered by (5.9), we can simply take a square root of the variance, which puts the units back into their original form:

Definition 5.21 (Standard Deviation)**.** Let X be a random variable. We define the standard deviation of X, denoted $\mathbf{SD}[X]$, as

$$\mathbf{SD}[X] = \sqrt{\mathbf{Var}[X]}.$$

5.3.4. Three Examples. In order to cement your conceptual understanding of the variance (and the expected value, for that matter), we now look at some examples and illustrations of the concept.

Example 5.22 (Sum of two dice)**.** Let X be a random variable that represents the sum of two fair and independent dice. The distribution of this random variable was computed in Example 5.8, yielding Figures 5.1 and 5.2. With this, we can compute the expected value and the variance. On the one hand, with Figure 5.1 we get

$$\begin{aligned}\mathbf{E}[X] &= \sum_{x=2}^{12} x\,\mathbf{P}[X=x]\\ &= 2\cdot\tfrac{1}{36}+3\cdot\tfrac{2}{36}+4\cdot\tfrac{3}{36}+5\cdot\tfrac{4}{36}+6\cdot\tfrac{5}{36}\\ &\qquad +7\cdot\tfrac{6}{36}+8\cdot\tfrac{5}{36}+9\cdot\tfrac{4}{36}+10\cdot\tfrac{3}{36}+11\cdot\tfrac{2}{36}+12\cdot\tfrac{1}{36}\\ &= 7.\end{aligned} \tag{5.10}$$

On the other hand, for the variance, we write

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2.$$

Having just computed the expected value of X, we can already solve half of this formula as follows:

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - 7^2 = \mathbf{E}[X^2] - 49.$$

Thus, it only remains to compute $\mathbf{E}[X^2]$. For this, we apply the definition of expected value, which yields (sparing you the detailed writeup as in (5.10))

$$\mathbf{E}[X^2] = \sum_{x=2}^{12} x^2\,\mathbf{P}[X=x] = \frac{329}{6}.$$

Putting everything together, we conclude that

$$\mathbf{Var}[X] = \frac{329}{6} - 49 = \frac{35}{6}$$

and that

$$\mathbf{SD}[X] = \sqrt{\frac{35}{6}} \approx 2.41.$$

X's expected value and standard deviation are illustrated against its distribution in Figure 5.7 below. The location of the expected value is marked by an

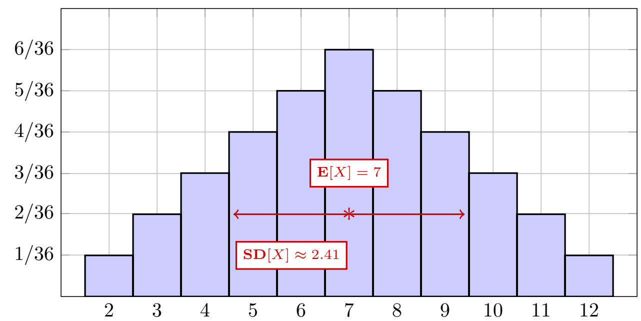

Figure 5.7. X's expected value and standard deviation visualized.

asterisk, and the standard deviation is represented by the length of the arrows extending to the right and left of the asterisk.

The fact that the expected value is equal to 7 makes sense intuitively, because 7 is the center point of the range of X (i.e., its set of possible values), and the distribution is symmetric around that value. In particular, for any value $x \in R_X$ smaller than 7, the value $y \in R_X$ bigger than seven that is symmetric to x across 7 occurs with the same probability. Therefore, it makes sense that if we were to carry out the experiment of observing X's output a great number of times, then the values smaller and bigger than 7 should more or less balance out, giving an average value close to 7.

Regarding the standard deviation, its precise value does not have an intuitive interpretation that is as obvious as the expected value, but it is nevertheless interesting to visualize the average distance between the value of X and $\mathbf{E}[X]$.

Example 5.23 (Uniform distribution)**.** Recall the uniform random variable on the set $\{2, 3, \ldots, 12\}$, which we called Y in Example 5.9, as well as the more general definition of uniform random variable in Definition 5.10. Using the fact that every element in the range of a uniform random variable has the same probability, it is not too difficult to show the following using the definition of expected value and the variance formula in (5.9):

Proposition 5.24. *If* $U \sim \mathrm{Unif}\{a, b\}$*, then,*

$$\mathbf{E}[U] = \sum_{x=a}^{b} x \cdot \frac{1}{b-a+1} = \frac{a+b}{2}$$

and since

$$\mathbf{E}[U^2] = \sum_{x=a}^{b} x^2 \cdot \frac{1}{b-a+1} = \frac{1}{6}\left(2a^2 + 2ab - a + 2b^2 + b\right),$$

we have that

$$\mathbf{Var}[U] = \mathbf{E}[U^2] - \mathbf{E}[U]^2 = \frac{(b-a+1)^2 - 1}{12}.$$

(I encourage you to keep a mental note that this formula is available here; you can henceforth use it without justification in the homework and exams.) Using this formula for $Y \sim \text{Unif}\{2, 12\}$, we obtain that

$$\mathbf{E}[Y] = \frac{2+12}{2} = \frac{14}{2} = 7,$$

$$\mathbf{Var}[Y] = \frac{(12-2+1)^2 - 1}{12} = \frac{120}{12} = 10,$$

and

$$\mathbf{SD}[Y] = \sqrt{10} \approx 3.16.$$

We illustrate Y's expected value and standard deviation in Figure 5.8 below. Once again, it makes sense that Y' expected value should be 7: The latter is at the

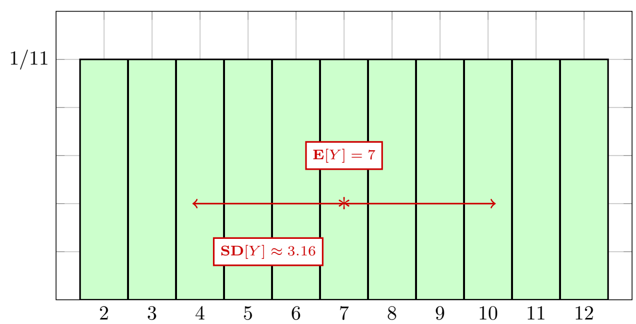

Figure 5.8. Y's expected value and standard deviation visualized.

center of R_Y's range, and the distribution is symmetric around that value. What is interesting to do here is compare the standard deviation of Y with that of X in Figure 5.7.

More specifically, we observe that $\mathbf{SD}[Y] > \mathbf{SD}[X]$. Following-up on our discussion of the intuitive interpretation of the variance and standard deviation as a measure of how random a random variable is, we conclude the following: The uniform random variable Y is in some sense more random than the sum of two dice X. Indeed, we note that the probabilities of X's outcomes in Figure 5.7 are highest for the values closer to the expected value of 7. Therefore, on average, we can expect that X will tend to be closer to 7. In contrast, the uniform random variable is just as likely to be any number in between 2 and 12. Consequently, before carrying out the uniform experiment, you cannot guess which outcome will come out with any degree of reliability; it could just as well be any of them. With the sum of two dice, if you guess that the value is somewhat close to 7, then you will be correct more often; while it is possible that you get an outcome far away from seven, it is less likely than outcomes close to 7.

Example 5.25 (Staircase distribution)**.** As a final example, let Z have the staircase distribution, which we recall we had defined as the random variable with range $R_Z = \{2, 3, \ldots, 12\}$ and distribution $\mathbf{P}[Z = z] = \frac{z-1}{66}$ for all $z \in R_Z$. Using this formula for the distribution, we can easily compute that

$$\mathbf{E}[Z] = \sum_{z=2}^{12} z \cdot \frac{z-1}{66} = \frac{26}{3} \approx 8.67.$$

Next, with

$$\mathbf{E}[Z^2] = \sum_{z=2}^{12} z^2 \cdot \frac{z-1}{66} = \frac{247}{3},$$

we obtain that

$$\mathbf{Var}[Z] = \mathbf{E}[Z^2] - \mathbf{E}[Z]^2 = \frac{247}{3} - \left(\frac{26}{3}\right)^2 = \frac{65}{9},$$

and

$$\mathbf{SD}[Z] = \sqrt{\frac{65}{9}} \approx 2.69.$$

We illustrate these facts against Z's distribution in Figure 5.9 below. Here, it is

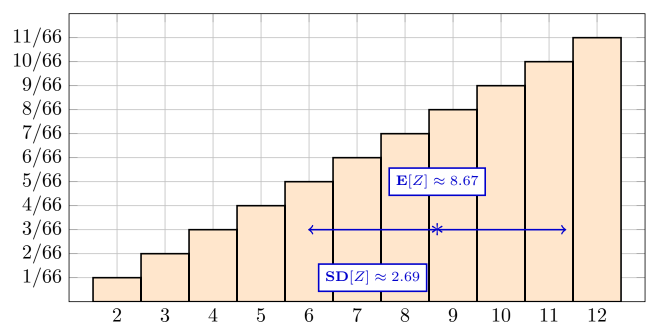

Figure 5.9. Z's expected value and standard deviation visualized.

interesting to note that $\mathbf{E}[Z]$ is greater than 7, unlike the previous two examples. This makes sense, because the most likely outputs of Z are all larger than 7. Thus, more often than not Z will output a number larger than 7, which skews the average in that direction.

With these three examples, you hopefully have a better grasp of the conceptual meaning of the expected value, variance, and standard deviation. I should emphasize that what is interesting about these examples is not the computations of the quantities themselves (most of which I have skipped anyway), but instead their illustration. Indeed, the computations in the above three examples amount to nothing more than a direct application of the definitions of the expected value in (5.4) and the variance in (5.9).

In the remainder of this chapter, we develop a few tools that allow us to compute the expected value of random variables in much more interesting cases, where an explicit formula for the distribution (i.e., the quantities $\mathbf{P}[X=x]$ for all $x \in R_X$) is not always available and/or practical to compute.

5.4. Conditioning and Independence

Much like the probabilities of events, random variables and expected values can get updated once we observe that some event has occurred. In the final section of this chapter, we discuss how conditioning and independence relate to random variables, the expected value, and the variance.

5.4.1. Conditional Distribution and Expected Value. We begin by discussing how the distribution and expected value of a random variable gets updated once we observe that some event occurs.

Definition 5.26 (Conditional distribution and expected value)**.** Let X be a discrete random variable on some sample space Ω, and let $A \subset \Omega$ be an event such that $\mathbf{P}[A] > 0$. Define the conditional range

$$R_{X|A} := \big\{x \in R_X : \mathbf{P}[X = x|A] > 0\big\}.$$

We define the conditional distribution of X given A as the set of probabilities

$$\big\{\mathbf{P}[X = x|A] : x \in R_{X|A}\big\}.$$

Then, given some function f, we define the conditional expected value of $f(X)$ given A as follows:

$$\mathbf{E}[f(X)|A] = \sum_{x \in R_{X|A}} f(x)\,\mathbf{P}[X = x|A].$$

Example 5.27 (Sum of two dice)**.** Let d_1 and d_2 be two fair and independence dice, and let $X = d_1 + d_2$, namely, the sum of the results of the two dice. What is the conditional distribution and expected value of X given $d_2 = 3$ (i.e., the conditional distribution and expected value of the sum of two dice, knowing that the second dice lands on three)?

For any number x, we have that

$$\mathbf{P}[X = x|d_2 = 3] = \mathbf{P}[d_1 + d_2 = x|d_2 = 3],$$

where we have simply used the fact that $X = d_1 + d_2$ by definition. If we observe that $d_2 = 3$, then the result of d_2 in the sum $d_1 + d_2$ is not random; we know it must be equal to three. Thus, we can write

$$\mathbf{P}[d_1 + d_2 = x|d_2 = 3] = \mathbf{P}[d_1 + 3 = x|d_2 = 3] = \mathbf{P}[d_1 = x - 3|d_2 = 3].$$

Next, since d_1 and d_2 are independent, knowing the result of d_2 provides no information on what should be the result of d_1. Therefore,

$$\mathbf{P}[d_1 = x - 3|d_2 = 3] = \mathbf{P}[d_1 = x - 3] = \begin{cases} \frac{1}{6} & \text{if } 1 \leq x - 3 \leq 6 \\ 0 & \text{otherwise} \end{cases}.$$

Given that $1 \leq x - 3 \leq 6$ translates to $4 \leq x \leq 9$, we therefore conclude that

$$R_{X|d_2=3} = \{4, 5, 6, 7, 8, 9\},$$

and that the corresponding conditional distribution is

$$\mathbf{P}[X = x|d_2 = 3] = \frac{1}{6}, \qquad \text{for all } 4 \leq x \leq 9.$$

Finally, for the conditional expectation, we have that

$$\mathbf{E}[X|d_2 = 3] = \sum_{x=4}^{9} x\,\mathbf{P}[X = x|d_2 = 3] = \frac{1}{6}\sum_{x=4}^{9} x = \frac{13}{2} = 6.5.$$

See in Figure 5.10 below for an illustration.

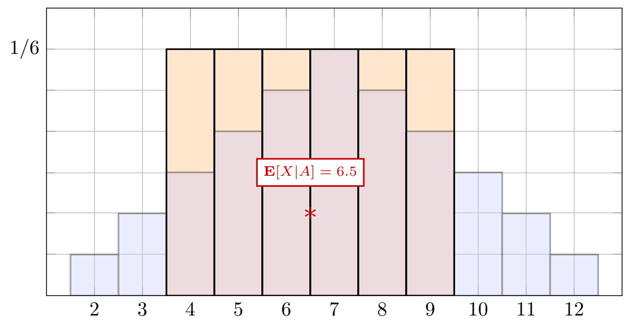

Figure 5.10. X's conditional distribution and expected value given A. Superimposed (in pale blue) is the original distribution of X, for comparison.

Thankfully, the very useful linearity properties of the expected value are not affected by conditioning on an event:

Proposition 5.28 (Linearity with conditioning)**.** *Let X and Y be discrete random variables on some sample space Ω, let $a \in \mathbb{R}$ be a nonrandom constant, and let $A \subset \Omega$ be an event such that $\mathbf{P}[A] > 0$. Then,*

$$\mathbf{E}[X + Y|A] = \mathbf{E}[X|A] + \mathbf{E}[Y|A]$$

and

$$\mathbf{E}[aX|A] = a\mathbf{E}[X|A].$$

5.4.2. Independence. Just like events, random variables can also be independent. As it turns out, the definition of independent random variables is a straightforward extension of the definition of independent events:

Definition 5.29 (Independent random variables)**.** We say that a collection of random variables $X_1, X_2, \ldots, X_n$ are independent if for every collection of subsets $A_1 \subset R_{X_1}$, $A_2 \subset R_{X_2}, \ldots$, $A_n \subset R_{X_n}$, the events

$$\{X_1 \in A_1\}, \{X_2 \in A_2\}, \ldots, \{X_n \in A_n\}$$

are independent.

In words, if $X_1, \ldots, X_n$ are independent, then any knowledge about the output of some of the variables has no impact on the behavior of the other variables in the collection. Stated in a different way, independence has the following consequence:

Proposition 5.30. *If X and Y are independent random variables, then for every $y \in R_Y$, we have that*

(1) $R_X = R_{X|Y=y}$*;*
(2) the conditional distribution of X given $Y = y$ is the same as the original distribution of X; and
(3) $\mathbf{E}[X|Y = y] = \mathbf{E}[X]$.

Thus, if X and Y are independent, then knowing about the outcome of Y has no impact on X's possible outputs, distribution, or expected value. Finally, we have the following two properties involving the expected value and variance of independent random variables:

Proposition 5.31. *If $X_1, X_2, \ldots, X_n$ are independent random variables, then*

(1) $\mathbf{E}[X_1X_2\cdots X_n] = \mathbf{E}[X_1]\mathbf{E}[X_2]\cdots\mathbf{E}[X_n]$*; and*
(2) $\mathbf{Var}[X_1 + X_2 + \cdots + X_n] = \mathbf{Var}[X_1] + \mathbf{Var}[X_2] + \cdots + \mathbf{Var}[X_n]$.

From the purely mathematical point of view, these two properties are not particularly deep; their proofs are relatively straightforward (and can easily be found online with a bit of research). Nevertheless, these two properties turn out to be extremely important due to the fact that they permit to simplify substantially computations that would otherwise be very difficult. Here are two examples of this phenomenon:

Example 5.32 (Product of one million dice)**.** Let $m = 1\,000\,000$, and let $d_1, d_2, \ldots, d_m$ be a million fair and independence dice. What is the expected value of

$$X = d_1 \cdot d_2 \cdots d_m,$$

that is, the value of the product of the one million dice? Computing the range and distribution of this random variable would be nothing short of a nightmare. However, we can get around that completely by noting that

$$\mathbf{E}[X] = \mathbf{E}[d_1 \cdot d_2 \cdots d_m] = \mathbf{E}[d_1] \cdot \mathbf{E}[d_2] \cdots \mathbf{E}[d_m],$$

by Proposition 5.31, since the d_i's are assumed independent. Since $d_i \sim \mathrm{Unif}\{1, 6\}$, we know from Proposition 5.24 that $\mathbf{E}[d_i] = \frac{7}{2} = 3.5$. Thus,

$$\mathbf{E}[X] = (3.5)^m = (3.5)^{1\,000\,000},$$

which is a very large number.

Example 5.33 (Sum of one million dice)**.** Let m and d_i be as in the previous example, but this time let

$$X = d_1 + d_2 + \cdots + d_m$$

be the sum of the results of the one million dice. Previously, in Example 5.17, we computed that

$$\mathbf{E}[X] = 3\,500\,000$$

by using linearity of the expected value, i.e., without needing to compute X's distribution. Thanks to Proposition 5.31, we can do more: Indeed, we know that

$$\mathbf{Var}[X] = \mathbf{Var}[d_1 + d_2 + \cdots + d_m] = \mathbf{Var}[d_1] + \mathbf{Var}[d_2] + \cdots + \mathbf{Var}[d_m].$$

Since $d_i \sim \mathrm{Unif}\{1,6\}$, we know from Proposition 5.24 that $\mathbf{Var}[d_i] = \frac{35}{12}$. Thus,

$$\mathbf{Var}[X] = m \cdot \frac{35}{12} = \frac{8\,750\,000}{3} \approx 2\,916\,000.$$

These two examples may seem to you to be a bit contrived, in that they suspiciously look like they were deliberately designed to illustrate the usefulness of Proposition 5.31. As we progress through the course, however, you will no doubt gain an appreciation of how often they arise in various natural problems.

Referring back to Remark 5.20, the property of the variance in Proposition 5.31-(2) further helps explain why the variance is superior to other measures of the distance between a random variable and its expected value, such as (5.8). In fact, the property in Proposition 5.31-(2) specifically will turn out to play a fundamental role in our study of the law of large numbers.

5.4.3. The Law of Total Expectation. We now close this section with one of the most important results regarding expected values, which is a direct analogue of the law of total probability:

Proposition 5.34 (Law of total expectation)**.** *Let $A_1, A_2, \ldots, A_n \subset \Omega$ be a collection of events that satisfy the following three conditions:*

(1) $\mathbf{P}[A_i] > 0$ *for every* $1 \leq i \leq n$*,*

(2) $A_i \cap A_j = \varnothing$ *whenever* $i \neq j$*, and*

(3) $\displaystyle\bigcup_{i=1}^{n} A_i = \Omega$.

Then, for any random variable X on Ω, one has

$$\mathbf{E}[X] = \sum_{i=1}^{n} \mathbf{E}[X|A_i]\mathbf{P}[A_i]. \tag{5.11}$$

Referring back to the statement of the law of total probability in (4.9), we see that (5.11) is essentially the same statement, but in the context of expected values instead. The intuition here is exactly the same as what it was for the law of total probability, that is, if we are given mutually exclusive events A_i that account for every possible outcome of a random experiment, then we can compute the expected value of X by first looking at what the expected value becomes if we observe that one of the A_i's occur, and then account for every possibility by summing over all events in the collection.

In similar fashion to the law of total probability, in order for (5.11) to be useful, it better be the case that $\mathbf{E}[X|A_i]$ and $\mathbf{P}[A_i]$ are easier to compute than $\mathbf{E}[X]$, or at the very least provide new useful information. Otherwise, we are simply wasting our time writing the expectation as a more complicated expression. In order to illustrate the usefulness of the law of total expectation, as well as some of the other concepts discussed in this section, we consider a challenging example:

Example 5.35 (Escape the mine; first-step analysis)**.** Suppose that a miner is lost in a mine. In front of them are three tunnels, call them tunnel 1, tunnel 2, and tunnel 3. As illustrated in Figure 5.11 below,

(1) if the miner goes through tunnel 1, then they escape the mine after two minutes of travel time;

(2) if the miner goes through tunnel 2, then they return to where they started after five minutes of travel time; and

(3) if the miner goes through tunnel 3, then they return to where they started after three minutes of travel time.

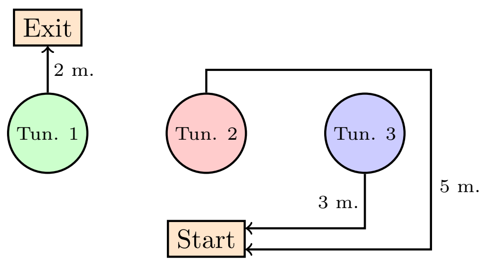

Figure 5.11. Three tunnels, one of which leads to the exit in two minutes, and two of which bring you back in three or five minutes.

Since the miner is lost, they do not know which tunnel leads to the exit. Thus, they first choose between the three tunnels uniformly at random. In the event that the miner does not choose the right tunnel, that is, tunnel 2 or 3, we assume that they do not realize that they are back in front of the same three tunnels. Therefore, they choose again between the three tunnels uniformly at random, and independently of their first choice. Going forward, we assume that the miner will keep choosing one of the three doors uniformly at random and independently of all previous choices until they finally choose tunnel 1, and thus escape. Let X be the random variable that denotes the amount of time that it takes the miner to escape (in minutes) using this procedure. What is $\mathbf{E}[X]$?

There are (at least) two methods that one could use to solve this problem. The first method, which is simple conceptually but very difficult computationally, is to compute the distribution of X and then use the formula that defines the expected value in (5.4) directly. The second method, which is very subtle conceptually but easy computationally, consists of a very clever use of the law of total expectation. We will use the second method to solve the problem. However, in order to illustrate why we even bother with a subtle solution that uses the law of total expectation, we begin by explaining what makes a direct computation difficult.

If we want to compute X's expected value using the formula

$$\mathbf{E}[X] = \sum_{x \in R_X} x\,\mathbf{P}[X = x],$$

then we need to understand two things. Namely, the range R_X of all possible values that X can take, as well as the probability $\mathbf{P}[X = x]$ of every of these values $x \in R_X$. In order to have any chance of doing this, we should probably think about what is the sample space that X is defined on. One way to define this sample space is as follows:

$$\Omega = \big\{(t_1, \ldots, t_k) : k \geq 1,\ t_i \in \{2, 3\} \text{ for all } 1 \leq i \leq k-1,\ t_k = 1\big\}.$$

In words, this sample space contains all possible sequences of tunnels that the miner could go through. This sequence always ends with 1, because the miner will exit as soon as they choose tunnel 1. Before the miner first goes through tunnel 1, they could go through an arbitrarily long sequence of choosing tunnels 2 or 3. This is what is represented by the numbers $t_1, \ldots, t_{k-1}$. In the case where $k = 1$, then the outcome is the one-tuple (1), indicating that the miner chose tunnel 1 at their first try and immediately got out. In the case where $k > 1$, then the miner chooses wrongly $k - 1$ times, giving an outcome of the form $(t_1, t_2, \ldots, t_{k-1}, 1)$, where the choices $t_1, \ldots, t_{k-1}$ are all equal to either 2 or 3.

Computing the probability of any given outcome in Ω is very easy. Because every choice of tunnel is uniform on $\{1, 2, 3\}$ and independent, we have that

$$\mathbf{P}\big[(t_1, \ldots, t_k)\big] = \frac{1}{3^k}.$$

However, that is not exactly what we want. Instead, we want the distribution of X, which represents the amount of time spent escaping the mine. For this, we note that for any outcome in Ω, we have that

$$\begin{aligned} X(t_1, \ldots, t_k) = 2 \cdot \#(\{1 \leq i \leq k : t_i = 1\}) \\ + 5 \cdot \#(\{1 \leq i \leq k : t_i = 2\}) + 3 \cdot \#(\{1 \leq i \leq k : t_i = 3\}). \end{aligned} \tag{5.12}$$

Indeed, the time to escape is 2 times the number of times the miner chose tunnel 1 (actually, this can only happen once because the miner escapes as soon as they choose that tunnel), plus 5 times the number of times that the miner chose tunnel 2, plus 3 times the number of times that the miner chose tunnel 3. At this point, we start to realize that computing the distribution of X will be rather hard: We somehow need to account for every possible sequence of decisions that the miner could make, and then group together all sequence of decisions that lead to the same escape time. This is by no means impossible, but certainly some extremely tedious work (I *dare* you to give it a try)!

As it turns out, there is a better way, which, as mentioned earlier, involves the law of total expectation. The method in question is called first-step analysis: Whenever you are confronted to a complex problem, you should always ask yourself: "Is there a piece of information about the randomness of this problem that, if you could know it in advance, would simplify the analysis in some way?" If the answer to that question is yes, then there is a good chance that the laws of total probability and expectation will be useful if we condition on knowing that particular piece of information.

Looking at the specific problem that we are thinking about here, we notice that if we could know in advance which tunnel the miner will first choose, then this provides some rather useful information. Indeed, if we let t_1 denote the first tunnel that the miner goes through, then we note the following:

(1) If $t_1 = 1$, then we automatically know that the miner will escape in two minutes. Thus, in this case, we have that $\mathbf{E}[X|t_1 = 1] = 2$.

(2) If $t_1 = 2$, then we know that the miner wastes at least 5 minutes going through tunnel 2. More specifically, if we write $X = X_1 + X_{2+}$, where X_1 is the time spent going through the first tunnel, and X_{2+} is the time spent going through

the tunnels after the first one, then we can write

$$\begin{aligned}\mathbf{E}[X|t_1 = 2] &= \mathbf{E}[X_1 + X_{2+}|t_1 = 2] \\ &= \mathbf{E}[X_1|t_1 = 2] + \mathbf{E}[X_{2+}|t_1 = 2] \\ &= 5 + \mathbf{E}[X_{2+}|t_1 = 2],\end{aligned}$$

where the second equality comes from linearity, and $\mathbf{E}[X_1|t_1 = 2] = 5$ comes from the fact that tunnel 2 takes 5 minutes to go through. As for the second term, we claim that

$$\mathbf{E}[X_{2+}|t_1 = 2] = \mathbf{E}[X].$$

Indeed, once the miner goes through tunnel 2, the amount of time left after that behaves in exactly the same way as the amount of time at the very beginning: The miner goes back to the starting point, and every future choice of tunnel from then on is made uniformly at random and independently of the first choice. Thus, in summary,

$$\mathbf{E}[X|t_1 = 2] = 5 + \mathbf{E}[X]. \tag{5.13}$$

(3) Finally, if $t_1 = 3$, then

$$\mathbf{E}[X|t_1 = 3] = 3 + \mathbf{E}[X].$$

The argument for this follows exactly the same logic as (5.13): If we know that the miner first chooses tunnel 3, then they waste 3 minutes going through that, and then the remaining time after that behaves in the same way as the time at the very beginning.

At first glance, it may seem that the computations above are useless. While we have been able to compute $\mathbf{E}[X|t_1 = 1]$ exactly, $\mathbf{E}[X|t_1 = 2]$ and $\mathbf{E}[X|t_1 = 3]$ are written in terms of $\mathbf{E}[X]$ itself, which is what we want to compute in the first place. However, if we write $\mathbf{E}[X]$ using the law of total expectation and then apply our computations, then something magical happens: By the law of total expectation, we have that

$$\mathbf{E}[X] = \sum_{i=1}^{3} \mathbf{E}[X|t_1 = i]\mathbf{P}[t_1 = i].$$

Indeed, the events $\{t_1 = 1\}$, $\{t_1 = 2\}$, and $\{t_1 = 3\}$ are clearly mutually exclusive, and they account for every possible way to choose the first tunnel. Given that the first tunnel is chosen uniformly at random, $\mathbf{P}[t_1 = i] = \frac{1}{3}$ for all $1 \leq i \leq 3$. Thus,

$$\mathbf{E}[X] = \frac{1}{3}\sum_{i=1}^{3} \mathbf{E}[X|t_1 = i].$$

If we then apply the calculations we performed in the previous paragraph, we get

$$\begin{aligned}\mathbf{E}[X] &= \frac{1}{3}\big(2 + (5 + \mathbf{E}[X]) + (3 + \mathbf{E}[X])\big) \\ &= \frac{10}{3} + \frac{2\mathbf{E}[X]}{3}.\end{aligned}$$

At this point, if we solve for $\mathbf{E}[X]$ in the above equation, we obtain

$$\mathbf{E}[X] = 10.$$

In summary, we now see that the law of total expectation allowed us to solve the problem with very simple computations. From the purely mathematical point of view, the most difficult aspect of this computation consists of solving for $\mathbf{E}[X]$, which is easy. What is nontrivial about this argument is the identification of the facts that

$$\mathbf{E}[X_{2+}|t_1 = 2] = 5 + \mathbf{E}[X] \qquad \text{and} \qquad \mathbf{E}[X_{2+}|t_1 = 3] = 3 + \mathbf{E}[X].$$

This is a useful trick to keep in mind going forward.

Remark 5.36. In Example 5.35, we made a few claims that were not justified with 100% rigor. Most notably, the claims that

$$\mathbf{E}[X|t_1 = 1] = 2;$$

$$\mathbf{E}[X_1|t_1 = 2] = 5 \qquad \text{and} \qquad \mathbf{E}[X_1|t_1 = 3] = 3;$$

as well as

$$\mathbf{E}[X_{2+}|t_1 = 2] = \mathbf{E}[X] \qquad \text{and} \qquad \mathbf{E}[X_{2+}|t_1 = 3] = \mathbf{E}[X].$$

If you feel uncomfortable with the justifications of these facts presented in Example 5.35, and would like to see how these claims can be formulated in a fully rigorous way, you can look at Section 5.4.4 for the details.

That said, going forward in this class, you do not need to perform an analysis as detailed (and, arguably pedantic) as that which is carried out in Section 5.4.4. In the homework and exams, you can argue as we did in Example 5.35.

5.4.4. Fully Rigorous/Pedantic Examination Remark 5.36 (Bonus). First we look at $\mathbf{E}[X|t_1 = 1] = 2$. According to Definition 5.26, in order to compute $\mathbf{E}[X|t_1 = 1]$, we can work out the conditional range and distribution of X given $\{t_1 = 1\}$. Note that if we condition on $t_1 = 1$, then we have that

$$\mathbf{P}[(1)|t_1 = 1] = \frac{\mathbf{P}[\{(1)\} \cap \{t_1 = 1\}]}{\mathbf{P}[t_1 = 1]} = \frac{\mathbf{P}[t_1 = 1]}{\mathbf{P}[t_1 = 1]} = 1,$$

and for every other outcome of the sample space of the form $(t_1, \ldots, t_k)$ for some $k > 1$, we have

$$\mathbf{P}[(t_1, \ldots, t_k)|t_1 = 1] = \frac{\mathbf{P}[\{(t_1, \ldots, t_k)\} \cap \{t_1 = 1\}]}{\mathbf{P}[t_1 = 1]} = \frac{\mathbf{P}[\varnothing]}{\mathbf{P}[t_1 = 1]} = 0.$$

Indeed, given that elements in Ω can only end in 1, the only outcome that has $t_1 = 1$ is the 1-tuple (1); hence

$$\{t_1 = 1\} = \{(1)\} \subset \Omega.$$

Given the definition of the random variable X in (5.12), this means that the only possible output if $t_1 = 1$ is $X(1) = 2$, hence

$$R_{X|t_1=1} = \{2\} \qquad \text{and} \qquad \mathbf{P}[X = 2|t_1 = 1] = 1.$$

Thus, we conclude that

$$\mathbf{E}[X|t_1 = 1] = \sum_{x \in R_{X|t_1=1}} x \cdot \mathbf{P}[X = x|t_1 = 1] = 2 \cdot 1 = 2.$$

Next, we look at $\mathbf{E}[X_1|t_1 = 2] = 5$ and $\mathbf{E}[X_1|t_1 = 3] = 3$. Notice that we can formally write the random variable

$$X_1(t_1, \ldots, t_k) = \begin{cases} 2 & \text{if } t_1 = 1 \\ 5 & \text{if } t_1 = 2 \\ 3 & \text{if } t_1 = 3 \end{cases}.$$

Here, once again we see that if we condition on t_1, then the range of X_1 becomes a single number (i.e., whatever t_1 is equal to); the computation of the conditional expectation is thus essentially the same as in the previous paragraph.

Finally, we look at $\mathbf{E}[X_{2+}|t_1 = 2] = \mathbf{E}[X]$ and $\mathbf{E}[X_{2+}|t_1 = 3] = \mathbf{E}[X]$. We will only consider the case $t_1 = 2$, since the case $t_1 = 3$ is very similar. For this we note that we can formally define

$$X_{2+}(t_1, \ldots, t_k) = 0$$

if $(t_1, \ldots, t_k) = (1)$ (in other words, $t_1 = 1$), and otherwise we define

$$\begin{aligned} X_{2+}(t_1, \ldots, t_k) = 2 \cdot \#(\{2 \leq i \leq k : t_i = 1\}) \\ + 5 \cdot \#(\{2 \leq i \leq k : t_i = 2\}) + 3 \cdot \#(\{2 \leq i \leq k : t_i = 3\}). \end{aligned}$$

In order to show that $\mathbf{E}[X_{2+}|t_1 = 2] = \mathbf{E}[X]$, it suffices to establish that

$$R_{X_{2+}|t_1=2} = R_X \qquad \text{and} \qquad \mathbf{P}[X_{2+} = x|t_1 = 2] = \mathbf{P}[X = x] \text{ for all } x.$$

In order to establish this formally, we must appreciate the fractal or self-similar nature of the sample space Ω.

More specifically, consider the event that we are conditioning on, namely:

$$\{d_1 = 2\} = \big\{(2, t_2 \ldots, t_k) : k \geq 2,\ t_i \in \{2, 3\} \text{ for all } 2 \leq i \leq k - 1,\ t_k = 1\big\}.$$

Inside of each element in this event, we can find an element of Ω as follows:

$$(2,\ \underbrace{t_2 \ldots, t_k}_{\text{element of } \Omega}\ ).$$

Indeed, we note that

(1) the requirement that $k \geq 2$ in the elements of $\{t_1 = 2\}$ means that $(t_2 \ldots, t_k)$ has at least one element; and
(2) the conditions $t_i \in \{2, 3\}$ for all $2 \leq i \leq k-1$ and $t_k = 1$ state that $(t_2 \ldots, t_k)$ can be viewed as a sequence of decisions made by the miner, which ends as soon as tunnel 1 is crossed for the first time.

Thus, each element of the event $\{t_1 = 2\}$ can be associated to a unique element in Ω, namely, the element $(t_2, \ldots, t_k) \in \Omega$ that consists of the choices made by the miner after the first choice of $t_1 = 2$. Conversely, if we look at any element $(t_1, \ldots, t_k) \in \Omega$ in the original sample space, then we can associate it to a unique element in the event $\{d_1 = 2\}$ by adding a 2 on the left:

$$(2, t_1, \ldots, t_k) \in \{d_1 = 2\}.$$

That is, the above is the unique element in $\{d_1 = 2\}$ that starts with 2 and is then followed by the sequence of decisions $(t_1 \ldots, t_k)$. Therefore, there is a one-to-one correspondence between the elements of $\{t_1 = 2\}$ and Ω.

With this, we can immediately conclude that $R_{X_{2+}|t_1=2} = R_X$. Indeed, if $x \in \mathbb{R}$ is such that

$$\mathbf{P}[X_{2+} = x|d_1 = 2] = \frac{\mathbf{P}[\{X_{2+} = x\} \cap \{t_1 = 2\}]}{\mathbf{P}[t_1 = 2]} > 0$$

(that is, if $x \in R_{X_{2+}|t_1=2}$), then this means that there exists at least one outcome $(2, t_2, \ldots, t_k) \in \{t_1 = 2\}$ such that

$$X_{2+}(2, t_2, \ldots, t_k) = x.$$

But then, given that

$$X_{2+}(2, t_2, \ldots, t_k) = X(t_2, \ldots, t_k),$$

this means that x is also a possible output for X (hence $x \in R_X$). Similarly, we can argue that each possible outcome of X must also be a possible outcome of X_{2+} conditional on $t_1 = 2$. Thus, $R_{X_{2+}|t_1=2}$ and R_X are the same set.

At this point, all that now remains to do is prove that the conditional distribution of X_{2+} given $t_1 = 2$ is the same as the distribution of X. For this, it is sufficient to note that for every outcome $(2, t_2, \ldots, t_k) \in \{t_1 = 2\}$, we have that

$$\mathbf{P}[(2, t_2, \ldots, t_k)|t_1 = 2] = \frac{\mathbf{P}[\{(2, t_2, \ldots, t_k)\} \cap \{t_1 = 2\}]}{\mathbf{P}[t_1 = 2]}$$
$$\frac{\mathbf{P}[(2, t_2, \ldots, t_k)]}{\mathbf{P}[t_1 = 2]} = \frac{1/6^k}{1/6} = \frac{1}{6^{k-1}} = \mathbf{P}[(t_2, \ldots, t_k)].$$

Indeed, with this in hand we can write

$$\begin{aligned}
\mathbf{P}[X_{2+} = x|t_1 = 2] &= \sum_{\substack{(2,t_2,\ldots,t_k)\in\{t_1=2\} \\ X_{2+}(2,t_2,\ldots,t_k)=x}} \mathbf{P}[(2, t_2, \ldots, t_k)|t_1 = 2] \\
&= \sum_{\substack{(2,t_2,\ldots,t_k)\in\{t_1=2\} \\ X_{2+}(2,t_2,\ldots,t_k)=x}} \mathbf{P}[(t_2, \ldots, t_k)] \\
&= \sum_{\substack{(t_2,\ldots,t_k)\in\Omega \\ X(t_2,\ldots,t_k)=x}} \mathbf{P}[(t_2, \ldots, t_k)] \\
&= \mathbf{P}[X = x],
\end{aligned}$$

as desired.

Chapter 6

Some Important Examples of Discrete Random Variables

In this chapter, we introduce a number of random variables that are of fundamental importance in basic modelling questions. In short, this will lead us to the development and study of three well-known probability distributions in probability, namely, the binomial, geometric, and Poisson distributions.

Apart from enabling us to study a variety of interesting modelling questions, some of the random variables introduced in this chapter will help pave the way for our study of the law of large numbers (the subject of the next chapter), which will finally allow us to formulate the precise relationships between probability measures and expected values and their empirical counterparts in (2.6) and (5.2)/(5.6).

6.1. Indicator Random Variables and Processes

The fundamental building block of all the distributions that will be considered in this chapter consists of a very simple random variable called an indicator:

Definition 6.1 (Indicator random variable)**.** Let Ω be a sample space and $A \subset \Omega$ be an event. We define the indicator random variable of A, which we denote by $\mathbf{1}_A$, as the random variable

$$\mathbf{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A. \end{cases}$$

The name "indicator" comes from the fact that $\mathbf{1}_A$'s output indicates whether or not A has occurred (i.e., if $\mathbf{1}_A = 1$, then A has occurred, and if $\mathbf{1}_A = 0$, then it has not). Moreover, the distribution of an indicator random variable contains the same information as the probability that A occurs: The only two possible values that $\mathbf{1}_A$ could take are 0 and 1, and

$$\mathbf{P}[\mathbf{1}_A = 1] = \mathbf{P}[A] \qquad \text{and} \qquad \mathbf{P}[\mathbf{1}_A = 0] = 1 - \mathbf{P}[A]. \tag{6.1}$$

Using this distribution, it is also easy to check the interesting fact that

$$\mathbf{E}[\mathbf{1}_A] = 1 \cdot \mathbf{P}[\mathbf{1}_A = 1] + 0 \cdot \mathbf{P}[\mathbf{1}_A = 0] = \mathbf{P}[A]. \tag{6.2}$$

Remark 6.2. As per (6.1), we conclude that the range of an indicator is given by the following formula:

$$R_{\mathbf{1}_A} = \begin{cases} \{0\} & \text{if } \mathbf{P}[A] = 0 \\ \{0,1\} & \text{if } 0 < \mathbf{P}[A] < 1 \\ \{1\} & \text{if } \mathbf{P}[A] = 1. \end{cases}$$

That is, the range is typically $\{0, 1\}$, unless the event either never occurs (i.e, $\mathbf{P}[A] = 0$) or always occurs (i.e., $\mathbf{P}[A] = 1$).

In isolation, a single indicator random variable $\mathbf{1}_A$ is not terribly interesting, because the study of its distribution and expectation reduces completely to just understanding the probability that its associated event A occurs (or not). However, the study of indicators starts to become interesting once we consider multiple indicator random variables at the same time. This leads us to the formulation of a more general definition:

Definition 6.3 (Indicator process)**.** Let Ω be a sample space and $A_1, A_2, A_3, \ldots$ be a finite or countably infinite collection of events on Ω. The indicator process of the collection of events $A_1, A_2, A_3, \ldots$ is the sequence of indicator random variables

$$(\mathbf{1}_{A_1}, \mathbf{1}_{A_2}, \mathbf{1}_{A_3}, \ldots).$$

A realization of an indicator process will be a sequence of 0's and 1's, such as

$$(0, 1, 1, 0, 0, 1, 1, 0, \ldots).$$

This sequence indicates which of the events A_i have occurred; for instance, the above sequence says that the events A_2, A_3, A_6, and A_7 have occurred, whereas A_1, A_4, A_5, and A_8 have not.

In contrast to isolated indicator random variables, what makes indicator processes interesting is that there is a number of basic questions that we could ask about such random sequences of ones and zeroes, which, as it turns out, leads to a surprisingly rich theory. While there are many features of indicator processes that we could consider, in this chapter we will focus our attention on two types of questions, namely, the question of counting and the question of arrival times. We now discuss both of these questions in details.

6.2. Counting Random Variables and the Binomial Distribution

The first interesting question involving indicator processes is that of counting, namely: "How many of the events A_i have occurred?" Mathematically, this can be defined as follows:

Definition 6.4 (Counting random variable)**.** Let $A_1, A_2, \ldots, A_n$ be a finite collection of events. The counting random variable of those events is

$$X = \sum_{i=1}^{n} \mathbf{1}_{A_i}.$$

In words, this adds one whenever one of the A_i's occur and zero otherwise; hence X is equal to the number of events in the collection $A_1, \ldots, A_n$ that have occurred.

It is customary to assume that we have a finite collection in this case, so as to avoid $X = \infty$. A simple example of a counting random variable is as follows:

Example 6.5. Suppose that, on a given year, 35 000 students apply to the University of Chicago. Suppose that we assign to each applicant a unique number from 1 to 35 000. If we define the events

$$A_i = \text{“Was student } \#i \text{ offered admission?”}$$

then the number of students admitted for that year is the random variable

$$X = \sum_{i=1}^{35\,000} \mathbf{1}_{A_i}.$$

Now that we have defined counting random variables, the obvious follow-up question is: “What can we say about them?” For instance, we could ask what is the distribution, the expected value, or the variance of a counting random variable. As one might expect, the answer to these questions depends significantly on how exactly we define the events A_i. That said, interestingly enough, the expectation of counting random variables is fairly easy to characterize:

Proposition 6.6 (Expected value of counting random variables)**.** *Let $A_1, A_2, \ldots, A_n$ be a finite collection of events, and let*

$$X = \sum_{i=1}^{n} \mathbf{1}_{A_i}$$

be the associated counting random variable. We have that

$$\mathbf{E}[X] = \sum_{i=1}^{n} \mathbf{P}[A_i].$$

This follows directly from the linearity of the expected value:

$$\mathbf{E}[X] = \mathbf{E}\left[\sum_{i=1}^{n} \mathbf{1}_{A_i}\right] = \sum_{i=1}^{n} \mathbf{E}[\mathbf{1}_{A_i}],$$

and then we use the fact that $\mathbf{E}[\mathbf{1}_A] = \mathbf{P}[A]$ (see (6.2)). That being said, apart from the expected value, it is very difficult to say anything meaningful about counting random variables without making additional assumptions on the events A_i. In order to illustrate this, consider the following example:

Example 6.7 (Range)**.** Let $A_1, A_2, \ldots, A_n$ be a finite collection of events, and let

$$X = \sum_{i=1}^{n} \mathbf{1}_{A_i}.$$

What is the range of X? The answer depends on what the events A_i are.

On the one hand, if we assume that the events A_i are all the same, i.e.,

$$A_1 = A_2 = \cdots = A_n = A$$

for some event A, then R_X can only contains 0 and/or n. In other words, either none of the A_i's occur (when A does not occur), or they all occur simultaneously (when A occurs). Thus, X cannot ever give any output that is not either equal to 0 or n.

On the other hand, if we assume that the events A_i are all independent and such that $0 < \mathbf{P}[A_i] < 1$, then $R_X = \{0, 1, 2, \ldots, n\}$. Indeed, since the events A_i have a probability that is neither zero nor one, then each $\mathbf{1}_{A_i}$ can be either 0 or 1. Moreover, since the events are independent, observing that any collection of A_i's occur (or not) has no impact on the probability that the other A_i's occur (or not). Therefore, any permutation of which A_i's occur or not is possible, meaning that the sum of their indicators could take any value between 0 and n.

As the above example illustrates, we can expect to have very different distributions when the events A_i satisfy different assumptions. In this course, we will mainly focus our attention on a special case of the counting random variable called the binomial:

Definition 6.8 (Binomial random variable)**.** Let $A_1, A_2, \ldots, A_n$ be a finite collection of events, and let

$$X = \sum_{i=1}^{n} \mathbf{1}_{A_i}.$$

Let $0 < p < 1$. We say that X is binomial with parameters n and p, which we denote by $X \sim \mathrm{Bin}(n, p)$, if the following two conditions hold:

(1) The events $A_1, A_2, \ldots, A_n$ are independent.

(2) The events A_i all have the same probability $\mathbf{P}[A_i] = p$.

What makes binomial random variables interesting is a combination of two factors, that is, binomial random variables arise in a number of interesting examples, and their distribution can be computed explicitly with relative ease. In order to understand the first point, we consider two examples:

Example 6.9 (Number of heads in 100 coin tosses)**.** Recall the problem studied in Homework 1, wherein we considered the experiment of tossing 100 fair and independent coins (at that time, we formulated this by saying that each possible sequence of 100 heads or tails are equally likely, but we now know thanks to Example 4.9 that these two points of view are the same). Let X be the number of coin flips that come up heads. Then, $X \sim \mathrm{Bin}(100, 1/2)$.

More generally, the binomial random variable is useful whenever we carry out the same random experiment independently multiple times in a row, and we want to count how many of these experiments gave a particular outcome. In fact, thanks to this connection, binomial random variables are fundamental to our understanding of how to approximate theoretical probabilities with empirical frequencies:

Example 6.10 (Experimental frequencies)**.** Suppose that we have an event $A \subset \Omega$ such that $\mathbf{P}[A] = p$ for some unknown number $0 < p < 1$. In other words, we do not know what is the probability of A. In order to estimate this probability, we could try to carry out the experiment a large number of times (say, n), and then approximate $\mathbf{P}[A]$ by the empirical frequency

$$\frac{\#\text{ times } A \text{ has occurred}}{n}.$$

As it turns out, this is connected to the binomial random variable. Indeed, the quantity "# times A has occurred" is some kind of counting random variable, as it keeps track of how many times a certain event has occurred. More specifically, if we assume that our successive trials of the random experiment are all independent, then we have that

$$\#\text{ times } A \text{ has occurred} \sim \mathrm{Bin}(n,p).$$

The significance of this connection will be explored in details in the next chapter on the law of large numbers.

As the above examples illustrate, understanding the binomial distribution is fundamental to a number of interesting problems. For this purpose, we have the following result:

Proposition 6.11. *Let* $X \sim \mathrm{Bin}(n,p)$ *for some positive integer* n *and probability* $0 < p < 1$*. The following holds:*

(1) $R_X = \{0,1,2,\ldots,n\}$*, and for every* $0 \leq x \leq n$*,*

$$\mathbf{P}[X = x] = {}_nC_x\, p^x (1-p)^{n-x}. \tag{6.3}$$

(2) $\mathbf{E}[X] = np$*.*

(3) $\mathbf{Var}[X] = np(1-p)$*.*

Proof. (1) Since X is the sum of n numbers that are all either 0 or 1, the range cannot contain any numbers other than $0,1,2,\ldots,n$ (i.e., R_X cannot contain a number that is negative, bigger than n, or not an integer). Thus, to establish that the range is in fact $\{0,1,2,\ldots,n\}$, it suffices to prove that $\mathbf{P}[X = x] > 0$ for every $x \in \{0,1,2,\ldots,n\}$. In particular, since every number on the right-hand side of (6.3) is positive, it suffices to establish that (6.3) holds.

For this purpose, we can write the event

$$\{X = x\} = \bigcup_{\substack{a_1,a_2,\ldots,a_n \in \{0,1\} \\ \sum_{i=1}^n a_i = x}} \{\mathbf{1}_{A_1} = a_1\} \cap \{\mathbf{1}_{A_2} = a_2\} \cap \cdots \cap \{\mathbf{1}_{A_n} = a_n\}.$$

The numbers a_i determine which of the events A_i have occurred (i.e., when $a_i = 1$), and which ones have not (i.e., when $a_i = 0$). The condition that $\sum_{i=1}^n a_i = x$ in the union is simply saying that we want exactly x events to have occurred, which is consistent with $\{X = x\}$. Thus, this union enumerates every possible way to assign which of the x events A_i have occurred.

Clearly, this union is disjoint, and so by Axiom 3 we have that

$$\mathbf{P}[X = x] = \sum_{\substack{a_1,a_2,\ldots,a_n \in \{0,1\} \\ \sum_{i=1}^n a_i = x}} \mathbf{P}\big[\{\mathbf{1}_{A_1} = a_1\} \cap \{\mathbf{1}_{A_2} = a_2\} \cap \cdots \cap \{\mathbf{1}_{A_n} = a_n\}\big].$$

Next, since we assume the events A_i to be independent, we have that

$$\begin{aligned}\mathbf{P}\big[\{\mathbf{1}_{A_1}=a_1\}\cap\{\mathbf{1}_{A_2}=a_2\}\cap\cdots\cap\{\mathbf{1}_{A_n}=a_n\}\big]\\ =\mathbf{P}[\mathbf{1}_{A_1}=a_1]\mathbf{P}[\mathbf{1}_{A_2}=a_2]\cdots\mathbf{P}[\mathbf{1}_{A_n}=a_n].\end{aligned}$$

Because all of these events occur with probability p, we know that

$$\mathbf{P}[\mathbf{1}_{A_i}=a_i]=\begin{cases}p & \text{if } a_i=1\\ 1-p & \text{if } a_i=0\end{cases}.$$

Thus, if we know that exactly x of the a_i's are equal to 1, and the remaining $n-x$ a_i's are equal to zero, then this means that

$$\mathbf{P}\big[\{\mathbf{1}_{A_1}=a_1\}\cap\{\mathbf{1}_{A_2}=a_2\}\cap\cdots\cap\{\mathbf{1}_{A_n}=a_n\}\big]=p^x(1-p)^{n-x}.$$

In summary, we have that

$$\mathbf{P}[X=x]=\sum_{\substack{a_1,a_2,\ldots,a_n\in\{0,1\}\\ \sum_{i=1}^n a_i=x}}p^x(1-p)^{n-x}.$$

Since the quantity $p^x(1-p)^{n-x}$ in the above sum does not depend on how we choose the a_i's (so long as exactly x of them are equal to 1), we conclude that $\mathbf{P}[X=x]$ is simply equal to $p^x(1-p)^{n-x}$ times the number of ways to choose $a_1,\ldots,a_n\in\{0,1\}$ such that $\sum_{i=1}^n a_i=x$. In other words, this is the number of ways to choose x elements out of n, which is ${}_nC_x$. Thus, we obtain (6.3).

(2) The formula for the expectation of $\mathrm{Bin}(n,p)$ is a direct consequence of Proposition 6.6:

$$\mathbf{E}[X]=\sum_{i=1}^n\mathbf{P}[A_i]=\sum_{i=1}^n p=np.$$

(3) The formula for the variance of $\mathrm{Bin}(n,p)$ uses the interesting property of the variance stated in Proposition 5.31-(2): Since the events A_i are independent, the variance of the sum of their indicators is the sum of the individual variances:

$$\mathbf{Var}[X]=\mathbf{Var}\left[\sum_{i=1}^n\mathbf{1}_{A_i}\right]=\sum_{i=1}^n\mathbf{Var}[\mathbf{1}_{A_i}].$$

To compute the variance of the indicators, we use the formula (5.9):

$$\mathbf{Var}[\mathbf{1}_{A_i}]=\mathbf{E}[\mathbf{1}_{A_i}^2]-\mathbf{E}[\mathbf{1}_{A_i}]^2=\mathbf{E}[\mathbf{1}_{A_i}^2]-p^2,$$

and

$$\mathbf{E}[\mathbf{1}_{A_i}^2]=1^2\cdot p+0\cdot(1-p)=p,$$

hence

$$\mathbf{Var}[\mathbf{1}_{A_i}]=p-p^2=p(1-p). \tag{6.4}$$

If we then sum this variance n times, we obtain the claimed variance formula. □

To close off our discussion of binomial random variables, we provide

(1) three illustrations of the binomial distribution with parameters $n=10$ and $p\in\{\frac{1}{4},\frac{1}{2},\frac{3}{4}\}$ in Figures 6.1, 6.2, and 6.3 below; and

(2) two illustrations of the distribution of the number of heads in 100 fair and independent coin tosses in Figures 6.4 and 6.5 (i.e., $\mathrm{Bin}(100, 50)$, which clarifies the open-ended question concerning the small size of $\mathbf{P}[\text{exactly 50 heads}] \approx 0.0796$ in Homework 1.)

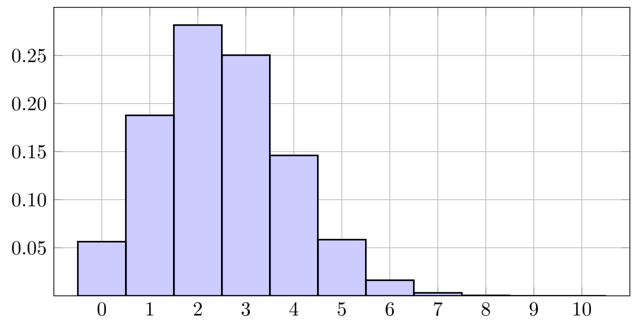

Figure 6.1. Binomial distribution with $n = 10$ and $p = 1/4$.

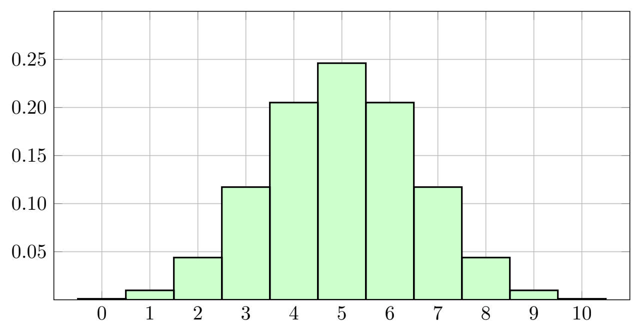

Figure 6.2. Binomial distribution with $n = 10$ and $p = 1/2$.

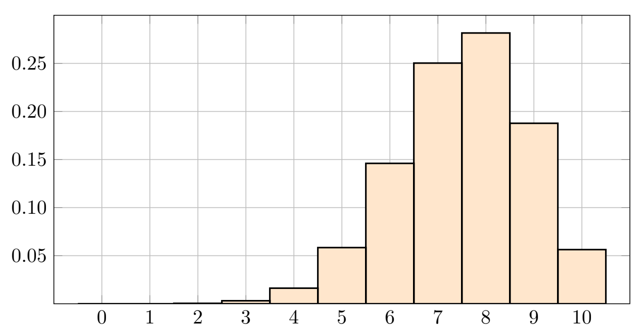

Figure 6.3. Binomial distribution with $n = 10$ and $p = 3/4$.

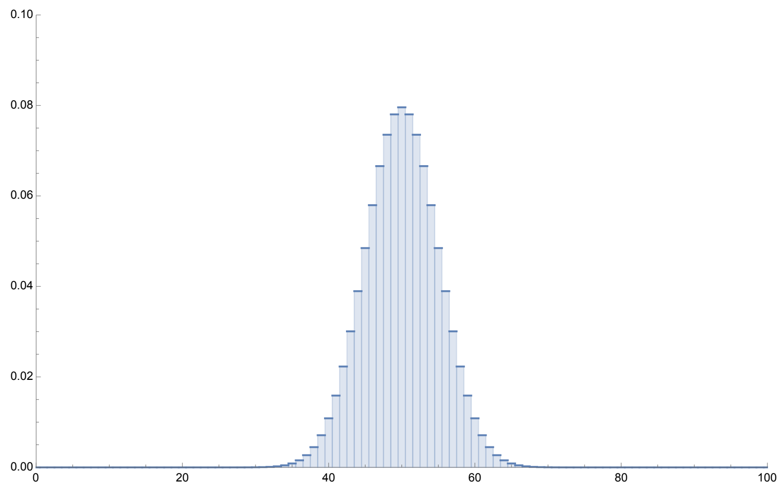

Figure 6.4. Number of heads in 100 fair and independent coin flips (full).

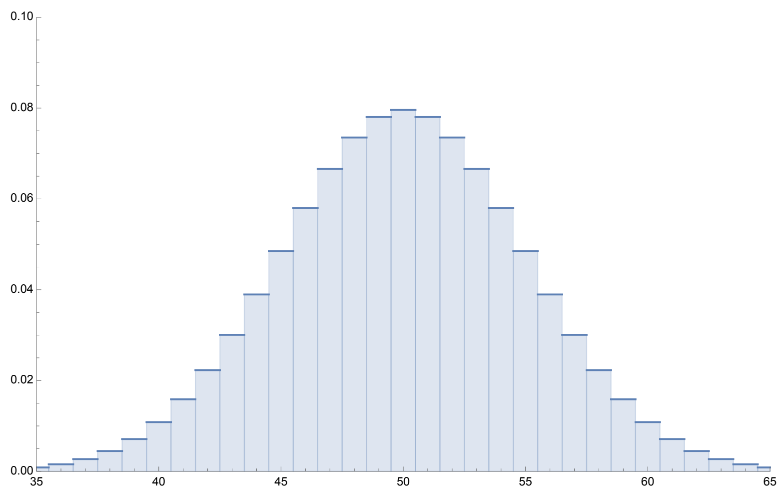

Figure 6.5. Number of heads in 100 fair and independent coin flips (closeup).

6.3. First Arrival Times and the Geometric Distribution

The second question that one could ask about indicator processes is that of the first arrival time:

Definition 6.12. Let $A_1, A_2, A_3, \ldots$ be a countable infinite sequence of events. The first arrival time of the A_i's is the random variable

$$X = \min\{i \geq 1 : \mathbf{1}_{A_i} = 1\}.$$

In words, we arrange the sequence of events A_i in order, and we count how many events we have to go through until we find one that has occurred. For example, suppose that we look at an outcome of an indicator process given by

$$(0, 0, 1, 0, 1, 0, 0, 0, \ldots).$$

This means that the events A_3 and A_5 have occurred, whereas A_1, A_2, A_4, A_6, A_7, and A_8 have not. If X is the first arrival time of this sequence of events, then $X = 3$, because A_3 is the first event that occurs in the ordered sequence $A_1, A_2, A_3, \ldots$.

Example 6.13 (Escape the mine)**.** Recall the problem of the miner lost in the mine, which we introduced in Example 5.35: A miner attempts to escape a mine with three tunnels by repeatedly choosing a tunnel uniformly at random.

Suppose that we modify the problem slightly, by saying that the miner will keep selecting numbers in $\{1, 2, 3\}$ at random forever. Consider the sequence of events defined as

$$A_i := \text{“Does the miner choose 1 at the } i^{\text{th}} \text{ attempt?”}$$

for every $i \geq 1$. If we let X denote the random variable that represents the number of attempts required to escape the tunnel, then X is the first arrival time of the sequence of events $A_1, A_2, A_3, \ldots$.

Remark 6.14. One crucial difference between counting random variables and first arrivals is that, for counting variables, the order in which we label the events A_i does not matter. Indeed, in that case all we care about is how many events occurred. In sharp contrast to that, in the case of first arrivals, the order of the A_i's is of crucial importance. For example, if we look at the sequence of indicators

$$(0, 0, 1, 0, 1, 0, 0, 0, \ldots)$$

and switch the order of the first and third events, then the sequence becomes

$$(1, 0, 0, 0, 1, 0, 0, 0, \ldots).$$

In doing so, the first arrival changes from 3 to 1. Therefore, first arrivals are most useful when there is a natural way to order the events A_i. As the word *time* in “first arrival time” suggests, this order most often comes from the fact that the events A_i represent a sequence of experiments that are carried out one after the other in time. This was the case, for instance, in the miner problem discussed in Example 6.13: Successive attempts to escape the mine have a natural order, in that they occur one after the other in time.

Just like counting random variables, we could ask basic questions about the behavior first arrival times, such as the expected value, distribution, etc. However, in the case of first arrivals, it is difficult to say anything meaningful without making some assumptions about how the events A_i depend on one another. This is, in large part, due to the fact that counting random variables are sums of indicator random variables, and that the sum interacts well with the expectation thanks to the linearity property. In contrast, the first arrival time is a minimum of an ordered list of indicators, which does not have a simple algebraic structure. For this reason, we immediately restrict our attention to a special case of first arrivals, which, like the binomial, concerns independent variables that all have the same probability:

Definition 6.15 (Geometric random variable)**.** Let $A_1, A_2, A_3, \ldots$ be events, and let X be their first arrival time. Let $0 < p < 1$. We say that X is geometric with probability p, denoted $X \sim \text{Geom}(p)$, if the A_i's are independent and $\mathbf{P}[A_i] = p$ for every $i \geq 1$.

In words, the geometric random variable consists of attempting the same experiment independently in successive trials, and then counting how many trials are required to get the first "success." (Here, we call the occurrence of one of the events A_i a success.)

Example 6.16 (Escape the mine)**.** In the formulation of the miner's problem in Example 6.13 (i.e., where the miner chooses a number among $\{1, 2, 3\}$ infinitely often, uniformly at random and independently), we have that the first arrival time X is geometric with probability $1/3$.

With the assumptions of independence and equal probability, the geometric distribution can be characterized thoroughly, in similar fashion to the binomial:

Proposition 6.17. *Let $X \sim \text{Geom}(p)$ for some probability $0 < p < 1$. The following holds:*

(1) $R_X = \{1, 2, 3, \ldots\}$*, and for every integer* $x \geq 1$*,*
$$\mathbf{P}[X = x] = (1 - p)^{x-1}p.$$
(2) $\mathbf{E}[X] = \frac{1}{p}$.

(3) $\mathbf{Var}[X] = \frac{1-p}{p^2}$.

Before proving this proposition, a couple of remarks are in order:

Remark 6.18. On the one hand, the formula for the expectation is fairly intuitive: If, for instance, $p = \frac{1}{10}$, then this means that about one in every ten events A_i will occur. Thus, it makes sense that we should have to wait for $\frac{1}{p} = 10$ attempts on average until we find the first A_i that has occurred. The same reasoning can be applied for any other value of p.

On the other hand, while the specific form of the variance (i.e., $\frac{1-p}{p^2}$) is arguably not very intuitive, the general shape of its plot as a function of p is nevertheless insightful; see Figure 6.6 below for an illustration. As shown in that figure, the variance blows up to $+\infty$ as p gets closer to zero, and it vanishes to zero as p gets closer to one. Thus, the output of a geometric random variable becomes increasingly uncertain as the probability of the event goes to zero.

Proof of Proposition 6.17. (1) For any positive integer x, we can write the event
$$\{X = x\} = \left(\bigcap_{i=1}^{x-1} \{\mathbf{1}_{A_i} = 0\}\right) \cap \{\mathbf{1}_{A_x} = 1\}.$$
Indeed, $X = x$ means that it took x attempts to observe the first occurrence of one of the events A_i; in other words the first $x - 1$ events did not occur, and A_x occurred. Since we assume that the A_i's are independent and all have probability p, this yields
$$\mathbf{P}[X = x] = \left(\prod_{i=1}^{x-1} \big(1 - \mathbf{P}[A_i]\big)\right) \mathbf{P}[A_x] = (1 - p)^{x-1}p.$$

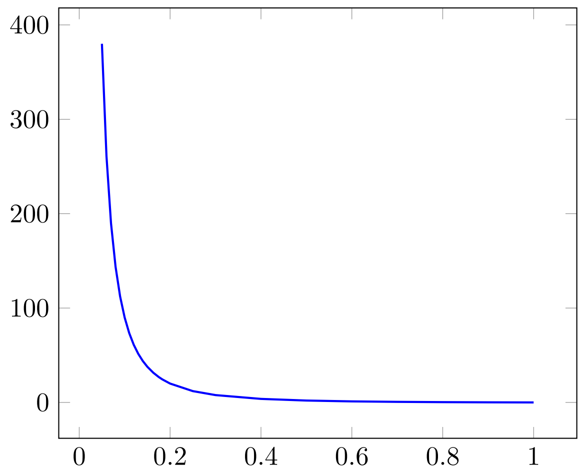

Figure 6.6. Plot of the variance of Geom(p) for $0 < p < 1$.

(2) This is something that we can easily solve by first-step analysis (i.e., recall the miner problem in Example 5.35, or the problem of counting the dice rolls until the first 6-1 or 6-6 in Homework 3). More specifically, by the law of total expectation we can write

$$\mathbf{E}[X] = \mathbf{E}[X|A_1]p + \mathbf{E}[X|A_1^c](1-p).$$

If the first event A_1 in the sequence occurs, then this immediately tells us that $X = 1$. Otherwise, if A_i^c occurs, then we waste one step, and the number of trials until we observe a first success in the events $A_2, A_3, A_4, \ldots$ is once again geometric with probability of success p (in particular, knowing that A_1 did not occur has no influence whatsoever on A_i for $i \geq 2$). Therefore,

$$\mathbf{E}[X] = p + (1 + \mathbf{E}[X])(1-p).$$

If we solve for $\mathbf{E}[X]$ in this expression, then we get

$$\mathbf{E}[X] = \frac{1}{p},$$

as desired.

(3) We now compute the variance:

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2.$$

We already know that $\mathbf{E}[X]^2 = 1/p^2$, and thus we only need to compute $\mathbf{E}[X^2]$. For this, we once again use first-step analysis. That is, we use the law of total expectation to write:

$$\mathbf{E}[X^2] = \mathbf{E}[X^2|A_1]p + \mathbf{E}[X^2|A_1^c](1-p).$$

On the one hand, if A_1 occurs, then $X^2 = 1$. On the other hand, if A_1^c occurs, then the conditional distribution of X^2 given A_1^c is the same as the distribution of $(1+X)^2$; as we argued for the expectation, this comes from the fact that we add one wasted trial because A_1 did not occur, and then the remaining number of trials has the same distribution as X itself. Thus,

$$\mathbf{E}[X^2] = p + \mathbf{E}[(1+X)^2](1-p).$$

If we expand the $(1+X)^2$ and use linearity of expectation, this gives

$$\mathbf{E}[X^2] = p + (1 + 2\mathbf{E}[X] + \mathbf{E}[X^2])(1-p).$$

We already know that $\mathbf{E}[X] = 1/p$, so if we substitute this value in the above expression, then we finally get

$$\mathbf{E}[X^2] = p + (1 + 2/p + \mathbf{E}[X^2])(1-p).$$

Solving for $\mathbf{E}[X^2]$ in this equation yields

$$\mathbf{E}[X^2] = \frac{2-p}{p^2}.$$

Putting everything together, we obtain

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2},$$

concluding the computation of the variance. $\square$

We now close this section with some illustrations of the geometric distribution for various values of p in Figures 6.7, 6.8, and 6.9 below. Therein, you can observe that the distribution becomes more and more "flat" as the probability p approaches zero, which is consistent with the fact that the variance blows up to $+\infty$ as $p \to 0$ (thus making the random variable more unpredictable).

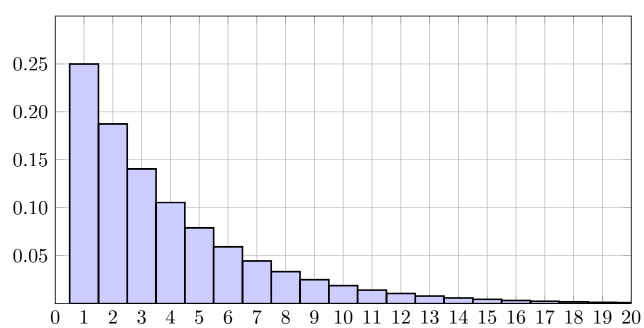

Figure 6.7. Geometric distribution with $p = 1/4$.

Remark 6.19 (Direct computation of the expectation and variance of the geometric distribution (optional))**.** While I personally find the computation of the expectation and variance of the geometric using first-step analysis more elegant, it is also possible to compute these quantities directly from the distribution. The latter method uses calculus, and is more in line with the standard approach in most textbooks. For your interest, here is how you can prove these statements directly:

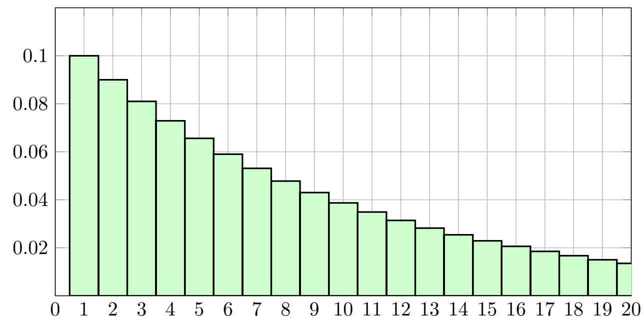

Figure 6.8. Geometric distribution with $p = 1/10$.

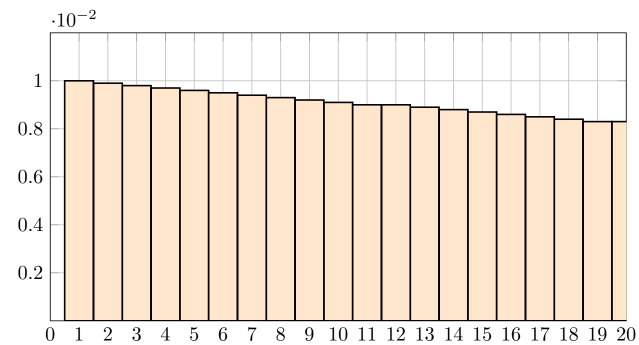

Figure 6.9. Geometric distribution with $p = 1/100$.

(2) In order to compute the expectation of Geom(p) directly, we first recall a fact from calculus, namely, the Taylor series of the function $y \mapsto \frac{1}{1+y}$ near zero:

$$\frac{1}{1-y} = \sum_{k=0}^{\infty} y^k \qquad \text{for every } |y| < 1. \tag{6.5}$$

The infinite sum in the above equation is typically called the geometric series, and its connection with the geometric random variable explains the name "*geometric* random variable." With this in hand, we now compute $\mathbf{E}[X]$. By definition of expected value, we have that

$$\mathbf{E}[X] = \sum_{x=1}^{\infty} x\,\mathbf{P}[X = x] = \sum_{x=1}^{\infty} x\,(1-p)^{x-1}p.$$

Since p does not depend on x, we can pull it out of the sum, which yields

$$\mathbf{E}[X] = p\sum_{x=1}^{\infty} x\,(1-p)^{x-1} = p\sum_{x=0}^{\infty} x\,(1-p)^{x-1}.$$

(In the second equality above, I have added $x = 0$ to the sum. When $x = 0$, we have that $x\,(1-p)^{x-1} = 0$; hence this does not change the value of the sum at all.

However, we will see in just a moment why it is useful to add this term to the sum.) For every $x \geq 0$, we note that

$$\frac{\mathrm{d}}{\mathrm{d}p}(1-p)^x = -x(1-p)^{x-1}.$$

Therefore, we can write

$$\mathbf{E}[X] = -p\frac{\mathrm{d}}{\mathrm{d}p}\sum_{x=0}^{\infty}(1-p)^x.$$

We now recognize that this is the derivative of a geometric series (which is why I added $x=0$)! Since $0<1-p<1$, we can apply the formula in (6.5), which yields

$$\mathbf{E}[X] = -p\,\frac{\mathrm{d}}{\mathrm{d}p}\left(\frac{1}{1-(1-p)}\right) = -p\,\frac{\mathrm{d}}{\mathrm{d}p}\frac{1}{p} = (-p)\left(-\frac{1}{p^2}\right) = \frac{1}{p}.$$

(3) We now compute the variance:

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2.$$

We already know that $\mathbf{E}[X]^2 = \frac{1}{p^2}$, and thus we only need to compute $\mathbf{E}[X^2]$. For this, we use the formula of the expected value:

$$\mathbf{E}[X^2] = \sum_{x=1}^{\infty} x^2\,\mathbf{P}[X=x] = \sum_{x=1}^{\infty} x^2\,(1-p)^{x-1}p = p\sum_{x=1}^{\infty} x^2\,(1-p)^{x-1}.$$

Inspired by the computation of the expected value, it is tempting to attempt to relate the above sum to some kind of derivative of the function $\frac{1}{1-y}$. While this is not entirely misguided, this time the argument is slightly more complicated. Indeed, if we compute the second derivative of the Taylor series, we get

$$\frac{2}{(1-y)^3} = \frac{\mathrm{d}^2}{\mathrm{d}y^2}\left(\frac{1}{1-y}\right) = \frac{\mathrm{d}^2}{\mathrm{d}y^2}\sum_{k=0}^{\infty} y^k = \sum_{k=2}^{\infty} k(k-1)y^{k-2}. \tag{6.6}$$

This is somewhat similar to the expression that we have for $\mathbf{E}[X^2]$, but still a bit different, because in the latter we have x^2 instead of $x(x-1)$. In order to get around this, consider the following:

$$\mathbf{E}[X(X-1)] = \sum_{x=1}^{\infty} x(x-1)(1-p)^{x-1}p = p\sum_{x=1}^{\infty} x(x-1)(1-p)^{x-1}.$$

This is already much more similar to (6.6) but there are still some minor differences. On the one hand, the sum for $\mathbf{E}[X(X-1)]$ starts at $x=1$, whereas (6.6) starts at $k=2$. However, this is not much of a problem. Indeed, given that

$$x(x-1)(1-p)^{x-1} = 0$$

when $x=1$, we can remove this term from the sum without changing its value. Thus, we can write

$$\mathbf{E}[X(X-1)] = p\sum_{x=2}^{\infty} x(x-1)(1-p)^{x-1}.$$

On the other hand, the exponent of $x-1$ in the sum for $\mathbf{E}[X(X-1)]$ does not match the exponent of $k-2$ in (6.6). To fix this, however, we can simply pull out a factor of $(1-p)$ in each term in the sum, which yields

$$\mathbf{E}[X(X-1)] = p(1-p)\sum_{x=2}^{\infty} x(x-1)(1-p)^{x-2}.$$

We can therefore apply (6.6), which gives us

$$\mathbf{E}[X(X-1)] = \frac{2p(1-p)}{p^3} = \frac{2(1-p)}{p^2}.$$

At this point, you may be thinking that this is all well and good, but that the quantity $\mathbf{E}[X(X-1)]$ is not actually what we want to compute; what we want is $\mathbf{E}[X^2]$. However, we can actually recover the latter from the former. Indeed,

$$\mathbf{E}[X(X-1)] = \mathbf{E}[X^2 - X] = \mathbf{E}[X^2] - \mathbf{E}[X];$$

solving for $\mathbf{E}[X^2]$ in the above equation, we then get

$$\mathbf{E}[X^2] = \mathbf{E}[X(X-1)] + \mathbf{E}[X] = \frac{2(1-p)}{p^2} + \frac{1}{p} = \frac{2-p}{p^2}.$$

Putting everything together, we obtain

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2},$$

concluding the computation of the variance.

6.4. The Poisson Distribution

6.4.1. Motivating Example and Informal Definition. We now finish this chapter with one last important example of random variables, namely, the Poisson distribution. Interestingly enough, the process of defining this random variable is different from what we did in the case of the Binomial and Geometric: For the latter two cases, the definitions of the random variables were more or less trivial, and then we needed to work a bit to uncover their basic properties. In contrast, in the case of the Poisson random variable, its very definition is nontrivial. In order to illustrate this, we begin with an example that motivates the introduction of the Poisson distribution:

Example 6.20 (Arrivals in a convenience store)**.** Suppose that you own a 24h convenience store somewhere in Hyde Park. You would like to construct a probabilistic model to predict the arrivals of customers in your store through time. After paying one of your employees to survey the arrivals of customers throughout several days, you come to the following conclusion:

$$\text{average number of arrivals during a one-hour period} \approx \mu,$$

where $\mu > 0$ is some positive real number. While this is better than nothing, you would like to know more than just the average number of arrivals during a one-hour period. For instance, if we define the random variable

$$X = \text{number of customers who enter the store within a one-hour period},$$

then we expect that $\mathbf{E}[X] = \mu$, but what is X's distribution?

Of course, in order to answer the question posed in the above example, we have to make more specific assumptions about how arrivals occur in time. More generally, knowing the expected value of a random variable alone is not enough to uniquely specify its distribution (as we have seen, e.g., in Figures 5.7 and 5.8; two random variables with the same expected value but very different distributions).

The Poisson distribution consists of one example of a random variable that is designed to model arrivals through time, similar to Example 6.20. It is based on the following simple assumptions:

Definition 6.21 (Informal definition of Poisson random variables)**.** Let X be a random variable that counts the number of arrivals of some process in a fixed time period.[1] Let us assume that $\mathbf{E}[X] = \mu$, where $\mu > 0$ is a positive number. We say that X is a Poisson random variable with mean μ, which we denote by $X \sim \mathrm{Poi}(\mu)$, if the following informal assumption holds: *"At every given time, an arrival occurs with equal probability, independently of all other times."*

The above assumptions seem natural enough for many situations: It essentially says that the rate at which arrivals occur is more or less constant through time, and that arrivals are independent of one another. The problem with the above formulation, however, is that it is not precise enough to turn into a precise mathematical definition (which is why I called it informal). In order to remedy this, we first consider a simpler model of arrivals in time.

6.4.2. A Simpler Model. Let us represent the fixed time period wherein we want to count the number of arrivals as a straight line. This might look like the illustration in Figure 6.10 below. Therein, we see that the time interval stretches

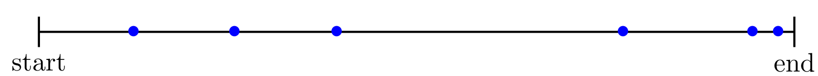

Figure 6.10. A fixed time period (marked by the time between start and end) with some arrivals (marked by blue dots). In this case $X = 6$.

between the start and end of the time period that we want to study. Arrivals in between the start and the end might look something like the blue dots in Figure 6.10. In that case, we would have that $X = 6$, because we see that six arrivals occurred during the time period.

In order to simplify the model, suppose that instead of considering the entirety of the time period, we pick some large number n, and then we subdivide the time period into n sub-intervals of equal length. That is, each sub-interval represents a duration equal to

$$\frac{1}{n} \times (\text{total duration of time period}).$$

This is illustrated in Figure 6.11 below. With this in hand, we consider the following simplification of the Poisson random variable: Inspired by the informal assumption stated in Definition 6.21, we assume the following:

[1] For instance, the number of arrivals of customers in a convenience store during a one-hour period; the number of insurance claims submitted to a company over a one-month period; the number of refund requests submitted to a customer service desk at a large retail store during a one-year period, etc.

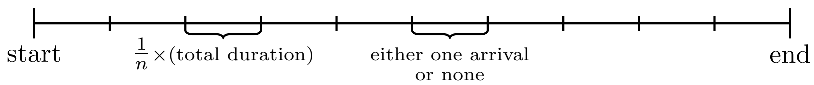

Figure 6.11. Spitting the time interval into a grid of n sub-intervals. The time length of each sub-interval is $1/n$ times the original duration. We assume that in each sub-interval, either one arrival occurs, or none.

(1) In each of the n sub-intervals, either one arrival occurs, or none.

(2) The probability that an arrival occurs is the same for each sub-interval.

(3) The occurrence of arrivals in different sub-intervals are independent.

(4) The average total number of arrivals in the sub-intervals is equal to $\mu > 0$.

Indeed, this is essentially the same as Definition 6.21, except that we have replaced "at every given time, an arrival occurs with equal probability independently" by "in each sub-interval, an arrival occurs with equal probability independently."

However, in contrast to Definition 6.21, there is no mathematical ambiguity whatsoever here: If we define

$$X_n = \text{"number of arrivals using our } n \text{ sub-interval approximation,"}$$

then $X_n \sim \text{Bin}(n, p)$, where p is the probability that an arrival occurs in one sub-interval. Indeed, X_n is nothing more than the counting random variable of the events

$$A_i = \text{"was there an arrival in the } i^{\text{th}} \text{ sub-interval?"}$$

and since these events are independent and all have the same probability by assumption, we get a binomial distribution.

That being said, we have not defined what should be the probability p that an arrival occurs in one sub-interval. Thankfully, we can determine this using the assumption that the average number of arrivals during the time period is equal to the number μ. Indeed, if $X_n \sim \text{Bin}(n, p)$, then we have computed in Proposition 6.11 that $\mathbf{E}[X_n] = np$. If we want this to be equal to μ, then we need to set $p = \frac{\mu}{n}$. Thus, we conclude the following:

$$X_n \sim \text{Bin}(n, \tfrac{\mu}{n}).$$

Remark 6.22. In order for $X_n \sim \text{Bin}(n, \frac{\mu}{n})$ to make sense, it must be the case that $\frac{\mu}{n} < 1$, since $\frac{\mu}{n}$ is the probability of the occurrence of the events counted by the binomial variable. Thus, in order for this model to make any sense, we must, at the very least, ensure that our grid size is small enough; more specifically, we must ensure that n is larger than μ.

In summary, in contrast to the vague assumptions formulated in Definition 6.21, the simplified model on the grid can be defined rather easily. However, the simplified model has a fatal flaw, that is, it assumes that *only one* arrival can occur in each sub-interval. For sake of illustration, consider Figure 6.12 below. Therein, we see that there is a mismatch between the real arrivals (blue dots) and arrivals as counted by the simplified model, because it just so happens that two arrivals occurred in the same sub-interval.

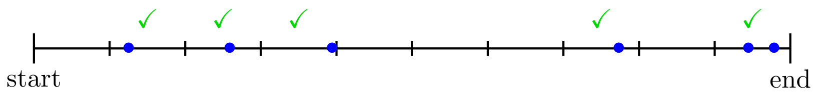

Figure 6.12. Discrepancy due to the low resolution of the approximation. The arrivals from the point of view of X_n are given by the green checkmarks. Thus, $X_n = 5$, whereas we know that actually $X = 6$.

If we want to make sense of the model introduced in Definition 6.21, then this is clearly not satisfactory. In most practical applications, there is in principle no reason to expect that arrivals can only occur one at a time in some discrete time sub-intervals. Looking back at Example 6.20, for instance, there is no reason to expect that two customers couldn't enter the store almost at the same time (e.g., one right after the other, the first customer holding the door open for the next one).

As it turns out, there is a relatively simple way to fix this problem. That is, we can simply take $n \to \infty$ in our approximation, which means that we are shrinking the size of the sub-intervals to zero. Indeed, no matter how close together in time the actual arrivals in X are, once our grid is small enough, there will only be one point in each sub-interval; see Figure 6.13. This then suggests that, in order to construct

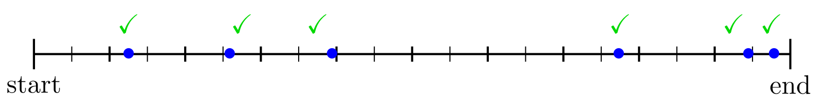

Figure 6.13. Looking at a fine enough grid eventually removes the discrepancy; X_n and X now agree on the number of arrivals.

the Poisson random variable, we should look at the limit of the approximation $X_n \sim \text{Bin}(n, \frac{\mu}{n})$ as $n \to \infty$.

6.4.3. Poisson as a Limit of Binomials. We can now formulate a precise definition of the Poisson random variable:

Definition 6.23 (Poisson random variable)**.** Let $\mu > 0$ be a positive number. For every positive integer $n > \mu$, let $X_n \sim \text{Bin}(n, \frac{\mu}{n})$. We say that X is a Poisson random variable with mean μ, denoted $X \sim \text{Poi}(\mu)$, if

$$\mathbf{P}[X = x] = \lim_{n\to\infty} \mathbf{P}[X_n = x] \qquad \text{for every } x \geq 0. \tag{6.7}$$

In words, the Poisson random variable is defined as what we could call the "continuous time limit" of the binomial approximations. We take the fact that the approximate random variables X_n satisfy assumptions very similar to Definition 6.21 as evidence of that (6.7) is the correct way to rigorously define the Poisson distribution. That being said, in order for (6.7) to be useful, we need to actually compute the limit therein. For this, we have the following result:

Proposition 6.24. *Let* $X \sim \text{Poi}(\mu)$ *for some* $\mu > 0$.

(1) *For every integer $x \geq 0$, we have that*

$$\mathbf{P}[X = x] = \frac{\mu^x \mathrm{e}^{-\mu}}{x!}. \tag{6.8}$$

(2) $\mathbf{E}[X] = \mathbf{Var}[X] = \mu$.

(Partial) Proof. We only prove the formula (6.8) for the distribution of the Poisson random variable. The computation that $\mathbf{E}[X] = \mathbf{Var}[X] = \mu$ can be carried out using (6.8) in a fairly straightforward manner (and is easily found in a variety of sources online).

By definition of binomial random variables,

$$\mathbf{P}[X_n = x] = {}_nC_x(\mu/n)^x(1-\mu/n)^{n-x}.$$

Thus, (6.8) amounts to computing that

$$\lim_{n\to\infty} {}_nC_x(\mu/n)^x(1-\mu/n)^{n-x} = \frac{\mu^x \mathrm{e}^{-\mu}}{x!}.$$

For this, we write out in details what the term ${}_nC_x$ is:

$${}_nC_x(\mu/n)^x(1-\mu/n)^{n-x} = \frac{n!}{x!(n-x)!}(\mu/n)^x(1-\mu/n)^{n-x}.$$

In order to make sense of the limit of this expression as $n \to \infty$, we split it into four parts:

$$\frac{n!}{x!(n-x)!}(\mu/n)^x(1-\mu/n)^{n-x} = \underbrace{\frac{\mu^x}{x!}}_{p_1} \cdot \underbrace{\frac{n!}{(n-x)!n^x}}_{p_2} \cdot \underbrace{(1-\mu/n)^{-x}}_{p_3} \cdot \underbrace{(1-\mu/n)^{n}}_{p_4}.$$

Indeed, we can take the limit as $n \to \infty$ of these four terms individually, and then take the product of the individual limits.

The limit of first term, which we called p_1, is trivial since it does not even depend on n. This gives us

$$\lim_{n\to\infty} \frac{\mu^x}{x!} = \frac{\mu^x}{x!}.$$

For the second term, which we called p_2, we need to think a bit more carefully: Looking at the ratio

$$\frac{n!}{(n-x)!}$$

therein, we recognize that this is equal to the counting number ${}_nP_x$. In particular, if we define the polynomial function

$$f(y) = y(y-1)(y-2)\cdots(y-x+1), \tag{6.9}$$

then we have that

$$f(n) = \frac{n!}{(n-x)!}.$$

Here, I call f a polynomial, because if we were to completely expand the products in (6.9), then we would obtain a polynomial expression of the form

$$f(y) = y^x + a_1y^{x-1} + a_2y^{x-2} + \cdots + a_x,$$

where $a_1, a_2, \ldots$ are some real numbers. As you may have learned in calculus, when we compute the limit of the ratio of two polynomials at infinity, only the dominant terms (i.e., the terms with the largest degrees) matter. In this particular case, we then get that

$$\lim_{n\to\infty}\frac{n!}{(n-x)!n^x}=\frac{f(n)}{n^x}=1;$$

indeed, the polynomials $f(y)$ and y^x both have the same dominating term (i.e., y^x), which cancel out in the limit, and all other terms vanish.

Looking at p_3, we know that μ/n goes to zero as $n\to\infty$. Thus,

$$\lim_{n\to\infty}(1-\mu/n)^{-x}=(1-0)^{-x}=1.$$

Finally, we look at p_4. This term is very similar to p_3, except that it has an exponent of n instead of an exponent of $-x$. Thus, it is not enough to simply say that $\mu/n\to 0$. To do this, we apply a clever manipulation: If we let log denote the natural logarithm function, then we can write any number z as

$$z=\mathrm{e}^{\log(z)}.$$

Indeed, the logarithm and exponential are inverses of one another. Recalling the basic property of logarithms that

$$\log(a^b)=b\cdot\log(a),$$

this then means that

$$(1-\mu/n)^n=\mathrm{e}^{\log((1-\mu/n)^n)}=\mathrm{e}^{n\cdot\log((1-\mu/n))}.$$

Next, we recall from calculus that the Taylor series of log close to 1 is

$$\log(1+y)=y-\frac{y^2}{2}+\frac{y^3}{3}-\frac{y^4}{4}+\cdots\qquad |y|<1.$$

In particular,

$$\log(1-y)=-y+R(y),\qquad |y|<1$$

where the remainder term R is such that $|R(y)|\leq Cy^2$ as $y\to 0$ for some constant $C>0$. Thus,

$$(1-\mu/n)^n=\mathrm{e}^{n\cdot(-\mu/n+R(\mu/n))}.$$

If we then distribute the n term in the sum, we get

$$(1-\mu/n)^n=\mathrm{e}^{-\mu+nR(\mu/n)}.$$

When you take $n\to\infty$, the only term that survives inside the exponential is $-\mu$, because $|nR(\mu/n)|\leq C\mu^2/n\to 0$. Thus, we conclude that

$$\lim_{n\to\infty}(1-\mu/n)^n=\mathrm{e}^{-\mu}.$$

If we then combine all of the computations that we have done for the limits of p_1, p_2, p_3, and p_4, we finally obtain (6.8). $\square$

Once again, we end this section with three illustrations of the Poisson distribution with mean $\mu=5,10,15$ in Figures 6.14, 6.15, and 6.16 below.

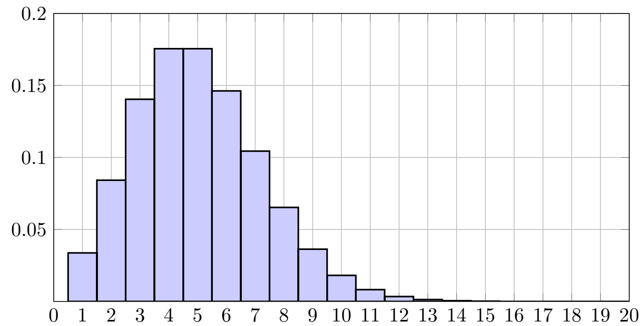

Figure 6.14. Poisson distribution with $\mu = 5$.

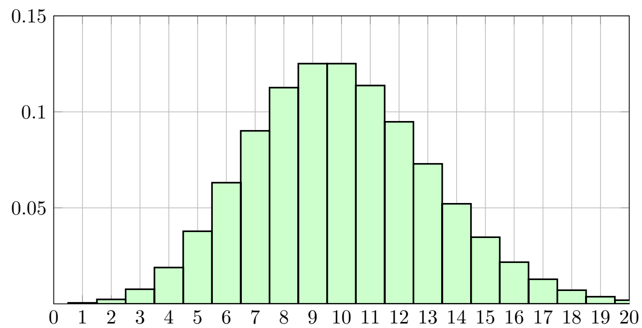

Figure 6.15. Poisson distribution with $\mu = 10$.

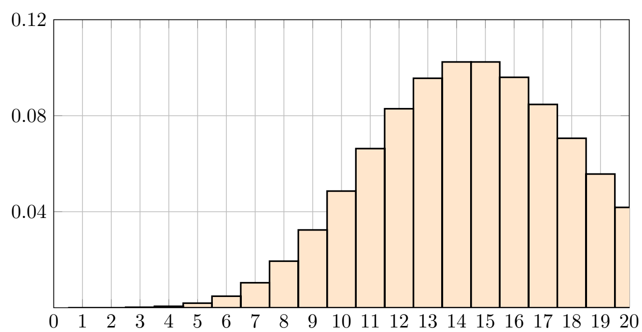

Figure 6.16. Poisson distribution with $\mu = 15$.

Chapter 7

The Law of Large Numbers

In this chapter, we begin our study of the the law of large numbers. As we have mentioned many times in previous chapters already, this is the result that will finally enable us to formulate a precise relationship between empirical frequencies and averages and their theoretical counterparts, namely, the probability measure and the expected value.

7.1. The Weak Law of Large Numbers

If you look through any standard textbook on probability (or do some research online) for a statement of the law of large numbers, then you are likely to find something similar to this:

Theorem 7.1 (Weak Law of Large Numbers)**.** *Let $X_1, X_2, X_3, \ldots$ be a countably infinite sequence of discrete random variables such that*

(1) $X_1, X_2, \ldots, X_n$ are independent for every $n \geq 1$;

(2) the X_i's all have the same distribution; and

(3) $\mathbf{E}[X_i] = \mu$ for some real number μ.

For every positive integer n, let us denote the empirical average of the first n random variables X_i by

$$\mathrm{EA}_n = \frac{X_1 + X_2 + \cdots + X_n}{n}. \tag{7.1}$$

Then, for every number $\varepsilon > 0$ (no matter how small), we have that

$$\lim_{n \to \infty} \mathbf{P}\big[|\mathrm{EA}_n - \mu| \leq \varepsilon\big] = 1.$$

Given the importance of this result, before we do anything, we should take the time to carefully parse every aspect of its statement.

7.1.1. The Assumptions. Assumptions (1) and (2) of Theorem 7.1 state that we are given a sequence of random variables that are all independent and have exactly the same range and probabilities (i.e., distribution).

Notation 7.2 (i.i.d. random variables)**.** The standard terminology for such a sequence of random variables is that they are "independent and identically distributed," which is abbreviated as i.i.d. Thus, going forward, whenever we say that a sequence of random variables are i.i.d., we mean that they are all independent and have the same distribution.

A natural interpretation of i.i.d. random variables is that their outputs represent multiple independent attempts of the same experiment. For example, we could imagine tossing the same die or coin multiple times, and then recording the sequence of numbers or faces thus obtained. In this interpretation, the random variable EA_n in (7.1) represents the empirical average obtained with the first n outcomes of the random experiment.

Notation 7.3. The number of trials n in an experiment is usually called the "sample size." We will be using this terminology going forward.

7.1.2. The Statement. If we accept the interpretation of the assumptions above, then we expect that, as n grows larger and larger, the empirical average EA_n should approach its theoretical value, which is $\mathbf{E}[X_i] = \mu$. In fact, this intuition formed the basis of our definitions of the probability measure and the expected value, as per (2.6) and (5.5)/(5.6). Theorem 7.1 allows to turn this intuition into a fully rigorous statement.

In specific terms, Theorem 7.1 says this: For any error threshold $\varepsilon > 0$, no matter how small, if we keep increasing the sample size n, then the probability that the difference between the empirical average EA_n and the theoretical expected value μ is less than ε tends to one. This process can be illustrated as in Figure 7.1 below. Therein, we have drawn a "buffer zone" of size ε on either side of EA_n in red. This

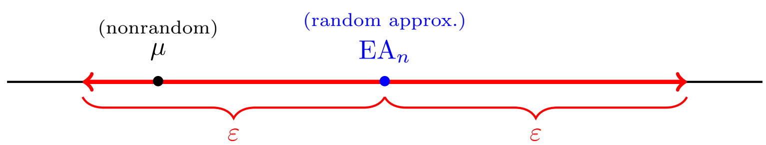

Figure 7.1. Surround the random empirical average EA_n by a "buffer zone" of size ε on either side, drawn in red. The nonrandom theoretical expected value μ happens to be within the buffer zone in this case, meaning that the event $|\mathrm{EA}_n - \mu| \leq \varepsilon$ has occurred.

red zone consists of all points on the real line that are at a distance of at most ε from EA_n. In this context, the event $\big\{|\mathrm{EA}_n - \mu| \leq \varepsilon\big\}$ states that the empirical average EA_n provides an estimate of μ within an error of ε. Theorem 7.1 states that, no matter how small this error ε is, if the sample size n grows to infinity, then the probability of $\big\{|\mathrm{EA}_n - \mu| \leq \varepsilon\big\}$ approaches 1.

Example 7.4 (Law of large numbers and indicators)**.** One of the most fundamental examples involving the law of large numbers is the case where the $X_i = \mathbf{1}_{A_i}$ are indicator random variables of some events. In this case, the assumption that the X_i's are i.i.d. translates to the following:

(1) the events $A_1, A_2, \ldots, A_n$ are independent for every $n \geq 1$; and

(2) the A_i's all have the same probability $\mathbf{P}[A_i] = p$.

Theorem 7.1 then says that for every number $\varepsilon > 0$, we have that

$$\lim_{n\to\infty} \mathbf{P}\left[\left|\frac{\mathbf{1}_{A_1}+\mathbf{1}_{A_2}+\cdots+\mathbf{1}_{A_n}}{n} - p\right| \leq \varepsilon\right] = 1. \tag{7.2}$$

Here, we note that

$$\mathbf{1}_{A_1}+\mathbf{1}_{A_2}+\cdots+\mathbf{1}_{A_n}$$

is the counting random variable of the events $A_1, \ldots, A_n$. Thus, if we think of $A_1, A_2, \ldots, A_n$ as representing multiple independent trials of the same event, then the ratio

$$\mathrm{EF}_n = \frac{\mathbf{1}_{A_1}+\mathbf{1}_{A_2}+\cdots+\mathbf{1}_{A_n}}{n}$$

represents the experimental frequency of the event, wherein we count the number of times that the event occurred and then divide by the sample size. The limit (7.2) then describes the precise relationship between this experimental frequency and the theoretical probability $\mathbf{P}[A_i] = p$ of the event, as the sample size n gets larger and larger.

In fact, knowing that

$$\mathbf{1}_{A_1}+\mathbf{1}_{A_2}+\cdots+\mathbf{1}_{A_n} \sim \mathrm{Bin}(n,p),$$

we can illustrate (7.2) using our knowledge of the exact distribution of the binomial. Suppose for instance that $p = 0.9$. In Figure 7.2 below, we plot the distributions of the empirical frequency $\mathrm{EF}_n = \frac{\mathbf{1}_{A_1}+\mathbf{1}_{A_2}+\cdots+\mathbf{1}_{A_n}}{n}$ for $n = 1\,000$ and $2\,000$. In both cases, we observe that the overwhelming majority of the probability is concentrated near 0.9. This makes sense: Since the sample size is in both cases fairly large, it is likely that the empirical average will give a number close to the theoretical probability 0.9. However, the distribution is even more concentrated for $n = 2\,000$, illustrating that the approximation given by the empirical frequency is more likely to be close to 0.9 as we increase the sample size.

In conclusion, the theoretical importance of the law of large numbers is that it provides a method of discovering what is the probability of an unknown event, and, more generally, discovering what is the expected value of an unknown random variable. Indeed, it provides a theoretical backing for the some of the most fundamental ideas in science, namely:

(1) If we want to discover what is the probability of a particular outcome, then we can assess this through experimentation; and

(2) the larger our sample size, the more confident we can be that our observations accurately describe reality.

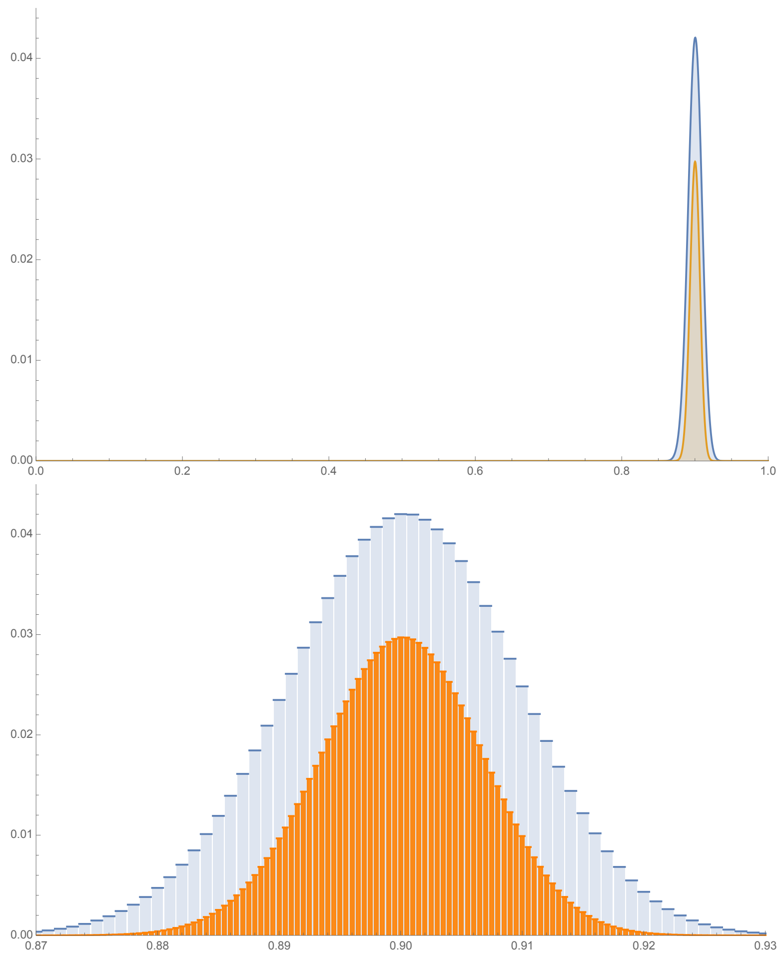

Figure 7.2. Empirical frequency distribution with $n = 1\,000$ and $p = 9/10$ (blue), and with $n = 2\,000$ and $p = 9/10$ (orange). The top figure includes the full range of the random variables between 0 and 1; In particular, the individual bins in the distribution are too small to make out, we can only see the general shape of the distribution. The bottom figure shows the same two distributions zoomed in on the interval $[0.87, 0.93]$, wherein we can easily make out individual bins.

7.1.3. A Remark on "Weak". You may be wondering why I called the statement of Theorem 7.1 the "weak" law of large numbers, instead of just the law of large numbers. The reason for this is that there exists a multitude of different statements of the law of large numbers, which differ in the technical details. The two most common statements are known as the weak and strong law of large numbers. As the same suggests, the strong law of large numbers is in some sense stronger than the statement in Theorem 7.1.

In this course, for a variety of reasons, we will only discuss the weak law of large numbers. This is in part due to time, but also because I think that the weak law of large numbers is actually more interesting. In any case, the specific aspects of the weak law of large numbers that we will study are more or less what you need in order to prove the so-called strong version of the result. That being said, for those of you who are interested in those details, I will briefly discuss the strong law of large numbers in the optional Section 7.6 in the notes.

7.2. A Problem with Theorem 7.1

In the previous section, we explained the importance of the law of large numbers from what could be viewed as the "purely philosophical" point of view. That is, the statement of Theorem 7.1 suggests that the theory of probability that we have been building up to this point is consistent with one of the most fundamental tenets of modern science, namely: The validity of certain hypotheses can be assessed with empirical evidence. In the case of the law of large numbers, this empirical evidence takes the form of observing the outcomes of repeated attempts of the same experiment.

However, I would argue that the statement of Theorem 7.1, by itself, is not necessarily what you should remember about the law of large numbers. Instead, I would encourage you to remember two other results, which in my view are much more fundamental, called Markov's and Chebyshev's inequalities. Before discussing what these are (we will do so in the next section), in this section I will provide an example that illustrates some of the shortcomings of the statement of the law of large numbers provided in Theorem 7.1.

Example 7.5 (New Vaccine)**.** Suppose that you are part of a team developing a new vaccine. In order for your vaccine to be approved by your local health authorities, you are required to run a clinical trial wherein a large number (call it n) of patients will receive your vaccine. To each of these patients, you associate an event A_i, defined as

$$A_i = \text{“Will the } i^{\text{th}} \text{ patient be immune after receiving the vaccine?”}$$

We assume that these events are all independent and have the same probability

$$\mathbf{P}[A_i] = p,$$

which we can think of as

$$p = \textit{“The probability that your vaccine creates immunity.”}$$

After you perform your clinical trial, you obtain the empirical frequency

$$\mathrm{EF}_n = \frac{\mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \cdots + \mathbf{1}_{A_n}}{n},$$

which is your estimate of p. With this in hand, the health authorities ask for specific requirements to be met:[1] You must show that there exists a number $\varepsilon > 0$ that satisfies the following:

(1) We can be at least 95% confident that, in a clinical trial with n individuals, EF_n approximates p within an error of at most $\varepsilon > 0$; in mathematical terms,

$$\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 0.95.$$

[1] This requirement is completely fictitious and only intended for the purpose of making the example concrete. It is not based on the requirements of any actual health authority.

(2) The empirical frequency EF_n that you obtain in your experiment is such that the interval

$$\big[\mathrm{EF}_n - \varepsilon, \mathrm{EF}_n + \varepsilon\big] = \big\{x \in \mathbb{R} : \mathrm{EF}_n - \varepsilon \leq x \leq \mathrm{EF}_n + \varepsilon\big\}$$

only contains probabilities above 0.9.

These requirements can be visualized as in Figure 7.3.

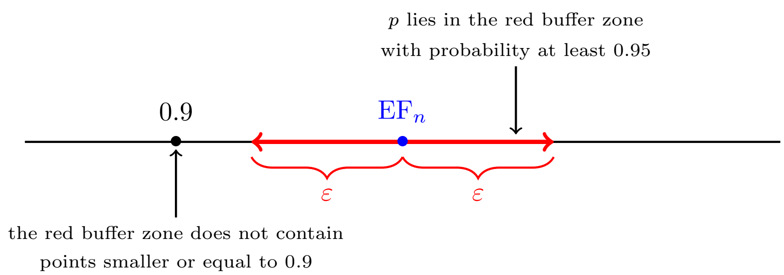

Figure 7.3. Requirements visualized.

At this point, all seems well and good: The weak law of large number states that for any choice of error threshold $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] = 1.$$

In particular, this means that there exists some n large enough so that, eventually,

$$\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 0.95, \tag{7.3}$$

no matter what ε is. However, problems arise when we attempt to implement this type of procedure in practice: If all you know is that there exists some purely abstract n such that (7.3) holds, then this is literally useless. Indeed, in that case you cannot dismiss the possibility that n could be arbitrarily large; maybe one billion, one trillion, or even a googol (which is 10^{100}).

While it may be possible in pure theory to always increase the sample size of an experiment to an arbitrarily large number, in the real world there are practical constraints that makes this impossible. In the specific example of a vaccine trial, there are hard limits on the sample size coming from the fact that there are only finitely many humans on which the vaccine can be tested. Moreover, having a larger sample size will not usually come for free. Administering a vaccine to more people takes more time and costs more money; time and money that could no doubt be allocated more productively elsewhere.

In summary, what is missing from Theorem 7.1 in the context of the present example is a quantitative control on how quickly the probability $\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big]$ tends to one. In other words, we would like to be able to answer the following:

For which value of n, exactly, can we guarantee that $\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 0.95$?

More generally, if we look back at the statement of the weak law of large numbers, our objective is the following:

Problem 7.6. Let $X_1, X_2, X_3, \ldots$ be a sequence of i.i.d. discrete random variables such that $\mathbf{E}[X_i] = \mu$ for some real number μ. Define EA_n as in (7.1). Find an explicit function $F(\varepsilon, n)$, which depends both on the error threshold ε and the sample size n, such that

$$\mathbf{P}\big[|\mathrm{EA}_n - \mu| \leq \varepsilon\big] \geq F(\varepsilon, n),$$

and such that solving for n or ε in an inequality of the form

$$F(\varepsilon, n) \geq c$$

is relatively easy.

Indeed, if we can find such an explicit function F, then guaranteeing that $\mathbf{P}[|\mathrm{EF}_n - p|] \geq 0.95$ amounts to solving for n in the inequality $F(\varepsilon, n) \geq 0.95$. The purpose of the remainder of this chapter is to develop the theory that will enable us to do this, and then look at a practical example.

7.3. Markov's and Chebyshev's Inequalities

7.3.1. Opening Remarks. To reiterate Problem 7.6, we wish to find an explicit estimate $F(\varepsilon, n)$ such that

$$\mathbf{P}\big[|\mathrm{EA}_n - \mu| \leq \varepsilon\big] \geq F(\varepsilon, n), \tag{7.4}$$

and for which it is relatively easy to solve for n or ε in an inequality of the form

$$F(\varepsilon, n) \geq c \tag{7.5}$$

Looking at this, some of you may be asking yourselves the following question: Why do we bother trying to find an *estimate* for the probability in (7.4)? Can we not compute it directly? Indeed, if R_{EA_n} is the range of the random variable EA_n, then we can write

$$\mathbf{P}\big[|\mathrm{EA}_n - \mu| \leq \varepsilon\big] = \sum_{\substack{s \in R_{\mathrm{EA}_n} \\ \text{s.t. } |s-\mu| \leq \varepsilon}} \mathbf{P}[\mathrm{EA}_n = s].$$

However, this is not a workable approach in practice. Indeed, computing the exact distribution of EA_n for arbitrary random variables yields extremely complicated formulas that get out of hand very quickly. Even in the simplest case imaginable, i.e., the empirical frequency

$$\mathrm{EF}_n = \frac{\mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \cdots + \mathbf{1}_{A_n}}{n}$$

for some independent events with $\mathbf{P}[A_i] = p$, the formulas get really nasty. In this case, we know that

$$\mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \cdots + \mathbf{1}_{A_n} \sim \mathrm{Bin}(n, p),$$

and therefore we can write

$$\mathbf{P}\big[|\mathrm{EF}_n - \mu| \leq \varepsilon\big] = \sum_{\substack{x \in \{0,1,\ldots,n\} \\ \text{s.t. } |x/n-\mu| \leq \varepsilon}} {}_nC_x p^x (1-p)^{n-x}. \tag{7.6}$$

However, solving for n or ε in the formula

$$\sum_{\substack{x\in\{0,1,\ldots,n\}\\ \text{s.t. } |x/n-\mu|\leq\varepsilon}} {}_nC_x p^x(1-p)^{n-x} \geq c$$

for some number c is nothing short of a nightmare. Moreover, if we are using EF_n to estimate p, the latter of which is unknown to us, then the expression (7.6) is useless because it depends on the unknown parameter p.

In light of the above remarks, attempting to carry out the program outlined in (7.4) and (7.5) with the exact distribution of EA_n is out of the question. The key to overcoming these difficulties, as it turns out, is to use the expected value and variance. On multiple occasions in the previous chapters, I have mentioned that part of what makes the expected value and variance so useful is that they can often be computed with relative ease, even when the distribution of the random variable in question is extremely hard. Just for sake of a reminder, here are some examples of problems for which we have computed the expected value (and sometimes also the variance) without computing the distribution:

(1) The sum of one million dice (Examples 5.17 and 5.33).
(2) The amount of time spent escaping a mine when tunnels are chosen uniformly and independently (Example 5.35).
(3) The number of die tosses to first observe a 6-6 or 6-1 (Homework 3, Problems 3.1-(b) and 3.2).
(4) The number of triangles in a random network (Homework 4, Problem 1.2).

This suggests that if we can somehow estimate the probability in (7.4) in terms of the expectation and/or variance of EA_n, then this should have the potential to yield substantial simplifications.

7.3.2. Markov's and Chebyshev's Inequalities.

In light of the previous section, the fundamental ingredient that makes a proof of the law of large numbers possible, as well as a practical implementation of it in the sense of (7.4) and (7.5), is a result called Markov's inequality:

Proposition 7.7 (Markov's Inequality). *Let X be a discrete random variable whose range R_X only contains nonnegative numbers. Then, for every positive number $t>0$, we have the upper estimates*

$$\mathbf{P}[X>t]\leq\mathbf{P}[X\geq t]=\sum_{\substack{x\in R_X\\ \text{s.t. } x\geq t}}\mathbf{P}[X=x]\leq\frac{\mathbf{E}[X]}{t}. \tag{7.7}$$

*(**Remark.** The first inequality in (7.7) follows from $\{X>t\}\subset\{X\geq t\}$ and Proposition 2.23-(1).)*

Proof. The proof of Markov's inequality amounts to a very simple but clever manipulation. We somehow want to relate the probability

$$\mathbf{P}[X\geq t]=\sum_{\substack{x\in R_X\\ \text{s.t. } x\geq t}}\mathbf{P}[X=x]$$

to the expectation divided by t, which we can write as

$$\frac{\mathbf{E}[X]}{t} = \frac{1}{t} \sum_{x \in R_X} x\,\mathbf{P}[X = x].$$

Given that the range R_X only contains nonnegative numbers, for any $x \in R_X$, the quantity $x\,\mathbf{P}[X = x]$ cannot be negative. Thus, if we remove some of these quantities from the sum in the expected value, then we are only making the sum smaller. In particular, if we take away all x's such that $x < t$, then we get that

$$\frac{\mathbf{E}[X]}{t} \geq \frac{1}{t} \sum_{\substack{x \in R_X \\ \text{s.t. } x \geq t}} x\,\mathbf{P}[X = x].$$

Next, for any $x \geq t$, we have that $x\,\mathbf{P}[X = x] \geq t\,\mathbf{P}[X = x]$. Therefore,

$$\frac{\mathbf{E}[X]}{t} \geq \frac{1}{t} \sum_{\substack{x \in R_X \\ \text{s.t. } x \geq t}} t\,\mathbf{P}[X = x] = \frac{t}{t} \sum_{\substack{x \in R_X \\ \text{s.t. } x \geq t}} \mathbf{P}[X = x] = \mathbf{P}[X \geq t],$$

which concludes the proof. □

To reiterate this result's claim and usefulness: In general, the quantities

$$\mathbf{P}[X > t] \qquad \text{and} \qquad \frac{\mathbf{E}[X]}{t}$$

will not be equal to one another. As a consequence, an application of the estimate (7.7) will typically cause a loss of accuracy. However, this is more than made up by the fact that $\mathbf{E}[X]$ is in many cases much easier to compute than $\mathbf{P}[X > t]$ or $\mathbf{P}[X \geq t]$. In other words, the usefulness of Markov's inequality is, for the most part, practical.

Let us now return to the task at hand, which is to provide an estimate for (7.4). We note that (7.4) asks for the probability that a random variable is smaller or equal to some quantity, and (7.7) gives an estimate of the probability that some variable is larger than some quantity. However, we can easily get around this by recalling the elementary fact that

$$\mathbf{P}[A] = 1 - \mathbf{P}[A^c]$$

for every event A. Indeed, with this in hand, we can write

$$\mathbf{P}\big[|\mathrm{EA}_n - \mu| \leq \varepsilon\big] = 1 - \mathbf{P}\big[|\mathrm{EA}_n - \mu| > \varepsilon\big]. \tag{7.8}$$

This now looks like something that we can apply Markov's inequality to.

If we apply Markov's inequality directly to (7.8), then we obtain that

$$\mathbf{P}\big[|\mathrm{EA}_n - \mu| > \varepsilon\big] \leq \frac{\mathbf{E}\big[|\mathrm{EA}_n - \mu|\big]}{\varepsilon}.$$

However, because of the absolute value, this expectation is not very easy to compute. As it turns out, there is a much better way: Firstly, we note that by linearity of the expected value,

$$\begin{aligned}\mathbf{E}[\mathrm{EA}_n] &= \mathbf{E}\left[\frac{X_1 + X_2 + \cdots + X_n}{n}\right] \\ &= \frac{\mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots + \mathbf{E}[X_n]}{n} \\ &= \frac{\mu + \mu + \cdots + \mu}{n} \\ &= \mu.\end{aligned}$$

Therefore, we have that

$$\mathbf{P}\big[|\mathrm{EA}_n - \mu| > \varepsilon\big] = \mathbf{P}\big[|\mathrm{EA}_n - \mathbf{E}[\mathrm{EA}_n]| > \varepsilon\big].$$

Secondly, we note that if we square both sides of the inequality

$$|\mathrm{EA}_n - \mathbf{E}[\mathrm{EA}_n]| > \varepsilon,$$

we obtain that

$$\mathbf{P}\big[|\mathrm{EA}_n - \mathbf{E}[\mathrm{EA}_n]| > \varepsilon\big] = \mathbf{P}\big[(\mathrm{EA}_n - \mathbf{E}[\mathrm{EA}_n])^2 > \varepsilon^2\big].$$

If we now apply Markov's inequality, then we get

$$\begin{aligned}\mathbf{P}\big[|\mathrm{EA}_n - \mu| > \varepsilon\big] &= \mathbf{P}\big[(\mathrm{EA}_n - \mathbf{E}[\mathrm{EA}_n])^2 > \varepsilon^2\big] \\ &\leq \frac{\mathbf{E}\big[(\mathrm{EA}_n - \mathbf{E}[\mathrm{EA}_n])^2\big]}{\varepsilon^2} = \frac{\mathbf{Var}[\mathrm{EA}_n]}{\varepsilon^2}.\end{aligned}$$

At this point, the third and final observation that we make is that, in contrast to the expectation $\mathbf{E}[|\mathrm{EA}_n - \mu|]$, the variance $\mathbf{Var}[\mathrm{EA}_n]$ is trivial to compute. Indeed, first by linearity of expected value, we have that

$$\mathbf{E}\left[\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right)^2\right] = \frac{\mathbf{E}[(X_1 + X_2 + \cdots + X_n)^2]}{n^2}$$

and

$$\mathbf{E}\left[\frac{X_1 + X_2 + \cdots + X_n}{n}\right]^2 = \frac{\mathbf{E}[X_1 + X_2 + \cdots + X_n]^2}{n^2};$$

therefore,

$$\mathbf{Var}[\mathrm{EA}_n] = \frac{\mathbf{Var}[X_1 + X_2 + \cdots + X_n]}{n^2}.$$

Next, we recall the property that we had stated in Proposition 5.31-(2), which is that the variance of a sum of independent random variables is the sum of the variances. Therefore,

$$\mathbf{Var}[\mathrm{EA}_n] = \frac{\mathbf{Var}[X_1] + \mathbf{Var}[X_2] + \cdots + \mathbf{Var}[X_n]}{n^2}.$$

Since the random variables X_i have the same distribution, they all have the same variance. Thus, we have that

$$\frac{\mathbf{Var}[X_1] + \mathbf{Var}[X_2] + \cdots + \mathbf{Var}[X_n]}{n^2} = \frac{n\mathbf{Var}[X_i]}{n^2} = \frac{\mathbf{Var}[X_i]}{n}.$$

Combining everything that we have done in this section, we therefore obtain the following quantitative version of the law of large numbers, which is often called Chebyshev's inequality:

Theorem 7.8 (Chebyshev's Inequality)**.** *Let $X_1, X_2, X_3, \ldots$ be a countably infinite sequence of i.i.d. discrete random variables such that*

$$\mathbf{Var}[X_i] = \sigma^2$$

for some positive number $\sigma > 0$. For every positive integer n, let us denote the empirical average

$$\mathrm{EA}_n = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

Then, for every integer $n \geq 1$ and positive number $\varepsilon > 0$, we have that

$$\mathbf{P}\big[|\mathrm{EA}_n - \mu| \leq \varepsilon\big] \geq 1 - \frac{\sigma^2}{n\varepsilon^2}. \tag{7.9}$$

Proof. If we reiterate the calculations that we have performed in the previous paragraphs, we get that

$$\begin{aligned}
\mathbf{P}\big[|\mathrm{EA}_n - \mu| \leq \varepsilon\big] &= 1 - \mathbf{P}\big[|\mathrm{EA}_n - \mu| > \varepsilon\big] && \text{(complement)}\\
&= 1 - \mathbf{P}\big[(\mathrm{EA}_n - \mu)^2 > \varepsilon^2\big] && \text{(square both sides)}\\
&\geq 1 - \frac{\mathbf{Var}[\mathrm{EA}_n]}{\varepsilon^2} && \text{(Markov's inequality)}\\
&= 1 - \frac{\sigma^2}{n\varepsilon^2},
\end{aligned}$$

concluding the proof. □

In summary, in contrast to an expression such as (7.6), the form of Chebyshev's inequality in (7.9) provides a very convenient estimate of $\mathbf{P}\big[|\mathrm{EA}_n - \mu| \leq \varepsilon\big]$, in the sense that it is very easy to solve for n or ε in an inequality of the form

$$1 - \frac{\sigma^2}{n\varepsilon^2} \geq c.$$

In order to showcase the usefulness of Chebyshev's inequality, we now discuss an example involving the estimate that it provides.

Remark 7.9. Given that Markov's inequality can provide a bound for both the probabilities $\mathbf{P}[X > t]$ and $\mathbf{P}[X \geq t]$, we also have the following manifestation of Chebyshev's inequality:

$$\mathbf{P}\big[|\mathrm{EA}_n - \mu| \geq \varepsilon\big] \leq \frac{\sigma^2}{n\varepsilon^2}, \tag{7.10}$$

i.e., a bound on the probability of the event $\{|\mathrm{EA}_n - \mu| \geq \varepsilon\}$ with an inequality, rather than the strict inequality $\{|\mathrm{EA}_n - \mu| > \varepsilon\}$.

7.4. An Example

We now discuss an example of how Chebyshev's inequality, as stated in (7.9), allows to provide quantitative statements on inference problems.

Example 7.10 (Unknown Event)**.** Inspired by the problem raised in Example 7.5 (i.e., testing the efficacy of a vaccine with a trial), consider the following general problem: Let A be an event with an unknown probability of success $p = \mathbf{P}[A]$. In order to get an empirical estimate of this probability, we assume that $A_1, A_2, A_3, \ldots$ are independent trials of the event A, and then look at the empirical frequency

$$\mathrm{EF}_n = \frac{\mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \cdots + \mathbf{1}_{A_n}}{n}$$

for a large sample size n. We may then ask: Given some threshold for error $\varepsilon > 0$, how large must the sample size n be in order to be at least 95% confident that EF_n approximates p with error at most ε? In other words, we want to find n large enough so that

$$\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 0.95.$$

Since $\mathbf{1}_{A_i}$ are indicator random variables of events with probability p, we know that $\mathbf{Var}[\mathbf{1}_{A_i}] = p(1-p)$ (we have computed this before, see (6.4)). Thus, Chebyshev's inequality, we have that

$$\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 1 - \frac{p(1-p)}{n\varepsilon^2}. \tag{7.11}$$

With this in hand, we can now find for which values of n do we have that the above probability is at least 0.95:

$$\begin{aligned} 1 - \frac{p(1-p)}{n\varepsilon^2} \geq 0.95 \iff & \frac{p(1-p)}{n\varepsilon^2} \leq 0.05 \\ \iff & n \geq \frac{20p(1-p)}{\varepsilon^2}. \end{aligned}$$

Thus, it suffices that our sample space is of size $20p(1-p)/\varepsilon^2$.

At this point, however, we run into a problem: On the one hand, ε is a known quantity that we specify ourselves in the model. Namely, it is our tolerance for error in our estimate of the theoretical probability p. On the other hand, $p(1-p)$ is not known. In fact, the number p is what we are actually trying to estimate with our empirical average EA_n in the first place!

However, not all is lost: Even though p is unknown, we can nevertheless estimate the quantity $p(1-p)$. Indeed, by virtue of being a probability, we know that p must be somewhere in between 0 and 1. If we then examine the parabola $p(1-p)$ for $0 \leq p \leq 1$, as done in Figure 7.4, we note that there appears to be a maximum

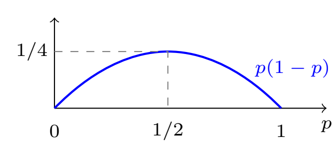

Figure 7.4. Plot of the parabola $p(1-p)$.

at $p = 1/2$, in which case $p(1-p) = 1/4$. We can confirm that this is the case by elementary calculus: Define the function $f(p) = p(1-p)$. We easily calculate that

$$f'(p) = 1 - 2p \qquad \text{and} \qquad f''(p) = -2.$$

Solving for $f'(p) = 0$ yields $p = 1/2$, confirming that there is a critical point at that location. Since $f'' < 0$, this critical point is a local maximum. If we then compare this with the boundary terms $f(0) = f(1) = 0$, we thus conclusively prove that $1/4$ is in fact the maximum of $p(1-p)$.

If we now apply our new insight that

$$p(1-p) \leq 1/4$$

no matter what the value of p is to (7.11), then we get the estimate

$$\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 1 - \frac{1}{4n\varepsilon^2}.$$

Solving for the values of n that make this at least 0.95, we obtain

$$n \geq \frac{5}{\varepsilon^2}.$$

There are now no more unknown terms; the minimal sample size now only depends on ε, which is our tolerance for error. To give a few concrete examples, here are what you would obtain with a few different values of ε:

ε	$5/\varepsilon^2$
0.1	500
0.01	50 000
0.001	5 000 000
0.0001	500 000 000

Table 7.1. Minimal sample sizes for different error thresholds, as per Chebyshev's inequality and the estimate $p(1-p) \leq 1/4$.

7.5. Closing Remarks

7.5.1. Error Threshold, Sample Size, and Confidence. The analysis of the law of large numbers performed in this section highlights the interactions between the following parameters:

(1) The error threshold ε, namely, the maximal distance between an empirical frequency EF_n and the actual probability p that we are willing to tolerate.

(2) The sample size n, namely, the number of observations that we are willing and/or can afford to include in our empirical estimate.

(3) The confidence in our estimate, namely, the probability

$$\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big].$$

Indeed, the estimate

$$\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 1 - \frac{1}{4n\varepsilon^2} \tag{7.12}$$

allows to quantify the tradeoff between these quantities:

(1) (Sample size vs. confidence). Increasing the sample size leads to an increased confidence in one's estimate. Looking specifically at (7.12), this manifests in the fact that the lower estimate $1 - \frac{1}{4n\varepsilon^2}$ converges to 1 as $n \to \infty$, making it increasingly certain that an estimate holds for larger sample sizes.

(2) (Error threshold vs. confidence). Increasing the tolerance for error ε leads to an increase of confidence (and vice versa), because $1 - \frac{1}{4n\varepsilon^2}$ increases as you make ε bigger. Intuitively this also makes sense: If you want to be extremely confident in your estimate, then you have to accept a larger degree of error; conversely, if you want a very small degree of error, then this necessarily decreases how confident you can be that your estimate actually holds.

(3) (Error threshold vs. Sample size) Decreasing the tolerance for error increases the minimal sample size, and vice versa. This is illustrated beautifully in Table 7.1, and is also fairly intuitive.

7.5.2. Independence. Among all of the technicalities and subtleties that we discussed regarding the law of large numbers, it is important not to lose sight of a crucial assumption that we made throughout the chapter, namely, that the sequence of random variables or events that we consider are independent.

Looking back at the details, this assumption manifested itself in the computation of the variance

$$\mathbf{Var}\left[\frac{X_1 + X_2 + \cdots + X_n}{n}\right].$$

Because the variance of a sum of independent random variables is the sum of the variances, we were able to obtain that this is equal to

$$\frac{\mathbf{Var}[X_i]}{n}.$$

If the random variables X_i are not independent, then it is not at all clear that we can compute this variance, or that it would have a similar behavior.

To give an extreme example, suppose that instead of being independent, the random variables X_i are in fact all exactly equal to one another, that is,

$$X_1 = X_2 = X_3 = \cdots.$$

Then, we have that

$$\mathbf{Var}\left[\frac{X_1 + X_2 + \cdots + X_n}{n}\right] = \mathbf{Var}\left[\frac{nX_i}{n}\right] = \mathbf{Var}[X_i].$$

This is very different from the independent case, namely, we have $\frac{\mathbf{Var}[X_i]}{n}$ when the variables are independent, which converges to zero as $n \to \infty$, whereas $\mathbf{Var}[X_i]$ does not converge to zero at all.

7.5.3. Optimality. Looking back at the quantities that we obtained in Table 7.1, it might be tempting to conclude the following:

In order to ensure that $\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 0.95$*, it is necessary to take* $n \geq 5/\varepsilon^2$.

However, this is not quite correct. A more accurate statement would be this:

According to Chebyshev's inequality and the estimate $p(1-p) \leq 1/4$*, in order to ensure that* $\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 0.95$*, it* **suffices** *to take* $n \geq 5/\varepsilon^2$.

Indeed, it is important to keep in mind that Markov/Chebyshev's inequality and $p(1-p) \leq 1/4$ are only estimates. In typical situations, the probability

$$\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big]$$

will actually be bigger than $1 - \frac{1}{4n\varepsilon^2}$.

Given that increasing the sample size typically costs time and resources in the real world, there is a strong incentive to provide guarantees that require the smallest possible sample size. With this in mind, it is thus natural to ask:

Can we improve on Chebyshev's inequality?

The short answer to this question is: In many cases, Chebyshev's inequality can be improved substantially. The business of providing the most optimal estimates of

$$\mathbf{P}\big[|\mathrm{EA}_n - \mu| \leq \varepsilon\big]$$

is the subject of a very deep and beautiful theory in mathematics called "concentration of measure."

Studying concentration of measure in any amount of detail is firmly outside the scope of this course. However, for those of you who are interested in learning more about this, I encourage you to take a look at Section 7.7, which is completely optional, wherein I discuss concentration of measure in a few more details.

7.6. The Strong Law of Large Numbers (Bonus)

In Section 7.1.3, I claimed that there exists different versions of the law of large numbers. In particular, I mentioned one result called the strong law of large numbers. The statement of this result is as follows:

Theorem 7.11 (Strong Law of Large Numbers)**.** *Let $X_1, X_2, X_3, \ldots$ be i.i.d. random variables with $\mathbf{E}[X_i] = \mu$ for some real number μ. For every positive integer n, let us denote*

$$\mathrm{EA}_n = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

Then, we have that

$$\mathbf{P}\left[\lim_{n\to\infty} \mathrm{EA}_n = \mu\right] = 1.$$

The weak law of large numbers states the following: Fix some error threshold $\varepsilon > 0$. No matter what that ε is, as we increase the sample size n, the probability that the empirical average EA_n is farther away than ε from μ goes to zero. In contrast, the strong law of large number says the following: The probability that the empirical average EA_n converges to μ as the sample space n goes to infinity is equal to one.

The difference between these two statements is admittedly a bit subtle, but nevertheless meaningful. Indeed, the weak and strong laws of large numbers are special cases of different modes of convergence in probability:

Definition 7.12. Let $X_1, X_2, X_3, \ldots$ be a sequence of random variables. We say that X_n converges to X in probability if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbf{P}\big[|X_n - X| \leq \varepsilon\big] = 1.$$

We say that X_n converges to X almost surely if

$$\mathbf{P}\left[\lim_{n\to\infty} X_n = X\right] = 1.$$

In this terminology, we see that the weak law of large numbers says that EA_n converges to μ in probability, whereas the strong law of large numbers says that EA_n converges to μ almost surely. In this generalized context, the appellations "strong" and "weak" can be justified by the following result:

Theorem 7.13. *If X_n converges to X almost surely, then X_n must also converge to X in probability. Conversely, if X_n converges to X in probability, then it is not always the case that X_n converges to X almost surely.*

Therefore, almost sure convergence is "stronger" than convergence in probability, in the sense that almost sure convergence automatically implies convergence in probability, but not vice versa. The proofs of Theorems 7.11 and 7.13 require a number of notions that are not covered in this course. However, the proofs in question are rather easy to find with a bit of googling, and those of you who will go on to study probability at the advanced (graduate) level will no doubt discuss these subtleties in great detail.

As a final remark, it is important to note that from the purely practical point of view, the statement of the strong law of large numbers provided in Theorem 7.11 is just as problematic as the statement of the weak law of large numbers in Theorem 7.1: Theorem 7.11 does not provide a quantitative control on how quickly EA_n converges to μ, it only says that it will eventually converge to μ as "n goes to ∞." Thus, if we have a fixed finite sample size n, Theorem 7.11 by itself provides no information whatsoever on how well EA_n approximates μ.

Interestingly enough, one of the keys to proving the strong law of large numbers is to provide a quantitative control of probabilities of the form $\mathbf{P}\big[|\mathrm{EA}_n - \mu| \leq \varepsilon\big]$, which is exactly what we did with Chebyshev's inequality (7.9). Combining this observation with the fact that quantitative estimates are what is actually useful in practice, it is my view that Markov's and Chebyshev's inequality are what you should try your best to remember about this chapter going forward.

7.7. Concentration of Measure (Bonus)

7.7.1. Intuition. Roughly speaking, the concentration of measure phenomenon refers to the following observation: Let $X_1, X_2, \ldots, X_n$ be independent random variables, where n is a large number. Let $f(x_1, x_2, \ldots, x_n)$ be a function of n variables that depends very little on any one of the components x_i; that is, if you only change the value of one component x_i, then the output of the function $f(x_1, x_2, \ldots, x_n)$ as a whole changes very little. Then, the value of the function evaluated in the random variables

$$f(X_1, X_2, \ldots, X_n),$$

which is itself a random quantity, is typically very close to its expected value

$$\mathbf{E}\big[f(X_1, X_2, \ldots, X_n)\big].$$

In this context, the law of large numbers can be viewed as a special case of the general principle of concentration. Indeed, in the case of the law of large numbers, the function that we are interested in is the sample average:

$$f(x_1, x_2, \ldots, x_n) = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

By linearity of expected value, if the X_i's are i.i.d., then we have that

$$\mathbf{E}[f(X_1, X_2, \ldots, X_n)] = \mathbf{E}[X_i].$$

Moreover, it is easy to see that this function depends very little on any of its components: If the sample size n is very large, then changing just one of the values x_i will have very little effect on the average as a whole (i.e., it will only change the average by a size of order of $\sim 1/n$, which goes to zero as $n \to \infty$).

The intuition behind the general principle of concentration can be explained thusly: The expected value

$$\mathbf{E}\big[f(X_1, X_2, \ldots, X_n)\big]$$

represents in some sense the "typical" value that the random variable

$$f(X_1, X_2, \ldots, X_n)$$

will output. In particular, in order for this random variable to be very far from its expectation, it must output an "atypical" or "unusual" value.

Because the function f changes its output very little if only one of its components is changed, in order for $f(X_1, X_2, \ldots, X_n)$ to take an unusual value, it is not enough that only one of the random variables X_i takes an unusual value; a large number of variables must simultaneously take unusual values. For instance, in order for the average

$$\frac{X_1 + X_2 + \cdots + X_n}{n}$$

to deviate from its typical value substantially, a large number of the random variables X_i must somehow "conspire" to simultaneously take atypical values.

However, because the random variables X_i are all independent of each other, there is no reason to expect that they should be able to conspire to simultaneously give unusual outputs. If a collection of random variables are independent, then they are completely unaffected by each other's behavior. Thus, even though it is *possible* that a large number of independent random variables somehow conspire to take very unusual values all at the same time, such an event is extremely improbable.

7.7.2. Example. With the intuition out of the way, one of the main purposes of the mathematical theory of concentration of measure is to provide the best possible quantitative bounds on probabilities of the form

$$\mathbf{P}\big[|f(X_1, X_2, \ldots, X_n) - \mathbf{E}[f(X_1, X_2, \ldots, X_n)]| \leq \varepsilon\big] \geq F(\varepsilon, n, f),$$

and understand how these bounds depend on the error threshold ε, the sample size n, and the function f. The version of Chebyshev's inequality that we stated in (7.9) is one particular result in this vein, restricted to the special case where the function f is the sample average.

If you ever study probability at the advanced level, then you will perhaps learn about results that improve on or generalize (7.9) in significant ways. To give a specific example, on such result is as follows:

Theorem 7.14 (Hoeffding's inequality)**.** *Let $A_1, A_2, A_3, \ldots$ be independent events, all with the same probability $\mathbf{P}[A_i] = p$, and let*

$$\mathrm{EF}_n = \frac{\mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \cdots + \mathbf{1}_{A_n}}{n}.$$

For every sample size $n \geq 1$ and error threshold $\varepsilon > 0$, one has

$$\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 1 - 2\mathrm{e}^{-2n\varepsilon^2}. \tag{7.13}$$

Proof Sketch. Interestingly enough, Hoeffding's inequality also uses Markov's inequality as a fundamental ingredient, but not in the same way as Chebyshev's inequality: First, we can write

$$\begin{aligned}\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] &= 1 - \mathbf{P}\big[|\mathrm{EF}_n - p| > \varepsilon\big]\\ &= 1 - \mathbf{P}\big[\{\mathrm{EF}_n - p < -\varepsilon\} \cup \{\mathrm{EF}_n - p > \varepsilon\}\big]\\ &\geq 1 - \mathbf{P}\big[\mathrm{EF}_n - p < -\varepsilon\big] - \mathbf{P}\big[\mathrm{EF}_n - p > \varepsilon\big].\end{aligned}$$

Next, if we look at the probability

$$\mathbf{P}\big[\mathrm{EF}_n - p > \varepsilon\big]$$

(the probability $\mathbf{P}\big[\mathrm{EF}_n - p < -\varepsilon\big]$ can be analyzed similarly), then we get that

$$\mathbf{P}\big[\mathrm{EF}_n - p > \varepsilon\big] = \mathbf{P}\big[\mathrm{e}^{\theta(\mathrm{EF}_n - p)} > \mathrm{e}^{\theta\varepsilon}\big]$$

for any choice of number $\theta > 0$. At this point, by Markov's inequality, we get

$$\mathbf{P}\big[\mathrm{EF}_n - p > \varepsilon\big] \leq \mathrm{e}^{-\theta\varepsilon}\mathbf{E}[\mathrm{e}^{\theta(\mathrm{EF}_n - p)}].$$

Then, by independence,

$$\mathbf{P}\big[\mathrm{EF}_n - p > \varepsilon\big] \leq \mathrm{e}^{-\theta\varepsilon}\prod_{i=1}^{n}\mathbf{E}[\mathrm{e}^{\theta(\mathbf{1}_{A_i} - p)/n}].$$

Finally, with a bit of hard work, it can be shown that if we choose the parameter $\theta > 0$ wisely, then we get

$$\mathbf{P}\big[\mathrm{EF}_n - p > \varepsilon\big] \leq \mathrm{e}^{-2n\varepsilon^2},$$

which yields the result. □

As a final remark, it may not be immediately obvious to you that Hoeffding's inequality improves on Chebyshev's inequality. For sake of making a clearer comparison, we can go through the solution of Example 7.10, but this time using (7.13) instead of (7.11). Using (7.13), if we want to ensure that

$$\mathbf{P}\big[|\mathrm{EF}_n - p| \leq \varepsilon\big] \geq 1 - 2\mathrm{e}^{-2n\varepsilon^2} \geq 0.95,$$

then it is enough to choose ε and n such that

$$1 - 2\mathrm{e}^{-2n\varepsilon^2} \geq 0.95.$$

Solving for n in this equation yields

$$n \geq \frac{\log(40)}{2\varepsilon^2} \geq \frac{1.85}{\varepsilon^2}.$$

If we then look at the minimal sample sizes that this gives us for different error thresholds in Table 7.2, we see that this is much better than what we had obtained

ε	$1.85/\varepsilon^2$
0.1	185
0.01	18 500
0.001	1 850 000
0.0001	185 000 000

Table 7.2. Minimal sample sizes for different error thresholds, as per Hoeffding's inequality.

in Table 7.1 earlier. This illustrates how Hoeffding's inequality provides a much better quantitative estimate than Chebyshev's inequality.

Chapter 8

Introduction to Continuous Random Variables

In the previous sections, we defined random variables as functions X that assign to every possible outcome $\omega \in \Omega$ in the sample space an output $X(\omega) \in \mathbb{R}$ that is a real number. Up to this point, every random variable that we have considered has been discrete, meaning that the set of all possible values that X can output is either finite or countably infinite (i.e., enumerated in an ordered list; see Notation 5.2 for a reminder of the definition).

In this chapter, our purpose is to introduce continuous random variables. That is, random variables X whose set of all possible outputs is uncountably infinite. For example, this could be the situation where X can output any real number $\mathbb{R}$, or any number in an interval $[a, b]$, etc. As we will soon see, the mathematical treatment of continuous random variables is substantially more involved than that of discrete random variables. In short, essentially every aspect of the description of random variables (e.g., the range and distribution, the expected value and variance, conditioning and independence, etc.) needs to be substantially reworked.

However, before getting on with this program (which is the subject of the next chapter), our purpose in this chapter is to explain

(1) how continuous random variables naturally arise in modelling problems; and

(2) why continuous random variables are so delicate to describe mathematically.

In order to do this, we will introduce two examples of continuous random variables that can be obtained as limits of discrete variables. Once this is done, we will discuss some of the peculiar properties of the variables that we have constructed.

8.1. A Model of Continuous Arrivals

Recall the scenario introduced in Example 6.20, which served as the motivation for the definition of the Poisson random variable: You own a convenience store in

Hyde Park, and would like to construct a probabilistic model to predict the arrivals of customers in your store through time. As stated in Definition 6.21, to simplify matters, we had made the assumptions that

(1) the average number of customers that enter the store during a one-hour period is equal to some positive number $\mu > 0$; and
(2) at every given time, an arrival occurs with equal probability, independently of all other times.

Back then, we were interested in the random variable

$$X = \text{number of customers who enter the store within a one-hour period.}$$

In order to make sense of this random variable, we introduced a discrete approximation of the problem, wherein we split the one-hour period into a large number of smaller time intervals of equal size $1/n$ (see Figure 6.11). Under this approximation, we assumed that in each sub-interval of size $1/n$, either one arrival occurs or none. The probability of having one arrival was equal to μ/n (assuming $n > \mu$) so that the average number of arrivals in the whole one-hour period would be μ, and arrivals in distinct sub-intervals were independent. We then obtained the Poisson distribution with parameter μ by taking the $n \to \infty$ limit of these approximations, which were Binomial with parameters n and μ/n (see equation (6.8)).

Now suppose that instead of the number of arrivals in some time period, we are interested in the following random variable:

$$X = \begin{array}{c}\text{amount of time (in hours) until the first customer}\\ \text{enters the store, starting from 12:00 AM}\end{array}. \tag{8.1}$$

In contrast to the number of people who enter the store within one hour, which has to be a nonnegative integer, it is not at all clear that we should expect the random variable (8.1) to be discrete. Indeed, it seems intuitive that the set of all possible outcomes of this variable should be the interval $[0, \infty)$ because, in principle, the first customer could enter the store at any time at or after midnight. Our first objective in this section is to provide a rigorous definition of this random variable.

8.1.1. Geometric Approximation. Our intention is to construct the first arrival (8.1) by using essentially the same assumptions and procedures as for the Poisson random variable. For this, we once again partition all time after 12:00 AM into infinitely many sub-intervals, each of which represents a duration of time of $\frac{1}{n}\times$(one hour) for some large number n (larger than μ at least).

Unlike the Poisson variable, this time we do not restrict the time interval under consideration to a one-hour period, because it is possible that it will take more than one hour for the first customer to show up. In fact, it could take an arbitrarily long amount of time, at least in principle, before the fist customer shows up. Thus, in order to account for every possible amount of time that it could take until you get the first customer, we let time after 12:00 AM extend all the way to infinity. This is illustrated in Figure 8.1 below.

Just like in the construction of the Poisson random variable, we assume that an arrival occurs in each sub-interval with probability μ/n (so that the average

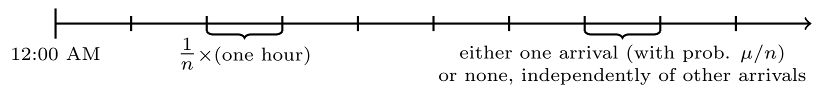

Figure 8.1. Spitting time into a grid of n sub-intervals. The time length of each sub-interval is $1/n$ times the original duration. We assume that in each sub-interval, either one arrival occurs, or none.

number of arrivals during an hour is μ), and independently of all other arrivals. Thus, if we let

$$X_n = \text{number of sub-intervals until we get a first arrival},$$

then $X_n \sim \text{Geom}(\mu/n)$. Consequently, the actual time of the first arrival, X, should be some kind of limit as $n \to \infty$ of some geometric random variables, in direct analogy to how the Poisson random variable was a limit of Binomials.

However, the process of taking the limit in this case is much less straightforward. In order to illustrate why that is, suppose that we want to compute the probability

$$\mathbf{P}[X > x],$$

where $x > 0$ is a fixed positive number. In words, this is the probability that it takes more than x hours after midnight until we observe the first customer. We would like to relate this to the probability of some event involving the geometric approximation. We could do this as is illustrated in Figure 8.2 below. That is, we

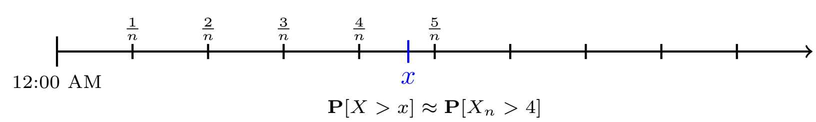

Figure 8.2. Finding the number of discrete sub-intervals that best corresponds to an arbitrary time x. In this particular case, x lies somewhere in between $\frac{4}{n}$ and $\frac{5}{n}$, so we approximate $\mathbf{P}[X > x] \approx \mathbf{P}[X_n > 4]$.

count the number of intervals of size $\frac{1}{n}$ that are before x, and then approximate $\mathbf{P}[X > x]$ with the probability that the first arrival counted by the geometric approximation X_n does not occur in any of the intervals before x. As we send $n \to \infty$, which is equivalent to shrinking the size of each interval down to zero, we expect that the amount of error incurred by this approximation should vanish.

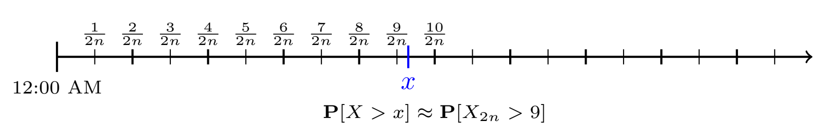

Figure 8.3. If the size of sub-intervals is half as large as what it was in Figure 8.2, then the approximation of $\mathbf{P}[X > x]$ changes. Now there are nine sub-intervals before x; hence now $\mathbf{P}[X > x] \approx \mathbf{P}[X_{2n} > 9]$.

While this procedure seems like a compelling way to approximate $\mathbf{P}[X > x]$ using the geometric variables X_n, it poses a nontrivial problem. That is, for different values of n, the number of intervals before x will change. To illustrate this, suppose that we halve the size of the sub-intervals from what it was in Figure 8.2. In doing so, we obtain Figure 8.3 above, wherein the number of sub-intervals before x has increased from 4 to 9.

In short, the geometric approximation method above suggests that for every grid size n, there exists some number x_n such that

$$\mathbf{P}[X > x] \approx \mathbf{P}[X_n > x_n].$$

More specifically,

$$x_n = \text{number of sub-intervals of size } \tfrac{1}{n} \text{ before } x. \tag{8.2}$$

Moreover, the accuracy of this approximation should improve as we take $n \to \infty$. This suggests that we can define

$$\mathbf{P}[X > x] = \lim_{n\to\infty} \mathbf{P}[X_n > x_n] = \lim_{n\to\infty} \left(1 - \frac{\mu}{n}\right)^{x_n}, \tag{8.3}$$

where the last equality comes from the fact that $X_n \sim \text{Geom}(\mu/n)$. In order to complete this program, however, we need to be able to say something meaningful about the number x_n in (8.2).

8.1.2. Sandwiches and the Exponential Limit. In principle, in order to compute the limit (8.3), we would need to find an exact formula for the number x_n. However, thanks to a beautiful result in calculus known as the sandwich theorem (or the much less amusing squeeze theorem), we can get around this issue:

Theorem 8.1 (Sandwich Theorem). *Let a_n, b_n, and c_n be three sequences of real numbers such that*

$$a_n \leq b_n \leq c_n.$$

(In words, the sequence b_n is sandwiched between a_n and c_n). If the limits

$$\lim_{n\to\infty} a_n = \ell = \lim_{n\to\infty} c_n$$

exist and are the same, then it must also be the case that

$$\lim_{n\to\infty} b_n = \ell.$$

You can easily convince yourself that this theorem is true by drawing an illustration of it, such as Figure 8.4 below. Armed with this knowledge, we see that

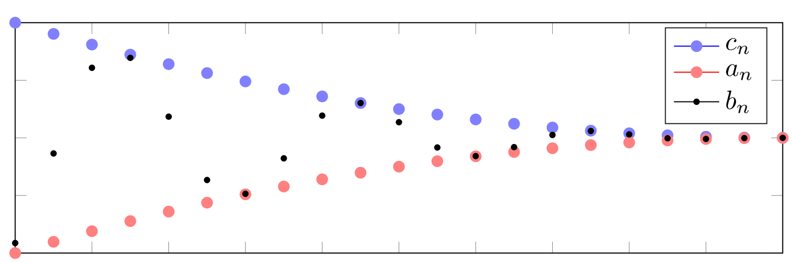

Figure 8.4. The sequence b_n (black dots) is sandwiched between a_n and b_n (red and blue dots, respectively). If a_n and c_n both converge to the same limit, then so must b_n.

we do not necessarily need to compute the numbers x_n in (8.2) exactly in order to solve the limit in (8.3); in principle, it is enough to sandwich the sequence in (8.3) in between two simpler sequences.

For this purpose, we claim that the number x_n satisfies

$$nx - 1 \leq x_n \leq nx. \tag{8.4}$$

To see why that is the case, we look at both inequalities one at a time. By definition, x_n counts the number of intervals of size $1/n$ before x. In particular, x_n/n is the largest fraction of the form k/n (with k being an integer) such that $k/n \leq x$ (see that this is true by inspecting Figures 8.2 and 8.3). This immediately implies the upper estimate $x_n \leq nx$. To see why the lower estimate in (8.4) is also true, we note that the number x is in the sub-interval $[x_n/n, (x_n + 1)/n)$. Because this sub-interval's total length is $1/n$, the maximal amount of distance between x and the lower edge of the interval x_n/n is at most $1/n$. In particular, $x - x_n/n \leq 1/n$, which gives the lower estimate in (8.4) by multiplying both sides by n and a trivial rearrangement.

The usefulness of (8.4) from the perspective of computing the limit (8.3) is that it implies the following: Whenever $n > \mu$, one has

$$\underbrace{\left(1 - \frac{\mu}{n}\right)^{nx}}_{a_n} \leq \underbrace{\left(1 - \frac{\mu}{n}\right)^{x_n}}_{b_n} \leq \underbrace{\left(1 - \frac{\mu}{n}\right)^{nx-1}}_{c_n}$$

If we can somehow prove that the sequences a_n and c_n above have the same limit, then we will also have computed the limit of b_n by the sandwich theorem. To this effect, we note that

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(1 - \frac{\mu}{n}\right)^{nx} = \mathrm{e}^{-\mu x},$$

and similarly,

$$\lim_{n\to\infty} c_n = \lim_{n\to\infty} \left(1 - \frac{\mu}{n}\right)^{nx} \cdot \left(1 - \frac{\mu}{n}\right)^{-1} = \mathrm{e}^{-\mu x} \cdot 1^{-1} = \mathrm{e}^{-\mu x}.$$

Therefore,

$$\mathbf{P}[X > x] = \lim_{n\to\infty} \mathbf{P}[X_n > x_n] = \lim_{n\to\infty} \left(1 - \frac{\mu}{n}\right)^{x_n} = \lim_{n\to\infty} b_n = \mathrm{e}^{-\mu x}.$$

8.1.3. A First Problem. To reiterate what we have done in the last few sections, we have seen that if we define

$$X = \begin{array}{c}\text{amount of time (in hours) until the first customer}\\ \text{enters the store, starting from 12:00 AM}\end{array},$$

then under the assumptions that

(1) the average number of customers that enter the store during a one-hour period is equal to some positive number $\mu > 0$; and

(2) at every given time, an arrival occurs with equal probability, independently of all other times,

we should have that for every $x > 0$,

$$\mathbf{P}[X > x] = \mathrm{e}^{-\mu x}.$$

As a first sanity check, we can verify that the quantities $\mathrm{e}^{-\mu x}$ behave in the way that we would expect. If we plot $\mathrm{e}^{-\mu x}$ as a function of x, then we obtain something like Figure 8.5 below. Looking at this picture, we see that $\mathrm{e}^{-\mu x}$ is always between

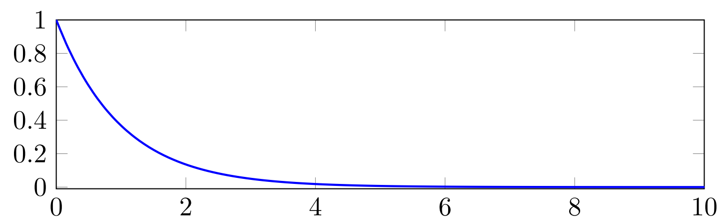

Figure 8.5. Plot of $\mathrm{e}^{-\mu x}$ as a function x for a fixed value of $\mu > 0$.

zero and one, and thus it is a bona fide probability.

With this result in hand, it is natural to wonder if we can compute other probabilities involving X. For instance, can we compute the distribution of X? If we apply the same definition of distribution that we had for discrete random variables, then this means that we must compute

$$\mathbf{P}[X = x]$$

for any choice of $x > 0$. However, if we try to compute this using our discrete approximation, then we run into a curious problem. Indeed, by arguing exactly in the same way as in the last few sections, we expect that

$$\mathbf{P}[X = x] = \lim_{n \to \infty} \mathbf{P}[X_n = x_n],$$

recalling that x_n is the number of sub-intervals of size $1/n$ before x, and X_n is a geometric approximation of X, counting how many such sub-intervals we have to wait for until we get an arrival. Since $X_n \sim \mathrm{Geom}(\mu/n)$, this then yields

$$\mathbf{P}[X = x] = \lim_{n \to \infty} \left(1 - \frac{\mu}{n}\right)^{x_n - 1} \frac{\mu}{n} = \mathrm{e}^{-\mu x} \cdot 0 = 0.$$

This is a very curious observation that, at first glance, seems paradoxical: On the one hand, the fact that $\mathbf{P}[X > 0] = \mathrm{e}^{-\mu \cdot 0} = 1$ seems to indicate that the set of all possible outputs of X is contained in the interval $(0, \infty)$. On the other hand, the fact that $\mathbf{P}[X = x] = 0$ for every $x > 0$ seems to indicate that X cannot output any such number, in the sense that the probability of observing the event $\{X = x\}$ is zero. At this point, you may be worried that there is something wrong with the way we carried out our construction of X. However, it turns out that this apparent paradox is actually inevitable when dealing with continuous random variables. In order to further convince ourselves of this fact, we briefly look at one more example.

8.2. Uniform Random Number on the Interval $[0,1]$

Suppose that we try to define the following random variable:

$$X = \text{uniform random number on the interval } [0,1].$$

That is, we wish to pick a real number in between zero and one at random in such a way that each number is "equally likely" to be picked. Here, I put "equally likely" in quotes, because if we think about it for a time, then it is not at all clear what this actually means. Indeed, it does not make sense to say that

$$\mathbf{P}[X = x] = \frac{1}{\#([0,1])}$$

in this case, because the number of points inside the interval $[0,1]$, which is the quantity $\#([0,1])$, is infinity, and $1/\infty$ is not a number.

In order to get around this, we can once again use a discrete approximation. That is, for a large integer n, we split the interval $[0,1]$ into n sub-intervals of size $1/n$, and then let X_n be uniformly chosen among the points on the grid formed by these sub-intervals, that is, the fractions

$$\left\{0, \frac{1}{n}, \frac{2}{n}, \ldots, 1\right\}.$$

(See Figure 8.6 for an illustration.) Then, it stands to reason that we should have

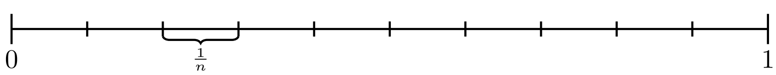

Figure 8.6. Spit the unit interval into a grid of n sub-intervals. Then, let X_n be chosen uniformly among all the grid points, namely, $\{0, \frac{1}{n}, \frac{2}{n}, \ldots, 1\}$.

that $X_n \approx X$ for large values of n, since the spacing between grid points goes to zero. More specifically, if we fix some number $x \in [0,1]$ and once again let x_n denote the number of sub-intervals before x, then we expect that

$$\mathbf{P}[X \leq x] = \lim_{n\to\infty} \mathbf{P}[X_n \leq x_n/n] \qquad \text{and} \qquad \mathbf{P}[X = x] = \lim_{n\to\infty} \mathbf{P}[X_n = x_n/n].$$

By virtue of X_n being uniform on a set with $n+1$ elements, the probability of each of its outcomes is $\frac{1}{n+1}$, hence

$$\mathbf{P}[X \leq x] = \lim_{n\to\infty} \frac{x_n}{n+1} \qquad \text{and} \qquad \mathbf{P}[X = x] = \lim_{n\to\infty} \frac{1}{n+1}.$$

Using once again the sandwich theorem and the fact that $nx - 1 \leq x_n \leq nx$, it is easy to conclude from this that

$$\mathbf{P}[X \leq x] = x \qquad \text{and} \qquad \mathbf{P}[X = x] = 0.$$

Thus, we are once again confronted to a seemingly paradoxical situation. On the one hand, the probability

$$\mathbf{P}[X \leq x] = x$$

for $x \in [0,1]$ makes sense: We always get a number between zero and one, and the probability increases with x, consistent the fact that the probability of observing the outcome of the uniform random number in the interval $[0,x]$ should get bigger as we increase x. On the other hand, we have that $\mathbf{P}[X = x] = 0$ for every $x \in [0, 1]$, which seems to indicate that the uniform variable X cannot give an output in $[0, 1]$.

8.3. A Solution to the Apparent Paradox

In short, in order to get around the apparent paradox that we have seen appear in the last two examples, we must abandon the notion that the quantities

$$\mathbf{P}[X = x]$$

are meaningful for continuous random variables. Instead, the best that we can typically do is assign a nontrivial meaning to probabilities of the form

$$\mathbf{P}[X \in A],$$

where A is a subset of the real line $\mathbb{R}$ that has nonzero length. For example, A could be an interval of the form $[a, b]$ with $a < b$, or a a union of several intervals, but not a finite or countable collection of isolated points.

Looking back at the examples of the first arrival time and the uniform variable on $[0, 1]$, we see that we can easily provide meaningful answers to $\mathbf{P}[X \in A]$ when A is an interval. On the one hand, if X is the first arrival defined in (8.1), then for every numbers $0 < a < b$, we have that

$$\begin{aligned}\mathbf{P}\big[X \in (a, b]\big] &= \mathbf{P}[\{X > a\} \cap \{X \leq b\}]\\ &= \mathbf{P}[\{X > a\} \cap \{X > b\}^c]\\ &= \mathbf{P}[X > a] - \mathbf{P}[X > b] \qquad \text{(case-by-case property)}\\ &= \mathrm{e}^{-\mu a} - \mathrm{e}^{-\mu b},\end{aligned}$$

which is a positive probability. In similar fashion, if X is uniform on $[0, 1]$, then for any numbers $0 < a < b < 1$, it is easily checked that

$$\mathbf{P}\big[X \in (a, b]\big] = b - a,$$

which is also a positive probability.

In closing, although we can dismiss the fact that $\mathbf{P}[X = x] = 0$ as a mathematical technicality, and be reassured by the fact that $\mathbf{P}[X \in A]$ makes sense when A's length is nonzero, the fact that $\mathbf{P}[X = x] = 0$ interferes with essentially every aspect of the theory of random variables that we have defined so far. This, in a nutshell, explains why continuous random variables are more difficult to deal with than discrete random variables. For instance:

(1) The fact that $\mathbf{P}[X = x] = 0$ for continuous random variables renders the concept of distribution that we have defined up to this point completely meaningless in this case. Thus, we need to formulate a new notion of distribution in this setting.

(2) If we try to apply the definition of expected value for discrete random variables in (5.4) to a continuous variable, then we get

$$\mathbf{E}[f(X)] = \sum_x f(x)\mathbf{P}[X = x].$$

This sum is ambiguous for two reasons: On the one hand, $\mathbf{P}[X = x] = 0$ for all x; on the other hand, sums involving an uncountable infinity of terms are not something that have an obvious meaning.

(3) The conditional probability of an event A given $X = x$ is defined as

$$\mathbf{P}[A|X = x] = \frac{\mathbf{P}[A \cap \{X = x\}]}{\mathbf{P}[X = x]}.$$

If $\mathbf{P}[X = x] = 0$, then this creates a problem as we have a division by zero. Thus, if we want to define a notion of conditioning involving continuous variables, then we must find a new definition.

Solving the problems above is the subject of the next chapter. Before we do this, however, we end this chapter by briefly discussing some of the philosophical implications of the fact that $\mathbf{P}[X = x] = 0$.

8.4. A Brief Comment on Philosophical Implications

Much of the discomfort that people have with the fact that $\mathbf{P}[X = x] = 0$ for every x when X is continuous can be explained with the following thought experiment: Once we carry out a random experiment, we obtain an outcome $\omega \in \Omega$ in the sample space. Then, once we plug this outcome in our random variable, $X(\omega)$, we obtain some output, say, the number x. Because we have just observed that X gave the output x, it clearly is the case that x is among the possible outputs of X. Why, then, is its probability zero? Up until now, we have been thinking about a probability of zero as being equivalent to the claim that an event is impossible.

In light of this thought experiment, a few mathematicians and philosophers argue that there is a fatal flaw in the way that continuous random variables are usually defined in the mathematical probability of theory. However, the more mainstream view of mathematicians who specialize in the field, including myself, is that this is not such a big problem after all, provided we think about the issue carefully enough. In my personal view, the key observation to make to resolve this issue is the following: It is not clear that it is actually possible to manifest a continuous random variable in the real world.

To give an example of what I mean by this, suppose that X is a uniformly random number on the interval $[0, 1]$. Suppose that we make the claim that

$$X = 0.5,$$

that is, the outcome of the random number is exactly one half. In order to verify that this is actually the case, we need to be able to specify X's outcome with infinite accuracy. Indeed, there is no limit to how close a real number can be to 0.5 without actually being equal to it. For example, if we want to make the claim that $X = 0.5$, then we have to be able to differentiate 0.5 from all of the following numbers:

$$\begin{aligned}
&0.51\\
&0.501\\
&0.5001\\
&\cdots\\
&0.5\underbrace{000\ldots000}_{\text{a googol } 0's}1\\
&\cdots
\end{aligned}$$

However, any measurement technique that we can actually deploy in the real world will have a maximal accuracy. Therefore, my proposed solution to the apparent paradox would be as follows: One should not worry about the meaning of the probability of the event $\{X = x\}$ when X is a continuous random variable, because the event in question does not correspond to any kind of occurrence that we can actually manifest in the real world. Conversely, events of the form

$$\{X \in A\}$$

when A is an interval or a more general set with positive length do make sense. For example, if we have a measurement device that can detect the outcome of a random variable up to some maximal precision $\varepsilon > 0$, then the occurrence or non-occurrence of the event

$$\{X \in [x - \varepsilon, x + \varepsilon]\}$$

is something that we can, at least in principle, observe in real life. As it happens, probabilities of such events for continuous random variables do make sense!

Before wrapping things up, while my proposed solution allows to get rid of the discomfort coming from $\mathbf{P}[X = x] = 0$, it raises an obvious question or possible objection: If we dismiss the paradox involving the event $\{X = x\}$ by saying that it requires infinite precision in our measurements and is thus meaningless, then why do we even bother with continuous random variables at all? Why not instead use a finite-grid approximation (similar to those we invoked in our definitions of continuous arrival times and the uniform variable on $[0, 1]$), where the grid size is equal to the maximal degree of accuracy of our measurement device?

The answer to this objection is that continuous random variables are often much simpler to work with and much more elegant than their discrete approximations. For instance, compare the probability of the continuous arrival

$$\mathbf{P}[X > x] = \mathrm{e}^{-\mu x}$$

with its geometric approximation

$$\mathbf{P}[X_n > x_n] = \left(1 - \frac{\mu}{n}\right)^{x_n},$$

recalling that x_n is the number of sub-intervals of size $1/n$ before x. Not only is the first expression arguably more elegant and tidier, but if we want to compute the second expression we also have to figure out what x_n is, which involves additional and potentially tedious work! Thus, the usefulness of continuous random variables can be justified by making a direct analogy with calculus: Even though, in real life, we cannot actually compute an instantaneous rate of change, or the area of a disk-like object with radius r is not *exactly* πr^2 (since no object made of atoms is a perfect disk in the platonic sense anyway), it still makes sense to use calculus to compute derivatives and integrals. Indeed, in doing so, we get quantities that we can actually compute with relative ease, as opposed to nasty approximation schemes. The same principle justifies the existence and usefulness of continuous random variables.

Chapter 9

The Theory of Continuous Random Variables

In the previous chapter, we highlighted some of the difficulties in dealing with continuous random variables mathematically. Therein, it was argued that the principal difficulty is the fact that if X is a continuous random variable, then

$$\mathbf{P}[X = x] = 0$$

for every $x \in \mathbb{R}$. As argued in more details in Section 8.3, this interferes with essentially every aspect of the theory of random variables. In this context, our purpose in this chapter is to remedy this problem by making sense of the distributions, expected values, and conditionings of continuous random variables.

9.1. Continuous Distributions

The first problem that we tackle is that of making sense of the distribution of continuous random variables. As highlighted in Section 8.3, one of the keys to making sense of continuous random variables is to abandon the notion that the quantities $\mathbf{P}[X = x]$ are meaningful in this context. Instead, we look at quantities of the form $\mathbf{P}[X \in A]$, where $A \subset \mathbb{R}$ is a set that has nonzero length. Thus, we would like to be able to solve the following problem:

Problem 9.1. Let X be a continuous random variable. Describe the probability

$$\mathbf{P}[X \in A]$$

for any choice of subset $A \subset \mathbb{R}$ that has nonzero length.

Indeed, if we are able to characterize every such probability, then we have all of the information that we could ever need regarding the behavior of X.

9.1.1. Cumulative Distribution Function. As it turns out, providing an explicit formula for every probability of the form

$$\mathbf{P}[X \in A]$$

(that is, for any arbitrary set A) is a bit too ambitious. What we can do, however, is use simpler objects that can be used to compute any probability $\mathbf{P}[X \in A]$, at least in principle. In this context, the main object that we need to solve Problem 9.1 is called the cumulative distribution function:

Definition 9.2 (Cumulative Distribution Function)**.** The cumulative distribution function (CDF) of a continuous random variable X, which is typically denoted F_X, is the function defined as

$$F_X(x) = \mathbf{P}[X \leq x], \qquad x \in \mathbb{R}.$$

In words, $F_X(x)$ is nothing more than the probability that the random variable has an outcome that is smaller or equal to x.

Before we explain how the CDF can be used to solve Problem 9.1, we record a number of its basic properties and look at a few examples.

Proposition 9.3. *For any continuous random variable X, the following holds:*

(1) F_X *is nondecreasing.*

(2) $\lim_{x \to -\infty} F_X(x) = 0.$

(3) $\lim_{x \to \infty} F_X(x) = 1.$

An illustration of the properties listed in Proposition 9.3 can be found in Figure 9.1 below, wherein you will find a plot of a typical CDF. Intuitively, it is fairly

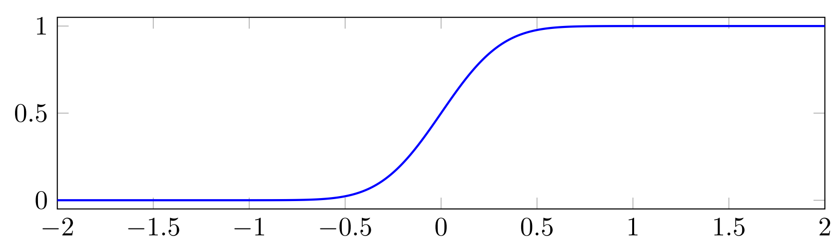

Figure 9.1. Typical plot of a CDF.

straightforward to understand why these properties hold:

(1) If $x < y$, then $\mathbf{P}[X \leq x] \leq \mathbf{P}[X \leq y]$, because it is easier for X to be smaller than a larger number (put another way, if $X \leq x$, then automatically $X \leq y$ also, but not necessarily the other way around).

(2) Since X only takes values on the real line, it cannot take the value $-\infty$. Thus, eventually, the probability that X gives extremely small values vanishes.

(3) Similarly, X cannot be equal to ∞, so eventually, the probability that X is smaller or equal to an extremely large number must increase to one.

We now look at two examples of CDFs:

Example 9.4 (Exponential)**.** We say that X is an exponential random variable with parameter $\mu > 0$, which we denote $X \sim \text{Exp}(\mu)$, if its CDF is equal to

$$F_X(x) = \begin{cases} 1 - \mathrm{e}^{-\mu x} & x \geq 0 \\ 0 & x < 0 \end{cases}. \tag{9.1}$$

See Figure 9.2 for an illustration.

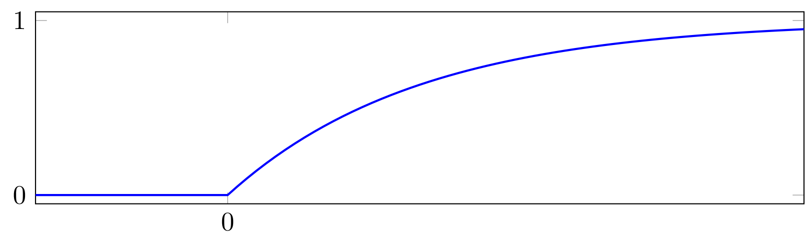

Figure 9.2. CDF of the exponential with parameter μ.

We note that this is nothing more than the first arrival random variable that we constructed in the previous chapter; however, you should know that this kind of random variable is more typically called the exponential in the literature. In that interpretation, μ was the average number of arrivals during one hour, and we had computed that

$$\mathbf{P}[X > x] = \mathrm{e}^{-\mu x}$$

for all $x > 0$. This easily gives the CDF in (9.1), as $\mathbf{P}[X \leq x] = 1 - \mathbf{P}[X > x]$. While we did not explicitly compute that $\mathbf{P}[X \leq x] = 0$ for $x < 0$ in the previous chapter, this is obvious from two points of view: On the one hand, it is intuitively clear that the time to get a first arrival after midnight cannot be negative. On the other hand, because $1 - \mathrm{e}^{-\mu \cdot 0} = 0$, we can conclude purely theoretically that $F_X(x)$ must be zero for $x < 0$ by combining Proposition 9.3 (1) and (2).

Example 9.5 (Continuous Uniform)**.** We say that X is a uniform random variable on the interval $[a, b]$ (where $a < b$ are real numbers), denoted $X \sim \text{Unif}[a, b]$, if its CDF is equal to

$$F_X(x) = \begin{cases} 1 & x > b \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 0 & x < a \end{cases}. \tag{9.2}$$

See Figure 9.3 below for an illustration. This definition extends to any arbitrary interval the uniform random variable on the unit interval $[0, 1]$ that we had constructed in the previous chapter.

9.1.2. Density Function. With these examples in hand, we now explain how CDFs can be used to solve Problem 9.1; this leads us the the notion of density function. First, we note that the CDF can be used to (more or less) directly evaluate $\mathbf{P}[X \in A]$ for a variety of simple sets A. For instance:

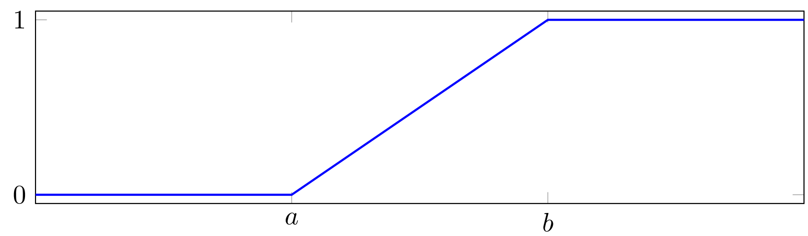

Figure 9.3. CDF of the uniform random variable on $[a, b]$.

(1) If $A = (-\infty, x]$, then
$$\mathbf{P}[X \in A] = \mathbf{P}[X \leq x] = F_X(x).$$
(2) If $A = (x, \infty)$, then
$$\mathbf{P}[X \in A] = \mathbf{P}[X > x] = 1 - \mathbf{P}[X \leq x] = 1 - F_X(x).$$
(3) If $A = (a, b]$, then
$$\mathbf{P}[X \in A] = \mathbf{P}[a < X \leq b] = F_X(b) - F_X(a).$$

At this point, we can make a clever observation that is very easy to miss: By the fundamental theorem of calculus,
$$F_X(b) - F_X(a) = \int_a^b \frac{\mathrm{d}}{\mathrm{d}x} F_X(x)\ \mathrm{d}x.$$
Moreover, because of Proposition 9.3 (2) and (3),
$$F_X(x) = F_X(x) - \lim_{y \to -\infty} F_X(y) = \int_{-\infty}^x \frac{\mathrm{d}}{\mathrm{d}x} F_X(x)\ \mathrm{d}x,$$
and
$$1 - F_X(x) = \lim_{y \to \infty} F_X(y) - F_X(x) = \int_x^\infty \frac{\mathrm{d}}{\mathrm{d}x} F_X(x)\ \mathrm{d}x.$$
In short, in all three cases discussed above, we have that
$$\mathbf{P}[X \in A] = \int_A \frac{\mathrm{d}}{\mathrm{d}x} F_X(x)\ \mathrm{d}x.$$
With a bit of effort (which is beyond the scope of this class), it can actually be proved that this is in fact the case for any arbitrary set:

Proposition 9.6. *Let X be a continuous random variable with CDF F_X. For any set $A \subset \mathbb{R}$, one has*
$$\mathbf{P}[X \in A] = \int_A \frac{\mathrm{d}}{\mathrm{d}x} F_X(x)\ \mathrm{d}x.$$

This then leads us to the following definition:

Definition 9.7 (Density Function)**.** Let X be a continuous random variable with CDF F_X. The density function of X, denoted f_X, is defined as
$$f_X(x) = \frac{\mathrm{d}}{\mathrm{d}x} F_X(x), \qquad x \in \mathbb{R}.$$

In particular,

$$\mathbf{P}[X \in A] = \int_A f_X(x)\, \mathrm{d}x.$$

Thus, the CDF does solve Problem 9.1 completely, albeit in a slightly roundabout way: That is, we have to compute its derivative, and then $\mathbf{P}[X \in A]$ is given by some integral formula, which one has to compute. Nevertheless, the power of this observation lies in the fact that, in order to compute $\mathbf{P}[X \in A]$ for *any arbitrary* A, then in principle all you need is to know how to compute $\mathbf{P}[X \le x]$ for all values of x, which is much simpler.

We may now revisit the examples of CDFs introduced above, and look at what their densities look like.

Example 9.8. The density function of the CDF in Figure 9.1 is illustrated in Figure 9.4 below. If $X \sim \mathrm{Exp}(\mu)$, then we obtain the density f_X by differentiating the CDF in (9.1), which yields

$$f_X(x) = \begin{cases} \mu \mathrm{e}^{-\mu x} & x \ge 0 \\ 0 & x < 0 \end{cases}. \tag{9.3}$$

This is illustrated in Figure 9.5 below. If $X \sim \mathrm{Unif}[a, b]$, then we obtain the density f_X by differentiating the CDF in (9.2), which yields

$$f_X(x) = \begin{cases} 0 & x > b \\ \frac{1}{b-a} & a \le x \le b \\ 0 & x < a \end{cases}. \tag{9.4}$$

This is illustrated in Figure 9.6 below.

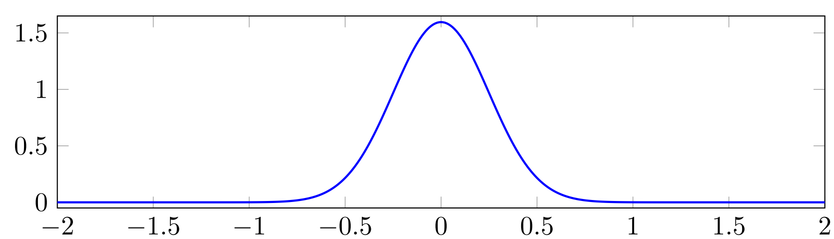

Figure 9.4. The density function of the CDF in Figure 9.1.

Before moving on to other things, it is worth taking the time to address a common point of contention regarding density functions. While the CDF's intuitive meaning as a probability (i.e., $F_X(x) = \mathbf{P}[X \le x]$) is fairly clear, the density function $f_X(x)$ evaluated at some point x is not the probability of any event; especially not $\mathbf{P}[X = x]$, which we know is zero. In particular, the density function does not need to be smaller than one, as illustrated in the above examples. In order to understand what the density function is, we can adopt two distinct points of view:

(1) From the purely pragmatic point of view, we can think of the density function as a computational tool that allows to compute $\mathbf{P}[X \in A]$ for any choice of A; provided one is actually able to solve the integral $\int_A f_X(x)\, \mathrm{d}x$.

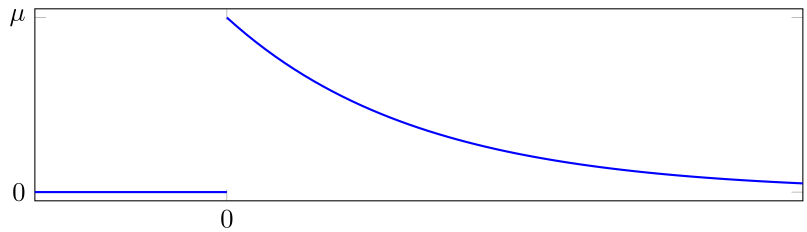

Figure 9.5. Density function of the exponential random variable with parameter $\mu > 0$.

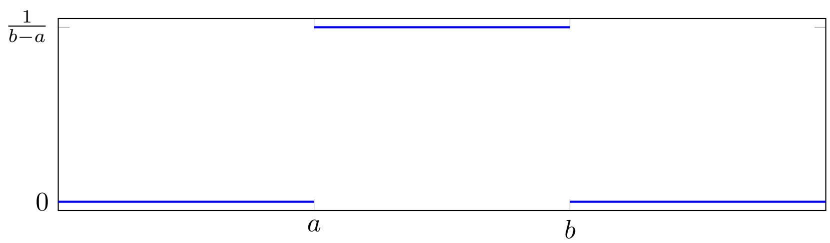

Figure 9.6. Density function of the uniform random variable on $[a, b]$.

(2) From the probabilistic point of view, while $f_X(x)$ is not a probability, it nevertheless encodes some useful information about probabilities. More specifically, knowing that the density is the derivative of the CDF, we can write

$$f_X(x) = \lim_{h \to 0} \frac{F_X(x+h) - F_X(x)}{h} = \lim_{h \to 0} \frac{\mathbf{P}[x < X \leq x+h]}{h}.$$

Looking carefully at this limit, we see that we can think of $f_X(x)$ as the probability that X will be in a small neighborhood of the point x (namely, the interval $(x, x+h]$ for small h), relative the the size of that interval (i.e., the quantity h; and we divide the probability of being in the interval by h).

Thus, the process of visually inspecting the density of a random variable carries meaningful insight into its behavior: Looking at Figure 9.4, for example, we see that it is most likely that X will have an outcome near zero, because the density is highest in that region. Otherwise, if we look at Figure 9.6, then we conclude that the random variable is equally likely to be in any region between a and b (because the density is flat over that region), and we will not get outcomes outside of the interval $[a, b]$ (because the density is zero outside the interval).

9.1.3. Ranges. To finish off this section, we discuss the range of continuous random variables.

Definition 9.9 (Ranges)**.** Let X be a continuous random variable with density function f_X. A range for X can be any set R_X such that

$$\int_{R_X} f_X(x) \, \mathrm{d}x = 1.$$

In words, much like in the case of discrete random variables, the range consists the set of all possible outcomes of X, in the sense that

$$\mathbf{P}[X \in R_X] = \int_{R_X} f_X(x) \, \mathrm{d}x = 1.$$

However, you will notice that in the above definition, we say *a range* for X, and not *the range* for X. This is because there will in general exist many different sets $A \subset \mathbb{R}$ such that

$$\int_A f_X(x) \, \mathrm{d}x = 1.$$

One way to see this is to note that, since the integral of a function over some region A consists of the area under the function over the set, then removing a single point from that set has no effect from the area, because the area of a line is zero. See Figure 9.7 for an illustration. Thus, given any set A that is a range for X, if we

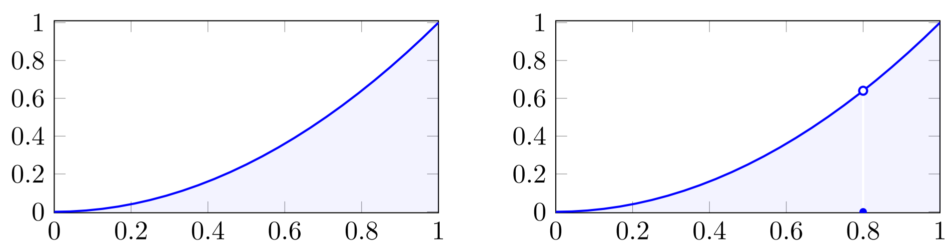

Figure 9.7. The integral of a function over a single point is zero, since there is no area in a line. Thus, the area under the curve on the left-hand side of the illustration is the same as the area under the curve minus the point 0.8, which is what is illustrated on the right.

add or remove any number of points from A, then the result is still a range for X.

Taking this one step further, we note that we can also change the value of the density at single points without changing its probabilistic behavior: If we have two functions f and g that are equal everywhere except at one point, then

$$\int_A f(x) \, \mathrm{d}x = \int_A g(x) \, \mathrm{d}x$$

for any choice of set A. Indeed, the region under the curves of f and g can only differ on a line, which has area zero. (This can also be illustrated as in Figure 9.7; imagine that f is the function on the left, and that g is equal to f everywhere except at $x = 0.8$, which would give it the graph on the right.)

This ambiguity in the range and density will often occur in situations where the CDF F_X is not differentiable everywhere. In the density computations that we carried out in the previous subsection, I neglected to mention that the CDF of the exponential and uniform random variables are actually not differentiable everywhere!

Example 9.10. Looking for example at the case of $X \sim \text{Unif}[0,1]$, we have the CDF

$$F_X(x) = \begin{cases} 1 & x > 1 \\ x & 0 \leq x \leq 1 \\ 0 & x < 0, \end{cases}$$

which is plotted in Figure 9.3 in the case $a = 0$ and $b = 1$. Looking at that plot, we see that the CDF has sharp corners at $x = 0$ and $x = 1$. At these sharp corners, there is not a unique choice of a tangent line; hence the derivative does not actually exist at those points. In the computation that we have done in the previous section, we have declared that

$$f_X(x) = \begin{cases} 0 & x > 1 \\ 1 & 0 \leq x \leq 1 \\ 0 & x < 0 \end{cases};$$

namely, we declared that $f_X(0) = f_X(1) = 1$. However, this was only for simplicity and cosmetic reasons. The value of $f_X(x)$ at $x = 0$ and $x = 1$ could be set to anything whatsoever, and this would never change the value of the integral

$$\int_A f_X(x) \, \mathrm{d}x$$

for any set A. Moreover, the sets

$$[0,1], \qquad (0,1], \qquad [0,1), \qquad (0,1), \quad \text{and} \quad [0,1/2) \cup (1/2,1]$$

are all equally legitimate choices for the range R_X, since the integral of f_X over all of these sets is equal to one.

Remark 9.11. In mathematics, it is customary to avoid having these kinds of ambiguities in the definitions of objects. Thus, mathematicians have come up with some notion of a standard range (called the support) and some notion of a standard density function (called an equivalence class of measurable functions). However, these notions are far beyond the scope of this course, and they are not at all needed to solve the problems that we are interested in here.

9.2. Continuous Expected Values

Now that we have a good handle on the distributions of continuous random variables, we can start looking into more advanced concepts. In this section, we define the expectation of continuous random variables.

In order to understand how this should be defined, we take inspiration from the following: If X is continuous with density function f_X, then

$$\mathbf{P}[X \in A] = \int_A f_X(x) \, \mathrm{d}x.$$

Looking at this expression more closely, we note that it is reminiscent of the fact that if Y is a discrete random variable, then we can always write

$$\mathbf{P}[Y \in A] = \sum_{y \in A} \mathbf{P}[Y = y].$$

This seems to suggest that, in going from the discrete to the continuous world, we

(1) replace sums by integrals; and

(2) replace $\mathbf{P}[X = x]$ (which is meaningless for a continuous variable) by $f_X(x)$.

Knowing that

$$\mathbf{E}[g(Y)] = \sum_{y \in R_Y} g(y)\mathbf{P}[Y = y]$$

whenever Y is discrete, this suggests that the following is the right way to define continuous expected values:

Definition 9.12 (Continuous expected value)**.** Let X be a continuous random variable with density function f_X and range R_X. Then, for any function g, we define

$$\mathbf{E}[g(X)] = \int_{R_X} g(x) f_X(x)\ \mathrm{d}x. \tag{9.5}$$

Given that the integral of a sum of two functions is the sum of the integrals of the individual functions, and that constants can be factored out of integrals, we immediately obtain that the continuous expected value satisfies the same convenient linearity properties as the discrete expected value:

Proposition 9.13 (General linearity of the expected value)**.** *Let X and Y be two continuous random variables, and let $a \in \mathbb{R}$ be a nonrandom constant. Then, we have the linearity properties*

$$\mathbf{E}[X + Y] = \mathbf{E}[X] + \mathbf{E}[Y]$$

and

$$\mathbf{E}[aX] = a\mathbf{E}[X].$$

In particular, this means that we can define the variance of a continuous random variable in the same way as before, and that the continuous variance can be written in the same convenient way that we are used to:

$$\mathbf{Var}[X] = \mathbf{E}\big[(X - \mathbf{E}[X])^2\big] = \mathbf{E}[X^2] - \mathbf{E}[X]^2.$$

In order to further reinforce the idea that the definition of the continuous expected value in (9.5) is the right one, we can perform the following sanity check:

Proposition 9.14. *If $X \sim \mathrm{Exp}(\mu)$, then*

$$\mathbf{E}[X] = \frac{1}{\mu} \qquad \textit{and} \qquad \mathbf{Var}[X] = \frac{1}{\mu^2}.$$

If $X \sim \mathrm{Unif}[a, b]$, then

$$\mathbf{E}[X] = \frac{a+b}{2} \qquad \textit{and} \qquad \mathbf{Var}[X] = \frac{(b-a)^2}{12}.$$

Indeed, if we think about the conceptual meaning of exponential and uniform random variables, we see that the expected values claimed in Proposition 9.14 make sense intuitively.

On the one hand, recall that we have constructed $X \sim \mathrm{Exp}(\mu)$ in the previous section by interpreting it as the amount of time needed to see the first arrival of a Poisson random variable, knowing that the average number of arrivals in the time interval $[0,1]$ is equal to μ. Since we fit on average μ arrivals in a time interval of size $[0,1]$, this suggests that the average spacing between any two arrivals (including between time zero and the first arrival) should be $1/\mu$. Thus, it makes sense that $X \sim \text{Exp}(\mu)$ is such that $\mathbf{E}[X] = 1/\mu$.

On the other hand, if $X \sim \text{Unif}[a,b]$, then the average value that X takes should be the midpoint of the interval $[a,b]$, which is $\frac{a+b}{2}$.

Now that you are hopefully more convinced that the definition in (9.5) is the correct way to define the continuous expected value, let us prove Proposition 9.14:

Proof of Proposition 9.14. Suppose first that $X \sim \text{Exp}(\mu)$. By combining the definition of continuous expected value (9.5) with the density formula (9.3), we get

$$\mathbf{E}[X] = \int_0^\infty x(\mu \mathrm{e}^{-\mu x})\,\mathrm{d}x.$$

Here, we are only integrating from 0 to ∞ because the density in (9.3) is zero when x is negative; hence these values do not contribute to the integral. If we apply integration by parts with

$$\begin{array}{rclcrcl} u & = & x & \qquad & \mathrm{d}u & = & \mathrm{d}x \\ \mathrm{d}v & = & \mu \mathrm{e}^{-\mu x}\mathrm{d}x & & v & = & -\mathrm{e}^{-\mu x} \end{array},$$

then we get that

$$\mathbf{E}[X] = [uv]_0^\infty - \int_0^\infty v\,\mathrm{d}u = [-x\mathrm{e}^{-\mu x}]_0^\infty + \int_0^\infty \mathrm{e}^{-\mu x}\,\mathrm{d}x.$$

Because $x\mathrm{e}^{-\mu x}$ vanishes both at $x=0$ and $x \to \infty$, this simplifies to

$$\mathbf{E}[X] = \int_0^\infty \mathrm{e}^{-\mu x}\,\mathrm{d}x = \left[-\frac{\mathrm{e}^{-\mu x}}{\mu}\right]_0^\infty = 0 - \left(-\frac{\mathrm{e}^{-\mu \cdot 0}}{\mu}\right) = \frac{1}{\mu}.$$

To compute X's variance, we write

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \int_0^\infty x^2\mathrm{e}^{-\mu x}\,\mathrm{d}x - \frac{1}{\mu^2}.$$

Using once again integration by parts (twice in a row, tedious but completely straightforward work), we can check that

$$\int_0^\infty x^2\mathrm{e}^{-\mu x}\,\mathrm{d}x = \frac{2}{\mu^2},$$

which confirms that $\mathbf{Var}[X] = \frac{1}{\mu^2}$.

Suppose now that $X \sim \text{Unif}[a,b]$. Using the density of this variable in (9.4), we get that

$$\mathbf{E}[X] = \int_a^b x\frac{1}{b-a}\,\mathrm{d}x;$$

here we use once again the fact that, because the density is zero outside $[a,b]$, we can ignore those points in the integral. We then get

$$\mathbf{E}[X] = \left[\frac{x^2}{2(b-a)}\right]_a^b = \frac{b^2-a^2}{2(b-a)}.$$

If we write the difference of squares $b^2 - a^2 = (a+b)(b-a)$, then we can simplify

$$\mathbf{E}[X] = \frac{(a+b)(b-a)}{2(b-a)} = \frac{a+b}{2},$$

giving us the desired answer.

Looking now at the variance, we have that

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \int_a^b x^2 \frac{1}{b-a}\ \mathrm{d}x - \frac{(a+b)^2}{4}.$$

The integral can be computed as

$$\begin{aligned}\int_a^b x^2 \frac{1}{b-a}\ \mathrm{d}x &= \left[\frac{x^3}{3(b-a)}\right]_a^b \\ &= \frac{b^3-a^3}{3(b-a)};\end{aligned}$$

hence

$$\mathbf{Var}[X] = \frac{b^3-a^3}{3(b-a)} - \frac{(a+b)^2}{4}.$$

If we put these two fractions on the same denominator, then we get

$$\begin{aligned}\mathbf{Var}[X] &= \frac{4b^3-4a^3}{12(b-a)} - \frac{3(a+b)^2(b-a)}{12(b-a)} \\ &= \frac{b^3-3ab^2+3a^2b-a^3}{12(b-a)} \\ &= \frac{(b-a)^3}{12(b-a)} \\ &= \frac{(b-a)^2}{12},\end{aligned}$$

concluding the proof. □

In short, computing the expected value and variance of continuous random variables is very similar to the discrete case. The main difference is that now we must use integration theory.

9.3. Conditioning and Independence with Continuous Variables

We now arrive at what is arguably one of the most complicated aspects of the theory of continuous random variables, namely, how to make sense of conditioning and independence.

9.3.1. Joint CDFs and Densities. In order to define conditioning and independence in the continuous setting, we must first develop a means of describing the behavior of two continuous random variables simultaneously. For this, we have the notion of joint CDFs and densities:

Definition 9.15 (Joint CDFs and Densities)**.** Let X and Y be two continuous random variables. The joint CDF of X and Y is the function

$$F_{X,Y}(x,y) = \mathbf{P}[\{X \leq x\} \cap \{Y \leq y\}], \qquad x, y \in \mathbb{R}.$$

The joint density of X and Y is the function

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y) = \frac{\partial^2}{\partial y \partial x} F_{X,Y}(x,y).$$

A joint range of X and Y is any subset $R_{X,Y} \subset \mathbb{R}^2$ such that

$$\iint_{R_{X,Y}} f_{X,Y}(x,y) \, \mathrm{d}x\mathrm{d}y = 1.$$

These objects are very similar to the CDF and density, but instead of characterizing the behavior of a single continuous random variable, they characterize the behavior of the *random vector* (X,Y), which is not a random variable but a random point in two-dimensional space $\mathbb{R}^2$:

Proposition 9.16. *For any subset $A \subset \mathbb{R}^2$ of two-dimensional space,*

$$\mathbf{P}[(X,Y) \in A] = \iint_A f_{X,Y}(x,y) \, \mathrm{d}x\mathrm{d}y.$$

For any function $g(x,y)$ from two-dimensional space to the real numbers,

$$\mathbf{E}[g(X,Y)] = \iint_{R_{X,Y}} g(x,y) f_{X,Y}(x,y) \, \mathrm{d}x\mathrm{d}y.$$

The reason why we are interested in the joint density is that, in general, if we only know the density functions f_X and f_Y of two continuous random variables X and Y, then this is not enough to characterize the *joint behavior* of the random variables X and Y. That is, how X and Y interact with one another. In order to illustrate this, consider the following property and example:

Proposition 9.17. *Let X and Y have joint density $f_{X,Y}$. Then, the density function of X, called in this context the marginal density of X, can be computed as*

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y) \, \mathrm{d}y.$$

Proof. We first compute the CDF of X:

$$F_X(z) = \mathbf{P}[X \leq z] = \mathbf{P}[\{X \leq z\} \cap \{Y \in \mathbb{R}\}] = \mathbf{P}[(X,Y) \in A_z],$$

where we define the set

$$A_z = \{(x,y) \in \mathbb{R}^2 : x \leq z\}.$$

The set in question is illustrated in Figure 9.8 below. Thus, by Proposition 9.16, we can write

$$F_X(z) = \iint_{A_z} f_{X,Y}(x,y) \, \mathrm{d}x\mathrm{d}y = \int_{-\infty}^{z} \left(\int_{-\infty}^{\infty} f_{X,Y}(x,y) \, \mathrm{d}y \right) \, \mathrm{d}x,$$

where the second equality is easily seen to be true by observing the illustration of the set A_z in Figure 9.8, and then performing an iterated integral with respect to

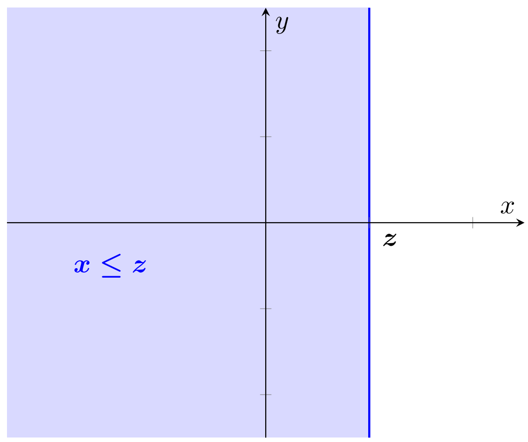

Figure 9.8. Illustration of the set A_z

y first, and then with respect to x. If we then compute the derivative of F_X, we obtain that

$$f_X(z) = \frac{\mathrm{d}}{\mathrm{d}z} F_X(z) = \frac{\mathrm{d}}{\mathrm{d}z} \int_{-\infty}^{z} \left(\int_{-\infty}^{\infty} f_{X,Y}(x, y)\ \mathrm{d}y \right)\ \mathrm{d}x = \int_{-\infty}^{\infty} f_{X,Y}(z, y)\ \mathrm{d}y,$$

where the last equality follows from the fundamental theorem of calculus. □

Example 9.18. Consider the joint densities

$$f_{X,Y}(x, y) = \begin{cases} 1 & 0 \leq x, y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$f_{W,Z}(w, z) = \begin{cases} 2 & \text{if } (w, z) \in S \\ 0 & \text{otherwise} \end{cases},$$

where the set S is defined as

$$S = \big([0, \tfrac{1}{4}] \times [\tfrac{1}{4}, \tfrac{3}{4}]\big) \cup \big([\tfrac{1}{4}, \tfrac{3}{4}] \times [0, \tfrac{1}{4}]\big) \cup \big([\tfrac{1}{4}, \tfrac{3}{4}] \times [\tfrac{3}{4}, 1]\big) \cup \big([\tfrac{3}{4}, 1] \times [\tfrac{1}{4}, \tfrac{3}{4}]\big),$$

and illustrated in Figure 9.9. Obviously, these joint densities are very different; hence the random vectors (X, Y) and (W, Z) have very different behaviors. For instance, the outcome of (X, Y) could be anywhere in the unit square, whereas (W, Z) can only outcome points in the region shaded in blue in Figure 9.9.

However, the random variables X, Y, W and Z are all uniform random variables on $[0, 1]$. To illustrate this, consider, for example, the random variable W. By Proposition 9.17, we know that

$$f_W(w) = \int f_{W,Z}(w, z)\ \mathrm{d}z$$

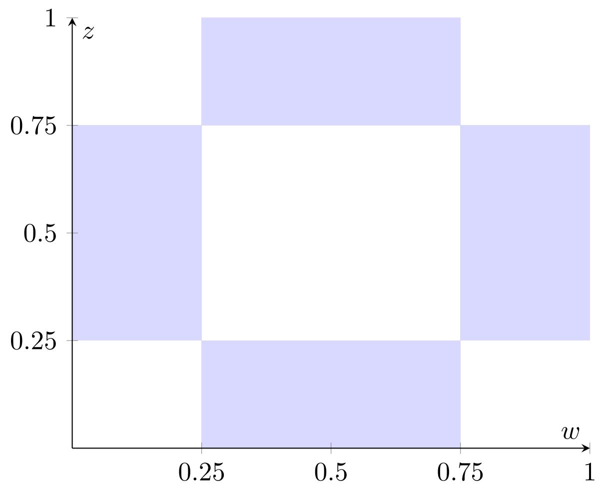

Figure 9.9. Illustration of the set S, being the region shaded in blue. $f_{W,Z}$ is equal to 2 on that set and zero elsewhere.

for any fixed w. Looking at the specific form of the joint density of W and Z (in particular, the set S illustrated in Figure 9.9), we then note that

$$f_W(w) = \begin{cases} 0 & w < 0 \\ \int_{1/4}^{3/4} 2\,\mathrm{d}z = 1 & 0 \le w \le 1/4 \text{ or } 3/4 \le w \le 1 \\ \int_0^{1/4} 2\,\mathrm{d}z + \int_{3/4}^{1} 2\,\mathrm{d}z = 1 & 1/4 \le w \le 3/4 \\ 0 & w > 1 \end{cases},$$

which simplifies nicely to

$$f_W(w) = \begin{cases} 0 & w < 0 \\ 1 & 0 \le w \le 1 \\ 0 & w > 1 \end{cases},$$

hence $W \sim \mathrm{Unif}[0,1]$. Similar arguments can be used in the case of X, Y, and Z.

9.3.2. Independence. Example 9.18 illustrates the fact that knowing the individual (i.e., marginal) densities of two random variables X and Y is not enough to uniquely specify the joint density $f_{X,Y}$. In general, the reason for this is that the joint density not only characterizes the behavior of X and Y individually, but also how the two variables depend on one another. Thus, the joint density is the key to characterizing independence and conditioning in the continuous case. We begin with the notion of independence:

Definition 9.19 (Continuous independence)**.** Two continuous random variables X and Y are independent if for every sets $A, B \subset \mathbb{R}$ such that $\mathbf{P}[X \in A] > 0$ and $\mathbf{P}[Y \in B] > 0$, one has

$$\mathbf{P}[X \in A | Y \in B] = \mathbf{P}[X \in A] \qquad \text{and} \qquad \mathbf{P}[Y \in B | X \in A] = \mathbf{P}[Y \in B],$$

or, equivalently,

$$\mathbf{P}[\{X \in A\} \cap \{Y \in B\}] = \mathbf{P}[X \in A]\mathbf{P}[Y \in B].$$

This definition is strongly reminiscent of the definition of independent events that we have seen previously. Thanks to the relationship between the joint density and probabilities of the form $\mathbf{P}[\{X \in A\} \cap \{Y \in B\}]$, we have the following convenient criterion for independence of two continuous random variables:

Proposition 9.20. *Two continuous random variables X and Y are independent if and only if*

$$F_{X,Y}(x,y) = F_X(x)F_Y(y),$$

for every $x, y \in \mathbb{R}$, or if and only if

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

for every $x, y \in \mathbb{R}$ (i.e., both of these conditions are equivalent).

Remark 9.21. Looking back at Example 9.18, the key difference between the couples (X,Y) and (W,Z) is that

(1) X and Y are independent uniform random variables; and
(2) W and Z are dependent. For instance, we see that if W takes a value in the interval $[0, \frac{1}{4}]$, then this forces Z to take a value in the interval $[\frac{1}{4}, \frac{3}{4}]$, so as to ensure that the couple (W,Z) is in fact in the shaded region in Figure 9.9.

It is interesting to note that the expected value and variance of independent continuous random variables has the same properties as its discrete counterpart:

Proposition 9.22. *Let X and Y be continuous and independent. Then,*

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

and

$$\mathbf{Var}[X+Y] = \mathbf{Var}[X] + \mathbf{Var}[Y].$$

We now end this section with an example that illustrates how independence of continuous random variables can be used in practice.

Example 9.23. Let $X \sim \mathrm{Exp}(\mu)$ and $Y \sim \mathrm{Exp}(\lambda)$, where $\mu, \lambda > 0$ are some positive parameters. Suppose that X and Y are independent. What is the CDF and Density of the random variable $Z = X + Y$?

Let us begin with the CDF. By definition, we have that

$$F_Z(z) = \mathbf{P}[Z \leq z] = \mathbf{P}[X + Y \leq z] = \mathbf{P}[(X,Y) \in A_z],$$

where we define the set

$$A_z = \{(x,y) \in \mathbb{R}^z : x + y \leq z\}.$$

This set is illustrated in Figure 9.10 below. By Proposition 9.16, we then have that

$$F_Z(z) = \iint_{A_z} f_{X,Y}(x,y)\,\mathrm{d}x\mathrm{d}y.$$

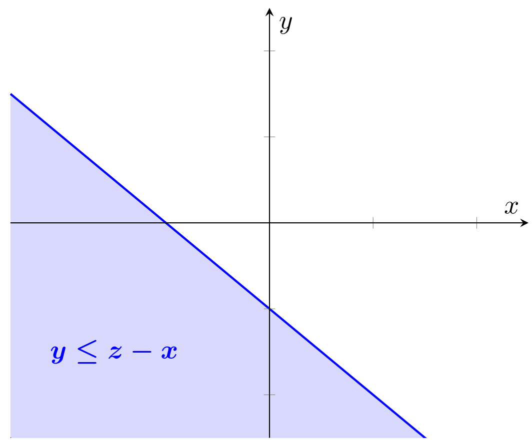

Figure 9.10. An illustration of the set A_z in blue.

A priori, our knowledge that $X \sim \text{Exp}(\mu)$ and $Y \sim \text{Exp}(\lambda)$ alone is not enough to determine the joint density $f_{X,Y}$. However, because X and Y are independent, we know that we can write

$$f_{X,Y}(x,y) = f_X(x)f_Y(y).$$

By definition of the exponential random variable, we have that

$$f_X(x) = \begin{cases} \mu e^{-\mu x} & x \ge 0 \\ 0 & x < 0 \end{cases} \quad \text{and} \quad f_Y(y) = \begin{cases} \lambda e^{-\lambda y} & y \ge 0 \\ 0 & y < 0 \end{cases}.$$

Therefore,

$$f_{X,Y}(x,y) = \begin{cases} \mu\lambda e^{-\mu x - \lambda y} & x, y \ge 0 \\ 0 & \text{otherwise} \end{cases}.$$

With this in hand, we may now compute the CDF of $Z = X + Y$:

Case 1. $z < 0$. Suppose first that $z < 0$. In this case, we claim that

$$F_Z(z) = \mathbf{P}[Z \le z] = \iint_{A_z} f_{X,Y}(x,y) \, dxdy = 0.$$

This can be easily understood thanks to an illustration, such as Figure 9.11 below. Indeed, we know that $f_{X,Y}(x,y)$ is only nonzero when $x, y \ge 0$, but in the case where $z < 0$, A_z has no intersection over that region. Thus, we are really integrating the zero function over the set A_z, which gives an integral of zero.

Case 2. $z \ge 0$. Suppose then that $z \ge 0$. In this case, there is an intersection between the set A_z and $x, y \ge 0$, which is illustrated as the magenta triangle in Figure 9.12 below. Thus, in this case $F_Z(z)$ will be equal to the integral of the joint density $f_{X,Y}$ over that triangular region. If we carry out an iterated integral with respect to y first (from $y = 0$ to the diagonal line $y = z - x$), and with respect to x (from $x = 0$ to $x = z$) second, then we get that

$$F_Z(z) = \int_0^z \left(\int_0^{z-x} \mu\lambda e^{-\mu x - \lambda y} \, dy \right) dx.$$

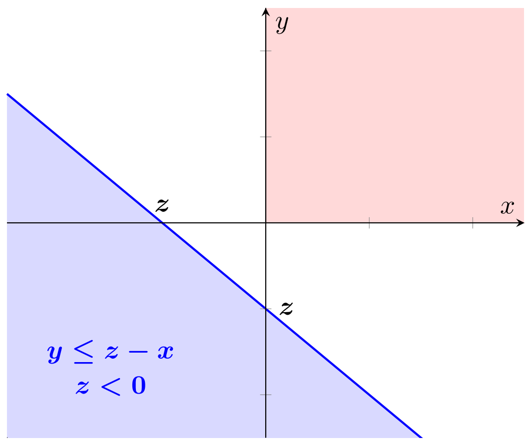

Figure 9.11. When $z < 0$, the set A_z (blue) does not intersect the first quadrant of the x-y axis (red).

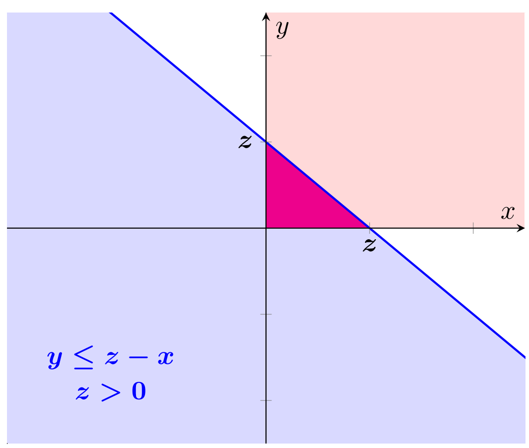

Figure 9.12. When $z \geq 0$, the set A_z (blue) does intersect the first quadrant (red). The intersection gives a triangle, which is colored magenta here.

If we then pull out the terms that do not depend on y out of the dy integral and them compute the dy integral, we get that

$$\begin{aligned}F_Z(z) &= \int_0^z \mu \mathrm{e}^{-\mu x} \left(\int_0^{z-x} \lambda \mathrm{e}^{-\lambda y} \, \mathrm{d}y \right) \, \mathrm{d}x \\ &= \int_0^z \mu \mathrm{e}^{-\mu x} \left(1 - \mathrm{e}^{-\lambda(z-x)} \right) \, \mathrm{d}x = \int_0^z \mu \mathrm{e}^{-\mu x} \, \mathrm{d}x - \mu \mathrm{e}^{-\lambda z} \int_0^z \mathrm{e}^{(\lambda-\mu)x} \, \mathrm{d}x.\end{aligned}$$

At this point, we note that the result of the rightmost integral actually depends on whether or not $\lambda = \mu$, and so we need to consider yet more sub-cases:

Case 2.1. $z \geq 0$ and $\mu = \lambda$. In this case, we have that $\mathrm{e}^{(\lambda-\mu)x} = 1$ and $\mu\mathrm{e}^{-\lambda z} = \mu\mathrm{e}^{-\mu z}$, hence

$$F_Z(z) = \int_0^z \mu\mathrm{e}^{-\mu x}\ \mathrm{d}x - \mu\mathrm{e}^{-\mu z}\int_0^z\ \mathrm{d}x = 1 - \mathrm{e}^{-\mu z} - \mu z\mathrm{e}^{-\mu z}.$$

Case 2.2. $z \geq 0$ and $\mu \neq \lambda$. In this case,

$$\begin{aligned} F_Z(z) &= \int_0^z \mu\mathrm{e}^{-\mu x}\ \mathrm{d}x - \mu\mathrm{e}^{-\lambda z}\int_0^z \mathrm{e}^{(\lambda-\mu)x}\ \mathrm{d}x \\ &= 1 - \mathrm{e}^{-\mu z} - \frac{\mu\mathrm{e}^{-\lambda z}(\mathrm{e}^{(\lambda-\mu)z} - 1)}{\lambda - \mu} \\ &= 1 + \frac{\mu\mathrm{e}^{-\lambda z} - \lambda\mathrm{e}^{-\mu z}}{\lambda - \mu}. \end{aligned}$$

Conclusion. Combining all cases, we obtain that

(1) If $\mu = \lambda$, then

$$F_Z(z) = \begin{cases} 1 - \mathrm{e}^{-\mu z} - \mu z\mathrm{e}^{-\mu z} & z \geq 0 \\ 0 & z < 0 \end{cases},$$

(2) If $\mu \neq \lambda$, then

$$F_Z(z) = \begin{cases} 1 + \frac{\mu\mathrm{e}^{-\lambda z} - \lambda\mathrm{e}^{-\mu z}}{\lambda - \mu} & z \geq 0 \\ 0 & z < 0 \end{cases}.$$

Finally, if we compute a derivative of each of these functions, we obtain the density:

(1) If $\mu = \lambda$, then

$$f_Z(z) = \begin{cases} \mu^2 z\mathrm{e}^{-\mu z} & z \geq 0 \\ 0 & z < 0 \end{cases},$$

(2) If $\mu \neq \lambda$, then

$$f_Z(z) = \begin{cases} \lambda\mu\frac{e^{-\mu z} - e^{-\lambda z}}{\lambda - \mu} & z \geq 0 \\ 0 & z < 0 \end{cases}.$$

Remark 9.24. Although the expression

$$\lambda\mu\frac{e^{-\mu z} - e^{-\lambda z}}{\lambda - \mu}$$

does not make sense when $\lambda - \mu$ (due to a division by zero), it is nevertheless interesting to note that

$$\lim_{\lambda \to \mu} \lambda\mu\frac{e^{-\mu z} - e^{-\lambda z}}{\lambda - \mu} = \mu^2 z\mathrm{e}^{-\mu z},$$

which is consistent with the above example (this limit is easily computed by l'Hôpital's rule, for example).

Remark 9.25. The above example serves as a nice illustration of the fact that much of the difficulties involved with the analysis of continuous random variables lies in the calculus problems that the latter induces.

9.3.3. Conditioning. We finally arrive at one of the most contentious aspects of the theory of continuous random variables, namely, how to condition with respect to the outcome of a continuous variable. The fundamental problem with the business of conditioning with respect to the outcome of some continuous random variable Y is that for every $y \in \mathbb{R}$, we have that

$$\mathbf{P}[Y = y] = 0.$$

Thus, it doesn't make sense to define

$$\mathbf{P}[A|Y = y] = \frac{\mathbf{P}[A \cap \{Y = y\}]}{\mathbf{P}[Y = y]}$$

for any event A.

As it turns out, the key to extending the notion of conditioning to continuous variables is not to try to interpret $\mathbf{P}[A|B] = \frac{\mathbf{P}[A\cap B]}{\mathbf{P}[B]}$ in this setting, but instead the laws of total probability and expectation. In this context, recall that if X and Y are discrete random variables, the law of total expectation says that

$$\mathbf{E}[g(X)] = \sum_{y \in R_Y} \mathbf{E}[g(X)|Y = y]\mathbf{P}[Y = y]$$

for any function g. Going off of the idea that, in going from discrete to continuous expectations

$$\text{discrete:} \qquad \mathbf{E}[g(X)] = \sum_{x \in R_X} g(x)\mathbf{P}[X = x]$$

$$\text{continuous:} \qquad \mathbf{E}[g(X)] = \int_{\mathbb{R}} g(x) f_X(x) \, \mathrm{d}x;$$

we replaced the sum by an integral and the probabilities $\mathbf{P}[X = x]$ by the density $f_X(x)$, the following seems natural: Suppose that Y is continuous. If it is at all possible to make sense of the conditional expectation $\mathbf{E}[g(X)|Y = y]$, then it should satisfy a continuous version of the law of total expectation, namely:

$$\mathbf{E}[g(X)] = \int_{\mathbb{R}} \mathbf{E}[g(X)|Y = y] f_Y(y) \, \mathrm{d}y.$$

Interestingly enough, this actually gives rise to a coherent notion of continuous conditioning:

Definition 9.26 (Conditional Density)**.** Let X and Y be continuous random variables. The conditional density of X given Y, which we denote as the function $y \mapsto f_{X|Y=y}$, is such that for every function g, one has

$$\mathbf{E}[g(X)] = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} g(x) f_{X|Y=y}(x) \, \mathrm{d}x \right) f_Y(y) \, \mathrm{d}y.$$

In particular, if we define $g_A(x) = \mathbf{1}_{\{x \in A\}}$ for some subset $A \subset \mathbb{R}$, then

$$\mathbf{E}[g_A(X)] = \mathbf{P}[X \in A] = \int_{\mathbb{R}} \left(\int_A f_{X|Y=y}(x) \, \mathrm{d}x \right) f_Y(y) \, \mathrm{d}y.$$

In light of the law of total expectation analogy mentioned above, it is customary to denote

$$\mathbf{E}[g(X)|Y = y] = \int_{\mathbb{R}} g(x) f_{X|Y=y}(x) \, \mathrm{d}x \tag{9.6}$$

and

$$\mathbf{P}[X \in A | Y = y] = \int_A f_{X|Y=y}(x) \, \mathrm{d}x. \tag{9.7}$$

Remark 9.27. The extent to which the notations in (9.6) and (9.7) should be taken seriously depends on the context. On the one hand, the continuous conditioning defined above satisfies many of the same properties as its discrete counterpart. Thus, many intuitions that you have regarding how to manipulate discrete conditional probabilities and expectations carry over to the continuous setting. The remainder of this section is devoted to studying these similarities.

On the other hand, from the conceptual point of view, there are good reasons why (9.6) and (9.7) should *not* be interpreted as "the expected value of $g(X)$ if we observe that $Y = y$" or "the probability that $X \in A$ if we observe that $Y = y$." We will discuss this point further in the forthcoming section on the infamous Borel-Kolmogorov paradox.

If one has access to the joint density of two continuous random variables, then the conditional density can be computed in a straightforward way. The formula in question is reminiscent of the identity

$$\mathbf{P}[X = x | Y = y] = \frac{\mathbf{P}[\{X = x\} \cap \{Y = y\}]}{\mathbf{P}[Y = y]}$$

in the case where X and Y are discrete, but it is instead formulated with density functions:

Proposition 9.28 (Conditional density formula)**.** *If X and Y have joint density $f_{X,Y}$ and Y has marginal density f_Y, then*

$$f_{X|Y=y}(x) = \begin{cases} \frac{f_{X,Y}(x,y)}{f_Y(y)} & \text{if } f_Y(y) > 0 \\ 0 & \text{otherwise} \end{cases}.$$

Proof. For any function g, we have that

$$\mathbf{E}[g(X)] = \int_{\mathbb{R}} g(x) f_X(x) \, \mathrm{d}x.$$

Next, if we use the fact that the marginal density of X can be obtained by integrating out the joint density of X and Y with respect to the y variable, then we get that

$$\mathbf{E}[g(X)] = \int_{\mathbb{R}} g(x) \left(\int_{\mathbb{R}} f_{X,Y}(x,y) \, \mathrm{d}y \right) \, \mathrm{d}x = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} g(x) f_{X,Y}(x,y) \, \mathrm{d}x \right) \, \mathrm{d}y,$$

where in the second equality we have permuted the order of the $\mathrm{d}x$ and $\mathrm{d}y$ integrals. At this point, if we write

$$f_{X,Y}(x,y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} f_Y(y)$$

whenever $f_Y(y)$ is positive, then we obtain the desired result. Indeed, if $f_Y(y) = 0$, then this automatically implies that $f_{X,Y}(x,y) = 0$ for every $x \in \mathbb{R}$ (except possibly a few isolated points that have no contribution to integrals), because

$$f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y) \, \mathrm{d}x$$

and $f_{X,Y}(x,y)$ is always nonnegative. Thus, we can write

$$\begin{aligned}\mathbf{E}[g(X)] &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} g(x) f_{X,Y}(x,y) \, \mathrm{d}x \right) \, \mathrm{d}y \\ &= \int_{\{y: f_Y(y) > 0\}} \left(\int_{\mathbb{R}} g(x) f_{X,Y}(x,y) \, \mathrm{d}x \right) \, \mathrm{d}y \\ &= \int_{\{y: f_Y(y) > 0\}} \left(\int_{\mathbb{R}} g(x) \frac{f_{X,Y}(x,y)}{f_Y(y)} \, \mathrm{d}x \right) f_Y(y) \, \mathrm{d}y \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} g(x) \frac{f_{X,Y}(x,y)}{f_Y(y)} \mathbf{1}_{\{y: f_Y(y) > 0\}} \, \mathrm{d}x \right) f_Y(y) \, \mathrm{d}y,\end{aligned}$$

as desired. □

As alluded to in Remark 9.27, continuous conditional expectations and probabilities satisfy many of the same properties as their discrete counterparts. One such property, which is very useful in all sorts of problems, is the following:

Proposition 9.29 (Replacement identity)**.** *Let X and Y be continuous random variables, and let g be a two-dimensional function and $A \subset \mathbb{R}^2$ be a two-dimensional set. For every $y \in \mathbb{R}$, one has*

$$\mathbf{E}[g(X,Y)|Y = y] = \mathbf{E}[g(X,y)|Y = y]$$

and

$$\mathbf{P}[(X,Y) \in A|Y = y] = \mathbf{P}[(X,y) \in A|Y = y].$$

In particular, if X and Y are independent, then

$$\mathbf{E}[g(X,Y)|Y = y] = \mathbf{E}[g(X,y)]$$

and

$$\mathbf{P}[(X,Y) \in A|Y = y] = \mathbf{P}[(X,y) \in A].$$

The replacement identity is very pleasing intuitively: If we "condition" on $Y = y$, then we can replace any appearance of the random variable Y by the number y. This is very similar to a manipulation that we can do with discrete conditionings. We now showcase an example that illustrates how the replacement identity can be used in practice:

Example 9.30. Let $X, Y \sim \mathrm{Exp}(\mu)$ be independent. As an alternative to Example 9.23, we can compute the distribution of $Z = X + Y$ using the replacement identity. That is, because of the independence of X and Y, in this case the replacement identity yields

$$\begin{aligned}F_Z(z) = \mathbf{P}[X + Y \leq z] = \int_{\mathbb{R}} &\mathbf{P}[X + Y \leq z|Y = y] f_Y(y) \, \mathrm{d}y \\ &= \int_{\mathbb{R}} \mathbf{P}[X + y \leq z] f_Y(y) \, \mathrm{d}y.\end{aligned}$$

Knowing that $Y \sim \mathrm{Exp}(\mu)$, this gives

$$F_Z(z) = \int_0^\infty \mathbf{P}[X + y \leq z] \mu \mathrm{e}^{-\mu y} \, \mathrm{d}y;$$

we can get rid of the integration over $y < 0$ because the density is zero on that region. Next, since $X \sim \text{Exp}(\mu)$, then we have that

$$\begin{aligned}\mathbf{P}[X + y \leq z] &= \mathbf{P}[X \leq z - y] \\ &= \begin{cases} 1 - \mathrm{e}^{-\mu(z-y)} & z - y \geq 0 \\ 0 & z - y < 0 \end{cases} = \begin{cases} 1 - \mathrm{e}^{-\mu(z-y)} & y \leq z \\ 0 & y > z \end{cases}.\end{aligned}$$

Thus, we only need to integrate values of y up to z, which yields

$$F_Z(z) = \int_0^z \left(1 - \mathrm{e}^{-\mu(z-y)}\right) \mu \mathrm{e}^{-\mu y} \, \mathrm{d}y = 1 - \mathrm{e}^{-\mu z} - \mu z \mathrm{e}^{-\mu z}.$$

I will let you be the judge of which of these two methods (i.e., the joint density in Example 9.23, or the conditioning/replacement trick used here) you find preferable; personally I prefer to use the replacement trick.

More generally, the replacement identity leads to a convenient computational tool when dealing with any sum of two independent random variables:

Proposition 9.31 (Convolution Identity)**.** *Let X and Y be independent continuous random variables with respective marginal density functions f_X and f_Y. Then, the density function of $X + Y$ is equal to*

$$f_{X+Y}(z) = \int_{\mathbb{R}} f_X(z - y) f_Y(y) \, \mathrm{d}y.$$

Proof. We begin, as always, with the CDF.

$$F_{X+Y}(z) = \mathbf{P}[X + Y \leq z].$$

By the replacement identity and independence, this can be written as

$$\begin{aligned}F_{X+Y}(z) &= \int_{\mathbb{R}} \mathbf{P}[X + Y \leq z | Y = y] f_Y(y) \, \mathrm{d}y \\ &= \int_{\mathbb{R}} \mathbf{P}[X + y \leq z] f_Y(y) \, \mathrm{d}y \\ &= \int_{\mathbb{R}} \mathbf{P}[X \leq z - y] f_Y(y) \, \mathrm{d}y.\end{aligned}$$

Then, to get the density, we take a derivative, which yields

$$\begin{aligned}f_{X+Y}(z) &= \frac{\mathrm{d}}{\mathrm{d}z} \int_{\mathbb{R}} \mathbf{P}[X \leq z - y] f_Y(y) \, \mathrm{d}y \\ &= \int_{\mathbb{R}} \frac{\mathrm{d}}{\mathrm{d}z} \mathbf{P}[X \leq z - y] f_Y(y) \, \mathrm{d}y \\ &= \int_{\mathbb{R}} f_X(z - y) f_Y(y) \, \mathrm{d}y,\end{aligned}$$

concluding the proof. $\square$

As a final example in this section, we discuss how continuous conditioning can be used to construct sophisticated models that involve multiple interacting sources of randomness:

Example 9.32 (Winter traffic)**.** Suppose that we want to model the amount of time that a random commuter in the greater Chicago area spends driving to work during winter. For this, we define the following random variable:

$$T = \text{Amount of time (in hours) to commute to work on a given morning.}$$

It is natural to expect that this travel time might depend on a number of external factors. One such example could be the amount of snowfall on a given morning:

$$S = \text{Amount of snowfall (in inches per hour) on a given morning.}$$

We would like to construct a probability model such that the commute time to work depends on the amount of snowfall; more specifically, the commute time is more likely to be greater if there is more snowfall.

Here is one way in which such a model could be constructed: Suppose that $S \sim \text{Unif}[0,2]$, and that

$$f_{T|S=s}(t) = \begin{cases} (1-s/4)\mathrm{e}^{-(1-s/4)t} & t \geq 0 \\ 0 & t < 0 \end{cases}.$$

In words, the conditional density of T given $S = s$ is exponential with parameter $\mu = (1 - s/4)$. This parameter becomes smaller when s increases. Since the expectation of $X \sim \text{Exp}(\mu)$ is $1/\mu$, then this means that the expected commute time increases with the amount of snowfall, which is consistent with the kind of model that we want to construct.

With this basic assumption in hand, what are T's CDF, density, and expectation? For this, we can apply the definition of continuous conditional expectation and probability: First, for the CDF we have that

$$\mathbf{P}[T \leq u] = \int_{\mathbb{R}} \mathbf{P}[T \leq u|S = s] f_S(s)\,\mathrm{d}s = \frac{1}{2}\int_0^2 \mathbf{P}[T \leq u|S = s]\,\mathrm{d}s.$$

For any $0 \leq s \leq 2$, and $u \geq 0$ one has

$$\mathbf{P}[T \leq u|S = s] = \int_0^u (1-s/4)\mathrm{e}^{-(1-s/4)t}\,\mathrm{d}t = 1 - \mathrm{e}^{-(1-s/4)u}.$$

If we then integrate this expression from zero to 2 and multiply by $\frac{1}{2}$, we have

$$\mathbf{P}[T \leq u] = \frac{1}{2}\int_0^2 \left(1 - \mathrm{e}^{-(1-s/4)u}\right)\,\mathrm{d}s = 1 + \frac{2(\mathrm{e}^{-u} - \mathrm{e}^{-u/2})}{u}$$

for $u \geq 0$, which yields

$$F_T(t) = \begin{cases} 1 + \frac{2(\mathrm{e}^{-t} - \mathrm{e}^{-t/2})}{t} & t > 0 \\ 0 & t \leq 0 \end{cases}.$$

Taking the derivative, we get the (rather horrible looking) density

$$f_T(t) = \begin{cases} \frac{\mathrm{e}^{-t}\left(\mathrm{e}^{t/2}(t+2) - 2(t+1)\right)}{t^2} & t > 0 \\ 0 & t \leq 0 \end{cases}.$$

See Figure 9.13 below for an illustration of this CDF and density function.

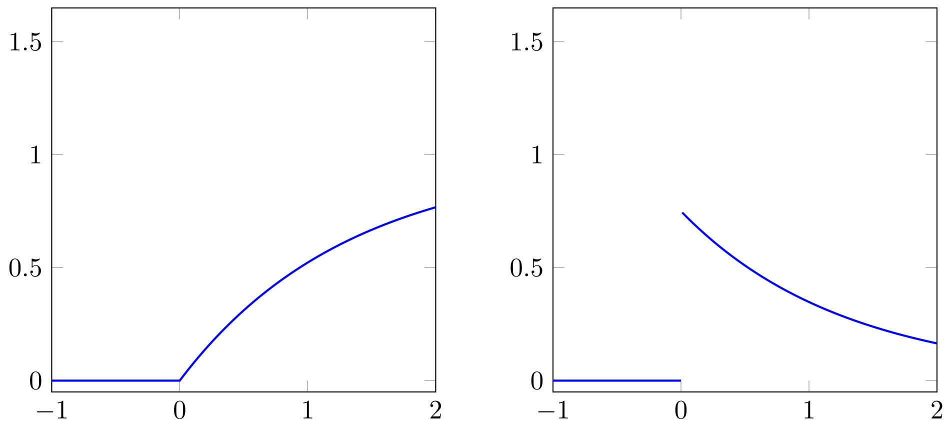

Figure 9.13. CDF (left) and density function (right) of T.

Finally, we compute the expected value. We could (if we really wanted to) compute this using the density, that is,

$$\mathbf{E}[T] = \int_{\mathbb{R}} t f_T(t) \, \mathrm{d}t.$$

However, the density of T is rather nasty, and so it would be nice to be able to not have to deal with it. Thankfully, there is a way to get around it, namely, the law of total expectation:

$$\begin{aligned}
\mathbf{E}[T] &= \int_{-\infty}^{\infty} \mathbf{E}[T|S = s] f_S(s) \, \mathrm{d}s \\
&= \frac{1}{2} \int_0^2 \mathbf{E}[T|S = s] \, \mathrm{d}s \\
&= \frac{1}{2} \int_0^2 \left(\int_0^{\infty} t f_{T|S=s}(t) \, \mathrm{d}t \right) \, \mathrm{d}s \\
&= \frac{1}{2} \int_0^2 \left(\int_0^{\infty} t(1 - s/4) \mathrm{e}^{-(1-s/4)t} \, \mathrm{d}t \right) \, \mathrm{d}s.
\end{aligned}$$

Looking at the inner $\mathrm{d}t$ integral, we note that this is nothing more than the expectation of a random variable with distribution $\mathrm{Exp}(1 - s/4)$; hence

$$\mathbf{E}[T] = \frac{1}{2} \int_0^2 \frac{1}{1 - s/4} \, \mathrm{d}s = \log(4) \approx 1.386,$$

where the last equality follows from a simple change of variables.

9.4. Change of Variables

In this section, we study the problem of change of variables involving continuous random variables. That is, given a continuous random variable X and some function g, how can we determine the distribution of $g(X)$?

In the discrete setting, this question is essentially trivial. Indeed, for any number y, we have that

$$\mathbf{P}[g(X) = y] = \sum_{x \text{ s.t. } g(x)=y} \mathbf{P}[X = x];$$

in words, the probability that $g(X) = y$ is the sum of the probabilities $\mathbf{P}[X = x]$ for every choice of number x such that $g(x) = y$.

In sharp contrast to the discrete case, the computation of $g(X)$'s distribution when X is continuous involves some nontrivial calculus. In order to illustrate this phenomenon, we begin by looking at some examples.

Example 9.33 (Boxes)**.** Suppose that you work at a warehouse. The warehouse contains a number of cubic boxes with different side lengths. Let L be a random variable that represents the side-length of a randomly selected cubic box, measured in inches. We assume that $L \sim \text{Unif}[12, 24]$, which we recall has CDF and density

$$F_L(x) = \begin{cases} 1 & x > 24 \\ \frac{x-12}{12} & 12 \leq x \leq 24 \\ 0 & x < 12 \end{cases} \qquad \text{and} \qquad f_L(x) = \begin{cases} 0 & x > 24 \\ \frac{1}{12} & 12 \leq x \leq 24 \\ 0 & x < 12 \end{cases}.$$

Suppose that we are interested in understanding the distribution of the volume (in cubic inches) of a box selected at random. That is, we would like to understand the CDF and density of the random variable $V = L^3$. Looking first at the CDF, we have

$$\begin{aligned} \mathbf{P}[V \leq v] &= \mathbf{P}[L^3 \leq v] && (9.8) \\ &= \mathbf{P}[L \leq v^{1/3}] \\ &= F_L(v^{1/3}) \\ &= \begin{cases} 1 & v^{1/3} > 24 \\ \frac{v^{1/3}-12}{12} & 12 \leq v^{1/3} \leq 24 \\ 0 & v^{1/3} < 12 \end{cases} \\ &= \begin{cases} 1 & v > 13\,824 \\ \frac{v^{1/3}-12}{12} & 1\,728 \leq v \leq 13\,824 \\ 0 & v < 1\,728 \end{cases}. && (9.9) \end{aligned}$$

If we then compute the derivative,

$$f_V(v) = \begin{cases} 0 & v > 13\,824 \\ \frac{1}{36v^{2/3}} & 1\,728 \leq v \leq 13\,824 \\ 0 & v < 1\,728 \end{cases}$$

At first glance, with this example, it may seem to you that there isn't much to the business of change of variables: We can just look at the CDF of the function of our random variable, then apply an inverse function on both sides of the inequality to isolate the initial random variable, and then directly plug the CDF of that initial random variable (i.e., the sequence of steps from (9.8) to (9.9)).

While this is true for that example, we should keep in mind that there is a bit of a subtlety here that we didn't really mention, which is that, sometimes, if you apply a function on both sides of the inequality, it can flip the order of the inequality. For this, we consider the following example:

Example 9.34 (Another example)**.** Suppose that $X \sim \text{Unif}[0,1]$. What are the CDF and density of $Y = 1/X$? Looking first at the CDF, if $y > 0$, then

$$\begin{aligned} F_Y(y) &= \mathbf{P}[1/X \le y] \\ &= \mathbf{P}[X \ge 1/y] \\ &= 1 - F_X(1/y) \\ &= \begin{cases} 0 & 1/y > 1 \\ 1 - 1/y & 0 \le 1/y \le 1 \\ 1 & 1/y < 0 \end{cases} \\ &= \begin{cases} 1 - 1/y & y \ge 1 \\ 0 & y < 1 \end{cases}. \end{aligned}$$

Looking at the density, by computing the derivative of the above we get

$$f_Y(y) = \begin{cases} 1/y^2 & y \ge 1 \\ 0 & y < 1 \end{cases}.$$

Comparing the previous two examples, we see that in the latter one when we applied the inverse of the function being applied to X, the inequality was flipped, and thus we got $1 - F_X(1/y)$ instead of $F_X(1/y)$. As it turns out, the previous two examples illustrate the two types of behaviors that can occur when computing the CDFs of changes of variables. We now state a general result that explains how to perform a change of variables in any situation:

Definition 9.35 (Invertible)**.** Let g be a function that assigns to every element a in some set A another element $g(a)$ in some set B. Let

$$g(A) = \{b \in B : \text{there exists } a \in A \text{ such that } g(a) = b\}.$$

That is, $g(A)$ is the set of all possible outputs that g can give when evaluated in some element in A. We say that g is invertible on A if for every $b \in g(A)$, there exists a unique element $a \in A$ such that $g(a) = b$. We then use the notation

$$g^{-1}(b) = a$$

to denote this relationship.

Example 9.36. Consider the function $g(x) = x^2$. If $A = [-1, 1]$, then $g(A) = [0, 1]$. However, g is not invertible on A. Indeed, $g(-1) = g(1) = 1$, hence there is not a unique element $a \in A$ such that $g(a) = 1$. Conversely, if $A = [0, 1]$, then once again $g(A) = [0, 1]$, but this time g is invertible on A. See Figure 9.14 for an illustration.

We may now state a general methodology for obtaining changes of variables:

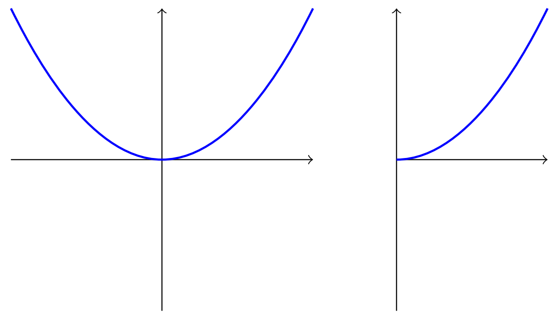

Figure 9.14. Although both plots represent the same parabola x^2 the first is not invertible, whereas the second is.

Proposition 9.37 (General 1-d change of variables)**.** *Let X be a continuous random variable and g be a function. Suppose that there exists a range for X, R_X, on which the function g exists and is invertible. If we denote the random variable $Y = g(X)$, then*

(1) $R_Y = g(R_X)$ is a range for Y;

(2) for every $y \in g(R_X)$, we have that

$$F_Y(y) = \begin{cases} F_X\big(g^{-1}(y)\big) & \text{if } g^{-1} \text{ is an increasing function} \\ 1 - F_X\big(g^{-1}(y)\big) & \text{if } g^{-1} \text{ is a decreasing function} \end{cases};$$

(3) for every $y \in g(R_X)$,

$$f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\frac{\mathrm{d}}{\mathrm{d}y}g^{-1}(y)\right|,$$

where we recall that $|\cdot|$ denotes the absolute value of a number. For $y \notin g(R_X)$, we can set $f_Y(y) = 0$.

In the next example, we revisit the two changes of variables that we have performed earlier, and discuss how they can be obtained in a more streamlined manner thanks to the above result:

Example 9.38. Returning to the example of $L \sim \text{Unif}[12, 24]$, and $V = L^3$, we can choose the range

$$R_L = [12, 24]$$

for L, and note that the function $g(x) = x^3$ is invertible on that set. The inverse in question is given by $g^{-1}(x) = x^{1/3}$, which is an increasing function. Therefore, we immediately recover that

$$R_V = g(R_L) = g([12, 24]) = [1\,728, 13\,824],$$
$$F_V(v) = F_L\big(g^{-1}(v)\big) = F_L(v^{1/3}),$$

as well as

$$f_V(v) = f_L\big(g^{-1}(v)\big)\left|\frac{\mathrm{d}}{\mathrm{d}v}g^{-1}(v)\right| = \frac{1}{12}\cdot\left|\frac{1}{3v^{2/3}}\right| = \frac{1}{36v^{2/3}}$$

for every v inside $g(R_L)$, namely, $[1\,728, 13\,824]$.

Now consider $X \sim \mathrm{Unif}[0,1]$ and $Y = 1/X$. In this case, we note that there is a slight issue with the range. That is, if we choose $R_X = [0,1]$, then the function $g(x) = 1/x$ is actually not defined at $x = 0$. To get around this issue, however, we can simply delete that point from the range and instead look at $R_X = (0,1]$. In this case, the function g is in fact invertible on R_X, and we have

$$R_Y = g\big((0,1]\big) = [1,\infty).$$

Moreover, since $g^{-1}(x) = 1/x$ is decreasing on $(0,1]$, we have that

$$F_Y(y) = 1 - F_X(1/y)$$

and

$$f_Y(y) = f_X(1/y)\left|\frac{\mathrm{d}}{\mathrm{d}y}g^{-1}(y)\right| = 1 \cdot \left|-\frac{1}{y^2}\right| = \frac{1}{y^2}$$

for every y inside $g(R_X) = [1,\infty)$.

9.4.1. Two Dimensions. Before moving on, we also discuss how to perform a change of variables on a pair of real random variables (X,Y). The general formula for such a change of variables is very similar as the case of one dimension, but involves the usual additional complexity due to the multivariate setting.

Proposition 9.39. *Let X and Y be two continuous random variables with joint range $R_{X,Y}$ and joint density $f_{X,Y}$. Suppose that we are given an invertible function $g : R_{X,Y} \to \mathbb{R}^2$ that we can write in the form*

$$g(x,y) = \big(g_1(x,y), g_2(x,y)\big).$$

For any $(u,v) \in g(R_{X,Y})$, we denote the inverse of this function for

$$g^{-1}(u,v) = \big(g_1^{-1}(u,v), g_2^{-1}(u,v)\big).$$

If we define the random vector $(U,V) = g(X,Y)$, then

(1) $R_{U,V} = g(R_{X,Y})$ is a range for (U,V); and

(2) the joint density of U and V is given by

$$f_{U,V}(u,v) = f_{X,Y}\big(g_1^{-1}(u,v), g_2^{-1}(u,v)\big)\,\big|J(u,v)\big|, \qquad (u,v) \in R_{U,V},$$

where

$$J(u,v) = \frac{\partial g_1^{-1}(u,v)}{\partial u} \cdot \frac{\partial g_2^{-1}(u,v)}{\partial v} - \frac{\partial g_1^{-1}(u,v)}{\partial v} \cdot \frac{\partial g_2^{-1}(u,v)}{\partial u}.$$

For $(u,v) \notin g(R_{X,Y})$, we can set $f_{U,V}(u,v) = 0$.

Remark 9.40. Recall that the expression $|J(u,v)|$ above, called the Jacobian, is the standard transformation that one applies when doing a change of variables in the multivariate setting (e.g., changing from cartesian to polar/spherical, etc.). Thus, this change of variables formula is nothing more than the usual change of variables applied to the setting of joint probability densities.

Remark 9.41. We will see an example of multivariate change of variables in the next section on the Borel-Kolmogorov paradox.

9.5. The Borel-Kolmogorov Paradox

9.5.1. Statement of the Paradox. We now arrive at the final section in this chapter, which concerns the Borel-Kolmogorov paradox. In Remark 9.27, I formulated the warning that, despite the fact that continuous conditional probabilities and expectations satisfy many of the same computational/intuitive properties as their discrete counterparts (e.g., the replacement identity of Proposition 9.29), one should *not* interpret (9.6) and (9.7) as "the expected value of $g(X)$ if we observe that $Y = y$" or "the probability that $X \in A$ if we observe that $Y = y$." We now explore why that is the case.

Consider two continuous uniform random variables $X, Y \sim \text{Unif}[0,1]$ that are independent of one another. In particular, the joint density of X and Y is the product of the marginal densities, which gives

$$f_{X,Y}(x,y) = \begin{cases} 1 & \text{if } 0 \leq x, y \leq 1 \\ 0 & \text{otherwise} \end{cases}.$$

One of the simplest incarnations of the famed Borel-Kolmogorov paradox consists of the following problem:

Problem 9.42. What is $\mathbf{P}[X \leq 1/2 | X = Y]$?

In words, that is the probability that $X \leq 1/2$ if we "observe" that $X = Y$; see Figure 9.15 for an illustration.

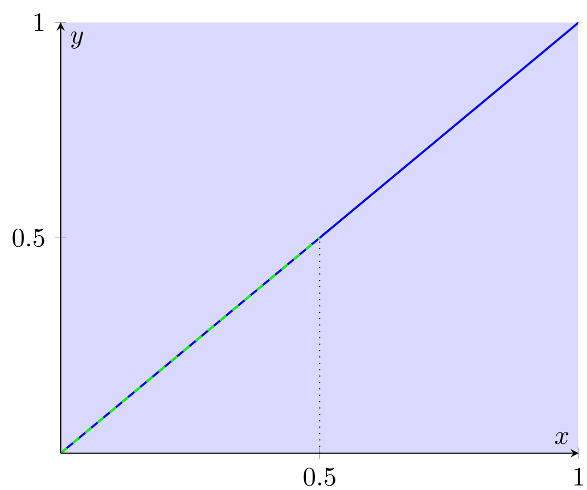

Figure 9.15. The light blue square is a range for (X, Y). The "event" that $X = Y$ is illustrated as the diagonal line in dark blue. The "conditional probability" that we are looking for in Problem 9.42 is illustrated in the green dashed segment. That is, the conditional probability that $X \leq 1/2$ having "observed" that $X = Y$.

At first glance, it may seem that Problem 9.42 has no answer: If we try to use the definition of conditional probability, then we have that

$$\mathbf{P}[X \leq 1/2|X = Y] = \frac{\mathbf{P}[\{X \leq 1/2\} \cap \{X = Y\}]}{\mathbf{P}[X = Y]}.$$

But then, we are confronted to the fact that

$$\mathbf{P}[X = Y] = \int_{x=y} f_{X,Y}(x, y) \, \mathrm{d}x\mathrm{d}y = 0,$$

where the last equality follows from the fact that the integral of any two-dimensional function over a single line is zero. Thus, the conditional probability in Problem 9.42 does not make sense from the point of view of how we defined conditional probabilities earlier in the course (i.e., (4.1)).

With this said, the fact that Problem 9.42 is considered to lead to a paradox is that it is tempting to give a meaning to $\mathbf{P}[X \leq 1/2|X = Y]$ using the notion of conditional density. Indeed, given that $\{X = Y\}$, $\{X - Y = 0\}$, and $\{X/Y = 1\}$ are all seemingly equivalent, one might be tempted to say that

(1) $\mathbf{P}[X \leq 1/2|X = Y] = \mathbf{P}[X \leq 1/2|X - Y = 0] = \int_0^{1/2} f_{X|X-Y=0}(x) \, \mathrm{d}x$; or

(2) $\mathbf{P}[X \leq 1/2|X = Y] = \mathbf{P}[X \leq 1/2|X/Y = 1] = \int_0^{1/2} f_{X|X/Y=1}(x) \, \mathrm{d}x$.

Then, the source of the paradox—as we will prove in a moment—is that

$$\int_0^{1/2} f_{X|X-Y=0}(x) \, \mathrm{d}x = \frac{1}{2} \neq \frac{1}{4} = \int_0^{1/2} f_{X|X/Y=1}(x) \, \mathrm{d}x. \tag{9.10}$$

Thus, two seemingly equivalent ways to parametrize the event $\{X = Y\}$ lead to a completely different answer for $\mathbf{P}[X \leq 1/2|X = Y]$.

Before we explain the source of the paradox, we take a moment to make sure that it does not contradict the definition of continuous conditional probability in Definition 9.26. Therein, we have said that, by definition, continuous conditioning must satisfy the continuous version of the law of total probability; that is,

$$\mathbf{P}[X \leq 1/2] = \int_{\mathbb{R}} \mathbf{P}[X \leq 1/2|X - Y = v] f_{X-Y}(v) \, \mathrm{d}v \tag{9.11}$$

and

$$\mathbf{P}[X \leq 1/2] = \int_{\mathbb{R}} \mathbf{P}[X \leq 1/2|X/Y = v] f_{X/Y}(v) \, \mathrm{d}v. \tag{9.12}$$

Later in this section, we will prove that these probabilities are both 1/2. Thus, despite the fact that

$$\mathbf{P}[X \leq 1/2|X - Y = 0] \neq \mathbf{P}[X \leq 1/2|X/Y = 1],$$

as per (9.10), when we integrate the conditional probabilities given $X - Y$ and X/Y on the entirety of the range of these two random variables, we nevertheless get $\mathbf{P}[X \leq 1/2]$ as a result.

With this out of the way, the source of the paradox is the idea that the equalities

$$\mathbf{P}[X \leq 1/2|X = Y] = \mathbf{P}[X \leq 1/2|X - Y = 0]$$

and

$$\mathbf{P}[X \leq 1/2 | X = Y] = \mathbf{P}[X \leq 1/2 | X/Y = 1]$$

are legitimate. A small number of philosophers and mathematicians are very uncomfortable with this fact, and believe that it indicates that there is something fundamentally wrong with the way that conditioning is done with continuous random variables. The more mainstream view among mathematicians specializing in probability, which I happen to share, is that (9.10) is not at all paradoxical if we adopt the correct perspective: Following-up on our discussion in Section 8.4, the solution to the apparent paradox is simply to abandon the notion that

$$\mathbf{P}[X \leq 1/2 | X = Y]$$

has any meaning. Indeed, since $\mathbf{P}[X = Y] = 0$, we can never actually "observe" that the outcome of two uniform random variables on $[0, 1]$ coincide, because this would require specifying the outcomes of X and Y to infinite precision. Put another way, given that continuous random variables are purely abstract idealizations, the Borel-Kolmogorov paradox will never actually manifest in real-life applications.

In conclusion, while the notions of continuous conditioning introduced in (9.6) and (9.7) share many of the same properties as discrete conditionals, and thus are useful as a computational tool (as illustrated in Examples 9.30 and 9.32), one should resist the temptation to interpret them too literally. In particular, expressions such as $\mathbf{P}[X \in A | Y = y]$ are meaningless in isolation (i.e., for a single value of y), and only make sense in the context of the continuous law of total probability

$$\mathbf{P}[X \in A] = \int_{\mathbb{R}} \mathbf{P}[X \in A | Y = y] f_Y(y) \, \mathrm{d}y,$$

wherein we combine the conditional probabilities over *all* possible values y in an integral with the marginal density of Y.

9.5.2. Proof of the Paradox. We split the proof of the paradox into two steps.

9.5.2.1. *Step 1.* $X - Y$. We begin by showing that

$$f_{X|X-Y=v}(u) = \begin{cases} \frac{1}{1-|v|} & 0 \leq u \leq 1 \text{ and } u - 1 \leq v \leq u \\ 0 & \text{otherwise} \end{cases}, \tag{9.13}$$

and thus, in particular,

$$\int_0^{1/2} f_{X|X-Y=0}(u) \, \mathrm{d}u = \int_0^{1/2} \frac{1}{1-0} \, \mathrm{d}u = \frac{1}{2}. \tag{9.14}$$

Recall that we can compute

$$f_{X|X-Y=v}(u) = \begin{cases} \dfrac{f_{X,X-Y}(u,v)}{f_{X-Y}(v)} & \text{if } f_{X-Y}(v) > 0 \\ 0 & \text{if } f_{X-Y}(v) = 0 \end{cases}.$$

Thus, we need only compute the joint density of X and $X - Y$, and the marginal density of $X - Y$. However, what we know instead is the joint density of X and Y. Nevertheless, we can get the joint density that we are looking for by using a change of variables.

Consider the two-dimensional function

$$g(x,y) = \big(g_1(x,y), g_2(x,y)\big) = (x, x-y).$$

If we let $R_{X,Y} = [0,1] \times [0,1]$ be the unit square, then g is invertible on $R_{X,Y}$ and has inverse function

$$g^{-1}(u,v) = \big(g_1^{-1}(u,v), g^{-1}(u,v)\big) = (u, -v+u).$$

(Indeed, if we want to recover (x,y) from $(u,v) = (x, x-y)$, then we have that $x = u$, and $y = -(x-y) + x = -v + u$.) The Jacobian of this function is

$$\begin{aligned}
|J(u,v)| &= \left| \frac{\partial g_1^{-1}(u,v)}{\partial u} \cdot \frac{\partial g_2^{-1}(u,v)}{\partial v} - \frac{\partial g_1^{-1}(u,v)}{\partial v} \cdot \frac{\partial g_2^{-1}(u,v)}{\partial u} \right| \\
&= \left| \frac{\partial(u)}{\partial u} \cdot \frac{\partial(-v+u)}{\partial v} - \frac{\partial(u)}{\partial v} \cdot \frac{\partial(-v+u)}{\partial u} \right| \\
&= |1 \cdot (-1) - 0 \cdot 1| = 1.
\end{aligned}$$

Therefore, by the two-dimensional change of variables formula,

$$R_{X,X-Y} = \{(u,v) \in \mathbb{R}^2 : 0 \le u \le 1 \text{ and } u-1 \le v \le u\}$$

(see Figure 9.16 below for an illustration of this range), and

$$f_{X,X-Y}(u,v) = f_{X,Y}\big(g_1^{-1}(u,v), g_2^{-1}(u,v)\big)\,|J(u,v)| = 1$$

for every $(u,v) \in R_{X,X-Y}$; otherwise $f_{X,X-Y}(u,v) = 0$.

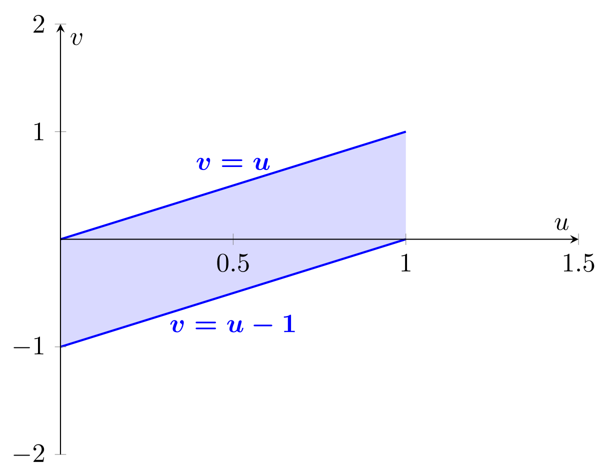

Figure 9.16. The range of $(X, X-Y)$ in $\mathbb{R}^2$.

With this in hand, we can now compute the marginal density of $X-Y$ as well, using Figure 9.16 as a guide for the bounds of integration:

$$
\begin{aligned}
f_{X-Y}(v) &= \int_{-\infty}^{\infty} f_{X,X-Y}(u,v)\,\mathrm{d}u \\
&= \begin{cases} 0 & v<-1 \\ \int_0^{v+1} 1\,\mathrm{d}u = 1+v & -1\le v\le 0 \\ \int_v^1 1\,\mathrm{d}u = 1-v & 0\le v\le 1 \\ 0 & v>1 \end{cases} = \begin{cases} 0 & v<-1 \\ 1-|v| & -1\le v\le 1 \\ 0 & v>1 \end{cases}.
\end{aligned}
$$

With all of these results in hand, we now finally arrive at a formula for the conditional density: For every $0\le u\le 1$ and $u-1\le v\le u$, one has

$$f_{X|X-Y=v}(u) = \frac{f_{X,X-Y}(u,v)}{f_{X-Y}(v)} = \frac{1}{1-|v|}.$$

This then confirms (9.13), from which we obtain (9.14).

9.5.2.2. *Step 2. X/Y.* We now show that

$$
f_{X|X/Y=v}(u) = \begin{cases} \frac{2u}{v^2} & 0<u\le v\le 1 \\ 2u & 0<u\le 1\le v<\infty \\ 0 & \text{otherwise} \end{cases}, \tag{9.15}
$$

and thus, in particular,

$$\int_0^{1/2} f_{X|X/Y=1}(u)\,\mathrm{d}u = \int_0^{1/2} 2u\,\mathrm{d}u = \frac{1}{4}. \tag{9.16}$$

We use the same strategy as in the previous step; namely:

$$
f_{X|X/Y=v}(u) = \begin{cases} \dfrac{f_{X,X/Y}(u,v)}{f_{X/Y}(v)} & \text{if } f_{X/Y}(v)>0 \\ 0 & \text{if } f_{X/Y}(v)=0 \end{cases}.
$$

Consider the two-dimensional function

$$g(x,y) = \big(g_1(x,y), g_2(x,y)\big) = (x, x/y).$$

If we let $R_{X,Y} = (0,1]\times(0,1]$ be the unit square that excludes the cases $x,y=0$, then g is invertible on $R_{X,Y}$ (we have to exclude $y=0$, because otherwise x/y is undefined), and has inverse function

$$g^{-1}(u,v) = \big(g_1^{-1}(u,v), g_2^{-1}(u,v)\big) = (u, u/v).$$

(Indeed, if we want to recover (x,y) from $(u,v)=(x,x/y)$, then we have that $x=u$, and $y = x(y/x) = u/v$; note that for this to make sense, we also have to exclude $x=0$, which we have done in our choice of range $R_{X,Y}$). The Jacobian of this function is

$$\begin{aligned}
|J(u,v)| &= \left|\frac{\partial g_1^{-1}(u,v)}{\partial u}\cdot\frac{\partial g_2^{-1}(u,v)}{\partial v}-\frac{\partial g_1^{-1}(u,v)}{\partial v}\cdot\frac{\partial g_2^{-1}(u,v)}{\partial u}\right|\\
&= \left|\frac{\partial(u)}{\partial u}\cdot\frac{\partial(u/v)}{\partial v}-\frac{\partial(u)}{\partial v}\cdot\frac{\partial(u/v)}{\partial u}\right|\\
&= |1\cdot(-u/v^2)-0\cdot(1/v)| = \frac{u}{v^2}.
\end{aligned}$$

Therefore, by the two-dimensional change of variables formula,

$$R_{X,X/Y} = \{(u,v)\in\mathbb{R}^2 : 0<u\leq 1 \text{ and } u\leq v<\infty\}$$

(see Figure 9.17 below for an illustration of this range), and

$$f_{X,X/Y}(u,v) = f_{X,Y}\big(g_1^{-1}(u,v), g_2^{-1}(u,v)\big)\,|J(u,v)| = \frac{u}{v^2}$$

for every $(u,v)\in R_{X,X/Y}$; otherwise $f_{X,X/Y}(u,v)=0$.

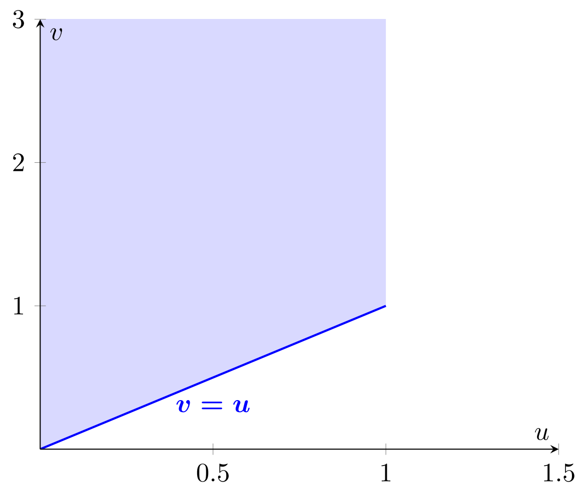

Figure 9.17. The range of $(X, X/Y)$ in $\mathbb{R}^2$.

We can now compute the marginal density of X/Y using Figure 9.17 as a guide for the bounds of integration:

$$f_{X/Y}(v) = \int_{-\infty}^{\infty} f_{X,X/Y}(u,v)\,\mathrm{d}u = \begin{cases} 0 & v\leq 0\\ \int_0^v \frac{u}{v^2}\,\mathrm{d}u = \frac{1}{2} & 0<v\leq 1\\ \int_0^1 \frac{u}{v^2}\,\mathrm{d}u = \frac{1}{2v^2} & 1\leq v<\infty\end{cases}.$$

Therefore, for every $0\leq u\leq 1$ and $u\leq v<\infty$, one has

$$f_{X|X/Y=v}(u) = \frac{f_{X,X/Y}(u,v)}{f_{X/Y}(v)} = \begin{cases}\frac{2u}{v^2} & 0\leq u\leq v\leq 1\\ 2u & 0\leq u\leq 1\leq v<\infty\end{cases};$$

hence (9.15) and (9.16) hold.

9.5.2.3. *Step 3. Law of Total Probability.* We have now proved the essence of the paradox, that is, equation (9.10), which was the claim that

$$\mathbf{P}[X \le 1/2|X-Y=0] = \frac{1}{2} \neq \frac{1}{4} = \mathbf{P}[X \le 1/2|X/Y=1].$$

We now conclude this section (and chapter) by showing that the paradox does not contradict the definition of continuous conditional probability in terms of the law of total probability. On the one hand, the calculations in Section 9.5.2.1 yield

$$\begin{aligned}
\mathbf{P}[X \le 1/2] &= \int_{\mathbb{R}} \mathbf{P}[X \le 1/2|X-Y=v] f_{X-Y}(v)\,\mathrm{d}v \\
&= \int_{\mathbb{R}} \left(\int_0^{1/2} f_{X|X-Y=v}(u)\,\mathrm{d}u \right) f_{X-Y}(v)\,\mathrm{d}v \\
&= \int_0^{1/2} \left(\int_{\mathbb{R}} f_{X|X-Y=v}(u) f_{X-Y}(v)\,\mathrm{d}v \right) \mathrm{d}u \\
&= \int_0^{1/2} \left(\int_{u-1}^{u} \frac{1-|v|}{1-|v|}\,\mathrm{d}v \right)\,\mathrm{d}u \\
&= \int_0^{1/2} \,\mathrm{d}u = \frac{1}{2}.
\end{aligned}$$

On the other hand, the computations in Section 9.5.2.2 yield

$$\begin{aligned}
\mathbf{P}[X \le 1/2] &= \int_{\mathbb{R}} \mathbf{P}[X \le 1/2|X/Y=v] f_{X/Y}(v)\,\mathrm{d}v \\
&= \int_{\mathbb{R}} \left(\int_0^{1/2} f_{X|X/Y=v}(u)\,\mathrm{d}u \right) f_{X/Y}(v)\,\mathrm{d}v \\
&= \int_0^{1/2} \left(\int_{\mathbb{R}} f_{X|X/Y=v}(u) f_{X/Y}(v)\,\mathrm{d}v \right) \mathrm{d}u \\
&= \int_0^{1/2} \left(\int_{u}^{\infty} \frac{u}{v^2}\,\mathrm{d}v \right)\,\mathrm{d}u \\
&= \int_0^{1/2} \,\mathrm{d}u = \frac{1}{2}.
\end{aligned}$$

Thus, there is no contradiction here with the theory of continuous conditioning as we have introduced it in Definition 9.26.

Chapter 10

The Central Limit Theorem

In this final chapter of the course, we discuss the central limit theorem. Along with the law of large numbers, the central limit theorem is often considered to be one of the two most fundamental results in the theory of probability.

The plan for this chapter is as follows: First, we introduce the main object of study in the central limit theorem, which is the Gaussian distribution. Then, we discuss the content of the central limit theorem, that is, the universality of the fluctuations of empirical averages. Next, we discuss how the central limit theorem is applied in practice, and finally explain some of the elements of its proof.

10.1. The Gaussian Distribution and Universality

10.1.1. The Gaussian Distributions. The main star of the central limit theorem is the Gaussian distribution:

Definition 10.1 (Gaussian Distribution)**.** Let $\mu \in \mathbb{R}$ and $\sigma > 0$. We say that a continuous random variable X has the Gaussian distribution with mean μ and variance σ^2, which we denote by $X \sim N(\mu, \sigma^2)$, if it has the following density function:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}, \tag{10.1}$$

where we recall the notation $\exp(z) = \mathrm{e}^z$.

Remark 10.2. Since the Gaussian density function is positive for every x, it is natural to choose $R_X = \mathbb{R}$.

The Gaussian distribution is also sometimes called the normal distribution, and is widely known by its colloquial name "the bell curve." The latter comes from the fact that the Gaussian density function has a distinctive bell shape, as illustrated in Figure 10.1 below.

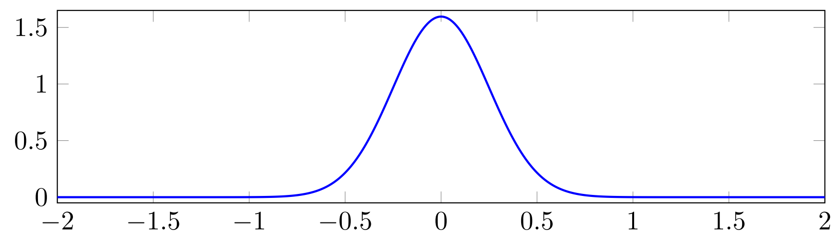

Figure 10.1. The bell-curve shape of the Gaussian density function.

In this context, the parameters μ and σ^2 control the shape of the bell curve. For example, in Figure 10.2, we illustrate three examples of bell curves with the same variance σ^2 and different means. Therein, it can be observed that the shape

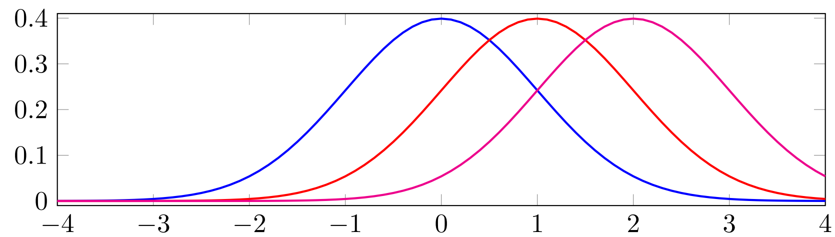

Figure 10.2. The density functions of $N(0,1)$ (blue), $N(1,1)$ (red) and $N(2,1)$ (megenta). We observe the same bell curve shape, but shifted more and more to the right. The center point of the bell curve (i.e., the highest point on the curve) is aligned with the mean μ; the Gaussian density function is symmetric about that point.

of the bell curve remains the same when we change μ; the only difference is that the bell curve will be shifted to the right or left.

Next, in Figure 10.3, we illustrate the effect of changing the variance parameter σ^2. With these examples, we see that changing this parameter does have an impact

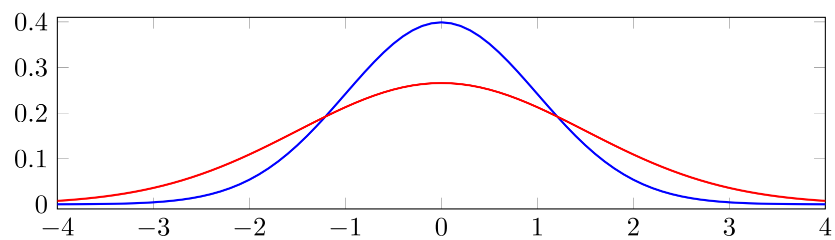

Figure 10.3. The density functions of $N(0,1)$ (blue) and $N(0,3/2)$ (red). We observe that both densities are bell curves centered at zero, but the red curve is flatter and wider. This is consistent with the fact that the red curve has a higher variance of 3/2.

on the shape of the bell curve.

Among all bell curves, we distinguish one as the standard Gaussian:

Definition 10.3. We say that X has the standard Gaussian distribution if it is Gaussian with mean zero and variance one, that is, $X \sim N(0,1)$. In particular, X has the density function

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), \qquad x \in \mathbb{R}.$$

10.1.2. Gaussian Universality. Upon being presented with the above definition, it is natural to ask: Why do we care so much about the Gaussian distribution? Why is the bell curve so widely known? Unlike distributions like the Poisson or exponential, the Gaussian distribution does not arise from a single model. Instead, the importance of the Gaussian distribution comes from a phenomenon in science that is known as universality.

In short, the universality phenomenon refers to the observation that the macroscopic behavior of many complex systems is more or less independent of the microscopic details of the system. In particular, this means that many complex systems that, at first glance, may appear to be very different from one another, exhibit exactly the same behavior at large scales. The relevance of the Gaussian distributions in that context is that they describe the macroscopic behavior of a very impressive amount of very different complex systems.

Here are two illustrative examples of this Gaussian universality phenomenon. On the one hand, Figure 10.4 contains a histogram of the mathematical reasoning scores of all students who took the SAT in 2002, as reported by The College Board . Though the bins are not exactly symmetric, we nevertheless recognize the

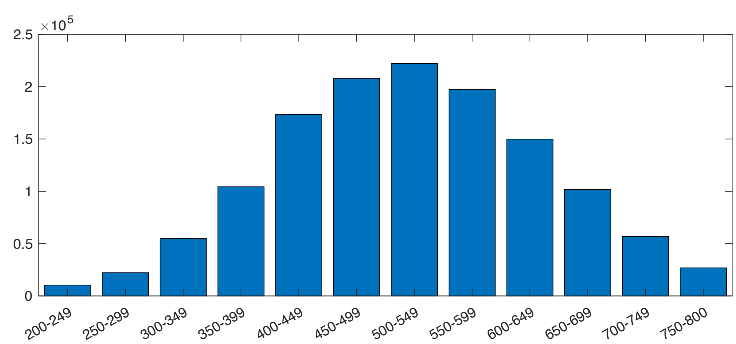

Figure 10.4. Histogram plot of the mathematical reasoning scores of the 1 327 831 students who took the SAT in 2002.

distinctive bell-curve shape in the distribution of scores.

On the other hand, Figure 10.5 features a histogram plot of the heights of a number of men who were arrested in the city of London in 1920, as per a database maintained by the University of Lyon . Despite the fact that the fluctuations of SAT scores and heights of humans come from (presumably) very different mechanisms, a similar bell-curve shape occurs.

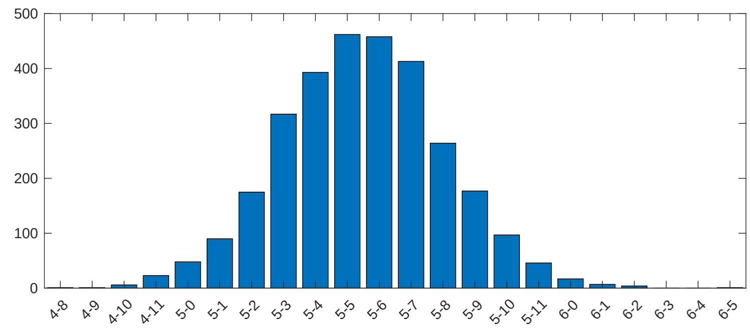

Figure 10.5. Histogram plot of the heights (in feet-inches) of a sample of 3 000 men who were arrested in London in 1902.

Despite the impressive generality of bell curves, it is important not to become unthinkingly enthusiastic about the Gaussian distribution, and to resist the temptation to assume that every complex system can be modelled accurately using a bell curve. For one thing, we have already seen two examples of continuous random variables that look nothing like the Gaussians, namely, the exponential and uniform distributions (e.g., compare Figures 9.5 and 9.6 with the plots of the Gaussian densities in this section). Moreover, apart from these two examples, there are many complex mechanisms that are well-known to generate distributions that are quite far from being Gaussian. One example of this is the value of insurance claims. Using the data provided by the insuranceData package for the R statistical software environment, we obtain Figure 10.6. Therein, a histogram plot of the value of

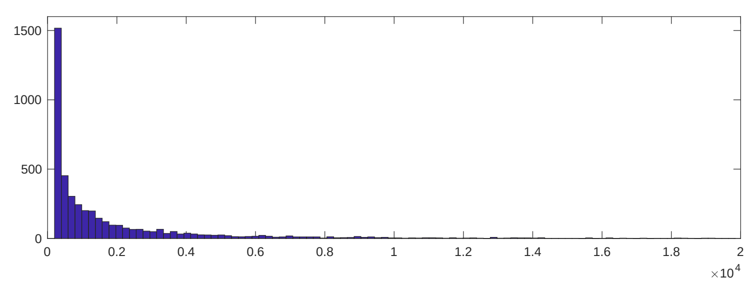

Figure 10.6. Histogram of the values (in USD) of claims submitted to a number of car insurance companies.

4 429 claims submitted to a number of car insurance companies in 2002 is compiled. A quick glance at the resulting plot quickly convinces us that this process is far from Gaussian, as the histogram does not at all resemble any kind of bell curve. In particular, any predictive analysis of future insurance claims based on the faulty assumption that the distribution is Gaussian would very likely lead to an utterly disastrous (and costly) result.

Consequently, an important problem in the theory of probability is to understand when the Gaussian distribution is or is not likely to be a good model for a given complex system. The main result of this chapter, the central limit theorem, studies one specific mechanism that gives rise to Gaussian distributions. We provide a statement of that result in the next section. Before we get to that, however, we finish this section with a few computations concerning the Gaussian density.

10.1.3. A Few Sanity Checks. The terminology used in Definition 10.1 tacitly assumes that the function f_X in (10.1) is in fact a density function, and that if $X \sim N(\mu, \sigma^2)$, then $\mathbf{E}[X] = \mu$ and $\mathbf{Var}[X] = \sigma^2$. However, these claims are not immediately obvious just by a single glance at the formula in (10.1). In fact, these computations are harder than they might appear at first glance. Thus, as a final result in this section, we go over a proof of these facts:

Proposition 10.4. *For every $\mu \in \mathbb{R}$ and $\sigma > 0$, one has*

$$\int_{\mathbb{R}} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \, \mathrm{d}x = 1. \tag{10.2}$$

Moreover, if $X \sim N(\mu, \sigma^2)$, then we in fact have that $\mathbf{E}[X] = \mu$ and $\mathbf{Var}[X] = \sigma^2$.

Proof. Let us begin with the claim in (10.2). First, we apply the change of variables

$$y = \frac{x-\mu}{\sqrt{2\sigma^2}} \qquad \mathrm{d}y = \frac{\mathrm{d}x}{\sqrt{2\sigma^2}}.$$

This yields

$$\int_{\mathbb{R}} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \, \mathrm{d}x = \pi^{-1/2} \int_{\mathbb{R}} \mathrm{e}^{-y^2} \, \mathrm{d}y. \tag{10.3}$$

The trouble with this integral is that the function e^{-y^2} does not admit a simple antiderivative. Thus, the integral cannot be solved directly. In order to get around this, mathematicians of the past have found a very elegant solution involving a clever change of variables: Suppose that, instead of the integral of e^{-y^2} itself, we look at the integral of its square

$$\left(\int_{\mathbb{R}} \mathrm{e}^{-y^2} \, \mathrm{d}y\right)^2 = \int_{\mathbb{R}} \mathrm{e}^{-y^2} \, \mathrm{d}y \cdot \int_{\mathbb{R}} \mathrm{e}^{-y^2} \, \mathrm{d}y.$$

Rewrite this in a slightly different way, by calling the variable in the second integral something other than y, let's say z:

$$\left(\int_{\mathbb{R}} \mathrm{e}^{-y^2} \, \mathrm{d}y\right)^2 = \int_{\mathbb{R}} \mathrm{e}^{-y^2} \, \mathrm{d}y \cdot \int_{\mathbb{R}} \mathrm{e}^{-z^2} \, \mathrm{d}z.$$

Then, write the product of these two integrals as a double integral over two-dimensional space:

$$\left(\int_{\mathbb{R}} \mathrm{e}^{-y^2} \, \mathrm{d}y\right)^2 = \iint_{\mathbb{R}^2} \mathrm{e}^{-(y^2+z^2)} \, \mathrm{d}y\mathrm{d}z.$$

At this point, all of this may seem to you like some pointless algebraic manipulations. However, here's the really clever bit: In cartesian coordinates $(y, z) \in \mathbb{R}^2$, the quantity $\sqrt{y^2+z^2}$ represents the distance between the origin and (y,z) (i.e., the radius). Thus, if we apply the polar change of variables

$$r=\sqrt{y^2+z^2},\ \theta=\operatorname{atan2}(y,z),\qquad \mathrm{d}y\mathrm{d}z=r\ \mathrm{d}r\mathrm{d}\theta,$$

then we get that

$$\left(\int_{\mathbb{R}}\mathrm{e}^{-y^2}\ \mathrm{d}y\right)^2=\int_{-\pi}^{\pi}\int_0^\infty r\mathrm{e}^{-r^2}\ \mathrm{d}r\mathrm{d}\theta.$$

First, since the integrand does not depend on θ, we can compute the integral with respect to $\mathrm{d}\theta$ trivially, which yields

$$\int_{-\pi}^{\pi}\int_0^\infty r\mathrm{e}^{-r^2}\ \mathrm{d}r\mathrm{d}\theta=\int_{-\pi}^{\pi}\ \mathrm{d}\theta\cdot\int_0^\infty r\mathrm{e}^{-r^2}\ \mathrm{d}r=2\pi\int_0^\infty r\mathrm{e}^{-r^2}\ \mathrm{d}r.$$

Now, the crucial difference between this integral and what we had at the beginning, namely $\int\mathrm{e}^{-y^2}\ \mathrm{d}y$, is that there is now an extra r factor in front of the exponential. This makes all the difference, and was the whole point of this complicated procedure: If we apply the change of variables

$$u=r^2\qquad \mathrm{d}u=2r\mathrm{d}r,$$

then we have that

$$\left(\int_{\mathbb{R}}\mathrm{e}^{-y^2}\ \mathrm{d}y\right)^2=2\pi\int_0^\infty r\mathrm{e}^{-r^2}\ \mathrm{d}r=\pi\int_0^\infty \mathrm{e}^{-u}\ \mathrm{d}u=\pi.$$

If we then finally look way back at what was our initial goal, namely, equation (10.3), then we obtain that

$$\int_{\mathbb{R}}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\ \mathrm{d}x=\pi^{-1/2}\sqrt{\left(\int_{\mathbb{R}}\mathrm{e}^{-y^2}\ \mathrm{d}y\right)^2}=\pi^{-1/2}\sqrt{\pi}=1,$$

finally concluding the proof of (10.2).

Next, we discuss the expectation. By definition,

$$\mathbf{E}[X]=\int_{\mathbb{R}}\frac{x}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\ \mathrm{d}x.$$

If we introduce the change of variables

$$y=\frac{x-\mu}{\sqrt{2\sigma^2}}\qquad \mathrm{d}y=\frac{\mathrm{d}x}{\sqrt{2\sigma^2}},$$

then this becomes

$$\begin{aligned}\mathbf{E}[X]&=\pi^{-1/2}\int_{\mathbb{R}}(\sqrt{2\sigma^2}y+\mu)\mathrm{e}^{-y^2}\ \mathrm{d}y\\&=\sqrt{\frac{2\sigma^2}{\pi}}\int_{\mathbb{R}}y\mathrm{e}^{-y^2}\ \mathrm{d}y+\mu\cdot\pi^{-1/2}\int_{\mathbb{R}}\mathrm{e}^{-y^2}\ \mathrm{d}y.\end{aligned}$$

On the one hand, since the function $y\mathrm{e}^{-y^2}$ is odd (i.e., $f(-y)=-f(y)$; see Figure 10.7), its integral over $[0,\infty)$ is cancelled out by its integral over $(-\infty,0]$. Thus, we simply have that

$$\sqrt{\frac{2\sigma^2}{\pi}}\int_{\mathbb{R}}y\mathrm{e}^{-y^2}\ \mathrm{d}y=0.$$

On the other hand, we have already computed in the previous step of this proof that

$$\pi^{-1/2} \int_{\mathbb{R}} \mathrm{e}^{-y^2} \, \mathrm{d}y = 1.$$

We therefore conclude that

$$\mathbf{E}[X] = 0 + \mu \cdot 1 = \mu.$$

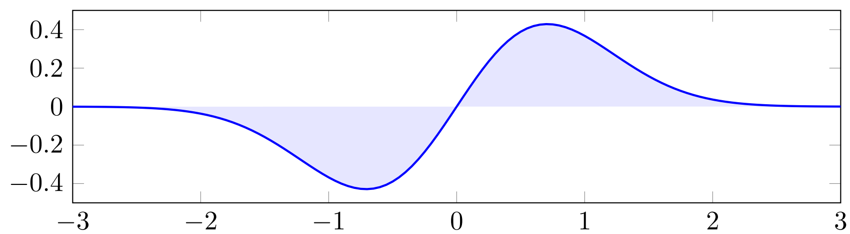

Figure 10.7. Plot of the function $y\mathrm{e}^{-y^2}$. The area under the curve over $[0, \infty)$ is cancelled out by the negative are under the curve over $(-\infty, 0]$.

Finally, we compute the variance. This is one of those rare cases where it is actually easier to compute $\mathbf{E}[(X-\mu)^2]$ directly:

$$\mathbf{Var}[X] = \int_{\mathbb{R}} \frac{(x-\mu)^2}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \, \mathrm{d}x.$$

If we apply once again the same change of variables we have used throughout this proof, this becomes

$$\mathbf{Var}[X] = \frac{2\sigma^2}{\sqrt{\pi}} \int_{\mathbb{R}} y^2 \mathrm{e}^{-y^2} \, \mathrm{d}y. \tag{10.4}$$

Next, if we apply integration by parts with the choices

$$u = y \qquad \qquad \mathrm{d}u = \mathrm{d}y$$

$$\mathrm{d}v = y\mathrm{e}^{-y^2} \, \mathrm{d}y \qquad \qquad v = \frac{-\mathrm{e}^{-y^2}}{2},$$

then we get

$$\int_{\mathbb{R}} y^2 \mathrm{e}^{-y^2} \, \mathrm{d}y = [uv]_{-\infty}^{\infty} - \int_{\mathbb{R}} v \, \mathrm{d}u = 0 + \frac{1}{2} \int_{\mathbb{R}} \mathrm{e}^{-y^2} \, \mathrm{d}y.$$

If we then refer back once again to the computation that we performed for (10.3), we are then led to

$$\int_{\mathbb{R}} y^2 \mathrm{e}^{-y^2} \, \mathrm{d}y = [uv]_{-\infty}^{\infty} - \int_{\mathbb{R}} v \, \mathrm{d}u = 0 + \frac{1}{2}\sqrt{\pi}.$$

Finally, if we plug this back into (10.4), then we obtain that $\mathbf{Var}[X] = \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \sigma^2$, concluding the proof. $\square$

10.2. The Central Limit Theorem

As alluded to in the previous section, the usefulness of the Gaussian distribution is that it describes the behavior of a large class of seemingly very different random processes. The central limit theorem provides part of an explanation for this phenomenon. Namely, we can expect a bell curve when we look at a large sum of independent and identically distributed random variables:

Theorem 10.5 (Central Limit Theorem)**.** *Let $X_1, X_2, X_3, \ldots$ be an infinite sequence of i.i.d. random variables with $\mathbf{E}[X_i] = \mu \in \mathbb{R}$ and $\mathbf{Var}[X_i] = \sigma^2 \in (0, \infty)$. For every positive integer n, let*

$$S_n = X_1 + X_2 + \cdots + X_n.$$

Then, as $n \to \infty$, we have the convergence

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \to N(0,1)$$

in distribution. More specifically, for any interval $I \subset \mathbb{R}$, we have that

$$\lim_{n\to\infty} \mathbf{P}\left[\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \in I\right] = \int_I \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \, \mathrm{d}x.$$

Remark 10.6. In the above statement, the interval I can be very general. On the one hand, it could be any finite interval of the form

$$[a,b], \qquad (a,b], \qquad [a,b), \quad \text{or} \quad (a,b)$$

for some $-\infty < a < b < \infty$. On the other hand, it could also be any half-infinite interval of the form

$$(-\infty, a], \qquad (-\infty, a), \qquad [a, \infty), \quad \text{or} \quad (a, \infty).$$

Remark 10.7. Note that, by linearity of expectation,

$$\mathbf{E}[S_n] = \mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots + \mathbf{E}[X_n] = n\mu.$$

Moreover, given that the variance of a sum of independent random variables is the sum of the variances,

$$\mathbf{Var}[S_n] = \mathbf{Var}[X_1] + \mathbf{Var}[X_2] + \cdots + \mathbf{Var}[X_n] = n\sigma^2.$$

Thus, we can reformulate

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} = \frac{S_n - \mathbf{E}[S_n]}{\sqrt{\mathbf{Var}[S_n]}}.$$

Before discussing applications of the central limit theorem and its proof, we should take some time to carefully parse its statement.

10.2.1. First Illustrations of the Central Limit Theorem. If we forget about the technical details of the statement of Theorem 10.5 for a moment, the main content of the result is the following: After some adjustment (i.e., subtraction by $n\mu$ and division by $\sqrt{n\sigma^2}$), the sum of a large number of i.i.d. random variables is approximately Gaussian (hence the limit as $n \to \infty$). Given that we have already studied a number of random variables that can be expressed as a sum of i.i.d. random variables, the CLT can easily be observed with simulations.

The first example that we will look at is the Binomial distribution. Indeed, if $X \sim \mathrm{Bin}(n,p)$, then this means that we can write

$$X = \mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \cdots + \mathbf{1}_{A_n},$$

where the events A_i are independent and have probability p of occurring. In particular, the indicators $\mathbf{1}_{A_i}$ are i.i.d. Thus, we expect by the central limit theorem that X should be approximately Gaussian when n is very large. This effect is illustrated in Figure 10.8 on the next page. More specifically, we plot the distribution of $\mathrm{Bin}(n, 9/10)$ for $n = 10, 20, 50, 100, 500$. In doing so, we observe that the distribution becomes increasingly similar to a Bell curve as n gets larger.

For a second example, consider the situation where $X_1, X_2, X_3, \ldots$ are i.i.d. uniform random variables on the interval $[0,1]$. That is, the density functions of the X_i are given by

$$f_{X_i}(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}.$$

Using the convolution identity stated in Proposition 9.31, we can compute the density function of the sum

$$S_n = X_1 + X_2 + \cdots + X_n$$

recursively for any n as follows:

(1) $f_{X_1+X_2}(x) = \displaystyle\int_{\mathbb{R}} f_{X_1}(x-y) f_{X_2}(y)\, \mathrm{d}y;$

(2) $f_{X_1+X_2+X_3}(x) = \displaystyle\int_{\mathbb{R}} f_{X_1+X_2}(x-y) f_{X_3}(y)\, \mathrm{d}y;$

(3) $f_{X_1+X_2+X_3+X_4}(x) = \displaystyle\int_{\mathbb{R}} f_{X_1+X_2+X_3}(x-y) f_{X_4}(y)\, \mathrm{d}y;$

(4) $\cdots$

(5) $f_{S_n}(x) = \displaystyle\int_{\mathbb{R}} f_{S_{n-1}}(x-y) f_{X_n}(y)\, \mathrm{d}y.$

While we do not attempt to compute this exactly here (as it amounts to extremely tedious but straightforward calculations; you will compute a similar distribution for $n = 3$ in Homework 8), it is nevertheless interesting to note that an exact formula for the density of f_{S_n} can be calculated. This is known as the Irwin-Hall distribution; see the Wikipedia page with that name.

The density functions of S_n for $n = 1, 2, 4, 8, 16$ are illustrated in Figure 10.9 on the next page. Therein, again, we see the appearance of the universal Gaussian shape as we make n larger.

Comparing between Figures 10.8 and 10.9 provides a compelling illustration of the power of the central limit theorem. At the "microscopic" level, the two examples are very different: In the case of the binomial, we are summing indicator random variables; in the case of Figure 10.9, we are summing continuous uniform random variables. One of these examples is discrete and can only take the values 0 and 1, whereas the other is continuous and could output any number in the interval $[0,1]$. Nevertheless, when we sum a large number of these variables, these microscopic differences become irrelevant; both distributions become similar to a bell-curve.

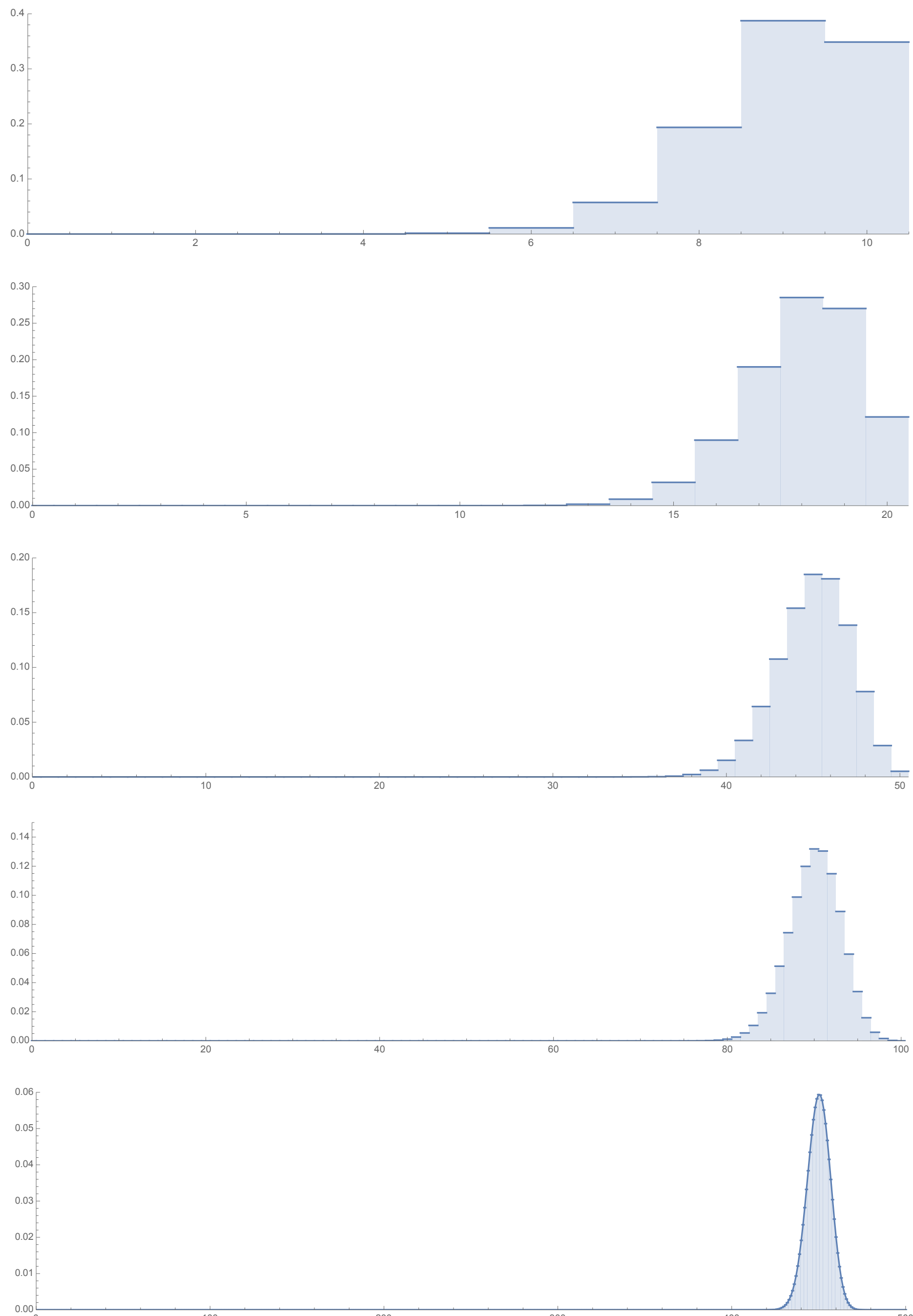

Figure 10.8. Illustration of the distribution of $\mathrm{Bin}(n, 9/10)$ with the choices $n = 10, 20, 50, 100, 500$ in increasing order from top to bottom. Consistently with the law of large numbers, the distribution concentrates more and more sharply around the value $n \cdot (9/10)$. However, the effect of the central limit theorem is that the curve not only concentrates around the expected value, but has a shape that becomes more and more like a bell curve.

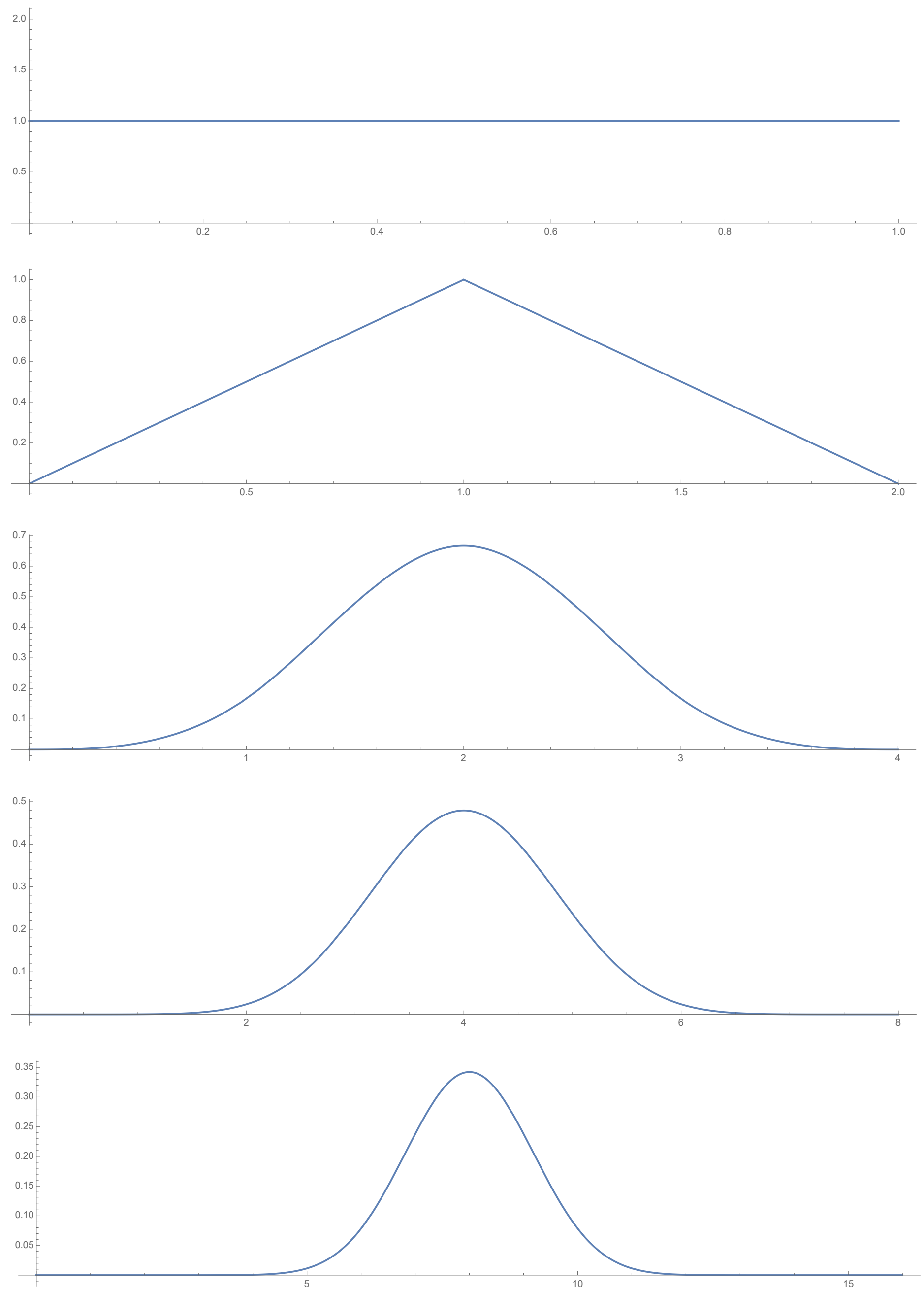

Figure 10.9. Illustration of the density functions of the sum of n i.i.d. uniform random variables on $[0, 1]$ for the choices $n = 1, 2, 4, 8, 16$ in increasing order from top to bottom. Although this distribution is very different from the Binomial, we nevertheless see the universal Gaussian shape appear as we sum more and more uniform random variables.

With this in mind, we can now have a better appreciation for the statement of the central limit theorem in Theorem 10.5. Therein, the only information that we have about the X_i's is their mean and variance. Apart from that, the X_i's could have any distribution with finite mean and variance. The central limit theorem then says that, no matter what this distribution is, the random behavior of

$$S_n = X_1 + X_2 + \cdots + X_n$$

is universal, in the sense that it is always approximately Gaussian.

Remark 10.8. Referring back to the examples of the SAT scores and heights illustrated in Figures 10.4 and 10.5, we now see that the central limit theorem provides *some* explanation for the appearance of bell curves in a variety of very different contexts. Here, I emphasize some, because a large sum of independent random variables is not the only context in which the Gaussian distribution appears.

In fact, if you want to challenge yourself, then you can try to decide if you think that the examples illustrated in Figures 10.8 and 10.9 can be interpreted as sums of i.i.d. random variables. Indeed, if one wants to explain the appearance of a bell curve in these two examples using the central limit theorem, then one better be able to justify that the process of getting a GRE score or a certain height can be expressed as a large sum of i.i.d. random variables. Otherwise, one cannot rule out the possibility that the appearance of the Gaussian in these two examples comes from a completely different mechanism than what is explained by Theorem 10.5.

10.2.2. A More Precise Comparison. Now that we have a basic understanding of what the central limit theorem states, we can begin discussing the details of Theorem 10.5. More specifically, the latter states that, for large n,

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \approx N(0,1).$$

If we isolate S_n in this approximate equality, then we can reformulate it as

$$S_n \approx \sqrt{n\sigma^2} \cdot N(0,1) + n\mu. \tag{10.5}$$

In order to understand the significance of this statement, we have the following technical proposition:

Proposition 10.9. *If $X \sim N(0,1)$, then for every constants $a > 0$ and $b \in \mathbb{R}$, one has $aX + b \sim N(b, a^2)$.*

Proof. This follows from a straightforward change of variables. Let

$$g(x) = ax + b.$$

This function is invertible on $\mathbb{R}$ with inverse

$$g^{-1}(y) = \frac{y-b}{a}.$$

Moreover, the inverse is increasing. Thus, by the one-dimensional change of variables formula, we have that

$$f_{aX+b}(y) = f_{g(X)}(y) = f_X\big(g^{-1}(y)\big) \cdot \frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y) = f_X\left(\frac{y-b}{a}\right) \cdot \frac{1}{a}.$$

Given that X is standard Gaussian, this means that

$$f_{aX+b}(y) = \frac{1}{a}\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{\left(\frac{y-b}{a}\right)^2}{2}\right) = \frac{1}{\sqrt{2\pi a^2}}\exp\left(-\frac{(y-b)^2}{2a^2}\right).$$

This is the density function of $N(b, a^2)$, thus concluding the proof. $\square$

With this result in hand, we may now reformulate (10.5) as

$$S_n \approx N(n\mu, n\sigma^2). \tag{10.6}$$

In particular, we see that Theorem 10.5 tells us not only that S_n is approximately Gaussian for large n, but also which Gaussian curve approximates S_n. This then allows us to formulate a more precise statement regarding the approximate-bell-curves that we saw appear in Figures 10.8 and 10.9.

Consider for instance $\text{Bin}(500, 9/10)$. In this case, we have $n = 500$, and

$$S_{500} = \sum_{i=1}^{500} \mathbf{1}_{A_i},$$

where the A_i are independent events each with probability $9/10$. Knowing that

$$\mathbf{E}[\mathbf{1}_A] = \mathbf{P}[A] \qquad \text{and} \qquad \mathbf{Var}[\mathbf{1}_A] = \mathbf{P}[A](1-\mathbf{P}[A]),$$

this means that in this case we have

$$\mu = \mathbf{E}[\mathbf{1}_{A_i}] = 9/10 \qquad \text{and} \qquad \sigma^2 = \mathbf{Var}[\mathbf{1}_{A_i}] = 9/10(1/10) = 9/100.$$

Therefore, (10.6) suggests that

$$S_{500} \approx N(500 \cdot 9/10, 500 \cdot 9/100) = N(450, 45).$$

If we plot the density function of a Gaussian with mean 450 and variance 45 against the distribution of $\text{Bin}(500, 9/10)$, as done in Figure 10.10 below, we see a very strong agreement between the distribution of S_n and its Gaussian approximation.

10.3. Applications of the Central Limit Theorem

We now discuss applications of the central limit theorem. The main problem that motivates using the central limit theorem in practice is the following: Suppose that we want to calculate the probability that the outcome of

$$S_n = X_1 + X_2 + \cdots + X_n$$

lies in some interval $I \subset \mathbb{R}$, assuming that n is large and that the X_i are i.i.d. In many such cases, even if the distribution of the X_i are very simple, it might be extremely difficult to compute the distribution of S_n exactly. You have seen (or will see) an example of this in Homework 8: Even in the case where the X_i are uniform on $[0, 2]$, already the computation of the distribution of S_3 is very tedious; you can imagine what it must be like for S_{10}, S_{20}, or S_{100}. Then, even if one has a formula for S_n's density or distribution, it is not guaranteed that the computation of the probability

$$\mathbf{P}[S_n \in J]$$

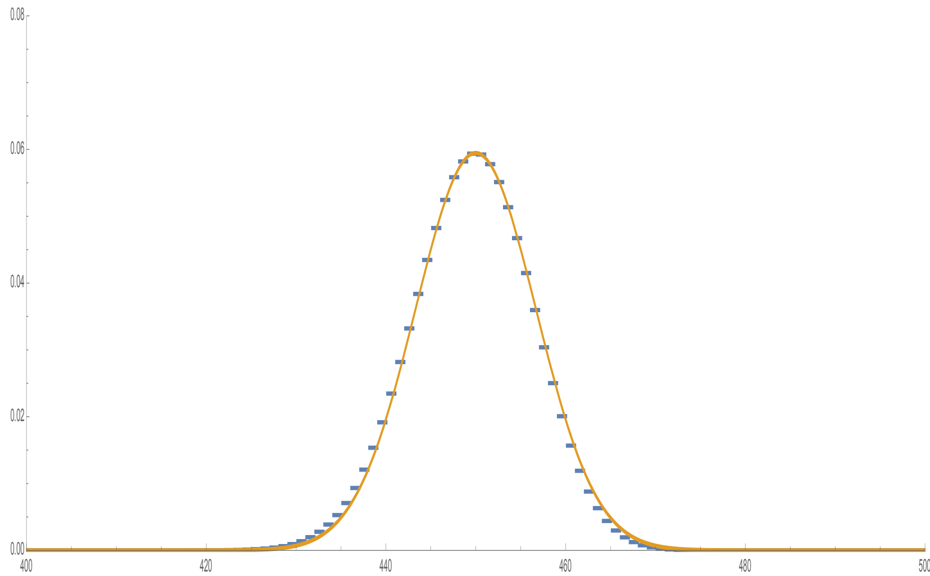

Figure 10.10. In blue are the heights of the bins corresponding to the distribution of Bin$(500, 9/10)$. Although the range of Bin$(500, 9/10)$ contains every integer from 0 to 500, only numbers from 400 to 500 are included to improve clarity of the illustration. Then, the density function of $N(450, 45)$ is plotted in orange. We see a very strong agreement between the two curves.

for an interval J will be practical. For instance, if S_n is a continuous random variable and the density function f_{S_n} is extremely complicated, then the integral

$$\mathbf{P}[S_n \in J] = \int_J f_{S_n}(x)\ \mathrm{d}x$$

might not be easy to compute.

In light of these observations, the usefulness of the central limit theorem can be illustrated using, for example, Figure 10.10. Indeed, despite the fact that the actual distribution of S_n in that picture is Bin$(450, 45)$, we know that S_n's distribution can be approximated rather well with a Gaussian. Therefore, the usefulness of the central limit theorem lies in the fact that integrals of the form

$$\int_I \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\ \mathrm{d}x, \tag{10.7}$$

though they can typically not be computed exactly, are much easier to approximate than the probability of $\mathbf{P}[S_n \in J]$ is to compute exactly. In fact, in large part thanks to the central limit theorem, Gaussian integrals of the form (10.7) are among the most well-studied integrals in terms of numerical approximation (many probability and/or statistics textbooks feature tables containing numerical approximations of (10.7) for a variety of intervals I; see, for instance, the Wikipedia page on the standard normal table).

In the remainder of this section, we provide a general blueprint on how to turn computations involving S_n into an integral involving the standard Gaussian, and then we go over two examples of such a computation.

10.3.1. A General Blueprint. Most applications of the central limit theorem follow the same general pattern. First, we are given some random variable S_n for which we want to compute a probability of the form

$$\mathbf{P}[S_n \in J] \tag{10.8}$$

where $J \subset \mathbb{R}$ is some interval. For the central limit theorem to be useful in approximating this probability, it must be the case that S_n can be written as sum of n i.i.d. random variables. Thus, the first step in the application of the central limit theorem is as follows:

Step 1. Find some i.i.d. random variables $X_1, X_2, \ldots, X_n$ such that

$$S_n = X_1 + X_2 + \cdots + X_n. \tag{10.9}$$

Remark 10.10. In some more challenging problems, such as Problem 3.1 in Homework 8, the random variable that you are interested in might not be exactly of the form (10.9). In such cases, if we let W_n denote the random variable that you are interested in, it might be possible that you will need to find a function f for which $S_n = f(W_n)$ is of the correct form for an application of the central limit theorem.

Next, in order to apply Theorem 10.5, we need to know what are the expected value and variance of the summands X_i:

Step 2. Compute the parameters

$$\mu = \mathbf{E}[X_i] \qquad \text{and} \qquad \sigma^2 = \mathbf{Var}[X_i].$$

With this in hand, we are now in a position to apply Theorem 10.5, that is, the fact that

$$\mathbf{P}\left[\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \in I\right] \approx \int_I \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \, \mathrm{d}x \tag{10.10}$$

for large n, where I can be any interval. The point of Step 3 is to keep track of how the initial interval J in the probability (10.8) is transformed into another interval I once we apply the renormalization by $n\mu$ and $\sqrt{n\sigma^2}$ in (10.10):

Step 3. Reformulate the probability (10.8) into one of the form (10.10):

$$\begin{aligned}
\mathbf{P}[S_n \in J] &= \mathbf{P}[S_n - n\mu \in J - n\mu] \\
&= \mathbf{P}\left[\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \in \frac{J - n\mu}{\sqrt{n\sigma^2}}\right] \\
&\approx \int_{\frac{J-n\mu}{\sqrt{n\sigma^2}}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \, \mathrm{d}x.
\end{aligned}$$

Here, the set

$$\frac{J - n\mu}{\sqrt{n\sigma^2}}$$

consists of all numbers of the form $\frac{y-n\mu}{\sqrt{n\sigma^2}}$, where y is an element of J. For instance, if $J = [a,b]$ for some finite numbers $a < b$, then we obtain the new interval

$$\frac{J-n\mu}{\sqrt{n\sigma^2}} = \left[\frac{a-n\mu}{\sqrt{n\sigma^2}}, \frac{b-n\mu}{\sqrt{n\sigma^2}}\right].$$

To give another example, if $J = (-\infty, a)$ for some number $a \in \mathbb{R}$, then

$$\frac{J-n\mu}{\sqrt{n\sigma^2}} = \left(-\infty, \frac{a-n\mu}{\sqrt{n\sigma^2}}\right).$$

We will see some practical examples in the next subsection.

10.3.2. Two Examples. We now go over two examples that showcase how to apply the general blueprint above in practical situations.

Example 10.11. A University of Chicago student would like to start their own student newspaper. They decide to launch a fundraising campaign to cover the cost of hosting the newspaper's website. Through questionable means, the would-be editor gains access to the emails of 2 500 of their fellow University of Chicago students. They then proceed to send a SPAM email to each of these 2 500 students, begging for a donation.

Each recipient of the email independently does the following: With probability 0.65 they ignore the email; with probability 0.2 they give \$5; with probability 0.1 they give \$10; and with probability 0.05 they give \$20. The would-be editor aims to raise at least \$8 000 (enough to host the website for a while, as well as a generous stipend for their hard work as chief editor). How can we estimate the probability that the would-be editor gets at least \$8 000 through their fundraising effort using the central limit theorem?

We apply the three steps outlined in the blueprint above. If we let $S_{2\,500}$ denote the amount of money raised from the 2 500 emails, then we are interested in the probability

$$\mathbf{P}[S_{2\,500} \geq 8\,000].$$

Step 1 is to write $S_{2\,500}$ as a sum of i.i.d. random variables. Given the problem statement, we can write

$$S_{2\,500} = X_1 + X_2 + \cdots + X_{2\,500},$$

where X_i is the amount of money donated by the i^{th} student who got the email. We know that each of these random variables have range $R_{X_i} = \{0, 5, 10, 20\}$, and have the distribution

$$\mathbf{P}[X=0] = 0.65, \quad \mathbf{P}[X=5] = 0.2, \quad \mathbf{P}[X=10] = 0.1, \quad \mathbf{P}[X=20] = 0.05.$$

With this information in hand, we can now carry out Step 2, which is to compute the expected value and variance of the X_i: For the expected value, we have that

$$\mu = \mathbf{E}[X_i] = 0 \cdot 0.65 + 5 \cdot 0.2 + 10 \cdot 0.1 + 20 \cdot 0.05 = 3.$$

For the variance, we first compute

$$\mathbf{E}[X_i^2] = 0^2 \cdot 0.65 + 5^2 \cdot 0.2 + 10^2 \cdot 0.1 + 20^2 \cdot 0.05 = 35,$$

from which we then obtain that

$$\sigma^2 = \mathbf{Var}[X_i] = \mathbf{E}[X_i^2] - \mathbf{E}[X_i]^2 = 35 - 9 = 26.$$

We now wrap things up with step 3. In this case, we have that $n = 2\,500$, hence

$$n\mu = 2\,500\mu = 7\,500$$

and

$$\sqrt{n\sigma^2} = \sqrt{2\,500\sigma^2} = 50\sqrt{26}.$$

Therefore, an approximation by the central limit theorem yields

$$\begin{aligned}
\mathbf{P}[S_{2\,500} \geq 8\,000] &= \mathbf{P}\left[\frac{S_{2\,500} - 2\,500\mu}{\sqrt{2\,500\sigma^2}} \geq \frac{8\,000 - 7\,500}{50\sqrt{26}}\right] \\
&= \mathbf{P}\left[\frac{S_{2\,500} - 2\,500\mu}{\sqrt{2\,500\sigma^2}} \geq \frac{10}{\sqrt{26}}\right] \\
&\approx \mathbf{P}\left[N(0,1) \geq \frac{10}{\sqrt{26}}\right] \\
&= \int_{10/\sqrt{26}}^{\infty} \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x.
\end{aligned}$$

If we then use computational software to approximate this last integral, we obtain the approximation

$$\mathbf{P}[S_{2\,500} \geq 8\,000] \approx 0.0249301.$$

Example 10.12. A teaching assistant at the University of Chicago has 100 homework to grade. The amount of time that they must spend on each individual homework (in minutes) is uniform on the interval $[0, 2]$. We assume that the time spent on different homework submissions are independent. How can we estimate the probability that the total time spent grading is in between one and two hours using the central limit theorem?

If S_{100} denotes the amount of time spent grading the 100 homework (in minutes), we want to estimate

$$\mathbf{P}\big[S_{100} \in [60, 120]\big].$$

Firstly, we write

$$S_{100} = X_1 + X_2 + \cdots + X_{100},$$

where X_i is the amount of time spent grading the i^{th} homework. Secondly, since the X_i are uniform on $[0, 2]$, we have that

$$\mu = \frac{2-0}{2} = 1 \qquad \text{and} \qquad \sigma^2 = \mathbf{Var}[X_i] = \frac{(2-0)^2}{12} = \frac{1}{3}.$$

Thirdly,

$$100\mu = 100 \qquad \text{and} \qquad \sqrt{100\sigma^2} = \sqrt{\frac{100}{3}} = \frac{10}{\sqrt{3}}.$$

Therefore, by the central limit theorem, we can approximate

$$\begin{aligned}
\mathbf{P}\big[S_{100} \in [60, 120]\big] &= \mathbf{P}\big[S_{100} - 100 \in [-40, 20]\big] \\
&= \mathbf{P}\left[\frac{S_{100} - 100}{10/\sqrt{3}} \in [-4\sqrt{3}, 2\sqrt{3}]\right] \\
&\approx \int_{-4\sqrt{3}}^{2\sqrt{3}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)\,\mathrm{d}x.
\end{aligned}$$

If we then use a scientific calculator to approximate this integral, we get

$$\mathbf{P}\big[S_{100} \in [60, 120]\big] \approx 0.999734.$$

10.4. A Sketch of the Proof

Now that we have a good understanding of what the central limit theorem says and how it can be used in practice, a fundamental question remains: Why is the central limit theorem true? In order to answer this question, we now go over a sketch of the proof of the central limit theorem.

Remark 10.13. I call the argument that is presented in this section a sketch, because some mathematical details are omitted. This is primarily because the details in question amount to mathematical technicalities, and thus are not essential to get a good high-level understanding of how and why the proof works. Secondarily, there is also the issue of time, in that an adequate coverage of the full details would require more time than we have left in the quarter.

Remark 10.14. The remainder of this chapter will not be tested on the final exam. Nevertheless, if you intend to study probability and statistics more deeply in the future, then I strongly encourage you to try your best to internalize what is being done here.

10.4.1. A First Simplification.

Much of what makes the central limit theorem useful in practice is its generality. That is, the expected value and variance of the variables X_i making up the sums

$$S_n = X_1 + X_2 + \cdots + X_n$$

could be any finite real and positive numbers. For the purposes of proving the result, however, a high degree of generality can be a burden. Fortunately, it turns out that proving the most general statement of the central limit theorem can be reduced to a simpler case.

To see how this works, consider the random variable

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}},$$

which is what we want to prove converges to a standard Gaussian. By redistributing $-n\mu$ in each random variable in the sum S_n, we can rewrite this as

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} = \frac{(X_1 + X_2 + \cdots + X_n) - n\mu}{\sqrt{n\sigma^2}} = \frac{(X_1 - \mu) + (X_2 - \mu) + \cdots + (X_n - \mu)}{\sqrt{n\sigma^2}}$$

Next, by distributing $1/\sqrt{\sigma^2}$ to each summand $(X_i - \mu)$, we are left with

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} = \frac{1}{\sqrt{n}}\left(\frac{X_1 - \mu}{\sqrt{\sigma^2}} + \frac{X_2 - \mu}{\sqrt{\sigma^2}} + \cdots + \frac{X_n - \mu}{\sqrt{\sigma^2}}\right).$$

Thus, if we define the new random variables

$$\tilde{X}_i = \frac{X_i - \mu}{\sqrt{\sigma^2}},$$

then we have the simpler expression

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} = \frac{\tilde{X}_1 + \tilde{X}_2 + \cdots + \tilde{X}_n}{\sqrt{n}}.$$

Moreover, since the X_i are i.i.d., the same is true of the $\tilde{X}_i$, and we note by linearity of expectation that

$$\mathbf{E}[\tilde{X}_i] = \frac{\mathbf{E}[X_i] - \mu}{\sqrt{\sigma^2}} = 0$$

and

$$\mathbf{Var}[\tilde{X}_i] = \mathbf{E}[\tilde{X}_i^2] = \frac{\mathbf{E}[(X_i - \mu)^2]}{\sigma^2} = \frac{\sigma^2}{\sigma^2} = 1.$$

Summarizing the above paragraph, we see that if our objective is to prove the central limit theorem, then there is no loss of generality in assuming that the X_i have mean zero and variance one. Indeed, if that is not the case, then we can always change our random variables from X_i into $\tilde{X}_i$ as done above, thus making the random variables have mean zero and variance one. In conclusion, in order to prove the general central limit theorem stated in Theorem 10.5, we actually only need to prove the following simpler version:

Theorem 10.15 (Simpler Central Limit Theorem)**.** *Let $X_1, X_2, X_3, \ldots$ be an infinite sequence of i.i.d. random variables with $\mathbf{E}[X_i] = 0$ and $\mathbf{Var}[X_i] = 1$. For every positive integer n, let*

$$S_n = X_1 + X_2 + \cdots + X_n.$$

Then, for any interval $I \subset \mathbb{R}$, we have that

$$\lim_{n\to\infty} \mathbf{P}\left[\frac{S_n}{\sqrt{n}} \in I\right] = \int_I \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)\, \mathrm{d}x.$$

10.4.2. Two Steps. The proof of Theorem 10.15 has two steps, which can be summarized as follows:

Lemma 10.16 (Step 1. Universality)**.** *Let $X_1, X_2, X_3, \ldots$ and $Y_1, Y_2, Y_3, \ldots$ be two infinite sequences of i.i.d. random variables with expected value 0 and variance 1. For each n, define*

$$S_n = X_1 + X_2 + \cdots + X_n \qquad \textit{and} \qquad T_n = Y_1 + Y_2 + \cdots + Y_n.$$

For every interval $I \subset \mathbb{R}$, one has

$$\lim_{n\to\infty} \big(\mathbf{P}[S_n/\sqrt{n} \in I] - \mathbf{P}[T_n/\sqrt{n} \in I]\big) = 0.$$

Lemma 10.17 (Step 2. Sum of Gaussians)**.** *Let $X_1, X_2, X_3, \ldots$ be i.i.d. standard Gaussian random variables. Then, for every n,*

$$\frac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}} \qquad \textit{and} \qquad X_1$$

have the same distribution. In particular,

$$\frac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}} \sim N(0,1).$$

Lemmas 10.16 and 10.17 nicely encapsulate the two main conclusions of the central limit theorem.

On the one hand, Lemma 10.16 contains the universality statement of the central limit theorem. That is, even if the distribution of the X_i is very different from the distribution of the Y_i, in the large n limit this difference vanishes completely when you take a large sum and divide by $\sqrt{n}$. In other words, the distribution of $S_n/\sqrt{n}$ always converges to the same distribution.

On the other hand, once Lemma 10.16 is proved, the work is not yet done. Indeed, with that result we might know that $S_n/\sqrt{n}$ always converges to the same distribution, but we do not necessarily know what that distribution is. This is where Lemma 10.17 comes into play, with a very clever observation: If the X_i are already standard Gaussian, then we do not need to take $n \to \infty$ to observe that

$$\frac{S_n}{\sqrt{n}} = \frac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}}$$

becomes standard Gaussian; the variable is *already* standard Gaussian.

In summary, if we know that $S_n/\sqrt{n}$ always converges to the same limit, and we know that this limit is Gaussian in the special case where the X_i are Gaussian, then this concludes the proof of the central limit theorem. We now discuss how Lemmas 10.16 and 10.17 can be proved.

Remark 10.18. The property stated in Lemma 10.17 explains why we see the Gaussian distribution appear in the central limit theorem. Indeed, the Gaussian distribution is the only distribution with finite mean and finite variance that has the property that $\frac{X_1+X_2+\cdots+X_n}{\sqrt{n}}$ and X_1 have the same distribution if X_i are i.i.d. Therefore, it is not at all a coincidence that the Gaussian distribution appears in the central limit theorem. Instead, it is a consequence of a very special algebraic property of the Gaussian density function.

10.4.3. Proof of Step 1. Universality. As it turns out, proving Lemma 10.16 directly in the way that it is stated is too difficult. In part, this is because the computation of the probability $\mathbf{P}[X \in I]$ for a random variable X and an interval I can be very different depending on what type of random variable X is: If X is discrete, then we have

$$\mathbf{P}[X \in I] = \sum_{x \in R_X \text{ s.t. } x \in I} \mathbf{P}[X = x],$$

and if X is continuous with density f_X, then

$$\mathbf{P}[X \in I] = \int_I f_X(x)\, \mathrm{d}x.$$

Then, there is an additional complexity from the fact that the random variables we are considering (i.e., $S_n/\sqrt{n}$ and $T_n/\sqrt{n}$) change with n in a way that is difficult to keep track of.

As it turns out, there is a different way to characterize the distribution of random variables using expected values of the form $\mathbf{E}[g(X)]$ for a certain class of functions g. We have not discussed this idea until now because it is typically not very helpful in the kinds of applications of probability theory that we have seen up to this point. However, for the purpose of proving the central limit theorem, it is very useful:

Definition 10.19. A function $g : \mathbb{R} \to \mathbb{R}$ is infinitely differentiable if for every $n \in \mathbb{N}$, the n^{th} derivative $\frac{\mathrm{d}^n}{\mathrm{d}x^n} g(x)$ exists and is differentiable. We say that g has bounded derivatives if for every integer $n \geq 0$, there exists two finite numbers $a_n < b_n$ such that

$$a_n \leq \frac{\mathrm{d}^n}{\mathrm{d}x^n} g(x) \leq b_n$$

for every $x \in \mathbb{R}$.

Proposition 10.20. *Let X and Y be two random variables. If*

$$\mathbf{E}[g(X)] = \mathbf{E}[g(Y)]$$

for every function g that is infinitely differentiable with bounded derivatives, then X and Y have the same distribution.

Proof Sketch. This is the one element in our proof of the central limit theorem that we only sketch. In order to specify the distribution of a random variable X, either continuous or discrete, we need to be able to compute probabilities of the form $\mathbf{P}[X \in A]$ for arbitrary choices of set $A \subset \mathbb{R}$. In particular, if we know that

$$\mathbf{P}[X \in A] = \mathbf{P}[Y \in A]$$

for every set A, then this means that X and Y must have the same distribution. The key to understanding the present proposition is to note that if we know the value of $\mathbf{E}[g(X)]$ for every function g that is infinitely differentiable with bounded derivatives, then this allows us to determine (at least in principle) the value of $\mathbf{P}[X \in A]$ for any set A.

In order to see why that is the case, consider the simple example where the set $A = (0, 1)$ is the open unit interval. Then, define the function $h(x) = \mathbf{1}_{\{x \in (0,1)\}}$; this is a step function which is equal to one inside the interval $(0, 1)$, and zero outside of it. See the first illustration in Figure 10.11 on the next page for a plot of that function. We note that

$$\mathbf{E}[h(X)] = \mathbf{P}[X \in A].$$

Indeed, if X is discrete, then we have that

$$\mathbf{E}[h(X)] = \sum_{x \in R_X} h(x)\mathbf{P}[X = x] = \sum_{x \in R_X \text{ s.t. } x \in A} \mathbf{P}[X = x] = \mathbf{P}[X \in A],$$

and if X is continuous with density f_X, then

$$\mathbf{E}[h(X)] = \int_{\mathbb{R}} h(x) f_X(x) = \int_A f_X(x) = \mathbf{P}[X \in A].$$

At this point, one might be under the impression that we are done, as we have reformulated $\mathbf{P}[X \in A]$ in terms of the expectation $\mathbf{E}[h(X)]$. However, h is not infinitely differentiable; in fact, it is not even continuous! Nevertheless, we can always find a sequence of functions g_n that are infinitely differentiable with bounded derivatives such that $g_n \to h$, and thus

$$\lim_{n \to \infty} \mathbf{E}[g_n(X)] = \mathbf{E}[h(X)] = \mathbf{P}[X \in A].$$

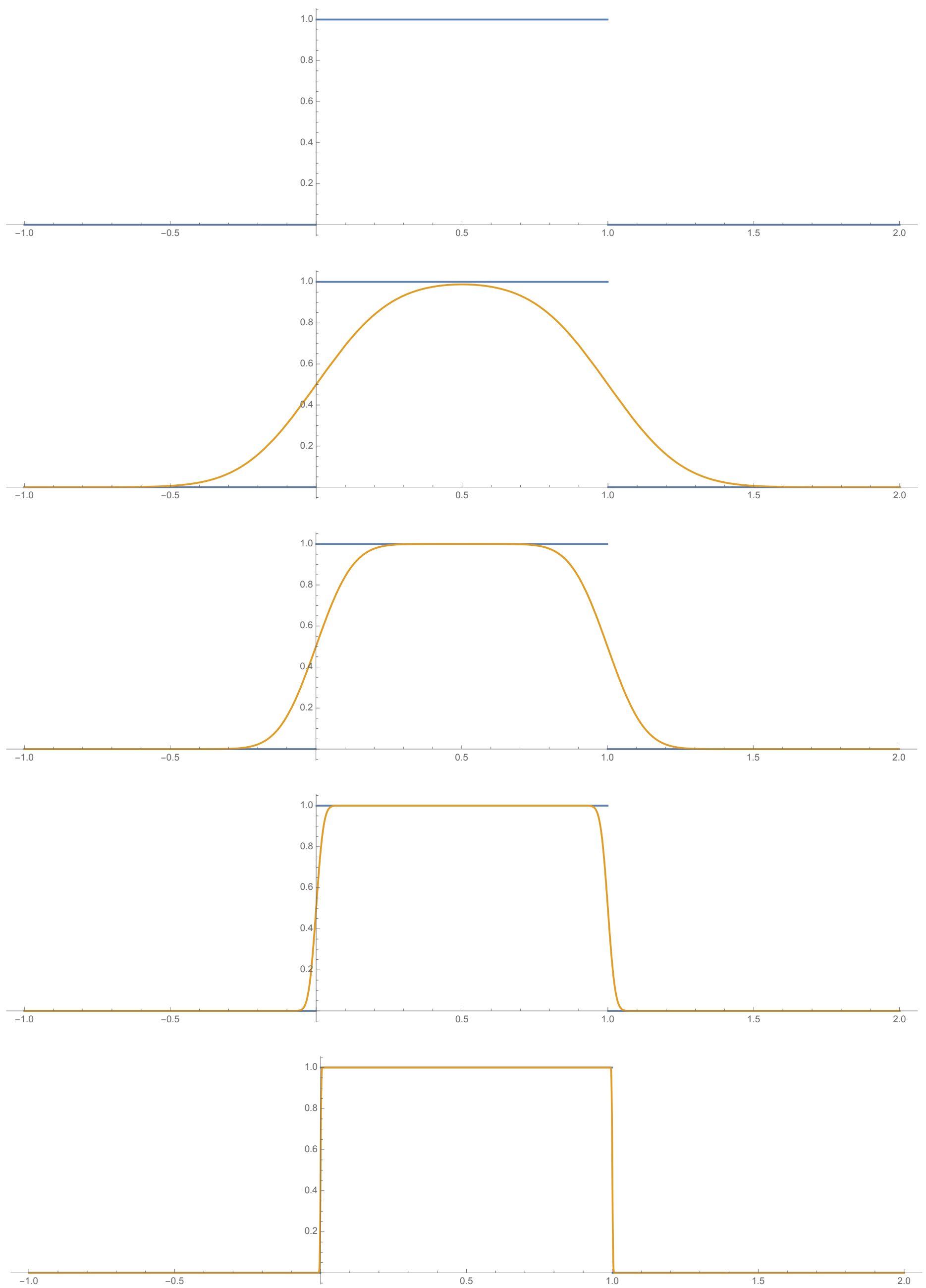

Figure 10.11. In blue is a plot of the function $h(x) = \mathbf{1}_{\{x \in (0,1)\}}$, that is, the indicator of the interval $(0,1)$. In particular, for any random variable X, one has $\mathbf{E}[h(X)] = \mathbf{P}[x \in (0,1)]$. In orange are increasingly accurate approximations of h with functions that are infinitely differentiable with bounded derivatives. Thus, if one knows the value of $\mathbf{E}[g(X)]$ for every g that is infinitely differentiable with bounded derivatives, then it stands to reason that we can recover from that the distribution of X by approximation. That is, the value of the probability $\mathbf{P}[X \in (0,1)]$ for any set $A \subset \mathbb{R}$ can be approximated by $\mathbf{E}[g_n(X)]$ with an appropriate sequence of functions g_n that are all infinitely differentiable with bounded derivatives.

This process is illustrated in Figure 10.11.

In conclusion, we therefore see that

$$\mathbf{P}[X \in A] = \lim_{n\to\infty} \mathbf{E}[g_n(X)] = \lim_{n\to\infty} \mathbf{E}[g_n(Y)] = \mathbf{P}[Y \in A],$$

where the equality in the middle comes from our assumption that $\mathbf{E}[g(X)] = \mathbf{E}[g(Y)]$ for every g that is infinitely differentiable with bounded derivatives. $\square$

With this result in hand, we can now reformulate the goal of Lemma 10.16 into a statement that is more amenable to analysis:

Corollary 10.21. *If we want to prove that*

$$\lim_{n\to\infty} \big(\mathbf{P}[S_n/\sqrt{n} \in I] - \mathbf{P}[T_n/\sqrt{n} \in I]\big) = 0$$

for every interval $I \subset \mathbb{R}$, then it is enough to prove that

$$\lim_{n\to\infty} \big(\mathbf{E}[g(S_n/\sqrt{n})] - \mathbf{P}[g(T_n/\sqrt{n})]\big) = 0$$

for every function g that is infinitely differentiable with bounded derivatives.

The reason why this corollary is useful for our purposes, and also why we insisted on the functions g that we consider to be infinitely differentiable, is the following well-known proposition from calculus:

Proposition 10.22 (Taylor's Theorem, Mean-Value Remainder)**.** *Let g be infinitely differentiable. Then,*

$$g(x) = g(0) + g'(0)x + \frac{g''(0)}{2}x^2 + \frac{g'''(\zeta_x)x^3}{6},$$

where $\zeta_x \in [0, x]$ is some number that depends on x.

The benefit of this proposition is that it provides us with a tool with which we can compare $\mathbf{E}[g(S_n/\sqrt{n})]$ and $\mathbf{E}[g(T_n/\sqrt{n})]$ in terms of quantities that we know. To see how this works, consider the following computations: If we plug Taylor's theorem in $g(S_n/\sqrt{n})$, then we get that

$$g(S_n/\sqrt{n}) = g(0) + \frac{g'(0)}{\sqrt{n}}S_n + \frac{g''(0)}{2n}S_n^2 + \frac{g'''(\zeta_{S_n/\sqrt{n}})}{6n^{3/2}}S_n^3$$

Because $g(0)$, $g'(0)$, and $g''(0)$ are nonrandom constants, linearity of expectation then tells us that

$$\mathbf{E}[g(S_n/\sqrt{n})] = g(0) + \frac{g'(0)}{\sqrt{n}}\mathbf{E}[S_n] + \frac{g''(0)}{2n}\mathbf{E}[S_n^2] + \frac{1}{6n^{3/2}}\mathbf{E}[g'''(\zeta_{S_n/\sqrt{n}})S_n^3].$$

If we apply a similar computation to T_n, we see that

$$\mathbf{E}[g(T_n/\sqrt{n})] = g(0) + \frac{g'(0)}{\sqrt{n}}\mathbf{E}[T_n] + \frac{g''(0)}{2n}\mathbf{E}[T_n^2] + \frac{1}{6n^{3/2}}\mathbf{E}[g'''(\zeta_{T_n/\sqrt{n}})T_n^3].$$

We now see the immense benefit of all the technicalities that we have deployed until now: We have a very convenient means of directly comparing the distributions of $S_n/\sqrt{n}$ and $T_n/\sqrt{n}$ via the two expectations above.

We now analyze the terms appearing in the above expectations. Firstly, we note that

$$\mathbf{E}[S_n] = \sum_{i=1}^n \mathbf{E}[X_i] = 0,$$

and similarly $\mathbf{E}[T_n] = 0$. Secondly, by independence of the X_i and the fact that the variance of a sum of independent random variables is the sum of the individual variances,

$$\mathbf{E}[S_n^2] = \mathbf{E}[S_n^2] - \mathbf{E}[S_n]^2 = \mathbf{Var}[S_n] = \sum_{i=1}^n \mathbf{Var}[X_i] = n;$$

similarly $\mathbf{E}[T_n^2] = n$. If we plug this into the expectations above, we get the simplifications

$$\mathbf{E}[g(S_n/\sqrt{n})] = g(0) + \frac{g''(0)}{2} + \frac{1}{6n^{3/2}}\mathbf{E}[g'''(\zeta_{S_n/\sqrt{n}})S_n^3].$$

and

$$\mathbf{E}[g(T_n/\sqrt{n})] = g(0) + \frac{g''(0)}{2} + \frac{1}{6n^{3/2}}\mathbf{E}[g'''(\zeta_{T_n/\sqrt{n}})T_n^3].$$

Thus, if we look at the difference between $\mathbf{E}[g(S_n/\sqrt{n})]$ and $\mathbf{E}[g(T_n/\sqrt{n})]$, then the only term that is not cancelled out is the one with the third derivative of g, whence

$$\big(\mathbf{E}[g(S_n/\sqrt{n})] - \mathbf{E}[g(T_n/\sqrt{n})]\big) = \frac{\mathbf{E}[g'''(\zeta_{S_n/\sqrt{n}})S_n^3] - \mathbf{E}[g'''(\zeta_{T_n/\sqrt{n}})T_n^3]}{6n^{3/2}}.$$

In particular, in order to establish Lemma 10.16, it suffices to prove that

$$\lim_{n\to\infty} \frac{1}{6n^{3/2}}\Big|\mathbf{E}[g'''(\zeta_{S_n/\sqrt{n}})S_n^3] - \mathbf{E}[g'''(\zeta_{T_n/\sqrt{n}})T_n^3]\Big| = 0. \tag{10.11}$$

In order to get a feeling for why this quantities vanish, we first note that

$$\mathbf{E}[S_n^3] = \mathbf{E}\left[\left(\sum_{i=1}^n X_i\right)^3\right] = \mathbf{E}\left[\sum_{i,j,k=1}^n |X_iX_jX_k|\right] = \sum_{i,j,k=1}^n \mathbf{E}[X_iX_jX_k].$$

In the above sum, if i, j, and k are all different, then by independence we have that

$$\mathbf{E}[X_iX_jX_k] = \mathbf{E}[X_i]\mathbf{E}[X_j]\mathbf{E}[X_k] = 0.$$

If two of the indices are equal, say $i = j$ and $k \neq i, j$, then once again by independence we have that

$$\mathbf{E}[X_iX_jX_k] = \mathbf{E}[X_i^2X_k] = \mathbf{E}[X_i^2]\mathbf{E}[X_k] = 0.$$

Consequently, the only terms that contribute to the sum are those such that all three indices are equal (i.e., $i = j = k$), which yields

$$\mathbf{E}[S_n^3] = \sum_{i=1}^n \mathbf{E}[X_i^3] = n\mathbf{E}[X_1^3].$$

In particular, this implies that

$$\lim_{n\to\infty} \frac{\mathbf{E}[S_n^3]}{n^{3/2}} = \lim_{n\to\infty} \frac{n\mathbf{E}[X_1^3]}{n^{3/2}} = \mathbf{E}[X_1^3]\lim_{n\to\infty}\frac{1}{n^{1/2}} = 0.$$

A similar result holds with T_n, with essentially the same calculation. At this point, in order to conclude the limit, we note that since g has bounded derivatives, we know that there exist constants $a < b$ such that

$$a \leq g'''(x) \leq b \qquad \text{for all } x \in \mathbb{R}.$$

This suggests (though requires a bit more effort establish rigorously) that the values of $g'''(\zeta_{S_n/\sqrt{n}})$ and $g'''(\zeta_{T_n/\sqrt{n}})$ do not affect the expected values in (10.11) by a substantial amount. With this in hand, we have now completed a sketch of the proof of Lemma 10.16.

10.4.4. Proof of Step 2. Sum of Gaussians. Lemma 10.17 is a consequence of the following result:

Proposition 10.23. *Let $X \sim N(0,\sigma^2)$ and $Y \sim N(0,\tau^2)$ be independent. Then, $X+Y \sim N(0,\sigma^2+\tau^2)$.*

Indeed, once we know this, we can compute the distribution of a sum of i.i.d. standard Gaussians by induction: If $X_1, X_2, X_3, \ldots$ are i.i.d. $N(0,1)$ random variables, then

(1) we know by Proposition 10.23 that $X_1 + X_2 \sim N(0, 1+1) = N(0,2)$;

(2) we know by combining the previous step with Proposition 10.23 that

$$X_1 + X_2 + X_3 = (X_1 + X_2) + X_3 \sim N(0, 2+1) = N(0,3);$$

(3) $\cdots$

(4) we know by combining the $(n-1)^{\text{th}}$ step with Proposition 10.23 that

$$X_1 + \cdots + X_{n-1} + X_n = (X_1 + \cdots + X_{n-1}) + X_n \sim N(0, n-1+1) = N(0,n).$$

With this in hand, we then conclude the statement of Lemma 10.17 with

$$\frac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}} \sim \frac{N(0,n)}{\sqrt{n}} = N(0,1),$$

where the last equality follows from Proposition 10.9. We can therefore now conclude the proof of the central limit theorem by establishing Proposition 10.23:

Proof of Proposition 10.23. By the convolution identity,

$$f_{X+Y}(z) = \int_{\mathbb{R}} f_X(z-y) f_Y(y) \, \mathrm{d}y.$$

Knowing that X and Y are Gaussians with mean zero and respective variances σ^2 and τ^2, this becomes

$$f_{X+Y}(z) = \int_{\mathbb{R}} \frac{1}{2\pi\sigma\tau} \exp\left(-\frac{(z-y)^2}{2\sigma^2} - \frac{y^2}{2\tau^2}\right) \, \mathrm{d}y.$$

If we then expand the square $(z-y)^2 = z^2 - 2zy + y^2$ in the above, we get

$$f_{X+Y}(z) = \int_{\mathbb{R}} \frac{1}{2\pi\sigma\tau} \exp\left(-\frac{z^2 - 2zy + y^2}{2\sigma^2} - \frac{y^2}{2\tau^2}\right) \, \mathrm{d}y.$$

At this point, in order to guide further calculations, it can help to keep in mind our ultimate objective. That is, we aim to show that the above integral expression simplifies to the density of $N(0,\sigma^2+\tau^2)$, which is equal to

$$\frac{1}{\sqrt{2\pi(\sigma^2+\tau^2)}}\exp\left(-\frac{z^2}{2(\sigma^2+\tau^2)}\right).$$

If we multiply and divide the convolution integral that we have for f_{X+Y} by this density, we obtain that

$$\begin{aligned}f_{X+Y}(z) = {}&\frac{1}{\sqrt{2\pi(\sigma^2+\tau^2)}}\exp\left(-\frac{z^2}{2(\sigma^2+\tau^2)}\right)\\ &\cdot\int_{\mathbb{R}}\frac{\sqrt{2\pi(\sigma^2+\tau^2)}}{2\pi\sigma\tau}\exp\left(\frac{z^2}{2(\sigma^2+\tau^2)}-\frac{z^2-2zy+y^2}{2\sigma^2}-\frac{y^2}{2\tau^2}\right)\,\mathrm{d}y.\end{aligned}$$

Thus, we need to prove that

$$\int_{\mathbb{R}}\frac{\sqrt{2\pi(\sigma^2+\tau^2)}}{2\pi\sigma\tau}\exp\left(\frac{z^2}{2(\sigma^2+\tau^2)}-\frac{z^2-2zy+y^2}{2\sigma^2}-\frac{y^2}{2\tau^2}\right)\,\mathrm{d}y=1. \tag{10.12}$$

Though this is not immediately obvious at first glance, the reason why the above integral is equal to one is that it is in fact a Gaussian. Using this hint as a guiding principle, we can look at the constant in front of the exponential, which is

$$\frac{\sqrt{2\pi(\sigma^2+\tau^2)}}{2\pi\sigma\tau}=\frac{\sqrt{(\sigma^2+\tau^2)}}{\sqrt{2\pi}\sigma\tau}=\frac{1}{\sqrt{2\pi\frac{\sigma^2\tau^2}{\sigma^2+\tau^2}}}.$$

In particular, if the function inside the integral is in fact the density of a Gaussian, it must be a Gaussian with variance $\frac{\sigma^2\tau^2}{\sigma^2+\tau^2}$. Going forward, to simplify computations, let us denote $\upsilon^2=\frac{\sigma^2\tau^2}{\sigma^2+\tau^2}$. Thus, we want to show that

$$\int_{\mathbb{R}}\frac{1}{\sqrt{2\pi\upsilon^2}}\exp\left(\frac{z^2}{2(\sigma^2+\tau^2)}-\frac{z^2-2zy+y^2}{2\sigma^2}-\frac{y^2}{2\tau^2}\right)\,\mathrm{d}y=1.$$

At this point, we must rewrite the expression inside the exponential into something that we can recognize as a Gaussian density. Since this gaussian density must have variance $\upsilon^2=\frac{\sigma^2\tau^2}{\sigma^2+\tau^2}$, we expect that it should be possible to write

$$\frac{z^2}{2(\sigma^2+\tau^2)}-\frac{z^2-2zy+y^2}{2\sigma^2}-\frac{y^2}{2\tau^2}=-\frac{(y-\mu)^2}{2\upsilon^2}$$

for some number μ. For this purpose, a good first step would be to put the terms inside the exponential on the denominator $2\upsilon^2$. To this end, we note that

$$(\sigma^2+\tau^2)=\frac{(\sigma^2+\tau^2)^2}{\sigma^2\tau^2}\upsilon^2,\qquad \sigma^2=\frac{\sigma^2+\tau^2}{\tau^2}\upsilon^2,\qquad\text{and}\qquad \tau^2=\frac{\sigma^2+\tau^2}{\sigma^2}\upsilon^2.$$

Thus,

$$\begin{aligned}&\frac{z^2}{2(\sigma^2+\tau^2)}-\frac{z^2-2zy+y^2}{2\sigma^2}-\frac{y^2}{2\tau^2}\\ &\qquad=\frac{1}{2\upsilon^2}\left(\frac{\sigma^2\tau^2}{(\sigma^2+\tau^2)^2}z^2-\frac{\tau^2}{\sigma^2+\tau^2}(z^2-2zy+y^2)-\frac{\sigma^2}{\sigma^2+\tau^2}y^2\right).\end{aligned}$$

If we then expand the terms in the above and simplify to group together the coefficients in front of z^2, zy, and y^2, then we get

$$\frac{1}{2\upsilon^2}\left(-\frac{\tau^4}{(\sigma^2+\tau^2)^2}z^2+2\frac{\tau^2}{\sigma^2+\tau^2}zy-y^2\right)=-\frac{1}{2\upsilon^2}\left(y-\frac{\tau^2}{\sigma^2+\tau^2}z\right)^2.$$

Summarizing the argument, we see that the integral on the left-hand side of (10.12) can be rewritten as

$$\int_{\mathbb{R}}\frac{1}{\sqrt{2\pi\upsilon^2}}\exp\left(-\frac{1}{2\upsilon^2}\left(y-\frac{\tau^2}{\sigma^2+\tau^2}z\right)^2\right)\,\mathrm{d}y,$$

which we know integrates to one because it is the density function of a Gaussian with mean $\mu=\frac{\tau^2}{\sigma^2+\tau^2}z$ and variance $\upsilon^2=\frac{\sigma^2\tau^2}{\sigma^2+\tau^2}$. With this observation, the proof is complete. □

10.5. Accuracy of the Central Limit Theorem (Bonus)

Much like the law of large numbers, the central limit theorem cannot be of much use if we do not know how much error we are incurring by replacing a probability involving S_n with a Gaussian. That is, although we know that

$$\lim_{n\to\infty}\mathbf{P}\left[\frac{S_n-\mu n}{\sqrt{n\sigma^2}}\in I\right]=\int_I\frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x,$$

in practical implementations of the central limit theorem, n is a fixed finite number; we cannot just take it to ∞. Thus, the best that we can do is say that

$$\mathbf{P}\left[\frac{S_n-\mu n}{\sqrt{n\sigma^2}}\in I\right]\approx\int_I\frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x, \tag{10.13}$$

and hope that the error in this approximation is negligible if n is "large enough."

Results that quantify the error in the approximation (10.13) are collectively known as Berry–Esseen theorems.[1] Such results typically take the following form: There exists a constant $\mathrm{BE}_{X_i,I}>0$, which depends on the distribution of the X_i's and the interval I, such that

$$\left|\mathbf{P}\left[\frac{S_n-\mu n}{\sqrt{n\sigma^2}}\in I\right]-\int_I\frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x\right|\leq\frac{\mathrm{BE}_{X_i,I}}{\sqrt{n}}.$$

As expected, the error incurred becomes smaller as n increases. However, in order for the above estimate to be useful, we also need to know something about the constant $\mathrm{BE}_{X_i,I}$. Much of the work involved in the theory of Berry-Esseen theorems is to find the best possible upper bounds on the constant $\mathrm{BE}_{X_i,I}$ under various assumptions. The best known upper bounds on the constant are not simple to state and contain various sub-cases; for your interest, you can consult the Berry-Esseen Wikipedia page to get a first impression of what this looks like.

[1] These results are named after Andrew C. Berry and Carl-Gustav Esseen, who independently derived the first quantitative estimate of the kind; see the Wikipedia page of the same name for more details, both mathematical and historical.

As a final note, apart from the error incurred by the application of the central limit theorem itself, there is often an error incurred in the calculation of the integral

$$\int_I \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x.$$

In a few very exceptional cases, this integral can be computed exactly; for instance, it is not difficult to see that

$$\int_{-\infty}^0 \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x = \int_0^\infty \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x = \frac{1}{2}.$$

However, for more general intervals I, there is no simple formula for the integral. This is partly because the function $\frac{\mathrm{e}^{-x^2}}{\sqrt{2\pi}}$ does not admit a simple explicit antiderivative/primitive. Thus, the Gaussian integral must often be approximated.

In sharp contrast to the approximation in the central limit theorem, however, Gaussian integrals are much better understood, and for any interval I,

$$\int_I \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x$$

can be approximated to any arbitrary degree of precision desired. Thus, provided one is prepared to expend sufficient computational resources, the error incurred here can always be made to be negligible. To give a concrete example of how this can work in practice, it turns out that the standard Gaussian CDF

$$\Phi(t) = \int_{-\infty}^t \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x$$

has a known Taylor expansion:

$$\Phi(t) = \frac{1}{2} + \frac{1}{\sqrt{\pi}}\sum_{k=0}^{\infty} \frac{(-1)^k 2^{-1/2-k} t^{1+2k}}{(1+2k)k!}.$$

(For more details, you can consult the Wikipedia page on the so-called error function as a starting point.) Thus, so long as we are prepared to include enough terms in a finite-sum approximation of Φ, arbitrary precision is accessible.

Remark 10.24. Using the above Taylor series also requires the ability to approximate π. Thankfully, this is something that is very well-known. Many such techniques also use series expansions; see the Wikipedia page on approximations of π for more details.

10.6. Alternate Proofs of the Central Limit Theorem (Bonus)

Due to its importance, the central limit theorem is one of the most well-studied results in all of mathematical probability. As a consequence, a multitude of proofs have been discovered for it. The proof sketched earlier in this chapter is my personal favourite, because I find that it most clearly illustrates the two mechanisms that are at the heart of the central limit theorem: Namely, universality of sums of i.i.d. random variables (Lemma 10.16), and sums of independent Gaussians being Gaussian (Lemma 10.17). However, the other proofs of the central limit theorem are also interesting in their own right.

The most prominent such proof, which you will no doubt study in depth if you take more advanced courses in probability in the future, uses what are called moments and generating functions:

Definition 10.25. Let X be a random variable. The moments of X consist of the expected values

$$\mathbf{E}[X^p],$$

where $p \geq 0$ is a nonnegative integer. The moment generating function of X is defined as the function

$$\mathcal{L}_X(t) = \mathbf{E}[\mathrm{e}^{tX}] = \sum_{p=0}^{\infty} \frac{t^p \mathbf{E}[X^p]}{p!}, \qquad t \in \mathbb{R}. \tag{10.14}$$

The characteristic function of X is defined as the function

$$\varphi_X(t) = \mathbf{E}[\mathrm{e}^{\mathrm{i}tX}] = \sum_{p=0}^{\infty} \frac{(\mathrm{i}t)^p \mathbf{E}[X^p]}{p!}, \qquad t \in \mathbb{R}, \tag{10.15}$$

where i denotes the imaginary unit (i.e., $\mathrm{i}^2 = -1$).

The interest of moments, and thus by extension the moment generating functions and characteristic functions, is that in many cases they allow to characterize the distributions of random variables. Consequently, the usefulness of these objects from the point of view of proving the central limit theorem is similar to the usefulness of introducing expectations of the form

$$\mathbf{E}[g(X)]$$

for infinitely differentiable functions g with bounded derivatives, as we did earlier in the chapter. In particular, we have the following result:

Theorem 10.26. *If $Y_1, Y_2, Y_3, \ldots$ is a sequence of random variables such that*

$$\lim_{n\to\infty} \mathbf{E}[Y_n^p] = \mathbf{E}[N(0,1)^p]$$

for every integer $p \geq 0$, then it is also the case that

$$\lim_{n\to\infty} \mathbf{P}[Y_n \in I] = \int_I \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}} \, \mathrm{d}x$$

for every interval $I \subset \mathbb{R}$.

Remark 10.27. Intuitively, the fact that the moments can uniquely characterize the distribution of certain random variables can be explained by the fact that moments contain a lot of useful information about distributions. For example:

(1) The first moment $\mathbf{E}[X]$ tells us something about the typical or average value of a random variable. Geometrically this should more or less correspond to the "center point" of the distribution (though, the expected value should not be confused with the median, which is the actual center point of the distribution).

(2) By combining the first and second moments, $\mathbf{E}[X]$ and $\mathbf{E}[X^2]$, we obtain the variance of X. This tells us how "spread out" the distribution of X is about its "center point."

(3) Let μ denote X's expectation and σ^2 its variance. If we combine these two elements with the third moment $\mathbf{E}[X^3]$, then we obtain the skewness of X, which is defined as

$$\tilde{\mu}_3 = \mathbf{E}\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] = \frac{\mathbf{E}[X^3] - 3\mu\sigma^2 - \mu^3}{\sigma^3}.$$

This quantity measures the extent to which X's distribution is symmetric about its average, and the sign of the skewness (i.e., positive or negative, assuming it is nonzero) indicates the direction of this asymmetry. See, for example, the Wikipedia page on skewness for more details.

(4) Etc...

In short, the more moments of X we know, the more accurately we can picture what its distribution looks like. Thus, it stands to reason that if we have all the moments of X, then, in some cases, we can uniquely recover X's distribution.

From the point of view of proving the central limit theorem, Theorem 10.26 means that, in principle, all we need to do is the following:

(1) Compute the moments of the standard Gaussian distribution

$$\mathbf{E}[N(0,1)^p] = \int_{\mathbb{R}} x^p \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x.$$

(2) Prove that if $X_1, X_2, X_3, \ldots$ are i.i.d. random variables with mean 0 and variance 1, then

$$\lim_{n\to\infty} \mathbf{E}\left[\left(\frac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}}\right)^p\right] = \mathbf{E}[N(0,1)^p]. \tag{10.16}$$

While proving the central limit theorem using (10.16) is possible, it is arguably not the easiest way to proceed from the technical point of view. This is where the moment generating function and characteristic function come into play:

Theorem 10.28. *If $Y_1, Y_2, Y_3, \ldots$ is a sequence of random variables such that either*

$$\lim_{n\to\infty} \mathcal{L}_{Y_n}(t) = \mathcal{L}_{N(0,1)}(t)$$

for every $t \in \mathbb{R}$ or

$$\lim_{n\to\infty} \varphi_{Y_n}(t) = \varphi_{N(0,1)}(t)$$

for every $t \in \mathbb{R}$, then it is also the case that

$$\lim_{n\to\infty} \mathbf{P}[Y_n \in I] = \int_I \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}\,\mathrm{d}x$$

for every interval $I \subset \mathbb{R}$.

Intuitively, Theorem 10.28 can be justified by noting that it is essentially equivalent to Theorem 10.26. Indeed, if we know all of the moments of X, then we can recover $\mathcal{L}_X$ and φ_X through the power series in (10.14) and (10.15), and vice versa. With this in hand, the proof of the central limit theorem is now reduced to the following two steps:

(1) Compute either

$$\mathcal{L}_{N(0,1)}(t) = \int_{\mathbb{R}} \mathrm{e}^{tx} \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}} \, \mathrm{d}x \qquad \text{or} \qquad \varphi_{N(0,1)}(t) = \int_{\mathbb{R}} \mathrm{e}^{\mathrm{i}tx} \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}} \, \mathrm{d}x$$

for all $t \in \mathbb{R}$.

(2) Prove that if $X_1, X_2, X_3, \ldots$ are i.i.d. random variables with mean 0 and variance 1, then either

$$\lim_{n \to \infty} \mathcal{L}_{\frac{X_1+X_2+\cdots+X_n}{\sqrt{n}}}(t) = \mathcal{L}_{N(0,1)}(t) \tag{10.17}$$

or

$$\lim_{n \to \infty} \varphi_{\frac{X_1+X_2+\cdots+X_n}{\sqrt{n}}}(t) = \varphi_{N(0,1)}(t) \tag{10.18}$$

for all $t \in \mathbb{R}$.

The advantage of proceeding this way is that (10.17) and (10.18) turn out to be much easier to prove than (10.16). This is not just a coincidence; in a way, the moment generating function and characteristic function are specifically tailored to make the proof of the central limit theorem easier. This is because of the very useful property of exponential functions that

$$\mathrm{e}^{x+y} = \mathrm{e}^x \mathrm{e}^y.$$

If we apply this to (10.17), for example, then we note the following: First, write

$$\begin{aligned}\mathcal{L}_{\frac{X_1+X_2+\cdots+X_n}{\sqrt{n}}}(t) &= \mathbf{E}\left[\exp\left(t \frac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}}\right)\right] \\ &= \mathbf{E}\left[\mathrm{e}^{tX_1/\sqrt{n}} \mathrm{e}^{tX_2/\sqrt{n}} \cdots \mathrm{e}^{tX_n/\sqrt{n}}\right].\end{aligned}$$

Next, if we use the fact that the X_i are i.i.d., then this simplifies to

$$\mathcal{L}_{\frac{X_1+X_2+\cdots+X_n}{\sqrt{n}}}(t) = \mathbf{E}\left[\mathrm{e}^{tX_1/\sqrt{n}}\right] \mathbf{E}\left[\mathrm{e}^{tX_2/\sqrt{n}}\right] \cdots \mathbf{E}\left[\mathrm{e}^{tX_n/\sqrt{n}}\right] = \mathbf{E}\left[\mathrm{e}^{tX_1/\sqrt{n}}\right]^n.$$

Finally, using the series expansion in (10.14), and assuming that $\mathbf{E}[X_1] = 0$ and $\mathbf{E}[X_1^2] = 1$, we get that

$$\mathbf{E}\left[\mathrm{e}^{tX_1/\sqrt{n}}\right] = 1 + \frac{t \cdot 0}{\sqrt{n}} + \frac{t^2 \cdot 1}{2n} + \cdots = 1 + \frac{t^2}{2n} + \cdots.$$

Therefore, as $n \to \infty$, we have that

$$\mathcal{L}_{\frac{X_1+X_2+\cdots+X_n}{\sqrt{n}}}(t) = \left(1 + \frac{t^2}{2n} + \cdots\right)^n \approx \left(1 + \frac{t^2}{2n}\right)^n \to \mathrm{e}^{t^2/2}.$$

At this point, in order to prove the central limit theorem, it only remains to establish the following, which is not terribly difficult, but still requires some amount of careful contemplation:

Proposition 10.29. *For every $t \in \mathbb{R}$,*

$$\mathcal{L}_{N(0,1)}(t) = \mathbf{E}[\mathrm{e}^{tN(0,1)}] = \mathrm{e}^{t^2/2}.$$

The proof sketch of (10.18) is very similar, but has slightly different details.